Paper 3 – Subspecialities of Urology: All Answers by Section

Compiled from: 2015–2025 Paper 3 (12 papers) and Practice Sets 1–8 Paper 3, plus the earlier version of Practice Set 6 Paper 3 Q3c — 161 answers.

Repeats merged: questions asked more than once are printed once, with every wording and source listed under "Asked as"; their answers are combined so that no fact from any version is lost. 36 merged topics replace 85 original answers; 112 topics in all.

Layout: the 17 revision-book sections; in each, long answer questions (LAQ) first, then short answer questions (SAQ), then a Rapid Revision Sheet of that section's key points.

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1. Therapeutic energy sources in urology (including laparoscopic energy devices) *(merged: 2 questions)*

Section 1 · Basic Principles — Short answer questions (SAQ)

1. Asymptomatic haematuria and urinary biomarkers *(merged: 2 questions)*
2. Collin's knife.
3. Baceriuria and use of antibiotic policy in pregnancy
4. Hemostasis in laparoscopic surgery.
5. APOGEE
6. Describe principles of tissue engineering and what its status in managemet of urological diseases is.

Section 2 · Oncology, DSD, Gender & Ageing — Long answer questions (LAQ)

1. DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia *(merged: 2 questions)*

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

1. Prepucioplasty
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Section 3 · Infections & Inflammation — Long answer questions (LAQ)

1. Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management *(merged: 3 questions)*
2. Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria.

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

1. Prognostic indicators in emphysematous pyelonephritis.

Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

1. Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome *(merged: 4 questions)*
2. Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) *(merged: 2 questions)*
3. Embryology and management of ectopic ureter.
4. Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?

Section 4 · Paediatric Urology 1 — Short answer questions (SAQ)

1. Bilateral Wilms tumour: treatment strategies *(merged: 2 questions)*

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Section 5 · Paediatric Urology 2 — Long answer questions (LAQ)

1. Embryological development of urinary bladder and management of Hutch Diverticulum?
2. Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) *(merged: 3 questions)*
3. Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias) *(merged: 2 questions)*
4. Bladder and ureter development, exstrophy–epispadias complex and its management *(merged: 3 questions)*

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

1. HSG in neonates
2. Prune belly syndrome
3. Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty *(merged: 2 questions)*
4. DD of frequency and urgency in 3 yr male child and treatment
5. Embryology and Management of P.U. valve patient.
6. CSIC.
7. Enuresis
8. Discuss diagnostics and operative pediatric laparoscopy in urology

Section 6 · Female Urology — Long answer questions (LAQ)

1. Urological complications in OBS-GYN and uretero-vaginal fistula *(merged: 3 questions)*
2. Female pelvic floor, continence mechanism, classification and management of SUI *(merged: 3 questions)*
3. SUI: evaluation in females, role of UDS, and surgical therapy in males *(merged: 2 questions)*
4. Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management *(merged: 2 questions)*
5. Urological complications of OBGY surgery and urethrovaginal fistula *(merged: 2 questions)*
6. Vesicovaginal fistula (VVF): management options and early vs delayed repair *(merged: 2 questions)*

Section 6 · Female Urology — Short answer questions (SAQ)

1. Female sexual dysfunction
2. Duloxetin

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

1. Sacral neuromodulation *(merged: 3 questions)*
2. Urinary bladder compliance
3. Botulinum toxin in urology *(merged: 2 questions)*
4. OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron *(merged: 2 questions)*
5. Uroflowmetry

Section 8 · Nocturia & Male Incontinence — Short answer questions (SAQ)

1. Desmopressin (DDAVP) *(merged: 2 questions)*

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

1. Priapism: Diagnosis and Management *(merged: 2 questions)*

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

1. Types of penile prosthesis. What are biofilms in penile prosthesis?
2. Oligospermia and oligoasthenozoospermia *(merged: 2 questions)*
3. Sperm retrieval techniques
4. Genetic causes of male infertility. Enumerate and classify.
5. Premature ejaculation.
6. Spermatogenesis
7. Ejacultory duct obstruction
8. Tadalafil
9. Retrograde ejaculation
10. Immunobead test for semen
11. CBAVD
12. Sertoli cell only syndrome
13. Oxidative stress and male infertility
14. Dhat syndrome

Section 10 · Testis, Penis & Urethra — Long answer questions (LAQ)

1. Urethral stricture: classification, DVIU and surgery after failed VIU *(merged: 2 questions)*
2. Classify testicular tumors and describe management of seminoma

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

1. Uses of buccal mucosa in urology
2. Varicocele: controversies and present status of treatment *(merged: 2 questions)*
3. Inguinal Block Dissection?
4. Minimal invasive retroperitoneal lymph node dissection
5. Eponymous urethroplasty techniques and perineal urethrostomy
6. Prognostic index score in carcinoma penis

Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.
2. Laparoscopic pyeloplasty.
3. Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis.
4. Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury *(merged: 3 questions)*
5. Describe early and late management of PFUD injuries

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

1. Accessories for RIRS
2. Management of lower ureteric stone of size less than 1 cm in pregnancy?
3. Type 2 Renal tubular acidosis?
4. Renal Autotransplantation *(merged: 3 questions)*
5. Indication, procedure and complications of Uretero-Calycostomy?
6. Post traumatic bladder neck injury: acute and delayed management?
7. Mini Perc.
8. Boari flap
9. Newer methods (techniques) of stone analysis *(merged: 3 questions)*

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

1. Immediate post TURBT intravesical Therapy for TCC
2. Management of Haemorrhagic cystitis.
3. Methods of preoperative bowel preparation.
4. Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder.

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

1. Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management *(merged: 3 questions)*
2. Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy.
3. Nephron-sparing surgery for renal masses: methods, recent advances and neoadjuvant therapy *(merged: 2 questions)*

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

1. Management of AML.
2. Discuss the management of Metastatic Renal Cell Carcinoma.

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

1. Discuss the role of tissue typing and cross match in live and cadaveric
2. Vascular grafts and vascular surgery in urology *(merged: 2 questions)*

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

1. Marginal Donor
2. Tacrolimus.
3. Assessment criteria for kidney donor.
4. Urological Complications post Kidney Transplantation.

Section 16 · BPH — Short answer questions (SAQ)

1. TUR Syndrome.
2. IPSS score
3. Laparoscopic Trans vesical Prostatectomy
4. Lasers in BPH, including HoLEP *(merged: 2 questions)*
5. Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate

Section 17 · Prostate Cancer — Short answer questions (SAQ)

1. Management of newly Metastatic prostate cancer

Section 1 · Basic Principles — Long answer questions (LAQ)

Therapeutic energy sources in urology (including laparoscopic energy devices)

Asked as:

- enumerate various therapeutic energy applications in urology and briefly give their present status in various urological diseases. — *Practice Set 8 Paper III Q2 (LAQ)* (energy physics also answered in 2025-Winter Paper 4, Q2)
- Energy sources in lap uro. — *Practice Set 7 Paper III Q3a (SAQ)* (also answered in 2015-Summer Paper 2, Q3d)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction and principles

- Therapeutic energy is absorbed by tissue or stone and released as **heat, mechanical stress, cavitation or a chemical reaction**. Wavelength or frequency sets penetration; absorption sets the effect.
- All (haemostatic) devices achieve haemostasis by **denaturing protein**, by heat (current, light) or mechanical vibration.
- Heat effects: about **60 °C** coagulates protein; **100 °C** vaporises cell water; higher temperatures carbonise.

Classification

Group	Modalities
Electrical	Monopolar and bipolar electrosurgery, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration)
Electromagnetic	Lasers, MWA (microwave ablation), chemohyperthermia, PDT (photodynamic therapy), ionising radiation
Acoustic	SWL (shock wave lithotripsy), intracorporeal ultrasonic and pneumatic lithotripsy, HIFU (high-intensity focused ultrasound), histotripsy, burst wave lithotripsy, Li-ESWT (low-intensity shock wave therapy)
Direct contact	Cryoablation, water vapour (Rezum), waterjet (Aquablation)

1. Electrosurgery

- **Principle: Electrosurgery** uses high-frequency alternating current at about **500 kHz** that heats tissue (the resistance) without nerve stimulation (does not stimulate nerve or muscle). **Current density** decides heating (the effect): the small tip burns, the large pad does not. Cut = continuous low voltage; coagulation = interrupted high voltage.

Mode	Waveform	Effect
Cut (50–80 W)	Continuous, low voltage	Cell water flash-boils; clean cut
Coagulation (30–50 W)	Interrupted (about 6% duty cycle), high voltage	Desiccation, deeper spread
Fulguration (spray)	Higher voltage, sparks	Surface charring of oozing beds
Blend	Cut with interruptions	Cut plus coagulation

- **Status:**
  - **TURP (transurethral resection of prostate):** still the reference BPH (benign prostatic hyperplasia) operation. **Bipolar** in saline matches monopolar efficacy and avoids TUR syndrome (glycine hyponatraemia).
  - **TURBT (transurethral resection of bladder tumour):** standard; obturator reflex risk on lateral wall. En bloc resection is increasing.
  - Vessel sealers (up to 7 mm vessels) are routine in laparoscopic and robotic surgery.
- **EMDA–MMC (mitomycin C):** iontophoresis raises drug uptake in NMIBC (non-muscle-invasive bladder cancer); evidence low quality, not standard.

Laparoscopic energy devices

Device	Principle	Lateral spread	Notes
Monopolar (hook, scissors)	Tip → patient → return pad	**5–7 mm**	Versatile; stray-current risk
Bipolar	Current only between jaws	1–6 mm	No pad; no capacitive coupling
Vessel sealer (LigaSure)	Bipolar + compression + impedance feedback; built-in blade	1–3 mm	Seals vessels **≤7 mm**; one application
Ultrasonic (Harmonic)	Blade vibrating at **25–55 kHz**	**≈1 mm**	No current: safe with pacemakers; jaws **>200 °C**, slow to cool, so do not touch bowel
Argon beam	Current through argon, non-contact	Superficial	Liver, spleen, raw surfaces; gas embolism

- **Also know:** lasers (KTP [potassium titanyl phosphate], holmium) rarely used; laparoscopic RFA (radiofrequency ablation, 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles); robotic HotShears, bipolar Maryland, Vessel Sealer.

Electrosurgical injuries

Mechanism	Prevention
Direct activation out of view	Activate only with tip in view
Direct coupling to another metal instrument	Whole metal tip visible
Capacitive coupling (metal cannula in plastic retainer)	Never mix metal and plastic ports
Insulation failure	Inspect; **AEM (active electrode monitoring)**
Pad burns; surgical fire	Pad on muscle, away from implants; REM (return electrode monitoring)

- Bowel injuries present late (days) as peritonitis.

2. Lasers

Laser	nm	Penetration	Present status
Ho:YAG (holmium:yttrium-aluminium-garnet)	2120	0.31 mm	Gold standard lithotripsy; **HoLEP** (holmium laser enucleation of prostate) size-independent; strictures, UTUC (upper tract urothelial carcinoma) ablation
TFL (thulium fibre laser)	1940	0.08 mm	Stones: beats low-power holmium, equals high-power; less retropulsion; ThuLEP (thulium laser enucleation of prostate)

Laser	nm	Penetration	Present status
KTP/LBO (potassium-titanyl-phosphate/lithium triborate; GreenLight)	532	0.8 mm	Photovaporisation; can continue anticoagulation
Tm:YAG (thulium:YAG)	2000	0.2 mm	Vaporesection, enucleation
Nd:YAG (neodymium:YAG)	1064	10 mm	Deep coagulation (penile CIS [carcinoma in situ], warts); largely obsolete for BPH

- **Mechanism:** pulse duration decides: long pulses photothermal; short pulses photomechanical.

3. Acoustic energy

- **SWL:** shock waves break brittle stones by spallation, shear and cavitation. **Status:** first line for renal stones ≤2 cm and favourable upper ureteric stones; slower rate (about 1 Hz) works better.
- **Intracorporeal:** pneumatic, ultrasonic and combined probes in PCNL (percutaneous nephrolithotomy); electrohydraulic now rare (ureteric injury).
- **HIFU:** focused beam heats a rice-grain focus to necrosis. **Status:** focal or salvage prostate cancer (best for posterior tumours); investigational in guidelines.
- **Histotripsy, burst wave lithotripsy, ultrasonic propulsion:** investigational.
- **Li-ESWT:** for vasculogenic ED (erectile dysfunction) and Peyronie pain; evidence still limited.

4. Thermal ablation of tumours

Modality	Mechanism	Present status
Cryoablation	Argon expansion (Joule–Thomson); **<−40 °C, two freeze–thaw cycles**; visible ice ball	Small renal mass ≤3–4 cm in frail patients; focal or salvage prostate
RFA	375–900 kHz; target 60 °C; char at 150 °C	Small renal mass; oncological results similar to cryo; heat sink near vessels
MWA	915 MHz or 2.45 GHz; dielectric hysteresis	Renal mass; hotter, less heat sink
IRE	Microsecond DC (direct current) pulses open permanent membrane pores; non-thermal	Focal prostate; good continence and potency; trials ongoing

- **Guidelines:** AUA (American Urological Association) 2021: consider thermal ablation for **cT1a ≤3 cm**; biopsy before or at ablation. EAU (European Association of Urology): not routinely >3 cm (cryo >4 cm).

5. Light and chemical energy

- **PDT:** photosensitiser + light → singlet oxygen. **VTP (vascular-targeted photodynamic therapy, padeliporfin)** for low-risk prostate cancer: RCT (Azzouzi 2017) halved progression versus AS (active surveillance; 28% vs 58%); not widely adopted. **Photodynamic diagnosis** (blue light) aids TURBT.
- **Chemohyperthermia:** MMC with bladder wall heated to **42 °C** for NMIBC; used after BCG (bacillus Calmette-Guérin) failure or shortage; not standard.

6. Prostate (BPH) newer energies

- **Rezum:** steam (103 °C) releases latent heat by convection; no char, zone-confined; preserves ejaculation.
- **Aquablation:** robotic non-thermal waterjet; good for large glands; haemostasis needed.
- **Also know:** TUMT (transurethral microwave thermotherapy) and TUNA (transurethral needle ablation) largely abandoned.

7. Ionising radiation

- EBRT (external beam radiotherapy) and **brachytherapy** (LDR [low dose rate] or HDR [high dose rate]): standard for localised prostate cancer; trimodality bladder preservation; penile cancer (brachytherapy for small T1–T2 glans tumours). **Radioligand therapy** (^{177}Lu-PSMA [prostate-specific membrane antigen], VISION trial) and **radium-223** (ALSYMPCA) for mCRPC (metastatic castration-resistant prostate cancer).

Diagram to draw: monopolar vs bipolar circuits; laser penetration depths compared, with cryoprobe ice ball.

Key points to remember:

- Spread: monopolar 5–7 > bipolar 1–6 > sealer 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; ultrasonic blade hottest after use.
- Capacitive coupling: avoid hybrid trocars; AEM detects it.
- Holmium 2120 nm: gold standard for stones and HoLEP.
- Bipolar TURP in saline avoids TUR syndrome.
- Thermal ablation for cT1a ≤3 cm; cryo <−40 °C, two cycles.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

Section 1 · Basic Principles — Short answer questions (SAQ)

Asymptomatic haematuria and urinary biomarkers

Asked as:

- Asymptomatic Hematuria — *Summer 2016 Paper III Q3d (SAQ)*
- Urinary biomarkers in diagnostic evaluation of hematuria. — *Summer 2021 Paper III Q3f (SAQ)*

Type: SAQ (10 marks)

Introduction and definitions

- **MH (microhaematuria)**: **3 or more RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020). A dipstick alone is not enough (myoglobin, semen and peroxidases give false positives).
- **GH (gross haematuria)**: visible; exclude pigmenturia and menstrual blood.
- Prevalence about 6.5%. Malignancy: about 3% of MH; 14–26% of GH (DETECT I, IDENTIFY).
- Haematuria is evaluated by **cystoscopy and upper tract imaging**; urine tests are non-invasive **adjuncts** for bladder cancer.

Causes

- **Neoplasm**: bladder (over 85% of cancers found), UTUC (upper tract urothelial carcinoma), RCC (renal cell carcinoma) about 1 in 1000 each, prostate.
- **Benign**: BPE (benign prostatic enlargement), stones, UTI (urinary tract infection), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal** (IgA nephropathy): dysmorphic RBCs, casts, proteinuria.
- **Others**: exercise, endometriosis, sickle cell. Anticoagulants do not explain haematuria; evaluate as usual.

Initial assessment

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch), examination, blood pressure, creatinine, culture.
- UTI → treat → repeat urinalysis; persistence means full evaluation. Glomerular picture (dysmorphic RBCs (red blood cells), casts or proteinuria suggest a glomerular source) → nephrology **and** (plus) urological evaluation.

AUA/SUFU risk stratification

	Low (all)	Intermediate (any)	High (any)
Age	Women under 50, men under 40	Women 50–59, men 40–59	60 or over
Smoking (pack-years)	Under 10	10–30	Over 30
RBCs/HPF	3–10	11–25	Over 25
Gross haematuria history	No	No	**Yes**

	Low (all)	Intermediate (any)	High (any)
Cancer risk	0–0.4%	1–2.5%	About 6%
Work-up	Repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound)	Cystoscopy + RUS	Cystoscopy + **CT urography**

- CT (computed tomography) contraindicated → MR (magnetic resonance) urography, or retrograde pyelograms.
- Cytology and urine markers are **not** used initially; cytology if CIS (carcinoma in situ) is suspected.
- **Gross haematuria**: cystoscopy, CT urography and cytology at any age.

Urine cytology

- **Sensitivity 31–62%, specificity 94–100%**: good for **high-grade tumours and CIS (carcinoma in situ)**, poor for low grade.
- **Paris System:** negative for HGUC (high-grade urothelial carcinoma), atypical, suspicious, HGUC (by nuclear-to-cytoplasmic ratio and nuclear features).
- Positive cytology, normal cystoscopy → random and prostatic urethral biopsies, upper tract washings.

Urinary biomarkers (sensitivity/specificity, Chou 2015 meta-analysis)

Marker	Principle	Sens / spec	Note
NMP22 (nuclear matrix protein 22)	Nuclear protein; lab or point-of-care	0.69/0.77	False positives: inflammation, stones
BTA (bladder tumour antigen) stat / TRAK	Complement factor H–related protein; dipstick or ELISA (enzyme-linked immunosorbent assay)	0.64/0.77	**False positive with blood** (factor H in serum)
UroVysion FISH (fluorescence in situ hybridisation)	Gain of chromosomes 3, 7, 17; loss of 9p21	0.63/0.87	After atypical cytology
ImmunoCyt	Antibodies to CEA (carcinoembryonic antigen) + two mucins	0.78/0.78	Highest sensitivity; operator dependent
Cxbladder	RT-PCR (reverse transcription polymerase chain reaction), 5 mRNAs	about 0.82/0.85	Needs validation

- **Also know:** Xpert Bladder (mRNA); AssureMDx (FGFR3, TERT, HRAS mutations + methylation).
- **Pattern:** more sensitive, less specific than cytology; miss small low-grade tumours.

Role of markers in haematuria (guidelines)

- **AUA/SUFU 2020: no cytology or markers in the initial evaluation of microhaematuria**; cytology only after a negative work-up if CIS is suspected.
- **Gross haematuria:** cytology with cystoscopy and CT urography. A negative marker still leaves **≥10% cancer probability**, so cystoscopy is still needed.
- **No marker replaces cystoscopy** (AUA, EAU (European Association of Urology)); a negative test may one day help triage low-risk patients.
- **Recent advances:** urinary tumour DNA; methylation assays (Bladder EpiCheck).

Follow-up

- Negative evaluation → repeat urinalysis within 12 months; release if negative.

- Re-evaluate for GH, rising MH or new symptoms.

Diagram to draw: AUA/SUFU flowchart from MH to risk-based work-up (haematuria pathway), with markers as adjuncts.

> **Key points to remember:**
> - MH: 3 or more RBCs/HPF on microscopy, one sample.
> - Low: optional; intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.
> - Cytology: specific for high grade/CIS, insensitive for low grade.
> - BTA is falsely positive with blood.
> - No markers in initial microhaematuria work-up; none replaces cystoscopy.

Collin's knife.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 3, Q3b

*Taken to mean the **Collings knife**, the knife-shaped cutting electrode of the resectoscope (the reference chapters spell it both "Collins" and "Collings").*

Description

- A **hot (electrosurgical) knife electrode** fitted to the **resectoscope working element** in place of the loop.
- Thin **straight or angled (L- or hook-shaped) blade**; cuts a precise line with **cutting current** under irrigation and direct vision.
- **Monopolar** versions need **non-conducting irrigant** (glycine, sorbitol or mannitol); bipolar versions allow normal saline.
- Adult and **infant (9.5 Fr) resectoscope** sizes.
- **Principle:** it **incises rather than resects**; a deep, controlled cut releases a fibrous ring with little tissue loss.

Uses (from the chapters)

Use	Technique point
TUIP (transurethral incision of prostate)	Gland under 30 mL, no median lobe; incise at 5 and 7 o'clock from below the ureteric orifice to just proximal to the verumontanum, down to the capsule
Bladder neck contracture (after TURP (transurethral resection of prostate) or simple prostatectomy)	Incise until the scar springs open, to about 22 Fr; mitomycin C for recurrence
Ureterocele incision (older children)	Low transverse full-thickness incision near the bladder floor to reduce reflux
PUV (posterior urethral valve) ablation	Infant resectoscope only if urethral calibre is normal; incise at 5, 7 (± 12) o'clock; avoid hot-loop resection (stricture)
Bladder diverticulum neck	Incise the narrow neck so the diverticulum drains
PCNL (percutaneous nephrolithotomy) tract	Incise scarred fascia when rigid dilators fail

Alternatives

- **Cold knife (Sachse)** urethrotome: preferred for urethral stricture; hot knife unnecessary there.
- **Holmium or thulium laser**, Bugbee electrode, hook electrode, **Mohan's valvotome** (for valves).

Precautions and complications

- Stay away from the **ureteric orifices** and external sphincter; control depth (perforation, extravasation).
- TUR (transurethral resection) syndrome with monopolar irrigant if veins open.
- **Retrograde ejaculation** after bladder neck incision; stricture if the urethra is too small for the sheath.

Diagram to draw: resectoscope tip with L-shaped knife electrode, and the 5 and 7 o'clock TUIP incisions.

Key points to remember:

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

Baceriuria and use of antibiotic policy in pregnancy

Type: SAQ (10 marks)

Source: 2019-Summer Paper 3, Q3b

Read as: bacteriuria in pregnancy and the antibiotic policy for it.

Definition

- **ASB (asymptomatic bacteriuria):** $\geq 10^5$ CFU (colony-forming units)/mL of one organism without urinary symptoms.

Why pregnancy matters

- Prevalence is **not raised** (about 2–7%), but ASB **persists and ascends**.
- **Changes:** progesterone relaxes ureteric muscle and the uterus compresses the ureters at the pelvic brim (right more), causing stasis and hydroureteronephrosis. GFR (glomerular filtration rate) rises 30–50%, lowering drug levels.
- Untreated ASB led to **pyelonephritis** in 15–45% (historical), mostly in the third trimester, causing **preterm birth and low birth weight**.
- **Cochrane 2019:** treatment reduces pyelonephritis, preterm delivery and low birth weight.

Screening

- **Urine culture at 12–16 weeks** or first visit (USPSTF [US Preventive Services Task Force]); dipstick misses cases.

Antibiotic policy

Safe	Restricted or avoided
Penicillins (amoxicillin), cephalosporins: throughout	**Nitrofurantoin:** stop near term (from 35 weeks): neonatal haemolysis (G6PD [glucose-6-phosphate dehydrogenase] deficiency)
Fosfomycin: cystitis only	**TMP-SMX (trimethoprim-sulfamethoxazole):** avoid 1st trimester (folate, neural tube); caution near term (kernicterus)
Gentamicin: pyelonephritis	**Fluoroquinolones** (cartilage), **tetracyclines** (teeth, bone), chloramphenicol: avoid

- ACOG (American College of Obstetricians and Gynecologists) 2017: nitrofurantoin and sulfonamides first-line in 2nd–3rd trimester; 1st only if no alternative.
- **ASB or cystitis:** culture-directed, **4–7 days** (cures more than single dose); **repeat culture** after treatment.

Pyelonephritis

- **Admit**; IV (intravenous) ampicillin plus gentamicin, ceftriaxone or aztreonam, then oral cephalexin; 10–14 days. No improvement by 72 h: renal ultrasound.

Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery (Cochrane 2015: close surveillance equally effective).
- **Also know:** diversion, augmentation, SCI (spinal cord injury): treat ASB, consider prophylaxis.

Diagram to draw: right hydroureter at the pelvic brim.

Key points to remember:

- Screen by culture at 12–16 weeks; treat all ASB.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term; TMP not in 1st trimester.

Hemostasis in laparoscopic surgery.

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 3, Q3f (also answered in Practice Set 2 Paper 4, Q3f; partial nephrectomy haemostasis in 2019-Summer Paper 3, Q1)

Principles

- Devices seal by **denaturing protein**. Plan from imaging; keep a **haemorrhage tray** (laparoscopic Satinsky, 10-mm suction, vascular suture with Lapra-Ty, haemostatic agent).

Effect of pneumoperitoneum

- CO_2 at 12–15 mmHg **tamponades venous bleeding**. Before exit, **lower pressure to about 5 mmHg** and inspect the operative site and each port.

Energy devices

Device	Lateral thermal spread	Note
Monopolar	5–7 mm	Capacitive coupling, insulation failure, direct coupling
Bipolar	1–6 mm	Current only between the jaws
Vessel sealer (LigaSure)	1–3 mm	Seals vessels **≤7 mm**; one application only
Ultrasonic (Harmonic)	≈1 mm	No current; blade >200 °C, cools slowly
Argon beam	Superficial	Raw surfaces; gas embolism risk

Clips

- **Titanium clips:** no diathermy near them (slip).
- **Hem-o-lok** (locking polymer): circumferential dissection, see the tip beyond the vessel, feel the snap, at least two clips on the patient side with a 2-mm cuff. The manufacturer **contraindicates it on the living-donor renal artery** after stump haemorrhages.
- **Lapra-Ty** absorbable clips act as knots (renorrhaphy).

Staplers

- Endo GIA via a 12-mm port; **grey (2.0 mm) or white (2.5 mm)** loads for renal vessels; blue or green for thick tissue.
- **Never fire across clips.**

Sutures and haemostatic agents

- Intracorporeal or barbed sutures; fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

Managing bleeding

- **Venous injury:** raise pressure to 25 mmHg → suction to identify → pledget tamponade → sealant → extra 5-mm port for suturing or Satinsky → hand-assist → convert. Keep the patient well filled (gas embolism).
- **Arterial:** minor, tamponade; renal artery, staple; aorta, convert.
- **Port-site bleeding:** cantilever the trocar to find it; diathermy or transfascial suture. Blunt trocars reduce it.

Diagram to draw: Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

Key points to remember:

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

APOGEE

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 3, Q3c

Assumed meaning: the Apogee transvaginal mesh kit (American Medical Systems) for posterior vaginal wall and vault prolapse. It is not described in the chapter notes; facts below are general.

Definition

- A **prepackaged polypropylene mesh kit** placed transvaginally to support the **posterior compartment and vaginal apex** (DeLancey levels I and II). Anterior companion: **Perigee** (transobturator).
- Mesh: **type I macroporous monofilament polypropylene** (Amid classification), the only type that integrates into tissue.

Indications (historical)

- Symptomatic posterior and vault prolapse, mostly **recurrent** or with poor native tissue, avoiding abdominal surgery.

Technique (principle)

1. Midline posterior vaginal incision; dissect the rectovaginal space to the ischial spines.
2. Two small perianal incisions lateral and posterior to the anus.
3. Curved trocars pass through the **ischiorectal fossa**, piercing the iliococcygeus near the ischial spine, and deliver the mesh arms.
4. Mesh body fixed to the apex and laid over the rectovaginal septum **tension-free**; epithelium closed without trimming.

Complications

- **Mesh exposure** (IUGA/ICS [International Urogynecological Association/International Continence Society] term; "erosion" deprecated), extrusion, **perforation** into rectum or bladder.
- **Dyspareunia**, buttock and pelvic pain, mesh contraction, defecatory dysfunction, ischiorectal haematoma or abscess, rectal injury.
- **Management:** small exposure → vaginal oestrogen, then partial excision; pain or perforation → excision of mesh and arms.

Regulatory status

- **FDA (US Food and Drug Administration) 2011:** serious complications of transvaginal POP (pelvic organ prolapse) mesh "not rare", with no clear benefit over native repair. Reclassified high-risk in 2016.
- **2019:** FDA ordered remaining manufacturers to **stop selling transvaginal POP mesh**; Apogee-type kits are off the market. Synthetic mesh improves anatomy but complicates more (Cochrane 2016).
- This does **not** apply to mid-urethral slings or **sacrocolpopexy mesh**.

Current alternatives

- Native posterior colporrhaphy or site-specific repair; uterosacral suspension or **sacrospinous fixation**; **sacrocolpopexy** (abdominal, laparoscopic, robotic) for apical prolapse; colpocleisis in the frail.

Diagram to draw: posterior mesh with two arms through the ischiorectal fossae.

Key points to remember:

- Apogee: transvaginal posterior/apical polypropylene mesh kit.
- Exposure, dyspareunia and pain drove withdrawal.
- FDA 2019 stopped transvaginal POP mesh sales.

Describe principles of tissue engineering and what its status in managemet of urological diseases is.

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 3, Q3e (also answered in 2023-Winter Paper 1, Q1 and 2022-Summer Paper 4, Q2)

Definition

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore or improve tissue function (Langer and Vacanti, 1993).
- **Rationale:** bowel in the urinary tract causes mucus, stones, acidosis and cancer; grafts are scarce.

Principles

```
Biopsy -> cells expanded -> seeded on scaffold (+ signals)
       -> bioreactor conditioning -> implant -> host vascularises
```

Element	Details
Cells	Autologous urothelial and smooth muscle cells (no rejection); adult stem cells (bone marrow, adipose, urine-derived; mainly paracrine); embryonic and iPSCs (induced pluripotent stem cells): tumour risk
Scaffolds	Biocompatible, porous, degrade as tissue forms. Natural (collagen), **acellular matrices** (bladder matrix, SIS [small intestinal submucosa]), synthetic (PGA, PLGA [polyglycolic, poly(lactic-co-glycolic) acid])
Seeding	Urothelium inside, muscle outside. **Unseeded** matrix works only for short onlay defects; tubes need seeded scaffolds or they contract
Signals	Growth factors (e.g. VEGF [vascular endothelial growth factor]), matrix and mechanical cues
Bioreactor	Perfusion and cyclic stretch to mature the construct
Vascularity	Oxygen diffuses only **100–200 µm**: main barrier; omental wrap, angiogenic factors

Status in urology

Organ	Status
Bladder	**Atala 2006**: seeded constructs in 7 myelomeningocele patients improved compliance; phase II trial showed **no significant gain**. Experimental
Conduit	Neo-Urinary Conduit trial: **half developed stenosis**
Urethra	Acellular onlay for short strictures; **Raya-Rivera 2011**: seeded tubular urethras in **5 boys**, functional up to 6 years
Sphincter	Muscle-derived cell injection for SUI (stress urinary incontinence): investigational
Vagina	Engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome
Penis, kidney, testis	Corporal grafts, kidney organoids, spermatogonial stem cell banking: experimental

- **Recent advances:** 3D bioprinting, organoids, vascularised composite bladder allograft.
- **Challenges:** vascularity, innervation, fibrosis, cost, regulation.

Diagram to draw: biopsy → culture → scaffold → bioreactor → implant.

Key points to remember:

- Cells, scaffold, signals, bioreactor, blood supply.
- Diffusion limit 100–200 µm.
- Still experimental except short urethral onlays.

Rapid Revision Sheet – Section 1 · Basic Principles

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Therapeutic energy sources in urology (including laparoscopic energy devices) (LAQ)

- Spread: monopolar 5–7 > bipolar 1–6 > sealer 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; ultrasonic blade hottest after use.
- Capacitive coupling: avoid hybrid trocars; AEM detects it.
- Holmium 2120 nm: gold standard for stones and HoLEP.
- Bipolar TURP in saline avoids TUR syndrome.
- Thermal ablation for cT1a ≤3 cm; cryo <−40 °C, two cycles.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

Asymptomatic haematuria and urinary biomarkers (SAQ)

- MH: 3 or more RBCs/HPF on microscopy, one sample.
- Low: optional; intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
- Any gross haematuria = high risk.
- Cytology: specific for high grade/CIS, insensitive for low grade.
- BTA is falsely positive with blood.
- No markers in initial microhaematuria work-up; none replaces cystoscopy.

Collin's knife. (SAQ)

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

Baceriuria and use of antibiotic policy in pregnancy (SAQ)

- Screen by culture at 12–16 weeks; treat all ASB.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term; TMP not in 1st trimester.

Hemostasis in laparoscopic surgery. (SAQ)

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

APOGEE (SAQ)

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- Exposure, dyspareunia and pain drove withdrawal.
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Describe principles of tissue engineering and what its status in managemet of urological diseases is. (SAQ)

- Cells, scaffold, signals, bioreactor, blood supply.
- Diffusion limit 100–200 µm.
- Still experimental except short urethral onlays.

Section 2 · Oncology, DSD, Gender & Ageing — Long answer questions (LAQ)

DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia

Asked as:

- Management of DSD — *Summer 2019 Paper III Q3f (SAQ)*
- Evaluation and treatment of newborn with ambiguous genitalia — *Practice Set 4 Paper III Q2 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(DSD management also answered as an SAQ in 2019-Summer Paper 3, Q3f)

DSD read as: disorders/differences of sex development. Detrusor–sphincter dyssynergia is covered in 2016-Summer Paper 1, Q3a.

Introduction and definition

- Atypical ("ambiguous") genitalia point to a **DSD (difference/disorder of sex development)**: congenital conditions with atypical chromosomal, gonadal or anatomical sex (Chicago consensus 2006).
- It is a **medical emergency only for salt-wasting CAH (congenital adrenal hyperplasia)**; sex designation is not a "social emergency". Evaluate within 24–48 hours (neonatal evaluation, first 24–48 h).

Classification (by karyotype)

Group	Main causes (examples)
46,XX DSD	**CAH** (commonest; 21-hydroxylase 95%, 11β-hydroxylase about 5%, 3β-HSD (hydroxysteroid dehydrogenase) <1%); maternal androgens or aromatase deficiency; ovotesticular; testicular (SRY (sex-determining region Y) positive, SRY+)
46,XY DSD	Partial/complete gonadal dysgenesis; androgen synthesis defects (17β-HSD3, 5α-reductase); androgen insensitivity (CAIS/PAIS, complete/partial; PAIS = partial androgen insensitivity); LH (luteinising hormone) receptor defect
Sex chromosome DSD	45,X/46,XY MGD (mixed gonadal dysgenesis); 45,X Turner; 47,XXY Klinefelter; chimerism

History

- Consanguinity, **unexplained infant deaths** (CAH), infertility or amenorrhoea in relatives; maternal virilisation, progestogens or androgens; prenatal cfDNA (cell-free DNA) karyotype versus ultrasound discordance.

Examination (gender-neutral terms)

- Stretched phallic length, glans width, meatus, number of perineal openings, labioscrotal fusion and rugosity, pigmentation, anogenital distance. **Prader grade (stage) I–V** for virilisation.
- Dehydration, hypotension, hyperpigmentation (CAH); dysmorphism.
- **Gonads are the key sign.** Ovaries never descend, so **a palpable gonad means testicular tissue**. Bilateral impalpable gonads with atypical genitalia = CAH until proven

otherwise (impalpable gonads suggest CAH). Asymmetry suggests MGD or ovotesticular DSD.
- Impalpable testis with hypospadias, or proximal hypospadias, also needs DSD work-up (Kaefer: DSD in about 30%, 50% if the testis is impalpable).

Investigations (tests)

- **Karyotype** at once (preliminary 2–3 days); **FISH (fluorescence in situ hybridisation) for X/Y/SRY** within hours.
- **CAH screen: 17-OHP (17-hydroxyprogesterone) at about 36 hours** (after 36 h; earlier values are falsely high); electrolytes and glucose daily. **Emergency:** exclude **salt-wasting CAH** (low sodium, high potassium, shock); **salt-wasting appears around day 5–7** (usually after day 5–7), so normal early electrolytes do not exclude CAH.
- **At about 1 week:**

Test	Meaning
AMH (anti-Müllerian hormone)	Present = functioning Sertoli cells
Testosterone, LH, FSH (follicle-stimulating hormone)	Testicular function (mini-puberty)
Androstenedione, DHEA-S (dehydroepiandrosterone sulfate)	Adrenal enzyme block
Testosterone:DHT (dihydrotestosterone) >20:1	5α-reductase deficiency
hCG (human chorionic gonadotropin) stimulation	No rise = no Leydig tissue or LH receptor defect

- **Pelvic ultrasound:** uterus (well seen neonatally), gonads, kidneys, cerebriform adrenals (CAH); then genitography, endoscopy, laparoscopy ± biopsy, genetic panels.
- **Genitography / endoscopy:** urogenital sinus, confluence level relative to bladder neck, cervix.
- If unclear: MRI (magnetic resonance imaging), **laparoscopy ± deep longitudinal gonadal biopsy**, gene panels (genetic panels), microarray, exome. **No gonadectomy during diagnosis (while diagnosing).**

Diagnostic algorithm

```
Palpable gonads: 0 -> CAH likely; 1 -> MGD, PAIS, ovotesticular;
                 2 -> 46,XY DSD (PAIS, 5α-reductase, synthesis defect)
46,XX + uterus + high 17-OHP           -> CAH
46,XY, no uterus                       -> androgen synthesis/action defect
46,XY or 45,X/46,XY + uterus           -> gonadal dysgenesis / MGD
```

- A precise diagnosis is reached in nearly all 46,XX but under 50% of 46,XY DSD.

Treatment

1. Emergency and medical management (CAH)

- Salt-wasting crisis: IV (intravenous) saline, dextrose, stress-dose hydrocortisone.
- Maintenance (Endocrine Society 2018): hydrocortisone 10–15 mg/m²/day in divided doses, fludrocortisone, salt supplements in infancy; stress dosing during illness and surgery.
- **Puberty:** oestrogen or testosterone induction if gonads fail.

2. Team and family

- **MDT (multidisciplinary team):** urologist, endocrinologist, geneticist, psychologist, neonatologist, ethicist; gender-neutral language. Congratulate parents; say "your baby", never "it"; open disclosure, peer support.

3. Sex of rearing (gender assignment)

- Weighs diagnosis, prenatal androgen exposure, likely gender identity, fertility, sexual function, family wishes and culture.
- Usually female: 46,XX CAH, CAIS (complete androgen insensitivity), complete gonadal dysgenesis. Trend male: PAIS, 5α-reductase and 17β-HSD3 deficiency.
- Micropenis (stretched length under 2 cm) alone is no reason for female rearing.

4. Surgery: timing and consent

- Irreversible surgery is ideally deferred until the child can participate (consent); parents decide with the team after balanced counselling.
- **Feminising** surgery (clitoroplasty, vaginoplasty, labioplasty) is irreversible, so deferral until the child can consent is favoured unless obstructed; severe CAH is debated.
    - Clitoroplasty (ventral, nerve-sparing, glans preserved; Prader III–V); vaginoplasty by confluence (flap for low, partial/total urogenital mobilisation, pull-through for high); labioplasty. The vagina's position relative to the bladder neck, not common-channel length, decides the vaginoplasty. Stress steroids in CAH.
- **Masculinising:** proximal hypospadias repair, orchidopexy, often done in infancy (complication rates high).

5. Gonads (gonadal tumour risk)

Risk	Condition	Plan / management
High	Dysgenesis with Y, intra-abdominal (15–35%); PAIS non-scrotal (50%)	Gonadectomy
Intermediate	Turner with Y (12%)	Gonadectomy
Low	CAIS (2%); ovotesticular (3%)	Keep CAIS gonads until after puberty; ovotesticular: remove discordant tissue

Follow-up and recent advances

- Pubertal induction, psychology, fertility preservation (experimental gonadal tissue cryopreservation), transition to adult care.
- Prenatal cfDNA diagnosis; rapid exome/genome sequencing. Prenatal dexamethasone for CAH is experimental.

Diagram to draw: Prader stages I–V and the urogenital sinus with confluence level; karyotype-based DSD algorithm.

> **Key points to remember:**
>
> - Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
> - 17-OHP after 36 h; salt crisis around day 5–7.
> - Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
> - Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

Prepucioplasty

Type: SAQ (10 marks)

Source: 2019-Summer Paper 3, Q3a

Read as: preputioplasty, a foreskin-preserving operation for phimosis.

Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring: forced retraction, balanoposthitis or BXO (balanitis xerotica obliterans).

Indications

- Symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks with gentle retraction). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance. Treat active infection first.

Techniques

Technique	Method
Dorsal slit with transverse closure	Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz principle)
Triple incision	Three longitudinal cuts over the ring, each closed transversely
Y–V plasty	V-flap of outer skin advanced into a Y-cut

- **Principle:** cutting the ring lengthwise and closing it crosswise widens its circumference.
- General anaesthesia or penile block; incise only the band; confirm retraction; absorbable sutures.

Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.

- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.
- **Also know:** preputial reconstruction during hypospadias repair is also called preputioplasty; dehiscence is its risk.

Diagram to draw: dorsal incision through the ring, closed transversely.

> **Key points to remember:**
> - Steroid first; about 1 in 6 fail.
> - Cut lengthwise, close crosswise; daily retraction after.
> - BXO: circumcision preferred.

Mixed gonadal dysgenesis.

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 3, Q3e (DSD management in general in 2019-Summer Paper 3, Q3f)

Definition

- MGD (mixed gonadal dysgenesis): a **dysgenetic testis on one side and a streak gonad on the other**, with persistent Müllerian structures and variable virilisation.
- A sex chromosome DSD (difference of sex development), usually **45,X/46,XY mosaicism** (about 1 in 5,000 births). **Second commonest cause of neonatal atypical genitalia after CAH (congenital adrenal hyperplasia).**

Pathogenesis

- The Y-bearing line forms a testis; the 45,X line leaves a streak.
- AMH (anti-Müllerian hormone) and testosterone act **locally**, so anatomy is asymmetric: Wolffian duct on the **testis side**; tube and hemiuterus on the **streak side**.
- 45,X/46,XY spectrum: female (Turner + Y), atypical (MGD) or male with normal testes (about 85%).

Clinical features

- **Asymmetric genitalia:** phallus with chordee, perineal hypospadias, urogenital sinus; one testis in a hemiscrotum, the other side impalpable. Hernia may contain a uterus or tube.
- **Turner features:** short stature, coarctation, horseshoe kidney.

Evaluation (MDT, multidisciplinary team)

- **Karyotype** (many cells, for mosaicism); **17-OHP (17-hydroxyprogesterone)** to exclude CAH; LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, AMH.
- **Pelvic ultrasound** (uterus, asymmetric gonads); genitography, endoscopy; **laparoscopy with gonadal biopsy**. No gonadectomy during diagnosis.
- Echocardiography, renal ultrasound (Turner screening).

	MGD	Ovotesticular DSD
Karyotype	45,X/46,XY	Mostly 46,XX
Gonads	Dysgenetic testis + streak	Ovary + testis, or ovotestis
Tumour risk	15–35%	About 3%

Management

- **Sex of rearing:** individualised; more androgen exposure predicts male identity; dysphoria is commoner when raised female. Avoid irreversible infant surgery.
- **Gonads:** germ cell neoplasia 15–35% (gonadoblastoma, then dysgerminoma or seminoma) in streak and testis alike. **Remove streaks** (biopsy first if tissue uncertain).
  - Raised male: keep a scrotal testis under surveillance; bring down an intra-abdominal one if possible.
  - Raised female: remove or relocate the testis.
- **Surgery:** male: staged hypospadias repair, orchidopexy; female: feminising surgery deferred.
- **Hormones:** pubertal sex steroids; growth hormone. Infertility usual.

Diagram to draw: asymmetric anatomy: testis with vas; streak with tube and hemiuterus.

Key points to remember:

- 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
- Second commonest cause of atypical genitalia after CAH.
- Tumour risk 15–35%: remove streaks.
- Screen for Turner features.

Telomerase

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 3, Q3b

Telomeres

- **Telomeres** are repetitive sequences (TTAGGG repeats) at chromosome ends. They protect ends from degradation and fusion.
- DNA polymerase cannot copy the very end of the lagging strand (end-replication problem). So about **100 base pairs are lost per division**: the "mitotic clock".
- Critical shortening triggers **replicative senescence** (Hayflick limit).

Telomerase: structure and function

- A specialised **reverse transcriptase** that adds telomeric repeats.
- Two core parts: **hTERT** (human telomerase reverse transcriptase, catalytic protein) and **hTR** (telomerase RNA, the template).
- **Normal expression:** only rapidly dividing tissues: germ cells (testis), intestinal epithelium, endometrium, activated lymphocytes. Most somatic cells lack it.

Role in cancer

```
Proliferation -> moderate shortening -> senescence/apoptosis (barrier 1)
-> checkpoint lost (p53) -> severe shortening -> chromosomal instability (barrier 2)
-> telomerase activation -> immortal cancer cell
```

- **85–90% of human tumours** express high telomerase, giving unlimited replication.
- **ALT (alternative lengthening of telomeres):** telomerase-independent recombination; linked to ATRX or DAXX loss.
- **TERT promoter mutations** switch on hTERT.

Urological relevance

Organ	Finding
Bladder	TERT promoter mutation is the commonest mutation (over 60% even in PUNLMP, papillary urothelial neoplasm of low malignant potential). Urine markers: telomerase activity by **TRAP (telomeric repeat amplification protocol)**, a PCR (polymerase chain reaction)-based assay; hTERT messenger RNA; AssureMDx (FGFR3, TERT, HRAS mutations plus methylation). Inflammation gives false positives; not routine
Upper tract	hTR positive in almost all UTUC (upper tract urothelial carcinoma); rises with stage
Prostate	Short telomeres in cancer and premalignant lesions; short tumour telomeres predict recurrence; variable stromal telomeres predict cancer death

- **Also know:** senescent cells secrete SASP (senescence-associated secretory phenotype) cytokines in ageing and BPH (benign prostatic hyperplasia).

Therapy

- **Imetelstat** (telomerase inhibitor) is used in myelodysplastic syndrome.
- hTERT-based vaccines (UV1, GX301, INO-9012) are in early prostate, renal and bladder trials.

Diagram to draw: telomere-to-cancer flowchart with two barriers.

Key points to remember:

- Telomerase = hTERT (catalytic) + hTR (RNA template).
- About 100 base pairs lost per division; 85–90% of tumours express telomerase.
- TERT promoter mutation is the commonest mutation in bladder cancer.
- TRAP assay detects urinary telomerase; inflammation gives false positives.

Rapid Revision Sheet – Section 2 · Oncology, DSD, Gender & Ageing

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia (LAQ)

- Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
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Section 3 · Infections & Inflammation — Long answer questions (LAQ)

Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management

Asked as:

- Describe etiopathogenisis, clinical presentation, diagnosis and management of Renal TB? — *Summer 2022 Paper III Q2 (LAQ)* (also answered in 2018-Summer Paper 2, Q1)
- Laboratory Diagnosis of Genitourinary Tuberculosis. — *Winter 2023 Paper III Q3f (SAQ)* (also answered in 2022-Summer Paper 3, Q2, as part of renal TB)
- Describe newer diagnostic modalities for Gu Kochs. Briefly outline role of all present diagnostic tests and give diagnostic algorithm. — *Practice Set 6 Paper III Q3d (SAQ)* (also answered in 2023-Winter Paper 3, Q3f)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Assumed meaning: "Gu Kochs" = genitourinary tuberculosis.

Introduction

- Caused by the MTBC (*Mycobacterium tuberculosis* complex); *M. bovis* (and BCG [bacille Calmette–Guérin]) is innately resistant to pyrazinamide.
- UG (urogenital) TB (tuberculosis) is 15–20% of pulmonary TB in developing countries; second extrapulmonary site after lymph nodes. **Kidney is involved in 80%.**

Etiopathogenesis

- **Primary infection:** inhaled bacilli spread to hilar nodes, then by blood to the well-perfused kidney and epididymis, before immunity develops (up to 12 weeks).
- **Granuloma:** epithelioid cells and Langhans giant cells in a lymphocyte cuff, maintained by IFN-γ (interferon-γ). In 90–95% infection stays latent.
- **Reactivation** (lifetime 5–15%), even decades later: old age, diabetes, renal failure, HIV (human immunodeficiency virus), malnutrition, immunosuppression.
- **Routes:** haematogenous (principal); ascending (intravesical BCG, about 1%); contiguous (psoas, bowel).
- **Renal sequence:** bacilli lodge in **periglomerular capillaries** → cortical granulomas coalesce → **caseation** and cavities open into calyces → papillary necrosis, abscess, calcification → bacilluria seeds ureter and bladder.
- **Autonephrectomy:** caseo-cavernous type, or fibrotic calcified "putty" kidney.
- **Ureter and bladder (descending spread):** mural granulomas give strictures and a beaded or pipestem ureter. Cystitis starts around the ureteric orifices; after about a year of fibrosis the bladder contracts (<100 mL; thimble bladder <20 mL).
- **Key concept:** healing occurs by fibrosis, so infundibular and ureteric strictures (commonest at the **VUJ (vesicoureteric junction)**) and bladder contraction can worsen during treatment.
- **Also know:** granulomatous interstitial nephritis; miliary renal TB; amyloidosis; squamous metaplasia of the pelvis (squamous cell carcinoma risk); ESRD (end-stage renal disease) in about 7%.

Clinical presentation

- Renal TB is silent until it reaches the bladder; asymptomatic in about 8%.

- **Storage LUTS (lower urinary tract symptoms)** and dysuria (commonest), haematuria and flank pain (about one-third), colic <10% (sloughed papillae, clot, debris).
- Constitutional symptoms <20%. Recurrent "UTI (urinary tract infection)" not responding to antibiotics.
- Signs: renal angle tenderness, flank sinus, beaded vas, epididymal nodule; hypertension; renal failure if bilateral.

Diagnosis

Principle

- GUTB (genitourinary tuberculosis, TB) is paucibacillary with intermittent shedding, so **multiple samples** are needed: send 3–5 early-morning urine samples.
- **Hierarchy of proof:** positive culture or DNA → caseating granulomas on histology → clinical picture with response to treatment.

Specimen

- **Early-morning first-void urine (EMU), 3–5 samples on consecutive days**, sent (processed) at once (acidic urine kills bacilli). Avoid prior fluoroquinolones.
- Also tissue, pus, FNAC (fine-needle aspiration cytology) or semen.

Laboratory tests

Test	Method	Comment / Role
Urine routine	Microscopy	**Sterile (acid) pyuria** ± (microscopic) haematuria in >90%
ZN (Ziehl–Neelsen) smear	Carbol fuchsin, acid-alcohol, methylene blue: red bacilli. Auramine fluorescence is more sensitive	Quick; sensitivity <50% (under 50%); saprophytic mycobacteria give false positives
Culture (urine AFB [acid-fast bacilli] culture): gold standard	Decontaminate and concentrate. **LJ (Löwenstein–Jensen)** solid (medium): 4–6 weeks. **MGIT (mycobacteria growth indicator tube)** liquid: about 10 days (as little as 10 days)	80–90% sensitivity with ideal technique; diagnosis; gives DST (drug-susceptibility testing / drug susceptibility); **monitors treatment** (the only test that monitors treatment)
NAAT (nucleic acid amplification test): GeneXpert MTB/RIF	Automated cartridge PCR (polymerase chain reaction); about 2 hours (result in 1–2 days)	Rapid diagnosis; detects MTB and **rifampicin resistance** (MDR [multidrug-resistant] TB surrogate). **Xpert Ultra** is more sensitive. Stays positive after cure, so cannot monitor treatment
TrueNat	Indian chip-based NAAT	NTEP (National TB Elimination Programme)
LPA (line probe assay)	DNA strip hybridisation	Rifampicin, isoniazid or second-line drug resistance
LF-LAM (lateral-flow lipoarabinomannan)	Urine antigen strip, <30 min (under 30 minutes); point-of-care urine test in people with HIV	HIV (human immunodeficiency virus) with symptoms, CD4 <200 or advanced disease (WHO [World Health Organization] 2019)
Histopathology	Biopsy (bladder, epididymis)	Caseating granulomas; NAAT on tissue raises yield
TST (tuberculin skin test)	5 TU (tuberculin units) PPD intradermal; read induration at 48–72 h	Shows (proves; TST/IGRA prove) infection only, not active disease; BCG false positive

Test	Method	Comment / Role
IGRA (interferon-γ release assay)	ESAT-6/CFP-10 blood test (QuantiFERON, T-SPOT)	Not affected by (unaffected by) BCG (bacille Calmette–Guérin); cannot separate latent and active; shows infection, not active disease

Newer modalities

- **Liquid culture, MGIT (mycobacteria growth indicator tube):** about 10 days versus 4–6 weeks on LJ (Löwenstein–Jensen).
- **NAAT (nucleic acid amplification test): GeneXpert MTB/RIF** (Xpert Ultra more sensitive): automated cartridge PCR, about 2 hours; detects **rifampicin resistance**. Also TrueNat; **LPA (line probe assay)** for isoniazid resistance.
- **Urine LF-LAM:** strip, under 30 minutes; WHO 2019.
- **IGRA:** unaffected by BCG.
- **Imaging:** CT (computed tomography) urography; **MR (magnetic resonance) urography with DWI (diffusion-weighted imaging)** grades fibrosis.

Supportive tests

- ESR, blood count, renal function, HIV, blood sugar; sputum and **chest radiograph in all** (mandatory; pulmonary / lung TB coexists in up to one-fifth).

Imaging and endoscopy

Imaging	Findings / Role
KUB (kidney–ureter–bladder) film	Calcification (24–44%); lobar "putty" kidney
IVU (intravenous urography) / CT (computed tomography) urography	**Moth-eaten calyx**, **phantom calyx** (infundibular stenosis), cavities, hydrocalycosis, **Kerr's kink**, pipestem or beaded corkscrew ureter, small bladder; IVU replaced by CT
Ultrasound	Children, pregnancy, scrotum; follows hydronephrosis during treatment
MRU (magnetic resonance urography)	Children, pregnancy
Cystoscopy (and biopsy)	**"Golf-hole" ureteric orifice**, ulcers; biopsy if culture negative (yield 19–52%)
Chest radiograph	Mandatory (lung TB in up to one-fifth)

Diagnostic algorithm

```
Suspicion (sterile pyuria, storage symptoms, epididymal nodule)
-> 3-5 EMU: ZN + culture (LJ + MGIT) + Xpert
-> chest X-ray; CT urography
Positive -> treat; DST guides regimen
Negative -> cystoscopy/biopsy, aspiration + NAAT on tissue
Still negative -> empirical trial if clinically compatible
```

Management

Medical (all patients)

- **Multidrug therapy** because bacilli exist in different metabolic states and single drugs select resistance.
- Baseline liver and renal function, HIV, hepatitis, diabetes; DOT (directly observed therapy).
- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months; **pyridoxine** against isoniazid neuropathy. India's NTEP (National TB Elimination Programme) uses 2HRZE/4HRE.

- Complicated disease: some treat 12–24 months, or ≥6 months after surgery. The new 4-month rifapentine–moxifloxacin regimen is not used for UG TB.
- Toxicity: hepatitis (all except ethambutol); optic neuritis (ethambutol).
- **Monitor ultrasound** for worsening obstruction; steroids (anecdotal) for strictures.
- MDR-TB: second-line drugs (bedaquiline, pretomanid, linezolid).

Surgical (about half need it), after 4–6 weeks of ATT (antituberculous therapy)

Problem	Procedure
Obstruction	Early **DJ (double-J) stent** (succeeds in 41%) or **PCN (percutaneous nephrostomy)**
Localised polar lesion not responding	Partial nephrectomy
Non-functioning kidney + persistent disease or hypertension	**Nephrectomy** (laparoscopic preferred; ligate artery and vein separately)
PUJ (pelviureteric junction) stricture	Pyeloplasty; ureterocalicostomy
Lower ureteric stricture	Reimplantation, psoas hitch, Boari flap; ileal ureter if long
Contracted bladder	Augmentation; substitution if <20 mL

- Stricture not improving after 6 weeks of drugs → open repair.
- Contested: nephrectomy of a non-functioning kidney without active disease.

Follow-up

- Relapse 6.3–22% (mean about 5 years): urine culture and imaging for up to 10 years.

Diagram to draw: kidney showing caseous cavity, phantom calyx and VUJ stricture; stepwise diagnostic flowchart (algorithm above) from sterile pyuria to culture and DST.

Key points to remember:

- Kidney 80%; periglomerular seeding; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Relapse up to 22%; follow up 10 years.

Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria.

Type: LAQ (25 marks)

Source: 2024-Winter Paper 3, Q2 (also answered in 2018-Summer Paper 1, Q1, with lymphatic anatomy)

Definition

- **Chyluria** is the passage of **chyle** (intestinal lymph rich in chylomicrons and protein) in the urine, making it milky.
- It needs an abnormal **lymphourinary fistula**, usually at a renal calyceal fornix (pyelolymphatic fistula).

Relevant lymphatic anatomy

- Intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction.
- Lumbar (para-aortic) trunks from the kidneys join the cisterna. Lymph flows cephalad and right to left.
- Valves normally stop backflow from the cisterna into renal hilar lymphatics.

Aetiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; Culex mosquito in urban India)	Congenital: lymphangiectasia, renal lymphangioma
Rarely other parasites	Trauma or surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection])
	Retroperitoneal tumour, fibrosis, tuberculosis, pregnancy

- Genital and urinary filariasis is almost always *W. bancrofti*; Brugia species spare the genitalia.

Pathogenesis

- **Life cycle:** a mosquito deposits infective larvae, which mature in central lymphatics over 6–9 months. Adults live 5–7 years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- **Obstructive theory:** adult worms and the host reaction cause **lymphangitis and fibrosis** of lymphatics between the intestinal trunks and thoracic duct.
- Lymphatics below the block **dilate**, their **valves fail**, and chyle **refluxes** into renal hilar and peripelvic lymphatics.
- Rising pressure ruptures the thin-walled lymphatics into a calyceal fornix, creating a **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- Chyluria appears **earlier than elephantiasis**, is usually **intermittent** and may remit spontaneously.

Clinical features

- **Milky urine**, worse after fatty meals and on standing; may clear after fasting.
- **Chylous clots:** colic, retention.
- **Haematochyluria** (blood-stained chylous urine).

- **Systemic effects** of fat and protein loss: weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- Endemic residence, filarial fever, hydrocele or lymphoedema.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Evaluation

1. Confirm chyle

- **Ether (chloroform) test:** milky urine shaken with ether clears, because the fat dissolves.
- **Urine triglycerides** positive; Sudan III staining shows fat globules; urine protein raised.
- Differentiate from **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial tests

- Night (midnight) blood smear: often negative by this stage.
- **Circulating filarial antigen** (card test), eosinophilia, PCR (polymerase chain reaction); ultrasound "filarial dance sign" in scrotal lymphatics.

3. Lateralise and localise the fistula

- **Cystoscopy after a fatty meal:** milky efflux from one ureteric orifice; bilateral in some.
- **Retrograde pyelography:** pyelolymphatic backflow.
- **Lymphoscintigraphy:** non-invasive; tracer seen in the kidney and bladder. MR lymphangiography or CT; pedal lymphangiography is the historical gold standard.

4. Baseline: albumin, haemoglobin, renal function.

Management

```
Mild             -> diet + antifilarial drugs
Moderate/persist -> add endoscopic sclerotherapy (repeatable)
Severe or failed -> renal hilar lymphatic disconnection
```

1. Conservative

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids. MCTs are absorbed directly into portal blood, not lacteals, so less chyle forms.
- **DEC (diethylcarbamazine)** 2 mg/kg three times daily, traditionally for 12 days (a 1-day course is as effective). It kills microfilariae but only modestly kills adults.
- **Doxycycline** (anti-*Wolbachia*, 200 mg daily for 4–8 weeks); albendazole; ivermectin.

2. Endoscopic sclerotherapy (renal pelvic instillation)

- Ureteric catheter on the affected side, head-down position. Instil **silver nitrate (0.5–1%)** or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- Good initial response; recurrence common.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:**
  1. Full renal mobilisation (nephrolysis).
  2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every hilar lymphatic.

3. Strip the upper ureter; remove perinephric fat.
4. Nephropexy; optional **omental wrap** around the hilum.
5. On-table cystoscopy confirms clear efflux.

- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open (shorter operation and stay, less blood loss, fewer complications).
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy obsolete.

4. Prevention

- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: retroperitoneal lymphatics with obstructed thoracic duct, dilated hilar lymphatics and pyelolymphatic fistula at a calyceal fornix.

Key points to remember:

- Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
- Chyluria precedes elephantiasis and is often intermittent.

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

Prognostic indicators in emphysematous pyelonephritis.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 3, Q3c

Definition

- EPN (emphysematous pyelonephritis) is an acute **necrotising** infection of the renal parenchyma and perirenal tissue by **gas-forming organisms**. It is a urological emergency.
- Most patients are **diabetic**; obstruction in about a quarter.
- Organisms: *E. coli* (commonest), *Klebsiella*, *Proteus*. Proposed mechanism: high tissue glucose fermented to CO_2 in poorly perfused tissue.

Imaging-based prognostic classifications

1. Huang–Tseng CT classification (2000)

Class	Gas location
1	Collecting system only (emphysematous pyelitis)
2	Renal parenchyma, no extrarenal extension
3A	Extension into **perinephric** space
3B	Extension into **pararenal** space
4	**Bilateral** EPN or EPN in a **solitary kidney**

2. Wan classification (CT pattern)

Type	CT	Mortality
I	Streaky or mottled gas, **no fluid**; rapid parenchymal destruction	50–60%
II	Renal or perirenal **fluid**, bubbly or loculated gas, or gas in collecting system	<20%

- **Why:** fluid means a walled-off collection that can be drained. Streaky gas without fluid means necrosis faster than the host can contain it.

Clinical and laboratory prognostic factors

- **Huang–Tseng risk factors: thrombocytopenia**, **AKI (acute kidney injury)/raised creatinine**, **altered sensorium**, **shock**.
- **Other adverse factors:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

How prognosis guides management

```
All: resuscitation, glucose control, IV antibiotics, relieve obstruction (stent/PCN)
Class 1–2               -> antibiotics + PCD
Class 3–4, <2 risk factors -> antibiotics + PCD (usually succeeds)
Class 3–4, >=2 risk factors or PCD failure -> nephrectomy
```

- PCD (percutaneous catheter drainage); PCN (percutaneous nephrostomy).
- Reported mortality 19–43%; routine emergency nephrectomy still carried 40–50%, so drainage-first care is now standard.

- **Also know:** emphysematous pyelitis (gas in the collecting system only) is often non-diabetic and responds to antibiotics.

Diagram to draw: CT sketches of Huang–Tseng classes 1, 2, 3A/3B and 4.

Key points to remember:

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors.

Rapid Revision Sheet – Section 3 · Infections & Inflammation

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management (LAQ)

- Kidney 80%; periglomerular seeding; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
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- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Relapse up to 22%; follow up 10 years.

Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria. (LAQ)

- Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
- Chyluria precedes elephantiasis and is often intermittent.

Prognostic indicators in emphysematous pyelonephritis. (SAQ)

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors.

Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome

Asked as:

- Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR. — *Summer 2015 Paper III Q2 (LAQ)*
- Discuss mgt. of b/l VUR. — *Practice Set 2 Paper III Q2 (LAQ)*
- Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection — *Practice Set 2 Paper III Q3d (SAQ)*
- Discuss the management of B/l VUR and discuss its long term outcome — *Practice Set 6 Paper III Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(also answered in 2015-Summer Paper 3, Q2 and 2020-Summer Paper 1, Q2; long-term outcome added from Practice Set 6 Paper 3, Q1. The SAQ Practice Set 2 Paper III Q3d: **See Practice Set 2 Paper 3, Q2 (full answer).**)

Introduction: definition and pathophysiology

- **VUR (vesicoureteral reflux)**: retrograde flow of urine from the bladder to the upper tract. **Primary**: deficient UVJ (ureterovesical junction) flap-valve; short intravesical tunnel; tunnel length to ureteric diameter should be about **5:1** (Paquin). Early budding gives a lateral orifice and dysplasia.
- **Secondary**: high bladder pressure from PUV (posterior urethral valves), neurogenic bladder or BBD (bladder and bowel dysfunction).
- In about **30% of children with UTI (urinary tract infection)** and 30% of siblings.
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more.
- **Infection, not reflux (not sterile reflux), injures/damages the kidney.** Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia. So management treats/targets infection risk, not the reflux itself.

International grading (VCUG, voiding cystourethrogram)

Grade	Findings
I	Non-dilated ureter only (ureter only, not dilated)
II	Ureter, pelvis and calyces reached; no dilatation, sharp fornices
III	Mild–moderate dilatation of ureter and pelvis; minimal forniceal blunting
IV	Moderate dilatation and tortuosity; fornices obliterated but **papillary impressions kept**
V	Gross dilatation and tortuosity; **papillary impressions lost**; intrarenal reflux

- **Also know:** grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

Diagram to draw: grades I–V and a Cohen cross-trigonal reimplant.

Evaluation

- History of febrile UTIs and BBD; BP (blood pressure), growth, spine, urine protein (proteinuria), creatinine.
- **RBUS (renal-bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG**: grade and urethra (excludes PUV). **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid) scan**: scars and split function; a defect persisting 6 months is a scar.
- **UDR (ureteral diameter ratio)** predicts resolution and breakthrough UTI better than grade.

Natural history

- Resolution falls with grade and age: grade II 63–85% resolve, grade III about 50%, **grades IV–V and bilateral III rarely**. **Bilateral grade IV persists most** (AUA (American Urological Association) 1997: about 10% resolve in 5 years). Infants with high grade still resolve fairly well.

Risk stratification (Hidas)

Low risk	High risk
Male, >5 years	Female, <1 year (infant)
Unilateral, grade I–II	**Bilateral**, grade III–V
No BBD, no scars	BBD, scars, multiple febrile UTIs

- Two-year breakthrough UTI on CAP (continuous antibiotic prophylaxis): 8.6% low, 26% intermediate, 62.5% high risk.

Medical management

- **Aim**: prevent febrile UTI and scars while VUR resolves.
- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed. BBD raises breakthrough UTI and lowers resolution and endoscopic success.
- **CAP (continuous antibiotic prophylaxis)**: one low nightly dose at night, a quarter of the treatment dose.

Drug / Agent	Age	Dose	Note
Trimethoprim or amoxicillin	<2 months (young infants)	—	Avoid sulfonamide (jaundice; neonatal jaundice)
TMP-SMX (trimethoprim-sulfamethoxazole)	>2 months	2 mg/kg TMP	Avoid under 2 months (kernicterus); used in RIVUR
Nitrofurantoin	>2 months	1–2 mg/kg	Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung, liver toxicity

- **RIVUR (2014)**, 607 children: recurrent UTI **23.6% → 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD; benefit mainly in high risk.
- **Swedish Reflux Trial** (grade III–IV, age 1–2 years): girls had febrile UTI 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.
- **Also know:** IRSC (International Reflux Study in Children) and Birmingham trials: surgery equalled medical care for scarring (new scars); PRIVENT: 6% absolute UTI reduction.

- **Circumcision** or topical steroid for phimosis: about 4 boys with VUR circumcised prevent one UTI (NNT (number needed to treat) about 4).
- After stopping CAP, 8–10% have a febrile UTI.
- **Guidelines**: AUA 2010 recommends CAP for VUR with BBD and treating BBD first; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology) give high-risk children CAP and intervene early after breakthrough infection (correct after breakthrough infection).

Indications for correction

- Breakthrough febrile UTI despite good bladder and bowel habits.
- New scars or falling function.
- Persistent high-grade (IV–V) bilateral VUR (girls traditionally corrected before puberty; benefit unproven).
- Non-adherence (poor adherence) or intolerance to CAP; parental choice.
- **Controversy:** asymptomatic bilateral high-grade VUR in an older boy with normal kidneys can be observed (controversial).

Case: 8-year-old boy, bilateral grade IV

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars**: observation or CAP with BBD therapy. **Breakthrough UTI, scars or falling function**: correct surgically.
- **Choice here: bilateral Cohen.**

```
Exclude PUV/neurogenic bladder, treat BBD
 -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA)
                                     No  -> observe or CAP; correct if UTI/scar
```

Surgical options and techniques

Principles: exclude secondary VUR; gentle handling; tension-free; tunnel 5:1; straight course.

1. Endoscopic injection (Dx/HA, dextranomer/hyaluronic acid)

- **STING** (subureteric Teflon injection): 2–3 mm below the orifice. **HIT** (hydrodistension implantation): inside the orifice, better (HIT beats STING). **Double HIT** about 90%.
- Aim for a "volcano" mound; day case. Repeat once or twice, then reimplant.
- Success per injection (Elder 2006): grade I–II about 79%, III 72%, **IV 63%**, V 51%.
- Antibiotics 3 months, ultrasound at 2–6 weeks.
- Recurrence about 20% at 2 years (20% recurred by 2 years); late obstruction reported.

2. Open ureteral reimplantation (success **95.9%** overall, Elder 1997)

Technique	Method / Tunnel	Key point: strength / weakness
Cohen (intravesical)	Cross-trigonal	Commonest, success over 98%; one bladder opening (cystotomy) for both sides; later ureteroscopy difficult/hard
Politano–Leadbetter (intravesical)	New superomedial hiatus above the old one	Long tunnel; blind hiatus risks bowel injury
Glenn–Anderson (intravesical)	Advanced towards/to bladder neck	Least kinking; short tunnel

Technique	Method / Tunnel	Key point: strength / weakness
Lich–Gregoir (extravesical)	Detrusorotomy, ureter laid in	No haematuria or spasm; **retention up to 10% after bilateral repair** (pelvic plexus injury)
RALUR / RALUR-EV (robot-assisted laparoscopic ureteral reimplantation; robot-assisted extravesical)	Extravesical; Y-shaped detrusorotomy	Less pain; results approaching open

Criteria for selection (best option for each situation)

Situation	Preferred option
Grades II–IV, parents want least invasive	Dx/HA injection (lower success at higher grade)
Bilateral VUR, small or trabeculated bladder	**Cohen** (both sides through one cystotomy)
Grade V, dilated ureter	Intravesical reimplant ± tapering (Politano–Leadbetter or Cohen)
Unilateral, older child	Lich–Gregoir or RALUR-EV
Bilateral extravesical	Nerve-sparing dissection; warn about retention
Duplex system	Common sheath reimplant
Failed injection	Reimplant (not harder after Dx/HA)
Neurogenic or PUV bladder (secondary VUR)	Treat the bladder first (anticholinergics, CIC (clean intermittent catheterisation))

Complications and follow-up

- Ultrasound at 6–12 weeks. Complications: transient obstruction, contralateral VUR (about 1 in 10, usually resolves). Persistent reflux usually resolves by 1 year. Obstruction early (oedema): stent or nephrostomy; late (ischaemia, kink): redo.
- Lifelong BP checks; scarred women risk pre-eclampsia.

Long-term outcome

Outcome	Facts
Renal scarring	About 15% get a febrile UTI within 2 years; about 15% of those scar. Risk highest under 1 year. Much "scarring" in boys with high grade is **congenital dysplasia**, not preventable by any treatment
Hypertension	Reflux nephropathy may be the commonest cause of childhood hypertension after essential hypertension (renin–angiotensin). **34–38%** of scarred children are hypertensive as adults. Correcting VUR does not lower BP; nephrectomy of a small scarred kidney helps selected cases
CKD (chronic kidney disease) and ESKD (end-stage kidney disease)	Now rare: RRT (renal replacement therapy) from reflux nephropathy fell from 1.9 to 0.6 per million (USRDS (United States Renal Data System), 1996–2014). Bilateral scars cause hyperfiltration, proteinuria, concentrating defect, tubular acidosis
Pregnancy	VUR in late pregnancy: pyelonephritis about one-third versus under 5%. Scars: about **3× hypertension, 8× pre-eclampsia**; bilateral worse. Impaired function: fetal loss. Reimplanted women still get UTIs
Growth; offspring	Catch-up growth after infection control; about 43% of screened offspring have VUR

- **Concept:** in adult life **the scar, not the reflux, carries the risk**, so correction does not remove it. Scarred patients need lifelong BP, proteinuria and creatinine checks.

```
Treat BBD -> low risk: observe | high risk (bilateral): CAP
Breakthrough UTI / new scar / persistent IV-V -> Dx/HA or Cohen
Scars -> lifelong BP and proteinuria checks
```

Key points to remember (including SAQ angle):

- Grade V loses papillary impressions; bilateral grade IV rarely resolves (persists most); bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI but not scarring.
- Treat BBD (bladder and bowel dysfunction) and give CAP (continuous antibiotic prophylaxis) first; correct for breakthrough febrile UTI (urinary tract infection), new scars or persistent grade IV–V.
- Endoscopic Dx/HA (dextranomer/hyaluronic acid) by HIT (hydrodistension implantation): least invasive; lower success at higher grades.
- Open reimplantation 95.9%; Cohen cross-trigonal reimplant (over 98%) suits bilateral VUR (vesicoureteral reflux) and does both sides through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- 34–38% of scarred children become hypertensive; correction does not lower BP.
- Scars bring 8× pre-eclampsia risk: the scar, not the reflux, matters.

Embryology of the penis and management of proximal hypospadias (including Koyanagi repair)

Asked as:

- Describe embryology of penis. Describe management of proximal hypospadias. — *Winter 2017 Paper III Q1 (LAQ)*
- Koyanagi Repair: Indication, procedure and complications? — *Summer 2021 Paper III Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

A. Embryology of the penis

Indifferent stage (up to 8–9 weeks)

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

Hormonal control

- SRY (sex-determining region Y) → testis → **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

Primordium (germ layer)	Male	Female
Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce)	Glans, corpora	Clitoris
Urethral folds	Fuse → penile urethra, median raphe	Labia minora
Labioscrotal swellings	Scrotum	Labia majora
Urethral plate (endoderm)	Whole urethra, including glanular	Vestibule

Urethral formation: the "double zipper"

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe.

- The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated.
- Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks.
- **Prepuce** fuses dorsally first; ventral fusion awaits urethral closure, hence the **dorsal hood** in hypospadias.
- Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site).
- **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

B. Proximal hypospadias: evaluation

- Meatus proximal shaft, penoscrotal, scrotal or perineal (10–25%). Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone). A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- No routine VCUG (voiding cystourethrogram) or renal ultrasound; prostatic utricle commoner.

C. Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects transient; evidence mixed, but two RCTs (randomised controlled trials) showed fewer complications.

D. Curvature correction (decides one- vs two-stage)

```
Deglove + excise dysplastic dartos + artificial erection
   ├─ straight        → urethroplasty
   ├─ residual <30°   → dorsal midline plication (12 o'clock)
   └─ >30°            → divide urethral plate
                          └─ still >30° → ventral corporotomy ± graft → staged repair
```

	Dorsal plication (Nesbit, Baskin)	Ventral lengthening
Method	Shortens the convex dorsal side	1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft
Use	Mild (≤15–30°)	Moderate–severe
Recurrence	28%	9%

E. Urethroplasty

Approach	Technique	Comment
One-stage **proximal TIP (tubularised incised plate)**	Plate kept, incised and tubularised	Recurrent curvature, stricture: largely abandoned for severe cases
One-stage **onlay island flap**	Inner preputial flap on preserved plate	Diverticulum, fistula
One-stage **TPIF (transverse preputial island flap, Duckett tube)**	Tube after plate division	High complications
Two-stage Bracka / STAG (staged tubularised autograft)	Stage 1: correct curvature, divide plate, quilt inner-prepuce or buccal graft. Stage 2 (≥6 months): tubularise, barrier layer	**Preferred for most proximal cases**; STAG lowest complications (about 23%)
Two-stage Byars	Dorsal preputial flaps moved ventrally, tubularised later	Complications 43–81%

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; scrotoplasty for transposition.

E2. Koyanagi repair (one-stage meatus-based parameatal foreskin flap): indication, procedure and complications

(proximal hypospadias also answered in 2017-Winter Paper 3, Q1)

Definition

- **One-stage repair of proximal hypospadias** with a **meatus-based parameatal foreskin flap** (Koyanagi). Skin strips round the meatus continue into the inner prepuce on both sides; they are swung ventrally and tubularised.
- **Principle:** the neourethra stays attached to the native meatus, so there is **no circular proximal anastomosis**, and the urethral plate can be divided to correct severe chordee.

Indications

- **Proximal hypospadias** (penoscrotal, scrotal, perineal) with **severe curvature** needing plate transection, and an **ample, unoperated prepuce**.
- **Contraindications:** prior circumcision or failed repair; small glans (give preoperative testosterone); curvature needing a ventral corporal graft (stage instead).

Procedure

```
Stay suture; artificial erection
→ incise round the meatus, along both sides of the plate to the
  corona and round the inner prepuce (meatus-based "wings")
→ deglove; divide the plate; excise dysplastic dartos
→ residual curvature <30° → dorsal midline plication
→ split the dorsal prepuce; bring both wings ventrally on dartos pedicles
→ join inner edges, then outer edges, over a catheter → tube
→ glanuloplasty, meatus at the tip
→ dartos barrier layer; skin cover ± scrotoplasty
→ stent 7–10 days; compressive dressing
```

- Modified and "neo-modified" versions add flap fixation and barrier layers to reduce fistula and diverticulum.

Complications

Complication	Cause
Fistula (commonest)	Long suture line; repair ≥6 months later
Meatal stenosis, stricture	Distal flap ischaemia
Diverticulum	No spongiosum; flap not fixed to the shaft
Glans dehiscence	Tension
Residual or recurrent curvature	Plication alone in severe cases
Flap necrosis, infection, poor cosmesis	Pedicle injury

- Proximal repairs overall have **23–68%** complications, so many now prefer **two-stage repair** (STAG (staged tubularised autograft), Bracka).

Also know: Duckett TPIF (transverse preputial island flap); onlay island flap; Byars two-stage (see table above).

Diagram to draw (Koyanagi): meatus-based foreskin wings split dorsally and joined ventrally into a tube.

F. Complications and outcomes

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** commonest (repair ≥6 months later; exclude distal obstruction), glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, diverticulum (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans).
- A long, rigid neourethra raises resistance (Poiseuille), causing fistula and diverticulum.
- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

Diagram to draw: double-zipper urethral formation; curvature algorithm.

Key points to remember:

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias plus undescended testis → DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka preferred; fistula commonest complication.
- Koyanagi: one stage; meatus-based parameatal foreskin flap.
- Koyanagi: plate divided; no proximal circular anastomosis.
- Koyanagi: fistula commonest; diverticulum, meatal stenosis.

Embryology and management of ectopic ureter.

Type: LAQ (25 marks)

Source: 2018-Summer Paper 3, Q1

Definition

- An **ectopic ureter** is any ureter, single or duplex, that does **not open on the trigone**. True ectopia means an orifice **at or distal to the bladder neck**.
- Commoner in girls. In girls most ectopic ureters drain the **upper pole of a duplex kidney**; in boys most are **single-system**.

Embryology

- **Week 4–6:** the **ureteric bud** arises from the **mesonephric (wolffian) duct** and induces the metanephric blastema to form the kidney.
- The duct segment distal to the bud is absorbed into the **urogenital sinus**; the ureteric orifice then moves cranially and laterally, the wolffian orifice caudally to the verumontanum.
- **Mackie–Stephens bud theory:**

Bud position on wolffian duct	Absorption	Orifice	Result
Too caudal	Early	High and lateral, short tunnel	VUR (vesicoureteral reflux)
Too cranial	Late or never	Low and medial (bladder neck, urethra) or stays on wolffian duct	**Ectopic ureter**, often obstructed

- A bud from the wrong place meets poor blastema, so **renal dysplasia increases with the degree of ectopia**.
- **Duplex system:** the upper-pole bud is more cranial, absorbed later, and ends **lower and more medial** (**Weigert–Meyer law**): upper pole obstructs; lower pole refluxes.

Sites of ectopia (the key sex difference)

	Female	Male
Sites (commonest first)	Urethra, vestibule, vagina; rarely cervix, uterus, rectum (via **Gartner duct**)	Posterior urethra, seminal vesicle; also vas, ejaculatory duct
Relation to sphincter	Can open **below** the external sphincter	Always **above** the external sphincter
Continence	**Continuous dribbling with normal voiding**	**Never incontinent**
Typical presentation	Constant dampness, perineal pus, UTI (urinary tract infection)	**Epididymo-orchitis**, UTI, flank pain, hydronephrosis

- **Zinner syndrome:** ectopic ureter into the seminal vesicle with ipsilateral renal agenesis or dysplasia and a seminal vesicle cyst.

Clinical features

- **Prenatal hydronephrosis**: most are detected this way.
- **Girl:** wet day and night despite normal voids; she is never dry for 30–60 minutes. Often misdiagnosed as voiding dysfunction.
- **Infection:** low-grade fever with spikes; urine culture may be negative because the infected moiety does not drain into the bladder.

- **Boy:** recurrent epididymitis in a young boy demands an upper-tract ultrasound.
- **Also know:** pain then perineal discharge; bilateral single-system ectopia with small bladder and wide bladder neck.

Evaluation

Test	What it shows
Ultrasound (first)	Dilated upper pole and ureter behind the bladder; parenchyma. Bladder view separates it from ureterocele
VCUG (voiding cystourethrogram)	Lower-pole or contralateral VUR ("drooping lily"), reflux into the ectopic ureter
DMSA (dimercaptosuccinic acid)	Gold standard for moiety function ("some" versus "none"); MAG3 (mercaptoacetyltriglycine) if drainage needed
MR (magnetic resonance) urography	Non-dilated upper pole, small dysplastic kidney, complex anatomy
Examination / cystovaginoscopy	Perineal orifice between urethra and vagina, cannulated for contrast

- **Also know:** clear urine leaking despite methylene blue in the bladder suggests an ectopic ureter; CT for a small unseen kidney.

Management

Goals: preserve renal function; remove infection, obstruction and reflux; achieve continence; with least morbidity. Start antibiotic prophylaxis in the neonate with significant dilation.

```
Neonate with sepsis or massive dilation
   → end cutaneous ureterostomy (or nephrostomy); DMSA later
DUPLEX system
   Upper pole non-functioning → upper-pole heminephroureterectomy
   Upper pole worth keeping   → ureteroureterostomy (UU) / pyeloureterostomy
   + lower-pole VUR           → common sheath reimplant
                                (or lower-pole reimplant + distal UU)
SINGLE system
   Functioning     → ureteral reimplant (± tapering)
   Non-functioning → nephroureterectomy (laparoscopic)
   Bilateral       → bladder neck reconstruction ± augmentation
```

- **Upper-pole heminephroureterectomy** (open, laparoscopic or robotic): upper-pole ureter as a handle, passed behind the hilum; ligate polar vessels; excise the pole; close over fat. Dissect on the ureter wall to protect the **lower-pole blood supply**. The stump is rarely a problem.
- **Ureteroureterostomy (UU):** end-to-side upper into lower pole ureter, distal (inguinal incision) or proximal (preferred with big size mismatch). **Stent the recipient lower-pole ureter** cystoscopically for identification. Now used even for non-functioning poles; feared hypertension has not been shown.
- **Common sheath reimplantation:** do not separate the two ureters (they share blood supply).

Complications

- Loss of lower-pole function (most serious); urinoma (up to 20% after minimally invasive partial nephrectomy, mostly harmless); stump infection; new reflux; persistent incontinence.

Recent advances

- Robotic UU and heminephrectomy; single-site surgery; non-dismembered side-to-side ureterovesicostomy in infants.

Diagram to draw: duplex kidney showing Weigert–Meyer orifices, and female versus male ectopic sites relative to the external sphincter.

Key points to remember:

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?

Type: LAQ (25 marks)

Source: 2020-Summer Paper 3, Q2

Development (summary; for detail see 2020-Summer Paper 1, Q2)

- **Weeks 4–6:** the ureteric bud arises from the mesonephric (Wolffian) duct. The common excretory duct is absorbed into the UGS (urogenital sinus), so ureter and duct open separately. The ureteric orifice moves cranially and laterally; the ejaculatory duct ends caudally in the prostatic urethra.
- **Bud position (Mackie–Stephens):** a bud arising too low reaches the UGS early → high, lateral orifice → short tunnel → **VUR (vesicoureteral reflux)**, often with a dysplastic kidney. A bud arising too high → caudal, medial, ectopic orifice.
- **Weigert–Meyer** (duplex): lower-pole orifice lateral, refluxes; upper-pole caudal-medial, ectopic.

Clinical anatomy of the VUJ (vesico-ureteric junction)

- **Course:** juxtavesical ureter in **Waldeyer's sheath** → oblique intramural segment through the detrusor hiatus → submucosal tunnel → orifice (caudal and medial to the entry).
- **Intramural length 1.2–2.5 cm in adults** (0.5–0.8 cm in neonates). **Narrowest part** of the ureter (about 3–4 mm): commonest stone impaction site.
- **Trigone:** superficial trigone (ureteric longitudinal muscle, to the verumontanum); deep trigone (Waldeyer's sheath plus detrusor, to the bladder neck). Base is the interureteric ridge.
- **Antireflux mechanism:**
  - **Passive flap valve:** filling compresses the submucosal ureter against firm detrusor backing.
  - **Paquin:** tunnel length to diameter **5:1** in normal VUJs vs 1.4:1 in refluxing ones.
  - **Active:** ureteric peristalsis opens the tunnel for each bolus, then it closes.
  - **Bladder pressure:** high storage or voiding pressure overwhelms a normal valve.
- **Orifice (Lyon):** position A (medial trigone) to C (lateral); shape cone (normal) → stadium → horseshoe → **golf-hole**. The more lateral and open, the more reflux.
- **Relations:** vas (male) or uterine artery (female) crosses the ureter. **Pelvic plexus nerves lie dorsomedial to the VUJ**; wide extravesical dissection causes voiding dysfunction.
- **Clinical:** Hutch diverticulum at the hiatus; "fish-hook" ureter with BPH (benign prostatic hyperplasia).

Grade 3 VUR (International Reflux Study)

- Mild–moderate dilatation of ureter, pelvis and calyces, minimal forniceal blunting.

VUR in adults

- **Adults rarely lose function from reflux alone**, unless high bladder pressure, stasis and UTI (urinary tract infection) coexist. The **scar**, not the reflux, carries the risk. Spontaneous resolution is unlikely.
- **Causes:** persistent primary VUR; **secondary**: BOO (bladder outlet obstruction), neurogenic bladder, resection near the orifice, ureterocele incision, paraureteral diverticulum, transplant ureter, TB (tuberculosis).
- **Presentation:** recurrent febrile UTI, pyelonephritis (especially in pregnancy), loin pain on voiding, hypertension, proteinuria, CKD (chronic kidney disease).

Evaluation

- **VCUG (voiding cystourethrogram):** grade, bladder, diverticulum, outlet.
- US (ultrasound) and **DMSA (dimercaptosuccinic acid) scan** for scars and split function; MAG3 (mercaptoacetyltriglycine) renogram if obstruction is suspected.
- **UDS (urodynamics):** compliance, voiding pressure, residual. Cystoscopy; blood pressure, creatinine, urine protein.

Management

1. Treat secondary causes first (relieve BOO; CIC (clean intermittent catheterisation) and antimuscarinic for neurogenic bladder), then reassess.

2. Primary grade 3 VUR:

Situation	Management
Incidental; no UTI, no scars	Observe: timed voiding, bowel care, prompt UTI treatment
Recurrent febrile UTI, new scars, falling function	**Correct the reflux**
Woman planning pregnancy with scars or prior pyelonephritis	Consider correction before pregnancy
Poorly functioning scarred kidney with infection or hypertension	Nephrectomy

- Low-dose prophylaxis selectively; adult evidence is limited.

3. Correction:

- **Endoscopic injection** (Dx/HA (dextranomer/hyaluronic acid)): day case, repeatable, but less effective in adults (see 2020-Summer Paper 4, Q3c).
- **Ureteroneocystostomy** (open, laparoscopic or robotic): gentle handling, tension-free, straight course, tunnel **5:1**.

Technique	Type	Note
Lich–Gregoir / **RALUR-EV** (robot-assisted extravesical)	Extravesical detrusorotomy	Suits adults; no cystotomy; stay close to ureter (nerves)
Politano–Leadbetter	Intravesical, new hiatus	Long tunnel
Cohen cross-trigonal	Intravesical	Over 98% success; later retrograde access hard

- **Psoas hitch** if length is needed; tapering seldom needed in grade 3.

Complications and follow-up

- Persistent or contralateral reflux, obstruction, leak, retention after bilateral extravesical repair.
- US at 6–12 weeks; lifelong blood pressure and renal checks if scarred.

Diagram to draw: oblique intramural ureter in Waldeyer's sheath with submucosal tunnel and detrusor backing.

> **Key points to remember:**
>
> - Bud too low → lateral orifice → short tunnel → VUR.
> - Adult intramural ureter 1.2–2.5 cm; Paquin 5:1.
> - In adults the scar, not the reflux, carries the risk; exclude secondary causes first.
> - Correct for recurrent pyelonephritis, new scars or pregnancy plans.

Section 4 · Paediatric Urology 1 — Short answer questions (SAQ)

Bilateral Wilms tumour: treatment strategies

Asked as:

- Treatment strategies for bilateral Wilm's tumour — *Summer 2016 Paper III Q3c (SAQ)*
- Management of bilateral Wilms Tumor? — *Summer 2020 Paper III Q3e (SAQ)*

Type: SAQ (10 marks)

Introduction

- **Synchronous bilateral WT (Wilms tumour)** is **5–7%**: **COG (Children's Oncology Group) stage V**. Each kidney is staged separately.
- Linked to (syndromes) **WAGR (Wilms, aniridia, genitourinary anomalies, retardation)**, Denys–Drash (about 50% risk) and BWS (Beckwith–Wiedemann syndrome); **nephrogenic rests in over 90%**.
- Aim (goal): **cure while preserving renal tissue** (nephrons). Renal failure is about 15% at 15 years; ESRD (end-stage renal disease) mainly follows (from) bilateral nephrectomy.

Evaluation

- Ultrasound (US) → contrast CT (computed tomography) or MRI (magnetic resonance imaging) abdomen, chest CT; IVC (inferior vena cava) and coagulation checks; syndrome review.
- **No upfront biopsy and no upfront nephrectomy.**

COG approach (AREN0534)

```
VCR + AMD + DOX x 6 weeks (no upfront biopsy)
        |
   imaging at week 6
   /              \
NSS feasible     not feasible
 -> surgery       -> 6 more weeks chemotherapy
                   -> surgery by week 12
                   -> if poor response: open BILATERAL biopsy
```

```
VCR + AMD + DOX × 6 weeks (no biopsy)
→ imaging at week 6
   ├─ resectable → bilateral NSS
   └─ not → 6 more weeks → surgery by week 12
        poor response → bilateral OPEN biopsy
→ further therapy by each kidney's stage and histology ± flank RT
```

- VCR = vincristine, AMD = actinomycin D (dactinomycin), DOX = doxorubicin; NSS = nephron-sparing surgery, RT = radiotherapy.
- **Why stop at 12 weeks**: in NWTS-4 (National Wilms Tumor Study 4), prolonged chemotherapy in poor responders delayed surgery and led to anaplasia (some tumours became anaplastic). Non-shrinkage (non-shrinking tumour) usually means differentiation (is often differentiated), cured by excision.
- **Why open, bilateral biopsy**: anaplasia is discordant (differs) between kidneys in about 4 of 5.

Surgery

- **Bilateral NSS** (partial nephrectomy or enucleation); frozen section; lymph node sampling; avoid spill.
- Compressed normal parenchyma is often underestimated, so try NSS in every case. Expert centres achieve bilateral NSS in over 90%. Enucleation: positive margins 23–36%, good outcome with (flank) radiotherapy.
- Unresectable both sides: bilateral nephrectomy and dialysis only for refractory tumour; transplant after **1–2 years** tumour-free.

Adjuvant

- Guided by worst histology and margins: anaplasia or positive margin → intensified chemotherapy and flank radiotherapy.

Outcomes

- AREN0534: surgery by week 12 in 84%; bilateral NSS 39%; **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%**, against (vs) 56% and 80.8% in (on) NWTS-5.

SIOP (International Society of Paediatric Oncology)

- Same principle (no biopsy, NSS by week 12) but **two drugs, VCR + AMD**, if non-metastatic.

Follow-up: long term (lifelong): blood pressure, proteinuria, creatinine (renal function), imaging (late relapse occurs / possible).

Diagram to draw: bilateral tumours with planned partial nephrectomies; AREN0534 timeline (weeks 0, 6, 12).

> **Key points to remember:**
>
> - Stage V, 5–7%; no upfront nephrectomy or biopsy.
> - COG: three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12; SIOP: two drugs.
> - AREN0534: 4-year OS about 95%.
> - Bilateral NSS to avoid ESRD.

Congenital ureteral anomalies

Type: SAQ (10 marks)

Source: 2017-Summer Paper 3, Q3f

Embryology basis

- The **ureteric bud** arises from the **mesonephric (wolffian) duct** at about week 5 and induces the metanephric blastema.
- Anomalies arise from **abnormal budding**, **failed distal canalisation**, **abnormal muscle maturation** or **persistent embryonic vessels**.

Classification

Category	Anomaly	Key points
Number	**Duplication**: incomplete (bifid, Y-ureter) or complete	About 1 in 125; unilateral 6× bilateral. Lower pole: VUR (vesicoureteric reflux), PUJ (pelviureteric junction) obstruction. Upper pole: ectopia, ureterocele
Termination (orifice)	**Ectopic ureter**	Opens at or below the bladder neck. **Females**: bladder neck to vagina, uterus; **continuous dribbling with normal voiding**. **Males**: always above external sphincter (posterior urethra, seminal vesicle, vas); **never incontinent**; epididymitis. Zinner syndrome
Structure	**Ureterocele**	Cystic dilation of distal submucosal ureter (Chwalle membrane). Intravesical vs ectopic (Glassberg); cecoureterocele (Stephens)
	PUJ obstruction	Intrinsic aperistaltic segment (infants) or crossing lower-pole vessel (older children)
	Megaureter (≥7 mm)	Obstructed, refluxing, non-refluxing non-obstructed, refluxing-obstructed; primary obstructive = adynamic distal segment
	Primary VUR	Short submucosal tunnel, lateral orifice
Position	**Preureteral (retrocaval) vena cava**	Persistent subcardinal vein traps the right ureter; fishhook (S) shape at L3–L4; type I obstructs, type II does not
	Retroiliac ureter; herniation	Rare

Weigert–Meyer law

- In complete duplication the **upper-pole ureter opens lower and more medial** (ectopic, obstructs, ureterocele); the **lower-pole ureter opens higher and lateral** (refluxes).
- Reason: the upper-pole bud arises more cephalad on the wolffian duct, is incorporated later and ends more caudally.
- **Also know:** blind-ending ureter, inverted-Y ureter, triplication, congenital stricture, ureteral valves, fibroepithelial polyp.

Evaluation

- Ultrasound, VCUG (voiding cystourethrogram), **DMSA (dimercaptosuccinic acid)** scan, MR (magnetic resonance) urography.

Management principles

- **Ureterocele:** endoscopic incision, upper-pole heminephrectomy or ureteroureterostomy.
- **Ectopic ureter:** heminephroureterectomy if non-functioning; otherwise ureteroureterostomy or reimplantation.

- **PUJ obstruction:** pyeloplasty. **Megaureter:** observe most; taper and reimplant (5:1) if function falls.
- **Retrocaval ureter:** divide, bring anterior to the cava, excise the aperistaltic segment, reanastomose.

Diagram to draw: complete duplication showing Weigert–Meyer orifice positions.

Key points to remember:

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.

Retrocaval ureter

Asked as:

- Retrocaval ureter. — *Winter 2017 Paper III Q3d (SAQ)*
- Retrocecal ureter — *Practice Set 1 Paper III Q3c (SAQ)* (also answered in 2017-Winter Paper 3, Q3d)

Type: SAQ (10 marks)

Assumed meaning: retrocaval (circumcaval) ureter; "retrocecal" is taken as a typo.

Definition

- The **right ureter** passes **behind the IVC (inferior vena cava)**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed (called) "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.

Embryology

- Paired fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney: supracardinal veins dorsally, subcardinal veins ventrally.
- Normally the right **supracardinal** vein, **dorsal** to the ureter, forms the infrarenal IVC, so the ureter lies in front.
- If a vein **ventral** to the ureter persists as the IVC (the lumbar part of the ventral **subcardinal** vein in Campbell; "persistent posterior cardinal vein" in many texts), the ureter is trapped behind it.
- Variant: double right IVC (ureter between limbs).

Incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**.
- Presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis. Associations: horseshoe kidney, left renal anomalies.

Classification (Bateson & Atkinson)

	Type I (low loop), commoner	Type II (high loop)
Crossing	Lower, at L3–L4	Higher, at (near the) renal pelvis level
Shape	**"Fishhook" / reverse-J**, S-curve	Smooth horizontal **sickle** curve
Obstruction	Moderate–severe, at the iliopsoas edge	Mild or none (pelvis also crosses)

Investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) alternative.
- IVU (intravenous urography): fishhook, ureter lost (not seen) beyond the J hook. Retrograde pyelography: **S-curve** at L3–L4.
- **Diuretic renogram:** obstruction and split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**; pyeloplasty fails.

Management

- **Non-obstructed (often type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it anterior to (in front of) the IVC, spatulated tension-free anastomosis

over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; preserve blood supply.

- **Laparoscopic or robotic** repair (transperitoneal or retroperitoneal) now preferred: equal success, faster recovery. Nephrectomy if non-functioning.
- **Also know:** retroiliac ureter (L5–S1, vasal anomalies); mimics: spermatic / gonadal or lumbar vein, psoas (iliopsoas) tendon.

Diagram to draw: right ureter hooking behind the IVC with fishhook pelvis.

Key points to remember:

- Venous anomaly (persistent subcardinal vein); right side; 1 in 1500; male 3:1.
- Type I fishhook (obstructs); type II sickle often does not.
- CT urography; repair by anterior transposition and (with) ureteroureterostomy.

Maldescended (undescended) testes and treatment algorithm

Asked as:

- Treatment algorithm of undescended testis. — *Summer 2018 Paper III Q3a (SAQ)*
- Maldescended testes — *Practice Set 5 Paper III Q3b (SAQ)*

Type: SAQ (10 marks)
(algorithm also in 2018-Summer Paper 3, Q3a; bilateral cases in 2024-Winter Paper 1, Q2)

Taken as any testis not in a stable scrotal position: undescended, ectopic, retractile and acquired.

Definitions

Type	Description
Undescended (cryptorchid)	Arrested on the normal path: abdominal, canalicular, prescrotal
Ectopic	Off the normal path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, transverse (crossed)
Retractile	Scrotal testis pulled up by the cremaster; stays down after traction
Acquired (ascending)	Previously scrotal; presents at 8–11 years
Vanished / agenesis	Lost in utero (torsion) / never formed

Epidemiology and aetiology

- 1–4% of term boys, more in preterm; most descend by 6 months. 75–80% palpable; 60–70% unilateral.
- Descent needs **INSL3 (insulin-like peptide 3)** and **androgen** acting on the gubernaculum. Risk factors: prematurity, low birth weight, syndromes (prune belly, cerebral palsy).

Evaluation

- Warm room, supine and cross-legged; sweep from the anterior superior iliac spine to the scrotum to find the lowest position. Sustained traction fatigues the cremaster and separates retractile from undescended.

Management principles (AUA (American Urological Association) 2014; EAU (European Association of Urology) similar)

- **Observe for 6 months** (corrected age) for spontaneous descent; refer if still undescended.
- **Orchidopexy after (from) 6 months, preferably by 12 months, and by 18 months at the latest.** Reason: germ-cell and Leydig-cell loss starts in infancy; early surgery improves testicular growth and lowers cancer risk.
- **No ultrasound or MRI (magnetic resonance imaging)**: inaccurate (ultrasound sensitivity about 45% for non-palpable testes); never changes the operation; delays surgery.
- **No hormonal therapy** (hCG (human chorionic gonadotrophin) or LHRH (luteinising hormone-releasing hormone)): descent in about 20% only (LHRH works in about 20%); hCG may increase germ-cell apoptosis.
- **Bilateral non-palpable testes** (especially with hypospadias or small phallus; bilateral non-palpable or with hypospadias): urgent **karyotype and hormones** (17-OHP (17-

hydroxyprogesterone), AMH (anti-Müllerian hormone)) to exclude **CAH (congenital adrenal hyperplasia)** and other DSD (differences of sex development).

Algorithm

```
Boy with empty scrotum
  ├─ Bilateral non-palpable ± abnormal phallus → DSD work-up (karyotype, hormones,
electrolytes)
  └─ Otherwise → re-examine at 6 months corrected age
       Still undescended → refer → examination under anaesthesia (EUA)
         ├─ Retractile → yearly review (risk of ascent)
         ├─ Palpable (undescended or ectopic) → inguinal or scrotal (Bianchi) orchidopexy
         └─ Non-palpable → diagnostic laparoscopy
              ├─ Testis seen → laparoscopic orchidopexy
              │       short vessels → Fowler–Stephens (1 or 2 stage) / Shehata
              ├─ Vessels + vas enter ring → inguinal exploration (excise nubbin)
              └─ Blind-ending vessels → stop (vanished testis)
```

- **Scrotal nubbin with contralateral hypertrophy** may start with scrotal exploration.

Key operative points

- **Inguinal orchidopexy:** free the cord, ligate the processus, retroperitoneal mobilisation ± **Prentiss manoeuvre**, subdartos pouch.
- **Fowler–Stephens:** divides testicular vessels; testis survives on vasal collaterals; for testes not within 2–4 cm of the ring. Primary orchidopexy, when possible, is more successful.
- **Postpubertal intra-abdominal testis with normal contralateral testis: orchidectomy** (poor function, cancer risk); offer prosthesis.

Sequelae and follow-up

- Check position and size at 6 months; teach **self-examination** after puberty.
- **Infertility:** paternity about four-fifths after unilateral, one-third after bilateral.
- **Cancer:** 2–5-fold TGCT (testicular germ cell tumour) risk (cancer risk 2–5 times), higher with late surgery; teach self-examination.
- **Also know:** hernia, torsion, trauma.

Diagram to draw: laparoscopic findings at the internal ring (testis, vessels entering ring, blind-ending vessels); line of descent with ectopic sites.

> **Key points to remember:**
>
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Operate (orchidopexy) from 6 months, preferably by 12 months (6–12 months) and by 18 months at the latest.
> - No imaging, no hormones.
> - Non-palpable: EUA then diagnostic laparoscopy.
> - Bilateral non-palpable: exclude CAH/DSD first (DSD work-up first).

Autosomal recessive polycystic kidney disease.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 3, Q3c

Definition and genetics

- **ARPKD (autosomal recessive polycystic kidney disease)**: bilateral, symmetrical renal enlargement from **dilated collecting ducts**, **always with congenital hepatic fibrosis**.
- Gene **PKHD1 on chromosome 6p**; protein **fibrocystin (polyductin)** on the **primary cilium** of tubular and bile-duct cells, so it is a **ciliopathy**.
- Incidence about 1 in 10,000–50,000 live births. Siblings have a 25% risk.
- "Infantile" is a misnomer: a **spectrum**; earlier onset means more severe disease.

Pathology

- Radially arranged dilated collecting ducts (**microcysts**).
- **Liver:** periportal fibrosis, biliary ectasia; **Caroli syndrome** if intrahepatic ducts dilate.

Clinical features

Age	Features
Fetus	Large echogenic kidneys, **oligohydramnios** → Potter facies, limb deformity, **pulmonary hypoplasia**
Neonate	Huge, smooth, hard flank masses; respiratory distress; rising creatinine; **30–50% die early**
Child	Hypertension, CKD (chronic kidney disease), growth failure, anaemia, osteodystrophy
Older child	**Portal hypertension**: varices, splenomegaly, cholangitis; liver function stays normal

- No association with renal tumours.

Investigations

- **Ultrasound:** bilateral very large, **homogeneously hyperechoic** kidneys (brighter than liver) from countless microcyst interfaces; discrete cysts appear with age.
- **Parents' ultrasound** normal (excludes ADPKD); **PKHD1 genetic testing**; liver ultrasound.
- **Differential:** neonatal ADPKD (autosomal dominant PKD), bilateral hydronephrosis, MCDK (multicystic dysplastic kidney), glomerulocystic disease, bilateral tumours, renal vein thrombosis.

Management (no cure)

- **Neonatal:** ventilation, fluid and electrolyte care, antihypertensives (ACE (angiotensin-converting enzyme) inhibitors).
- **CKD care:** nutrition, growth hormone, anaemia and bone treatment; dialysis and **kidney transplantation**.
- **Nephrectomy** (breathing, feeding, blood pressure, room for peritoneal dialysis): cautiously; **very early bilateral nephrectomy** (first 3 months) had the worst neurological outcomes (ARegPKD registry).
- **Liver:** variceal sclerotherapy, portosystemic shunt; **combined liver–kidney transplant** for severe liver disease.
- **Genetic counselling** and prenatal diagnosis.

ARPKD versus ADPKD

	ARPKD	ADPKD
Gene	PKHD1 (chromosome 6)	PKD1 (16), PKD2 (4)
Age	Perinatal	30–50 years
Cysts from	Collecting duct	Whole nephron
Ultrasound	Symmetric, homogeneous, echogenic	Large discrete cysts, asymmetric
Liver	Fibrosis (always)	Cysts
Other	None	Berry aneurysm, mitral prolapse

Diagram to draw: cut kidney with radial dilated collecting ducts.

Key points to remember:

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

Endoscopic correction of VUR (STING procedure): indications and complications

Asked as:

- Endoscopic correction of VUR : indications and complications — *Summer 2019 Paper III Q3e (SAQ)*
- Sting procedure. — *Winter 2024 Paper III Q3f (SAQ)* (also answered in 2020-Summer Paper 4, Q3c)

Type: SAQ (10 marks)

Taken as STING (subureteric Teflon injection), the endoscopic treatment of VUR (vesicoureteral reflux).

Introduction / Definition and principle

- Endoscopic injection of a **bulking agent** beneath the ureteric orifice. The mound supports the intramural ureter and restores the **flap-valve**.
- A **bulking agent** injected under the ureteric orifice supports and lengthens the intramural tunnel, restoring the flap-valve in VUR (vesicoureteral reflux).
- First done with Teflon (1981). Popularised by **O'Donnell and Puri, 1986**. **Dx/HA (dextranomer/hyaluronic acid, Deflux)** is now standard.

Bulking agents

Agent	Particle / Property	Problem
Teflon (PTFE, polytetrafluoroethylene paste)	90% <40 µm (particles <40 µm), permanent	**Migration**; never FDA (US Food and Drug Administration) approved
Dx/HA (dextranomer/hyaluronic acid, Deflux)	Microspheres **80–250 µm**; gel resorbed	Volume loss about 23%; late recurrence; FDA approved 2001; standard agent
Polyacrylate (Vantris)	300 µm, permanent	Late ureteric obstruction

Indications

- Persistent primary VUR (grades II–IV) with **breakthrough febrile UTI** (urinary tract infection) on prophylaxis, **new scars**, poor adherence or parental preference.
- Older girls; residual VUR after reimplantation; duplex or secondary VUR (lower success).
- **Relative contraindications:** grade V, **UVJ (ureterovesical junction) obstruction**, untreated BBD (bladder and bowel dysfunction), large diverticulum.

Technique

1. Antibiotic at induction; offset-lens cystoscope; 3.7–5 Fr needle; bladder partly filled.
2. **STING:** needle enters **2–3 mm below the orifice at 6 o'clock**, bevel up, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.
4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding (implant extrudes).

Technique / Method	Needle entry	Result
STING (subureteric Teflon injection)	2–3 mm below orifice, at 6 o'clock	Baseline
HIT (hydrodistension implantation technique)	Inside the hydrodistended (distended) orifice	Better than STING, all grades
Double HIT	Two intraureteric sites	About 90% resolution

Results / Success

- Low-grade VUR about 90% after one injection.
- **Elder 2006 meta-analysis** (one injection): grade I–II about 79%, III 72%, IV 63%, V 51%; repeat injections raise success.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after initial success.

Complications

- **Failure or recurrence** (commonest): repeat injection (1–2), then reimplantation (reimplant).
- **New contralateral VUR** after unilateral injection.
- **Ureteric obstruction:** early or **late-onset**, so long-term ultrasound.
- Febrile UTI despite negative VCUG (voiding cystourethrogram).
- **Calcified mound** (over half) mimics a ureteric stone. Teflon migration.

Follow-up

- Day case; antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound ± VCUG (voiding cystourethrogram) at 3 months.

Diagram to draw: STING versus HIT needle positions — needle sites: STING below the orifice, HIT and double HIT inside the tunnel, with volcano mound.

Key points to remember:

- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
- Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction and calcified mound; calcified mound mimics a stone.

Snodgrass (TIP) repair for hypospadias, including extended indications

Asked as:

- Snodgrass repair for Hypospadias. — *Winter 2023 Paper III Q3c (SAQ)*
- Extended indications of Shodgrass repair in hypospadias — *Practice Set 5 Paper III Q3c (SAQ)*
- Snodgrass Repair for distal hypos. — *Practice Set 5 Paper III Q3d (SAQ)* (also answered in 2023-Winter Paper 3, Q3c)

Type: SAQ (10 marks)

"Shodgrass" is taken as Snodgrass TIP (tubularised incised plate) repair. Technique: see Practice Set 5 Paper 3, Q3d. "Distal hypos." is taken as distal hypospadias.

Introduction

- **TIP (tubularised incised plate) urethroplasty**, described by Snodgrass in 1994 (based on Rich's hinging incision, 1989).
- Designed for distal hypospadias, it now covers (is used for) **over 90% of distal and about 80% of midshaft repairs**.
- Distal hypospadias forms 70–85% of cases.

Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate, so (so that) it can be tubularised without tension.
- The incised raw area **heals by re-epithelialisation** (the incision re-epithelialises) without a graft, giving a **vertical slit-like meatus**.

Indications, prerequisites (for any extended use) and assessment

- Distal and midshaft hypospadias, and redo cases with a healthy plate, when curvature is mild (<30° after degloving) and corrected by degloving or (dorsal) plication **without dividing the plate**.
- Supple, well-vascularised plate of adequate width and depth; adequate plate and glans. **Glans width <14 mm** raises complications; give preoperative testosterone (to enlarge it) — i.e. **glans width ≥14 mm** needed.
- **POST (plate objective scoring tool):** a low ratio (AB/BC ratio) of neomeatus length to glans fusion length means a poor plate, which favours (favouring) an inlay graft.
- Timing: 6–12 months of age.
- Not for severe curvature needing plate division; proximal TIP is abandoned.

Extended indications

Indication	Rationale	Caveat
Midshaft hypospadias	Same principle	Good results
Proximal hypospadias with mild curvature, good plate	Single stage; no flaps	More fistula, stricture, recurrent curvature; Campbell-Walsh authors now prefer staging
Redo or failed repairs with preserved plate	Plate often survives	Worse than primary; inlay graft if scarred
MIP (megameatus intact prepuce)	Wide, deep plate	GAP (glans approximation procedure) alternative

Indication	Rationale	Caveat
Narrow or shallow plate: TIP + **dorsal inlay graft** ("Snodgraft")	Preputial or buccal graft fills the incision, so it cannot contract	Less meatal stenosis
Adults, circumcised boys	No preputial tissue needed	Success similar

- **Also know:** in redo cases with poor blood supply a dorsal buccal inlay beats onlay or tube; incise-and-graft is also used in adult urethral stricture (Asopa dorsal inlay) and female urethroplasty.

Contraindications

- Curvature >30° needing plate division; absent or scarred plate; penoscrotal transposition. Use staged repair (Bracka, STAG [staged tubularised autograft], Byars).

Operative steps

1. Glans stay suture; tourniquet.
2. U-incision around the plate and meatus; circumscribe and deglove; artificial erection.
3. Separate the glans wings deeply from the plate.
4. **Midline incision of the plate** from (inside) the meatus to the end of the plate, down to the corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, from distal to proximal, stopping 2–3 mm short (the first stitch about 3 mm from the plate end leaves an oval meatus) to leave an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier.
7. Two-layer glansplasty; mucosal collar; skin closure (Byars flaps) or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin for spasm.

Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, meatal stenosis 3.6%, stricture 1.3%.
- **Meatal stenosis is commoner than after Mathieu**; glans dehiscence also occurs.
- Distal repair success 83–95%.
- **Proximal and redo cases do (did significantly) worse**; a dartos (barrier) layer improves results.

Modifications

- **Dorsal inlay graft** (inner preputial skin) for a shallow plate; lateral plate incisions to 12 mm width.
- **Also know:** MAGPI (meatal advancement and glanuloplasty incorporated), Mathieu, Thiersch–Duplay, onlay flap.

Diagram to draw: cross-sections showing plate incision, inlay graft, tubularisation and dartos cover.

Key points to remember:

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate → dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

Ureterocele

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 3, Q3c (ectopic ureter also answered in 2018-Summer Paper 3, Q1)

Definition

- Cystic dilatation of the distal ureter, in the bladder or spanning bladder neck and urethra; **never wholly urethral**, never joined to Wolffian structures.
- Commoner in girls (about 4:1); about 80% drain the **upper pole of a duplex kidney** (Weigert–Meyer rule: upper pole orifice caudal-medial); about 10% bilateral.
- **Cause:** persistent **Chwalla membrane**; upper pole often dysplastic.

Classification

System	Basis	Types
Glassberg (most useful)	Position	**Intravesical** vs **ectopic (extravesical)**: part at bladder neck or urethra
Stephens	Orifice morphology	Stenotic, sphincteric, sphincterostenotic, **cecoureterocele** (cavity extends under urethra), blind, non-obstructed
Churchill	Function	Upper pole only / whole kidney / contralateral at risk

Clinical features

- Antenatal hydronephrosis (most); febrile UTI (urinary tract infection) or urosepsis; adult single-system type with stone.
- **Prolapse:** smooth interlabial mass in girls; BOO (bladder outlet obstruction).
- **No incontinence when untreated** (unlike female ectopic ureter).

Evaluation

- **Ultrasound:** thin-walled cyst in bladder, dilated upper pole and ureter.
- **VCUG (voiding cystourethrogram):** lower pole reflux ("drooping lily"), eversion (weak floor), prolapse.
- **DMSA (dimercaptosuccinic acid):** upper pole function; cystoscopy before incision.

Management

- **Goals:** preserve function; remove infection, obstruction, reflux; keep continence.
- **CAP (continuous antibiotic prophylaxis)** from birth if hydronephrotic.
- **Observation:** small, non-obstructing, low-grade reflux, non-functioning upper pole.

Option	Indication / result
TUI (transurethral incision)	Sepsis, neonates, intravesical type. Low transverse full-thickness incision; relieves obstruction in 78–97%; new reflux 0–50%. Intravesical without reflux: 70–80% cured; ectopic: secondary surgery up to 70%
Upper pole heminephrectomy	Non-functioning upper pole, no reflux; secondary surgery 40–50%
Ureteroureterostomy / pyeloureterostomy	Functioning upper pole
Excision + common sheath reimplant ± bladder neck repair	Persistent reflux, ectopic or cecoureterocele; persisting reflux 5–10%
Total reconstruction	Older child, non-functioning upper pole, lower pole reflux

- **Also know:** "watering-can" laser punctures (less reflux); pseudoureterocele; ureterocele disproportion.

Diagram to draw: duplex kidney with upper pole ureterocele and lower pole reflux.

> **Key points to remember:**
> - Upper pole of duplex; never wholly urethral; Chwalla membrane.
> - Intravesical vs ectopic predicts TUI success.
> - TUI relieves obstruction in 78–97%; new reflux 0–50%.

Primary obst. megaureter.

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 3, Q3f

Assumed meaning: primary obstructive megaureter (POM).

Definition

- **Megaureter:** ureter diameter **≥7 mm** (normal up to 16 years is 5.0–6.5 mm). It is a descriptive term.
- **Classification:** primary (intrinsic) or secondary (neurogenic bladder, outlet obstruction); and **obstructed, refluxing, non-refluxing non-obstructed, or refluxing and obstructed**.
- **POM:** an **adynamic juxtavesical segment** at the UVJ (ureterovesical junction) blocks peristalsis and causes functional obstruction above.

Pathogenesis

- Ureteric muscle matures cranially to caudally; the juxtavesical segment matures last, even after birth. This explains spontaneous resolution.
- Excess collagen, thick muscle sleeve, abnormal interstitial cells of Cajal.

Clinical features

- 10–23% of antenatal upper-tract dilatation. Boys, **left side**; **bilateral 25%**; contralateral dysplasia or obstruction 10–15%.
- Mostly asymptomatic; or UTI (urinary tract infection), pain, haematuria, stones.

Evaluation

- **Ultrasound:** hydroureteronephrosis.
- **VCUG (voiding cystourethrogram):** excludes reflux and PUV (posterior urethral valves).
- **MAG3 (mercaptoacetyltriglycine) renogram:** split function; washout T½ unreliable (false positive up to 44%).
- MRU (magnetic resonance urography) for anatomy.

Management

Conservative (most)

- CAP (continuous antibiotic prophylaxis) in infants; serial ultrasound and renograms; most resolve.
- Diameter <10 mm: virtually never needs surgery; ≥10 mm: a substantial minority do.
- Follow to early adulthood.

Indications for surgery

- Recurrent febrile UTI; progressive dilatation; **DRF (differential renal function) <40%**; **fall in DRF ≥5%**.

Infant (small bladder, huge ureter): temporise

- Cutaneous ureterostomy; refluxing reimplant; nephrostomy; **endoscopic balloon dilatation with double-J** for 2–6 months (70–80% success; complications up to a third).

Definitive repair (child)

- Excise stenotic segment → straighten → **taper** → **non-refluxing reimplant at 5:1** (Cohen or Politano–Leadbetter; extravesical for huge ureters). Stent afterwards.

Tailoring	Method	Use
Starr plication	Imbricate lateral wall	Moderate (<1.75 cm)
Kalicinski folding	Exclude and fold lumen	Moderate
Hendren excision	Excise lateral wall	Massive; risk of ischaemia

- Success over 90%.
- **Complications:** obstruction (stent first), VUR (vesicoureteral reflux; observe or inject), persistent dilatation.

Diagram to draw: dilated ureter above narrow adynamic segment; tapered reimplant.

> **Key points to remember:**
> - Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
> - Surgery: UTIs, progression, DRF <40% or fall ≥5%.
> - Excise, taper, reimplant 5:1; success >90%.

Paquin and Gil-Vernet techniques of ureteric reimplantation

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 3, Q3f

Principles of antireflux surgery

- Normal flap valve: intramural ureter compressed against detrusor during filling.
- **Paquin's rule:** tunnel length to ureteric diameter about **5:1** in non-refluxing junctions versus 1.4:1 in refluxing ones.
- Exclude secondary VUR (vesicoureteral reflux); tension-free, straight course.

Indications for surgery

- Breakthrough febrile UTI (urinary tract infection) on prophylaxis, new scars, persistent high-grade VUR.

1. Paquin technique

- **Combined extravesical–intravesical** reimplant described by Paquin (1959), with the 5:1 rule.
- **Steps:**
  1. Ureter mobilised **extravesically** and divided at the bladder.
  2. Bladder opened; a **new hiatus** created higher on the posterior wall (**suprahiatal**).
  3. Long **submucosal tunnel** from the new hiatus down to the original orifice site, length about 5 × diameter.
  4. Ureter drawn through, spatulated, sutured to mucosa and anchored to detrusor.
  5. Old hiatus closed.
- **Advantages:** long tunnel under direct vision; good for **dilated ureters and megaureter** (can combine with tapering).
- **Disadvantages:** kinking or obstruction at the new hiatus.

2. Gil-Vernet trigonoplasty

- **Intravesical, no tunnelling and no ureteric mobilisation.**
- **Steps:**
  1. Bladder opened.
  2. **Transverse mucosal incision** across the trigone between the two orifices.
  3. Intramural ureters with their muscle sheath **advanced medially** by one non-absorbable or absorbable stitch across the midline.
  4. Mucosa closed.
- **Effect:** lengthens and medialises the intravesical ureters; corrects both sides at once.
- **Indications:** low to moderate grade (I–III), usually bilateral VUR, with non-dilated ureters.
- **Advantages:** quick, simple, low obstruction risk; easy later retrograde access.
- **Disadvantages:** unsuitable for high grade, dilated ureters, paraureteral diverticulum.

Comparison

Feature	Paquin	Gil-Vernet
Approach	Extra + intravesical	Intravesical
Ureter divided	Yes	No
Tunnel	New long submucosal	None (advancement)
Best for	High grade, dilated	Low grade, bilateral

- **Also know:** Politano–Leadbetter, Cohen (commonest, >98% success), Glenn–Anderson, Lich–Gregoir. Open success benchmark 95.9% (AUA [American Urological Association] meta-analysis).

Diagram to draw: trigone showing Gil-Vernet transverse incision and medial advancement stitch.

> **Key points to remember:**
> - Paquin: tunnel 5:1; combined, suprahiatal reimplant.
> - Gil-Vernet: trigonoplasty, medial advancement, no tunnel.
> - Gil-Vernet for low-grade bilateral VUR.

Rapid Revision Sheet – Section 4 · Paediatric Urology 1

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome (LAQ)

- Grade V loses papillary impressions; bilateral grade IV rarely resolves (persists most); bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI but not scarring.
- Treat BBD (bladder and bowel dysfunction) and give CAP (continuous antibiotic prophylaxis) first; correct for breakthrough febrile UTI (urinary tract infection), new scars or persistent grade IV–V.
- Endoscopic Dx/HA (dextranomer/hyaluronic acid) by HIT (hydrodistension implantation): least invasive; lower success at higher grades.
- Open reimplantation 95.9%; Cohen cross-trigonal reimplant (over 98%) suits bilateral VUR (vesicoureteral reflux) and does both sides through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- 34–38% of scarred children become hypertensive; correction does not lower BP.
- Scars bring 8× pre-eclampsia risk: the scar, not the reflux, matters.

Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) (LAQ)

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias plus undescended testis → DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka preferred; fistula commonest complication.
- Koyanagi: one stage; meatus-based parameatal foreskin flap.
- Koyanagi: plate divided; no proximal circular anastomosis.
- Koyanagi: fistula commonest; diverticulum, meatal stenosis.

Embryology and management of ectopic ureter. (LAQ)

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult? (LAQ)

- Bud too low → lateral orifice → short tunnel → VUR.
- Adult intramural ureter 1.2–2.5 cm; Paquin 5:1.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first.
- Correct for recurrent pyelonephritis, new scars or pregnancy plans.

Bilateral Wilms tumour: treatment strategies (SAQ)

- Stage V, 5–7%; no upfront nephrectomy or biopsy.
- COG: three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12; SIOP: two drugs.
- AREN0534: 4-year OS about 95%.
- Bilateral NSS to avoid ESRD.

Congenital ureteral anomalies (SAQ)

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.

- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.

Retrocaval ureter (SAQ)

- Venous anomaly (persistent subcardinal vein); right side; 1 in 1500; male 3:1.
- Type I fishhook (obstructs); type II sickle often does not.
- CT urography; repair by anterior transposition and (with) ureteroureterostomy.

Maldescended (undescended) testes and treatment algorithm (SAQ)

- Ectopic = off the path; commonest site superficial inguinal pouch.
- Operate (orchidopexy) from 6 months, preferably by 12 months (6–12 months) and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy.
- Bilateral non-palpable: exclude CAH/DSD first (DSD work-up first).

Autosomal recessive polycystic kidney disease. (SAQ)

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

Endoscopic correction of VUR (STING procedure): indications and complications (SAQ)

- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
- Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction and calcified mound; calcified mound mimics a stone.

Snodgrass (TIP) repair for hypospadias, including extended indications (SAQ)

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate → dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

Ureterocele (SAQ)

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

Primary obst. megaureter. (SAQ)

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

Paquin and Gil-Vernet techniques of ureteric reimplantation (SAQ)

- Paquin: tunnel 5:1; combined, suprahiatal reimplant.
- Gil-Vernet: trigonoplasty, medial advancement, no tunnel.
- Gil-Vernet for low-grade bilateral VUR.

Section 5 · Paediatric Urology 2 — Long answer questions (LAQ)

Embryological development of urinary bladder and management of Hutch Diverticulum?

Type: LAQ (25 marks)

Source: 2021-Summer Paper 3, Q1 (bladder diverticulum also answered in 2016-Summer Paper 2, Q3c)

A. Embryology of the bladder

1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. The cloacal membrane perforates to form the urogenital and anal openings.
- The **Wolffian (mesonephric) duct entry** is the landmark:
  - **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
  - **Caudal (definitive) UGS** → phallic urethra in males; distal vagina and vestibule in females. The whole female urethra comes from the pelvic part.

2. Wall layers

- **Urothelium:** endoderm of the UGS.
- **Lamina propria, detrusor and adventitia:** surrounding splanchnic mesenchyme. Epithelium and mesenchyme induce each other (epithelial–mesenchymal signalling).

3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the mesonephric ducts.
- Ureteric orifices migrate **cranially and laterally**; the Wolffian ducts end caudally as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric (Wolffian) tissue. Current view (mouse lineage tracing): **mostly bladder smooth muscle**, minor ureteric contribution.
- **Bud position (Weigert–Meyer):** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis. The allantoic duct elongates and obliterates into the **urachus** (median umbilical ligament).

5. Function

- Urine production starts at **week 10**. Fetal bladder is seen from 10 weeks, almost always by 13 weeks, and empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

B. Anomalies explained by embryology

Anomaly	Embryological basis
Agenesis	Cranial UGS atrophy, or trigone not formed; survival only with ectopic ureters
Exstrophy	Cloacal membrane not reinforced by mesoderm

Anomaly	Embryological basis
Duplication	Complete or incomplete; sagittal or coronal
Urachal (patent urachus, sinus, cyst, vesicourachal diverticulum)	Urachus not obliterated at different levels
Congenital diverticulum	Focal detrusor deficiency, often at the ureteric hiatus

C. Hutch diverticulum: definition

- **Paraureteral diverticulum**: mucosa herniates through the **ureteric hiatus**, between the intravesical ureter and the roof of the hiatus, superolateral to the orifice, sparing the trigone. **Hutch** divided them into primary and secondary.

	Primary (congenital)	Secondary (acquired)
Cause	Congenitally deficient detrusor at the hiatus	High pressure: PUV (posterior urethral valves), BOO (bladder outlet obstruction), neurogenic bladder (described in paraplegics), dysfunctional voiding, prior surgery
Bladder	Smooth; single, often intermittent	Trabeculated; one of many

D. Pathophysiology

- No muscle backing → **VUR**; no muscle in the sac → cannot empty → stasis, UTI (urinary tract infection), stones; tumour in adults.
- An enlarging sac damages **Waldeyer's sheath** and pulls the ureter out → UVJ (ureterovesical junction) dysfunction, reflux, later **obstruction** and renal damage.
- Large sacs compress the bladder neck → retention (vicious circle).

E. Evaluation

- **US (ultrasound):** sac, residual urine, hydronephrosis.
- **VCUG (voiding cystourethrogram), gold standard:** oblique and post-void films show size, VUR and whether the ureter enters the sac. Repeat if suspicion persists (sac is dynamic).
- **DMSA (dimercaptosuccinic acid):** scars; **MAG3 (mercaptoacetyltriglycine) renogram** if obstruction.
- **Urodynamics** if secondary; cystoscopy (orifice, neck); in adults enter the sac, cytology.

F. Management

1. Treat the cause first (secondary): valve ablation, BOO surgery, CIC (clean intermittent catheterisation) and antimuscarinics. Remodelling may make excision unnecessary.

2. Primary diverticulum:

Situation	Management
Small, asymptomatic, no VUR	Observe
VUR, ureter beside the sac	As primary VUR (resolves like VUR of the same grade); many excise if it persists. Dx/HA (dextranomer/hyaluronic acid) injection possible
Ureter enters the sac	VUR will not resolve → surgery
Breakthrough febrile UTI, new scars, large sac with residual, stones, obstruction, retention	**Diverticulectomy + ipsilateral reimplantation**

3. Operative steps (open, laparoscopic or robotic):

```
Cystoscopy ± ureteric catheter → cystotomy → evert sac or dissect it
from its pseudocapsule (0.5 cm mucosal cuff round the neck)
→ close detrusor defect in two layers (firm backing)
→ reimplant ureter: Cohen cross-trigonal or Politano–Leadbetter
  (tunnel 5:1) → catheter ± suprapubic tube, drain
```

- Extravesical or combined approaches suit selected cases. In adults, endoscopic outlet surgery is done first.

G. Complications and follow-up

- Ureteric injury, leak, persistent VUR, obstruction, recurrence (syndromic patients).
- US at 6–12 weeks; VCUG for febrile UTI; urodynamics if voiding is abnormal.

Also know: posterolateral multiple diverticula (Ehlers–Danlos, Williams, Menkes); hypoplasia; megacystis.

Diagram to draw: cloacal division by the urorectal septum; Hutch sac beside the ureteric orifice.

> **Key points to remember:**
> - Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
> - Hutch: herniation through the ureteric hiatus → VUR.
> - Ureter entering the sac: VUR will not resolve.
> - Secondary: treat the outlet first; else diverticulectomy + reimplantation.

Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO)

Asked as:

- Define algorithm for evaluation of antenatally diagnosed unilateral HN in male fetus. What are the current rate of prenatal fetal anomalies — *Practice Set 1 Paper III Q2 (LAQ)*
- Antenatal management of Fetal Uropathy. — *Summer 2021 Paper III Q3b (SAQ)*
- ANH — *Practice Set 1 Paper III Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(antenatal LUTO management also answered in 2021-Summer Paper 3, Q3b)

See Practice Set 1 Paper 3, Q2 (full answer).

Assumed meaning: HN = hydronephrosis, now called UTD (urinary tract dilation). ANH (antenatal hydronephrosis) = UTD.

Introduction and rates of prenatal anomalies

- Major congenital anomalies occur in about **2–3% of births**. Urinary tract anomalies are among the commonest detected prenatally (commonly quoted as 20–30% of all prenatal anomalies).
- **UTD is the second commonest prenatal anomaly after cardiac defects, in 1–3% of pregnancies.**
- Mostly **UTD** on the 20-week scan, usually mild and transient. Most is mild: mild 57–88%, moderate 10–30%, severe 1.5–13% of UTD. The serious group is **LUTO (lower urinary tract obstruction)**: PUV (posterior urethral valves), urethral atresia, PBS (prune-belly syndrome).
- Screening US (ultrasound) at 18–20 weeks detects dilation well (sensitivity about 90%), but predicts diagnosis poorly. Fetal urine production begins at about 14 weeks; amniotic fluid is mainly fetal urine after 16 weeks (**after 16 weeks amniotic fluid is fetal urine**); oligohydramnios causes **pulmonary hypoplasia**.

Causes (SFU [Society for Fetal Urology] consensus)

Cause	Incidence
Transient/physiological dilation	50–70%
UPJ (ureteropelvic junction) obstruction pattern	10–30%
VUR (vesicoureteral reflux)	10–40%
UVJ (ureterovesical junction) obstruction/megaureter	5–15%
MCDK (multicystic dysplastic kidney)	2–5%
Duplex system, ureterocele, ectopic ureter	5–7%
LUTO (lower urinary tract obstruction: PUV [posterior urethral valves], prune-belly)	0.1–0.5%

- Chance of postnatal pathology rises with grade: about 12% (mild) to 88% (severe). UPJ obstruction and PUV track grade; **VUR does not** (VUR is not predicted by grade).

Grading systems

APD (anteroposterior renal pelvic diameter): transverse image at the hilum. APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.

Grade	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm

Grade	Second trimester	Third trimester
Severe	>10 mm	>15 mm

SFU (postnatal only): 1 pelvis barely split; 2 pelvis and major calyces; 3 minor calyces uniformly dilated; 4 plus parenchymal thinning.

UTD consensus 2014 (risk stratification): six parameters: APD, central/peripheral calyces, parenchymal thickness, parenchymal appearance, ureter, bladder (plus amniotic fluid prenatally).

Prenatal parameter	A1 low risk	A2–3 increased risk (any one)
APD 16–27 weeks	4 to <7 mm	≥7 mm
APD ≥28 weeks	7 to <10 mm	≥10 mm
Calyces	Central calyces only (central or none)	Peripheral
Others: parenchyma, ureter, bladder	Normal (all normal)	Abnormal
Amniotic fluid	Normal	Unexplained oligohydramnios

- **A1:** repeat US after 32 weeks (one scan). **A2–3:** US every 4 weeks to delivery. Both: postnatal US after 48 hours.

Prenatal algorithm (unilateral UTD, male fetus)

```
Anomaly scan: unilateral UTD
 → Detailed scan: contralateral kidney, ureter, bladder wall and emptying,
   posterior urethra (keyhole), amniotic fluid, other anomalies
 ├─ Bilateral / thick bladder / keyhole / oligohydramnios ─► LUTO pathway
 │    (fetal medicine; karyotype; Ruano staging; possible shunt)
 ├─ A1 ─► repeat US after 32 weeks
 └─ A2–3 ─► US every 4 weeks to delivery
 → Normal delivery at local centre; no fetal intervention for unilateral UTD
 → Postnatal US after 48 hours (earlier if solitary kidney or suspected LUTO)
```

- Repeat scans because the lesion may progress and a second anomaly is more likely. **Fetal MRI** only for complex anatomy.
- Counsel the parents: most cases resolve; prognosis with a normal opposite kidney is excellent.

Antenatal management of fetal uropathy: suspected LUTO

Evaluation

- Male fetus, thick-walled "keyhole" bladder, bilateral hydroureteronephrosis, oligohydramnios.
- Anomaly scan, **fetal echocardiogram, karyotype** (one-fifth of megacystis is aneuploid); fetal MRI (magnetic resonance imaging) if unclear.
- **Serial fetal urine** (3 samples, 48–72 hours apart). Favourable: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <6 mg/L; not validated.
- **Ruano staging:** I mild, normal fluid → observe. **II severe, function preserved** (oligohydramnios, favourable urine) → intervene. III cysts, dysplasia, unfavourable urine → for lungs only.

Options

Option	Notes
Observation, serial scans	Most cases
VAS (vesicoamniotic shunt)	Dislodgement, blockage, preterm labour
Fetal cystoscopy + laser valve ablation	Diagnoses and treats; fistula up to 10%
Serial amnioinfusion	Lungs in anhydramnios; under trial
Early delivery / termination	Mature lungs, falling fluid / poor prognosis

- **PLUTO (2013):** shunt vs conservative (31 fetuses): better perinatal survival, **no renal benefit**.

Counselling and delivery

- Multidisciplinary team: fetal medicine, paediatric urology, nephrology, neonatology.
- Intervention improves survival, not renal function; complications **over 40%**.
- Deliver where the bladder can be decompressed at once; prophylaxis, then VCUG (voiding cystourethrogram).

Postnatal algorithm

- **First US after 48 hours**, because neonatal dehydration under-reads dilation. Postnatal normal: APD <10 mm, no calyceal or ureteral dilation.

	P1 (SFU 1–2)	P2 (SFU 3)	P3 (SFU 4)
Findings	APD 10–15 mm, central calyces	APD ≥15 mm, peripheral calyces or dilated ureter	Abnormal parenchyma or bladder
Repeat US	3–6 months	Individualised	1 month
Antibiotic prophylaxis	No	Discretion; **yes if ureter >7 mm**	Yes
VCUG (voiding cystourethrogram)	No	Discretion	Yes
MAG3 (mercaptoacetyltriglycine) renogram	No	Discretion	If persistent

- **Male-specific points:** VCUG confirms a normal posterior urethra. **Circumcision** protects (NNT [number needed to treat] 10). UTI (urinary tract infection) risk: female sex, uncircumcised boys, high grade, dilated ureter.
- Prophylaxis: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin and trimethoprim only after 8 weeks.
- MAG3: DRF (differential renal function) <40%, T½ >20 minutes, or delayed retention suggest obstruction. MRU (magnetic resonance urography) for anatomy; DMSA (dimercaptosuccinic acid) for MCDK or duplex function.
- **Surgery indications (UPJ):** rising APD, DRF <40% with poor drainage, falling DRF, symptoms (UTI, stone, pain). Megaureter surgery: DRF <40%, febrile UTI; defer reimplant beyond 1 year.
- Outcome: SFU 1 resolves in nearly all by 2.5 years; SFU 2 about 95% by 4 years.

Bilateral dilation and PUV: how they differ

- Bilateral hydroureteronephrosis, thick bladder, keyhole sign, oligohydramnios = LUTO until proved otherwise.
- At birth: 5 Fr feeding tube, prophylaxis, creatinine, early VCUG; prenatal intervention improves survival but not renal function (PLUTO).

Diagram to draw: UTD A1/A2–3 prenatal and P1–P3 postnatal flowchart; keyhole bladder with bilateral hydroureteronephrosis and shunt.

Key points to remember (including SAQ angle):

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias)

Asked as:

- Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias — *Practice Set 6 Paper III Q2 (LAQ)*
- Isolated penopubic Epispadias. — *Summer 2022 Paper III Q3f (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(isolated penopubic epispadias also answered in 2022-Summer Paper 3, Q3f)

Development of the urethra

1. Division of the cloaca (weeks 5–6)

- The urorectal septum divides the cloaca into the dorsal anorectal canal and the ventral **UGS (urogenital sinus)**.
- Wolffian duct entry is the landmark. Above it, the **vesicourethral canal** forms the bladder and pelvic urethra. Below it, the **definitive UGS** forms the phallic urethra in males, and the urethra and vestibule in females.
- UGS also forms the prostate (buds at 10 weeks) and Cowper and Littre glands; the **utricle** is Müllerian.
- The epithelium of the **entire male urethra is endodermal** (UGS origin).

2. Penile urethra: the "double zipper" (androgen-dependent)

- Indifferent genital tubercle until 8–9 weeks. Testosterone (Leydig cells, 8 weeks) is converted to DHT (dihydrotestosterone) by **type 2 5α-reductase**.
- The **urethral plate** (endoderm) forms in the tubercle.
- **Opening zipper:** the plate canalises proximal to distal into a urethral groove bounded by the urethral folds.
- **Closing zipper:** the folds fuse in the ventral midline, forming the tubular urethra and median raphe.
- **Glanular urethra:** formed by direct canalisation of the plate inside the glans (current endodermal view; Glenister's ectodermal ingrowth theory is outdated).
- **Timetable:** coronal sulcus 11–12 weeks; fusion complete 16 weeks; urethra complete 18 weeks; fetal curvature gone 20 weeks.
- The prepuce forms with the glanular urethra, so hypospadias has a hooded foreskin.
- **Female:** the vestibular plate canalises but the folds do not fuse, giving an open vestibule and labia minora.

Classification of congenital urethral anomalies

Mechanism	Anomaly
Failed ventral fold fusion	**Hypospadias** (1 in 250 males; commonest at shaft–glans junction); congenital urethral fistula
Dorsal defect	**Epispadias** (exstrophy–epispadias complex)
Obstruction	**PUV (posterior urethral valves)**, Young type 1 (95%) and type 3; **anterior urethral valves** with diverticulum; urethral atresia or hypoplasia; meatal stenosis
Duplication	**Urethral duplication**: sagittal; ventral channel is functional; Effmann type I blind, II complete, III with caudal duplication
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); prune-belly association

Mechanism	Anomaly
Cystic or remnant	Cowper's syringocele; enlarged utricle (proximal hypospadias); parameatal cyst

Epispadias (including isolated penopubic epispadias): definition, embryology, classification and clinical features

- Part of the **exstrophy–epispadias complex**: a **dorsal urethral defect** (defect of the dorsal urethral wall): the roof is replaced by a broad (dorsal) mucosal strip (urethral plate). Cause (embryology): absent mesodermal reinforcement of the cloacal membrane; the genital tubercles fuse **caudal** to it, giving a dorsal urethral defect.
- **Male (1 in 117,000; incidence of isolated male epispadias about 1 in 117,000), classification by meatus:** glanular, penile, **penopubic** (subsymphyseal) (about 70% of male cases are complete (penopubic), with incontinence). **Female (1 in 484,000), Davis:** vesicular (patulous), subsymphyseal, retrosymphyseal (incontinent).
- **Features (clinical):** penopubic meatus; outlet may admit a finger → **total incontinence**. **Short, broad, upturned penis with dorsal chordee**, like exstrophy; pubic diastasis (less than exstrophy); small bladder from no outlet resistance; **VUR (vesicoureteral/vesicoureteric reflux) 30–40% (male), 30–75% (female)**. Female (Davis retrosymphyseal type): bifid clitoris, depressed mons, incontinence.

Evaluation

- (Renal) ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy and closure) if pubic diastasis is over 3 cm, the intrasymphyseal bar is poor, sphincter is absent or capacity is tiny (tiny bladder).

Management of epispadias

- **Aims:** continence, safe upper tracts, a straight functional penis of good length.
- **Preoperative testosterone** 2 mg/kg intramuscularly (intramuscular), 5 and 2 weeks before urethroplasty.

Stage 1: penile reconstruction and urethroplasty (6–12 months)

- Release chordee: divide suspensory ligaments and scar; free corpora from the inferior pubic rami.
- **Modified Cantwell–Ransley:** outline an 18-mm plate to the tip; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; free the plate from the corpora (distal 1 cm left on the glans); tubularise over 8 Fr; transpose it (the tubularised plate is moved) **ventral to the corpora**, which are rotated medially ± cavernocavernostomy; close glans wings; Z-plasty at the base; stent 10–12 days.
- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.
- Outlet resistance rises, so capacity grows.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **YDL (Young–Dees–Leadbetter)** bladder neck repair + Marshall–Marchetti–Krantz suspension + **ureteric reimplantation**. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

Female epispadias: at 6–12 months excise the urethral roof, close over 10 Fr, unite clitoral halves and labia minora, monsplasty; YDL at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula up to (about) one-fifth (at the penile base); long-term fistula and stricture about 5% (stricture rare); erections and paternity (usually) preserved.
- Better than exstrophy (bladder never exposed or scarred). Failures (failure): bulking injection (injectables), augmentation, continent stoma.

Diagram to draw: double zipper (canalisation then fusion); penopubic epispadias (dorsal penopubic meatus and plate).

Key points to remember:

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
- Diastasis over 3 cm (>3 cm) → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

Bladder and ureter development, exstrophy–epispadias complex and its management

Asked as:

- Discuss the developmental basis of extrophy epispadias complex. Briefly outline management — *Practice Set 7 Paper III Q2 (LAQ)* (theories also answered in Practice Set 6 Paper 1, Q2; single-stage repair in Practice Set 3 Paper 3, Q3e)
- Describe normal development of bladder and ureter and outline management of a child born with ectopia vesicae. — *Practice Set 8 Paper III Q1 (LAQ)* (bladder embryology and exstrophy theories also answered in Practice Set 6 Paper 1, Q2)
- Single stage repair of Extrophy of UB. — *Practice Set 3 Paper III Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Assumed meaning: ectopia vesicae = classic bladder exstrophy (CBE). Single stage repair of Extrophy of UB = single-stage (complete primary) repair of exstrophy of the urinary bladder, mainly CPRE (complete primary repair of exstrophy, Mitchell).

Introduction

- **EEC (exstrophy–epispadias complex)** is one spectrum: glanular epispadias → **CBE (classic bladder exstrophy)**, over 50% → cloacal exstrophy.
- CBE (incidence) about 1 in 46,000; male:female 2.3:1. Offspring of an affected parent: 1 in 70.

A. Development of the ureter

- **Weeks 4–6:** the **ureteric bud** arises from the distal **Wolffian (mesonephric) duct**. GDNF (glial cell line-derived neurotrophic factor) from the metanephric mesenchyme induces a single bud (extra buds → duplication).
- **Reciprocal induction:** the bud invades the metanephric blastema. Bud stalk → ureter; its branches → pelvis, calyces and collecting ducts; blastema → nephrons.
- **Wall:** urothelium, lamina propria, smooth muscle and serosa, the same plan as the bladder.
- **Chwalla membrane** at the future orifice normally ruptures; failure is the classic explanation of ureterocele.

B. Development of the bladder (normal development)

- The **cloacal membrane** (bilaminar ectoderm–endoderm) occupies the infraumbilical wall. **Mesenchyme grows in** between its layers and forms the lower abdominal muscles and pelvic bones.
- **Cloacal division (weeks 5–6):** the **urorectal septum** divides the cloaca into dorsal **anorectal canal** / rectum (posterior) and ventral **UGS (urogenital sinus)** / bladder (anterior), meets the membrane, which perforates into urogenital and anal openings. Paired genital tubercles fuse **cephalad** to the membrane.
- **Wolffian duct entry** is the landmark: cranial **vesicourethral canal** → bladder and pelvic urethra; caudal **definitive UGS** → phallic urethra (male), vestibule (female).
- **Layers:** urothelium from UGS endoderm; detrusor and adventitia from splanchnic mesenchyme by epithelial–mesenchymal signalling.
- **Urachus:** the bladder descends (months 4–5); the allantois obliterates to the urachus (median umbilical ligament).
- Urine from week 10; after week 16 amniotic fluid is mainly fetal urine.

C. Trigone and ureterovesical junction

- **Weeks 5–6:** the **common excretory duct** (Wolffian duct distal to the bud) is absorbed into the UGS; **apoptosis** separates ureter from Wolffian duct. The arms rotate (**Mackie–Stephens**): ureteric orifice moves cranially and laterally; Wolffian duct ends caudally as the ejaculatory duct.
- **Trigone muscle:** classic view, ureteric extension; current lineage tracing, mostly bladder smooth muscle.

Bud origin	Orifice	Result
Too low (early contact)	High, lateral, short tunnel	VUR (vesicoureteral reflux)
Too high (late contact)	Caudal, medial	Ectopic ureter, often obstructed
Either	Abnormal bud–blastema contact	Dysplasia (VUR and dysplasia coexist)

- **Weigert–Meyer:** in duplex systems the upper-pole orifice is caudal and medial.

D. Developmental basis of EEC / ectopia vesicae

1. Core defect (cause)

- The cloacal membrane is **not reinforced by mesodermal ingrowth** (by mesoderm), so it ruptures (**Marshall–Muecke** overdevelopment theory). **Timing and extent of rupture decide the variant:** before septum descent gives cloacal exstrophy; after gives CBE; limited rupture gives epispadias.

2. Theories

Theory	Proposed defect	Consequence
Cloacal membrane overdevelopment (Marshall–Muecke)	Abnormally large membrane in week 4 blocks medial mesenchymal migration	Timing of rupture gives the variant
Lateral body-wall fold failure	Folds fail to meet in midline	Abdomen and pelvis → cloacal; pelvis alone → classic
Caudal genital hillocks	Tubercles fuse **below** the membrane	Dorsal urethral defect (epispadias)
Abnormal body-stalk insertion	Caudal insertion stops mesenchyme interposing	Unstable membrane disintegrates
Bony pelvis first	Pelvic ring primordium fails to rotate	Bladder herniates

- All converge on **absent infraumbilical midline mesoderm**.

3. Aetiology

- Multifactorial (inconsistent twin concordance). **22q11.2 duplication** commonest variant; **ISL1** susceptibility gene; p63 under-expressed in exstrophic urothelium.
- **IVF (in vitro fertilisation)** about 7.5-fold and early high-dose progesterone about tenfold raise risk; maternal smoking linked to cloacal exstrophy; folate to the mildest form.

4. Resulting anatomy / defects (why each defect occurs)

- Exposed (open) bladder plate; epispadias; low umbilicus; anterior anus; hernias.
- **Pelvis:** externally rotated (posterior 12°, anterior 18° each side), **anterior pubic segment 30% short**, **pubic diastasis mean 4.8 cm**.
- **Male genitalia:** anterior corpora about 50% shorter; short, upturned penis with dorsal chordee.
- **Female:** bifid clitoris, short vagina, anterior stenotic introitus; later prolapse.

- **Bladder:** immature, not neuropathic (raised type III collagen falls after closure). Ureters enter with no obliquity, so **VUR (vesicoureteric reflux) in 100%** after closure.
- Anterior anus, wide levator hiatus, inguinal hernias, low umbilicus.

E. Management (outline)

1. Prenatal: prenatal diagnosis only about 25%: ultrasound shows absent (no) bladder filling, low umbilicus, wide pubic rami, small genitalia, enlarging lower abdominal mass. Counsel (counselling); deliver at an exstrophy centre.

2. At birth

- Tie cord with silk (no clamp); cover the plate with plastic wrap; saline irrigation at nappy change.
- Renal ultrasound; multidisciplinary team.
- **Assess:** template size and polyps (**≥3 × 3 cm** suits newborn closure), diastasis and pelvic malleability under anaesthesia, urethral plate and penile length, associated anomalies.
- Small template, penoscrotal duplication, hypoplastic bladder: **delayed closure at 6–9 months** (delay closure, not a bad early closure).

3. Pelvic osteotomy and immobilisation

- **Indications:** closure **after 72 hours**, diastasis **>4 cm**, stiff (poorly malleable) pelvis, failed closure, or cloacal exstrophy.
- **Combined bilateral anterior innominate + vertical iliac osteotomy** with external fixator and (Buck) traction 4–6 weeks. Under 72 hours with malleable pubes / without osteotomy: modified Bryant traction for 4 weeks. Spica casting fails more often.

4. MSRE (modern staged repair of exstrophy)

- Aim of closure: convert exstrophy into incontinent epispadias, with enough outlet resistance for bladder growth but not enough to harm the kidneys.

Stage	Age	Procedure	Purpose
1	Newborn	Bladder, posterior urethra and abdominal wall closure; ± osteotomy	Converts to complete epispadias; outlet resistance stimulates bladder growth
2	6–12 months	Epispadias repair (**modified Cantwell-Ransley**), after testosterone	Raises capacity; straightens penis
3	5–7 years (another source: 4–7 years) (capacity ≥100 mL)	BNR (bladder neck reconstruction): modified **Young-Dees-Leadbetter (YDL)** + bilateral **Cohen** reimplantation (reimplant)	Continence; needs capacity **≥100 mL** and a motivated child

- **Closure keys: radical division of the urogenital diaphragm fibres** so the bladder / vesicourethral unit sits deep in the pelvis (commonest error if missed); ureteric stents; suprapubic tube 4 weeks; pubic nylon mattress suture.
- **Alternatives: CPRE (complete primary repair of exstrophy, Mitchell)** with penile disassembly; Kelly radical soft-tissue mobilisation; Erlangen total repair (single stage); Mainz ureterosigmoidostomy. Most still need an outlet procedure.

5. Single-stage repair (CPRE and others) *Background:* classic bladder exstrophy: absent infraumbilical mesoderm; open bladder, epispadias, pubic diastasis (mean 4.8 cm).

- **MSRE (modern staged repair):** closure at birth, epispadias repair 6–12 months, bladder neck reconstruction 4–7 years. Single-stage repairs aim for fewer operations and continence without bladder neck repair.

Single-stage options:

Repair	Features
CPRE (Mitchell)	Newborn closure + penile disassembly + bladder neck tailoring
Erlangen ("total")	All in one at about 8 weeks, no osteotomy
Combined closure + epispadias repair	Delayed or failed closures, boys >6 months

- **Selection:** template ≥3 × 3 cm, elastic, few polyps; experienced team. Not for penoscrotal duplication, hypoplastic bladder, bilateral hydronephrosis.
- **CPRE technique:**
  1. Combined general and epidural anaesthesia; **osteotomy if >72 hours** (3 days) old or diastasis >4 cm or stiff pelvis.
  2. Incise around plate; free bladder; **divide urogenital diaphragm fibres** to levator hiatus so the unit sits deep.
  3. **Penile disassembly:** right and left hemicorporoglanular bodies plus urethral wedge on spongiosum (now often left attached to glans).
  4. Urethra placed ventrally between corpora; corpora joined; meatus usually **hypospadiac**.
  5. Bladder and neck closed and tapered; ± bilateral ureteric reimplantation (BUR-CPRE).
  6. Pubic No. 2 nylon mattress suture; suprapubic catheter 4 weeks, ureteric stents.
  7. Immobilise: external fixation plus traction (failure 7.5% vs spica 47%).
- **Complications:** same frequency as MSRE but **more serious**: dehiscence, prolapse, vesicocutaneous fistula, **glans or corporal loss** (ischaemia), early reimplantation for VUR (vesicoureteric reflux) or hydronephrosis; retention in girls.

6. Follow-up

- Prophylaxis; ultrasound; yearly cystogram and cystoscopy under anaesthesia.

7. Failures and salvage

- Dehiscence or prolapse: wait 4–6 months, reclose with osteotomy.
- Failed BNR (no 3-hour dry interval by 2 years): bulking, redo YDL, or **augmentation + bladder neck closure + Mitrofanoff** (continent catheterisable stoma).

8. Cloacal exstrophy (OEIS: omphalocele, exstrophy, imperforate anus, spinal defects)

- Stabilise; neurosurgery (spinal defects 85–100%); separate bowel from hemibladders; colostomy; osteotomy always; gender congruent with karyotype; continence mostly by augmentation with CIC (clean intermittent catheterisation).

F. Outcomes and long-term

- Daytime dryness (day continence) after MSRE about **70–74%** (another source: 70–75%); versus about 56% after CPRE with YDL (Young-Dees-Leadbetter). Most (after CPRE) still need bladder neck surgery.
- **Bladder capacity at neck repair is the main predictor**; each failed closure lowers it. **Best (continence) predictors: template size at birth and a successful first closure.**
- **Also know:** variants (superior vesical fissure, duplicate, pseudoexstrophy); adenocarcinoma risk (cytology, cystoscopy; colonoscopy after ureterosigmoidostomy); fertility (reduced male fertility) and uterine prolapse in adults; female epispadias; psychosocial support.

Diagram to draw: cloacal membrane with urorectal septum and failed mesodermal ingrowth; Wolffian duct with ureteric bud and its absorption into the UGS; MSRE timeline; penile disassembly into two hemicorpora and urethral wedge.

Key points to remember:

- Bud from Wolffian duct; position decides VUR or ectopia.
- Failed mesodermal reinforcement of the cloacal membrane; rupture timing decides variant.
- Diastasis 4.8 cm; anterior pubis 30% short; VUR in 100%.
- MSRE: (birth) closure → epispadias repair 6–12 months (mo) → YDL / BNR at 5–7 years, at ≥100 mL.
- Osteotomy if >72 h (closure after 72 h) or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL predicts continence; template size + first successful closure (first closure success) predict continence.

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

HSG in neonates

Type: SAQ (10 marks)

Source: 2015-Summer Paper 3, Q3c

HSG (hysterosalpingography) is not done in neonates, so this is taken as "USG (ultrasonography) in neonates".

Introduction

- First-line, radiation-free imaging. For antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies), febrile UTI (urinary tract infection), mass, poor stream.

Timing and technique

- **First scan after 48 hours**: earlier, relative dehydration under-reads dilatation.
- **Scan early if outlet obstruction is suspected**; catheterise.
- Measure **APD (anteroposterior pelvic diameter)** on a transverse mid-kidney view; assess calyces, parenchyma, ureters, bladder wall (thick if over 3 mm when full).
- Normal: **prominent hypoechoic pyramids** (not hydronephrosis); APD under 10 mm.

Grading

- **SFU (Society for Fetal Urology)**: 0 none; 1 pelvis only; 2 major calyces; 3 minor calyces too; 4 plus parenchymal thinning.
- **UTD 2014**: prenatal A1 (APD 4 to <7 mm before 28 weeks, 7 to <10 mm after) versus A2–3 (larger, or any other abnormality); postnatal:

Grade	Findings	Plan
P1	APD 10 to <15 mm or central calyces	USG at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis
P2	APD 15 mm or more, peripheral calyces or dilated ureter	VCUG, prophylaxis optional (give if ureter >7 mm)
P3	Abnormal parenchyma or bladder	USG at 1 month, VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan

Specific findings

Condition	USG
PUJO (pelviureteric junction obstruction)	Dilated pelvis, normal ureter; pyramids under 3 mm predict pyeloplasty
VUR (vesicoureteral reflux)	Normal first scan in up to 25%
PUV (posterior urethral valves)	Bilateral hydroureteronephrosis, thick bladder, dilated posterior urethra (**keyhole sign**); VCUG before discharge
Ureterocele	Thin-walled intravesical cyst
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, no central pelvis
ARPKD (autosomal recessive polycystic kidney disease)	Bilateral large, bright kidneys

Condition	USG
Renal vein thrombosis	Large echogenic kidney, no venous flow; haematuria, mass, thrombocytopenia
Adrenal haemorrhage	Avascular, evolving suprarenal mass (neuroblastoma is vascular)

- **Also know:** mesoblastic nephroma; twinkling artefact (stones).

Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

Diagram to draw: SFU grades 0–4.

> **Key points to remember:**
> - First scan after 48 hours, except suspected PUV.
> - UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
> - Keyhole sign suggests PUV.

Prune belly syndrome

Type: SAQ (10 marks)

Source: 2015-Summer Paper 3, Q3f

Definition

- **PBS (prune belly syndrome)**, Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **dilated poorly contracting urinary tract**, **bilateral intra-abdominal testes**.
- 1 in 29,000–44,000 births; **95% male**. **Renal dysplasia, not the abdominal wall, decides survival.**

Pathogenesis (unproven)

- Fetal **urethral obstruction**, though most have none after birth; **lateral plate mesoderm defect**; intrinsic ureteric defect; yolk sac defect. Mostly sporadic; smooth-muscle genes (MYOCD, FLNA) in families.

Features

- **Kidneys**: dysplasia in about 50%. **Ureters**: tortuous, worse distally; collagen replaces muscle, so peristalsis fails; **VUR (vesicoureteral reflux) about 75%**.
- **Bladder**: large, smooth, urachal diverticulum (urachus patent in 25–30%). **Prostatic hypoplasia**, dilated posterior urethra; atresia or megalourethra.
- **Testes**: intra-abdominal; azoospermia usual.
- **Extra-genitourinary (about 75%)**: pulmonary hypoplasia; cardiac 10%; gastrointestinal at least 30%; orthopaedic up to 65%.

Woodard classification

	Features	Outlook
I	Oligohydramnios, **pulmonary hypoplasia**, dysplasia, urethral atresia	Perinatal death
II	Full triad, moderate uropathy, **no pulmonary hypoplasia**	Risk of CKD (chronic kidney disease); debated
III	Mild, **stable function**	Good; majority

- **Also know:** RUBACE severity score (renal, ureter, bladder, abdominal wall, cryptorchidism, extra-genitourinary); "pseudo-prune" presents late.

Management

- **Neonatal**: lungs and heart first; creatinine trend, ultrasound, DMSA (dimercaptosuccinic acid) at 4–6 weeks, prophylaxis, circumcision. VCUG (voiding cystourethrogram) only if renal insufficiency or obstruction.
- Category III: surveillance. Sepsis or rising creatinine: wide **cutaneous vesicostomy**.
- **Reconstruction** after 3–6 months: ureteric tailoring and reimplantation using the healthier proximal ureter; reduction cystoplasty gives little lasting benefit.
- **Orchidopexy**: bilateral **transabdominal at about 6 months**; staged Fowler–Stephens later.
- **Abdominoplasty** (cosmesis, better cough and emptying): **Randolph** (transverse excision of the worst lower wall); **Ehrlich** (vertical incision, umbilicus kept, vest-over-pants overlap); **Monfort** (umbilicus kept, central fascial strip with lateral flaps overlapped; good exposure for combined surgery).

Prognosis

- 20% die perinatally; **up to 30% develop chronic renal failure**; nadir creatinine **under 0.7 mg/dL** is favourable.

Diagram to draw: VCUG showing large bladder and dilated posterior urethra.

> **Key points to remember:**
> - Triad; 95% male; dysplasia decides prognosis.
> - Woodard I–III.
> - Vesicostomy for drainage; orchidopexy at about 6 months.

Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty

Asked as:

- Prenatal and Neonatal assessment of neurogenic bladder dysfunction — *Summer 2016 Paper III Q3e (SAQ)*
- Bladder myeloplsty — *Summer 2017 Paper III Q3a (SAQ)*

Type: SAQ (10 marks)

Part A: Prenatal and Neonatal assessment of neurogenic bladder dysfunction

Introduction

- Commonest cause: **myelomeningocele** (open NTD (neural tube defect)); also occult dysraphism, sacral agenesis, anorectal malformation.
- **Bony or motor level does not predict bladder function**, so every newborn needs testing.

Prenatal assessment

- **Maternal serum AFP (alpha-fetoprotein)** raised in open defects; amniotic AFP and acetylcholinesterase confirm.
- **Ultrasound**: spinal defect; "lemon" and "banana" signs of Chiari II; ventriculomegaly; hydronephrosis, thick bladder. Fetal MRI (magnetic resonance imaging) for level.
- Multidisciplinary counselling at a fetal centre.
- **MOMS (Management of Myelomeningocele Study) 2011**: closure **before 26 weeks** reduced shunt need (about 40% vs 82%) and improved motor and cognitive scores, but caused prematurity (34 vs 37 weeks) and uterine thinning. **No proven bladder benefit**: CIC (clean intermittent catheterisation) and reconstruction rates unchanged.

Neonatal assessment (baseline)

Test	Timing / purpose
Examination	Spine, legs, anal tone, perineal sensation, bladder
Renal and bladder ultrasound	Before discharge, then 6-monthly for 2 years; hydronephrosis, wall thickness
Residual urine	Start CIC if not emptying
Creatinine	Within 3 months (cystatin C later)
VCUG (voiding cystourethrogram) or video-UDS	Reflux; dilating VUR (vesicoureteric reflux) in about 7%
UDS (urodynamics) with EMG (electromyography)	Within 3 months, once stable after closure
DMSA (dimercaptosuccinic acid)	Baseline scarring (EAU (European Association of Urology))

Urodynamic risk (UMPIRE (Urologic and Renal Protocol for the Newborn and Young Child with Spina Bifida))

Risk	DLPP (detrusor leak point pressure) or end-fill pressure
High (hostile)	40 cm H2O or more (McGuire 1981)
Intermediate	25–39, or detrusor overactivity
Low	Under 25

- Poor compliance and DSD (detrusor–sphincter dyssynergia) also predict upper-tract damage.

Action

- High risk → **early CIC plus oxybutynin**: hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) about 30% falls to under 2%.
- Repeat studies: the lesion can change (re-tethering).

Diagram to draw: neonatal UDS trace with DLPP over 40 cm H2O.

Part B: Bladder myeloplsty

Assumed to mean "bladder myoplasty" (a muscle procedure on the bladder wall).

Definition

- **Detrusor myoplasty**: wrapping a **vascularised, innervated skeletal muscle flap** round the bladder so that voluntary contraction of that muscle empties an **acontractile bladder**.

Indications

- **Acontractile or underactive detrusor** with catheter dependence: lower motor neuron lesions (spina bifida, cauda equina injury), myogenic failure.
- Needs good bladder capacity and compliance, and a competent but not obstructing outlet.

Latissimus dorsi detrusor myoplasty (LDDM; Stenzl)

- **Free latissimus dorsi flap**: artery and vein joined to the **inferior epigastric vessels**; motor nerve joined to an **intercostal nerve**.
- Muscle wrapped **spirally over more than three-quarters of the bladder** and anchored to pelvic floor fascia and ligaments.
- Patient voids by contracting the transferred muscle (abdominal effort).
- **Results:** most voided with low residual; bladder contractility index rose to the strong range.
- **Complications** in about one-third: thromboembolism, abscess, wound infection; none long term.
- **Rectus abdominis detrusor myoplasty**: simpler, less morbid; small spinal cord injury series with spontaneous voiding.
- **Status: investigational.** Standard care stays **clean intermittent catheterisation**.

Other "myoplasty" procedure: detrusor myotomy/myectomy (autoaugmentation)

- Purpose is the opposite: **increase capacity and compliance** in a small, poorly compliant or overactive bladder.
- **Technique (Cartwright and Snow, 1989):** strip and excise the detrusor over the dome, leave the mucosa intact, and anchor the detrusor edges (psoas hitch) so the mucosa bulges as a **wide-mouthed diverticulum**.
- **Pros:** urothelial lining, no mucus, no metabolic change, no bowel.
- **Cons:** seldom adds significant capacity; compliance gain may not last; **not recommended for adult detrusor overactivity**.
- **Also know:** seromuscular enterocystoplasty covers the bulge with demucosalised sigmoid.

Alternatives

- **Also know:** sacral neuromodulation, sacral anterior root stimulation, reduction cystoplasty (abandoned), muscle-derived stem cells (research).

Diagram to draw: latissimus flap spiralled round the bladder with vascular and nerve anastomoses.

Key points to remember:

- Level does not predict bladder; UDS within 3 months.
- DLPP 40 cm H2O or more = hostile; early CIC + antimuscarinic.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

DD of frequency and urgency in 3 yr male child and treatment

Type: SAQ (10 marks)

Source: 2017-Summer Paper 3, Q3c

Introduction

- **Frequency**: ≥8 voids a day (ICCS, International Children's Continence Society). **Urgency**: sudden, compelling need to void.
- At 3 years toilet training may be ongoing; decide first whether the pattern is abnormal.

Differential diagnosis

Group	Causes and clues
Infection	**UTI (urinary tract infection)**: fever, dysuria; balanoposthitis; viral haemorrhagic cystitis
Functional	**BBD (bladder and bowel dysfunction)**/OAB (overactive bladder); voiding postponement; **extraordinary daytime urinary frequency** (sudden very frequent small voids, no nocturia or wetting, often after stress; self-limiting)
Bowel	**Constipation**: rectal mass presses on bladder; commonest associated cause
Obstruction	**PUV (posterior urethral valves)**, mild late form; **meatal stenosis** (circumcised boys; thin deflected stream); severe phimosis
Neurogenic	Occult spinal dysraphism, tethered cord (sacral dimple, hair tuft, gait or foot abnormality)
Bladder irritants	Stone, foreign body, hypercalciuria, pinworms, rarely **rhabdomyosarcoma**
Polyuria	Diabetes mellitus, diabetes insipidus, chronic kidney disease
Behavioural	Anxiety, family stress, developmental disorders

Evaluation

- **History:** onset, nocturia, wetting, stream, fever, **bowel habit (Bristol chart)**, thirst, stressors.
- **Examination:** abdomen, meatus and foreskin, **lumbosacral spine**, lower-limb neurology.
- **Urinalysis and culture**; glucose; specific gravity >1.005 makes diabetes insipidus unlikely.
- **Ultrasound** of kidneys and bladder: wall thickness, residual, dilation, stones.
- **Bladder and bowel diary**; uroflow if the child cooperates.
- **Selective:** VCUG (voiding cystourethrogram) for suspected PUV or febrile UTI with abnormal ultrasound; spinal MRI (magnetic resonance imaging) for neurological signs; urine calcium/creatinine ratio.

Treatment (cause-directed)

- **UTI:** antibiotics per culture.
- **Constipation first:** disimpaction, then maintenance **polyethylene glycol**, diet and toilet routine.
- **Urotherapy:** timed voiding every 2–3 hours, relaxed posture, adequate fluids, avoid caffeine.
- **OAB persisting after urotherapy: oxybutynin** 0.3–0.6 mg/kg/day in 2–3 doses (maximum 15 mg/day); watch for constipation.
- **Extraordinary daytime frequency:** reassurance; resolves in weeks to months.

- **Meatal stenosis:** meatotomy. **PUV:** endoscopic valve ablation. **Tethered cord:** neurosurgical release.

Diagram to draw: flowchart: urinalysis → ultrasound with residual → targeted tests.

Key points to remember:

- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

Embryology and Management of P.U. valve patient.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 3, Q3d

Embryology

- **Weeks 5–6:** the **urorectal septum** divides the cloaca into rectum and **urogenital sinus**; wolffian ducts migrate to the verumontanum, leaving small folds.

Young type	Anatomy	Origin
I (about 95%)	Leaflets from the verumontanum, fused anteriorly	Abnormal anterior insertion of the wolffian duct ends
II	Folds from verumontanum to bladder neck	Not real (secondary hypertrophy)
III	Ring diaphragm with central hole	**Persistent urogenital membrane**

- **COPUM (congenital obstructing posterior urethral membrane):** one membrane, torn by the first catheter into type I leaflets.
- Fetal urine from 10–14 weeks: obstruction causes **renal dysplasia**, oligohydramnios, pulmonary hypoplasia.

Antenatal

- Bilateral hydroureteronephrosis, thick bladder, **keyhole sign**, oligohydramnios. **Vesicoamniotic shunt** helps survival (lungs), not kidneys (PLUTO trial, 2013).

Neonatal management

1. **Drain the bladder:** 5–7 Fr feeding tube (coudé if it curls); minimal balloon inflation.
2. **Stabilise:** ventilation; correct **post-obstructive diuresis**, hyponatraemia, acidosis, hyperkalaemia; antibiotics.
3. **Confirm:** VCUG (voiding cystourethrogram), renal ultrasound, MAG3 (mercaptoacetyltriglycine) at 2–3 months.
4. **Primary endoscopic valve ablation** (preferred): laser, Bugbee or hook at 5 and 7 o'clock ± 12; VCUG within 1 month.
5. **Vesicostomy** if the urethra is too small or drainage fails. **Upper-tract diversion** rarely: no proven renal benefit.
6. **Circumcision:** cuts febrile UTI (urinary tract infection) sharply.

- **VUR (vesicoureteral reflux)** (50–80%) is secondary; treat the bladder, not the ureter.

Prognosis

Nadir creatinine (first year)	ESRD (end-stage renal disease) risk
< 0.4 mg/dL	Nil
0.4–0.69	2%
0.7–0.99	27%
> 1.0	Up to 100%

- Lifetime ESRD risk 20–50%.

Long-term care

- **Valve bladder** (Mitchell): polyuria, poor sensation and residual urine keep the upper tracts dilated. Timed and double voiding, anticholinergic or α-blocker, **CIC (clean intermittent catheterisation)**, **overnight drainage**, Mitrofanoff, rarely augmentation.

- **CKD (chronic kidney disease) care; transplantation** after urodynamics and bladder optimisation; graft survival similar.

Diagram to draw: type I leaflets versus type III membrane, with keyhole VCUG.

Key points to remember:

- Type I (95%): wolffian duct insertion; type III: persistent urogenital membrane.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: overnight drainage and CIC.

CSIC.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 3, Q3e

*Taken to mean **CISC – clean intermittent self-catheterisation**.*

Definition and principle

- Regular, **clean (not sterile)** passage of a catheter by the patient (or carer) to empty the bladder, then removal.
- **Lapides (1971–72)**: infection comes from **stasis and overdistension** (poor bladder-wall blood flow), not from bacteria alone. Regular complete emptying protects the bladder wall and upper tracts, even though bacteriuria develops.
- Gold standard for chronic emptying failure.

Indications

- **Neurogenic bladder:** spina bifida, spinal cord injury, multiple sclerosis, cauda equina, diabetic cystopathy.
- **Underactive or acontractile detrusor**; chronic retention; post-operative retention.
- **Detrusor–sphincter dyssynergia**: CISC bypasses the obstructed sphincter.
- After **augmentation or continent diversion**; **valve bladder**; high residual after botulinum toxin.
- **Contraindications:** poor hand function or cognition without a carer, urethral false passage or stricture, unwillingness.

Technique

- Wash hands → clean meatus with water → lubricated catheter (or pre-lubricated **hydrophilic** catheter) → pass until urine flows → drain fully → withdraw slowly.
- Sizes: adults about 12–14 Fr; children by age (8–10 Fr).
- **Frequency:** 4–6 times a day, keeping each volume **below about 400–500 mL** (child: below expected bladder capacity).
- Combine with **antimuscarinics** for overactivity; bladder diary.

Catheters

Type	Note
Plain PVC (polyvinyl chloride), reusable after washing	Cheap; common in India
Hydrophilic-coated single-use	Less urethral trauma; longer time to first UTI (urinary tract infection)
Compact or closed-system	Convenience, outdoor use

Children

- Start **early (infancy)** in spina bifida with hostile bladder; families adapt best; child involved from 3–5 years.
- **Mitrofanoff appendicovesicostomy** (1980) if urethral catheterisation is difficult (wheelchair, sensate urethra, girls).

Complications

- **Asymptomatic bacteriuria** (common): **do not treat or give prophylaxis**; treat only symptomatic UTI.

- Urethral trauma, bleeding, **false passage**, stricture, epididymitis, bladder stones (hair, mucus), rarely lost catheter.

Outcome

- Less pyelonephritis, epididymo-orchitis and urosepsis than indwelling catheters; preserves the upper tracts; with drugs, continence in many.

Diagram to draw: neurogenic bladder management ladder: CISC + antimuscarinic → botulinum toxin → augmentation + channel.

> **Key points to remember:**
> - Lapides: overdistension, not bacteria, causes infection; clean is enough.
> - 4–6 times daily, under about 500 mL each.
> - Do not treat asymptomatic bacteriuria.
> - Mitrofanoff when urethral access is hard.

Enuresis

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 3, Q3e

Definition (ICCS [International Children's Continence Society])

- Intermittent urinary incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

Epidemiology

- About **5–10% at 7 years**; spontaneous resolution about 15% a year; about 2% of adults; familial.

Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (reduced night-time ADH [antidiuretic hormone] rise).
2. **Reduced nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep.

- Comorbidities: constipation, sleep-disordered breathing (atrial natriuretic peptide raises urine output), ADHD (attention deficit hyperactivity disorder).

Evaluation

- History: day symptoms, bowels, snoring, fluids, family history.
- **Bladder diary**: daytime voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL in the chapter (ICCS uses [age + 1] × 30); night urine output by weighing nappies.
- Examination: spine, genitalia, abdomen for faecal loading; urinalysis.
- Uroflowmetry, PVR (post-void residual) and ultrasound only for NMNE or before adding anticholinergics. No invasive tests in MNE.

Management

- **Treat daytime LUTS and constipation first** in NMNE.
- Education removes blame; standard urotherapy. Waking, fluid restriction alone and punishment do not work.

	Alarm	Desmopressin
Target	Arousal (conditioning)	Night urine output
Use	Nightly for months; parent wakes child; continue to 14 dry nights	0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed
Response	50–70%; durable	Rule of thirds; immediate but relapses
Caution	Needs motivated family	Restrict evening fluids (hyponatraemia)

- Overlearning (extra evening fluid once dry) reduces alarm relapse.
- **Failure:** switch or combine. **Second line:** add an anticholinergic (oxybutynin) after checking PVR and bowels. **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists; treat sleep apnoea (adenotonsillectomy).

Diagram to draw: triad diagram: nocturnal polyuria, small bladder capacity, arousal failure.

Key points to remember:

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Discuss diagnostics and operative pediatric laparoscopy in urology

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 3, Q3a

Evolution

- Diagnostic (non-palpable testis, 1976) → nephrectomy (1994) → pyeloplasty (1995) → robotic (2002).

Diagnostic laparoscopy

- **Non-palpable testis** (procedure of choice, after examination under anaesthesia): note vessels, vas, testis and internal ring.

Finding	Meaning
Blind-ending vessels and vas, closed ring	Vanishing testis: stop
Vessels and vas entering ring	Explore groin; excise remnant
Intra-abdominal testis	Orchidopexy (primary or Fowler-Stephens)

- **DSD (differences of sex development):** gonadal inspection and biopsy, Müllerian structures.

Operative laparoscopy

- **Testis:** primary orchidopexy (>90% success); Fowler-Stephens one- or two-stage; gonadectomy in DSD.
- **Kidney:** nephrectomy, heminephrectomy, **pyeloplasty** (commonest MIS (minimally invasive surgery) procedure).
- **Ureter and bladder:** extravesical or vesicoscopic (cross-trigonal) reimplantation, Mitrofanoff, augmentation, bladder neck reconstruction, utricle and urogenital sinus.
- **Also know:** RPLND (retroperitoneal lymph node dissection) for paratesticular rhabdomyosarcoma.

Technical principles

- **Access:** open Hasson or Veress at umbilicus (Palmer's point if scarred); equal safety.
- 3–5 mm ports; robots use 8 mm. Transperitoneal (most space), retroperitoneal (spares peritoneum, useful in peritoneal dialysis), vesicoscopic.
- **Pressures:** CO_2 absorption, reduced venous return, raised airway pressure; reversible oliguria (no fluid bolus).

Age	0–1 month	2–12 months	1–2 years	2–8 years
Max IAP (intra-abdominal pressure)	≤6 mmHg	≤8	≤10	≤12

- **Positioning:** modified flank for kidney; supine steep Trendelenburg for pelvis. Infants: minimise inhalational anaesthetic, rectal tube to reduce bowel distension.
- **Only absolute contraindication:** inability to tolerate pneumoperitoneum.

Complications

- Right common iliac vessel injury (behind umbilicus; commonest major), inferior epigastric (commonest minor), bowel injury, subcutaneous emphysema, **port-site hernia 0.5–3%** (close fascia).

Pros and cons

- Shorter stay, better cosmesis, deep pelvic access; pain benefit unproven. Robot: cost, learning curve; infant working space only about 1 L.

Diagram to draw: laparoscopic view of internal ring with blind-ending vessels.

> **Key points to remember:**
> - Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
> - Pressure ceiling 6–12 mmHg by age.
> - Right common iliac at Veress entry; close port fascia.

Rapid Revision Sheet – Section 5 · Paediatric Urology 2

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Embryological development of urinary bladder and management of Hutch Diverticulum? (LAQ)

- Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
- Hutch: herniation through the ureteric hiatus → VUR.
- Ureter entering the sac: VUR will not resolve.
- Secondary: treat the outlet first; else diverticulectomy + reimplantation.

Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) (LAQ)

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias) (LAQ)

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
- Diastasis over 3 cm (>3 cm) → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

Bladder and ureter development, exstrophy–epispadias complex and its management (LAQ)

- Bud from Wolffian duct; position decides VUR or ectopia.
- Failed mesodermal reinforcement of the cloacal membrane; rupture timing decides variant.
- Diastasis 4.8 cm; anterior pubis 30% short; VUR in 100%.
- MSRE: (birth) closure → epispadias repair 6–12 months (mo) → YDL / BNR at 5–7 years, at ≥100 mL.
- Osteotomy if >72 h (closure after 72 h) or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL predicts continence; template size + first successful closure (first closure success) predict continence.

HSG in neonates (SAQ)

- First scan after 48 hours, except suspected PUV.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign suggests PUV.

Prune belly syndrome (SAQ)

- Triad; 95% male; dysplasia decides prognosis.
- Woodard I–III.

- Vesicostomy for drainage; orchidopexy at about 6 months.

Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty (SAQ)

- Level does not predict bladder; UDS within 3 months.
- DLPP 40 cm H2O or more = hostile; early CIC + antimuscarinic.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

DD of frequency and urgency in 3 yr male child and treatment (SAQ)

- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

Embryology and Management of P.U. valve patient. (SAQ)

- Type I (95%): wolffian duct insertion; type III: persistent urogenital membrane.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: overnight drainage and CIC.

CSIC. (SAQ)

- Lapides: overdistension, not bacteria, causes infection; clean is enough.
- 4–6 times daily, under about 500 mL each.
- Do not treat asymptomatic bacteriuria.
- Mitrofanoff when urethral access is hard.

Enuresis (SAQ)

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Discuss diagnostics and operative pediatric laparoscopy in urology (SAQ)

- Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
- Pressure ceiling 6–12 mmHg by age.
- Right common iliac at Veress entry; close port fascia.

Section 6 · Female Urology — Long answer questions (LAQ)

Urological complications in OBS-GYN and uretero-vaginal fistula

Asked as:

- Mention in brief, urological complications in OBS-GYN. Describe in detail treatment of ureterovaginal fistula. — *Practice Set 7 Paper III Q1 (LAQ)* (OBGY complications also answered in Practice Set 3 Paper 3, Q1; ureterovaginal fistula in 2015-Summer Paper 3, Q3e and 2017-Summer Paper 3, Q3b)
- Uretero-vaginal fistula — *Summer 2015 Paper III Q3e (SAQ)*
- Pathogenesis and management of uretero-vaginal fistula — *Summer 2017 Paper III Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Assumed meaning: OBS-GYN = obstetric and gynaecological practice and surgery.

Part A: Urological complications in OBS-GYN

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding.

Setting	Complications
Obstetric	**Obstructed labour** (fetal head impacted for days): pressure necrosis, so large VVF (vesicovaginal fistula), urethral loss, ureteric involvement; over 90% of VVF in the developing world. **Caesarean**: bladder injury (about 0.3%), ureteric injury, **vesicouterine fistula** (Youssef syndrome: cyclical haematuria, amenorrhoea, continence preserved). Post-partum or post-epidural **retention**
Gynaecological surgery	**Bladder injury** (commonest organ; hysterectomy about 0.3–0.75%). **Ureteric injury** (hysterectomy 0.2–2.4%; over 90% distal third; ligation commonest). **Fistulae**: VVF (over 75% follow hysterectomy in the developed world), UVF (ureterovaginal fistula), urethrovaginal fistula
Radical hysterectomy	Pelvic plexus injury: retention, poor compliance; manage with CIC (clean intermittent catheterisation)
Prolapse and continence surgery	Ureteric kinking at apical suspension; mesh or sling exposure, perforation, outlet obstruction, pain
Radiotherapy	Radiation cystitis, VVF, ureteric stricture

- **Prevention:** know the four ureteric danger sites (know the danger sites); cystoscopy after (at) hysterectomy where indicated (recognise injury intraoperatively); prophylactic stents only selectively (selective prophylactic stents), only if high risk (they aid recognition but do not prevent injury).

Part B: Ureterovaginal fistula (UVF)

Definition

- **UVF (uretero-vaginal fistula)**: abnormal communication between the ureter and vagina (ureter–vagina communication), causing continuous leakage.

Aetiology / pathogenesis and presentation (brief)

- Usually **unrecognised injury to the distal third of the ureter** (distal ureteric injury) during pelvic surgery. About **three-quarters follow an uneventful abdominal hysterectomy** (often uneventful; ureteric injury in 0.2–2.4%): an unrecognised distal ureteric injury (ligation, crush, laceration, thermal, devascularisation / ischaemia) necroses and drains (leaks) into the vaginal cuff, which is the weakest point. Diagnosis is delayed in 18–88%; **delayed detection (recognition) raises fistula risk about fourfold**.
- Others: obstetric, cancer, radiation, endometriosis, foreign body.
- **Sites of injury (at risk):** pelvic brim (near the infundibulopelvic ligament); **under the uterine artery** in the cardinal ligament; lateral to (beside) the uterosacral ligament; anterolateral vaginal fornix near the UVJ (ureterovesical junction). Most in the **distal 6 cm**.
- **Mechanisms:** by ligation, crush, laceration, thermal injury, **devascularisation** or ischaemia. The injured segment necroses and leaks into the vaginal cuff. **Clinical features**
- **Continuous leakage with normal voiding** (the other kidney fills the bladder), usually (appears) **within 4 weeks**, often with fever, flank pain or ileus, as thermal or ischaemic injury necroses.

	VVF (vesicovaginal fistula)	UVF
Voiding	Reduced	Normal
Upper tract	Normal	Hydronephrosis common
First line	Catheter, repair	Stent

Confirming the diagnosis before treatment (investigations)

Test	Method and reading
Double dye test	Oral phenazopyridine 100–200 mg (orange urine); after 1 hour, blue dye instilled into the bladder; tampon; walk about. **Orange = UVF**; blue = VVF; both = both
Three-swab test	Blue dye in the bladder; three swabs stacked in the vagina. Blue upper swab = VVF; **wet, unstained swab = UVF**
CTU (computed tomography urography) / CT (computed tomography) urography, delayed phase	Hydronephrosis, extravasation; MR (magnetic resonance) urography if contrast is contraindicated
Cystoscopy + RGP (retrograde pyelogram)	Abrupt cut-off or contrast leak; stent (stenting) at the same sitting

- **Exclude a coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury.

Treatment (management)

Goals: dryness and preservation of renal function. Management has moved from nephrectomy, through reimplantation, to endourology.

1. Endourological (first line): stent first

- **Retrograde double-J stent as soon as possible** (if a wire passes); otherwise PCN and antegrade stent. Stenting **within 2 weeks** of detection works best: succeeds far more often than at 2–6 weeks, because the ureter heals over the stent before fibrosis.
- Wire will not pass: **PCN (percutaneous nephrostomy)** with antegrade stent, or rendezvous.
- Stent for **4–6 weeks**, then remove and re-image. Most stentable fistulae close; a stricture may follow.

2. Surgical repair (stent fails or impossible)

- **Timing:** no consensus. Early repair equals delayed: early repair took less time and blood loss with equal success; a systematic review found equal success before and after 6 weeks. Delay only if tissue is still inflamed.
- **Ureteroneocystostomy** (distal third, so usually feasible):
  1. Transperitoneal or extraperitoneal approach; identify ureter above the fistula.
  2. Divide above the diseased segment; dissect as distally as possible for length; **preserve periureteric adventitia**, especially laterally where vessels enter.
  3. Mobilise the bladder; spatulate; tension-free mucosa-to-mucosa anastomosis over a double-J stent; refluxing or tunnelled.
  4. Drain; catheter 7–14 days.
- **Adding length:**

Technique	Extra reach
Psoas hitch (sutures parallel to psoas fibres, spare genitofemoral nerve)	Up to about 5 cm
Boari flap (base ≥4 cm, length:base ≤3:1)	10–15 cm (up to 15 cm)
Downward nephropexy; ileal ureter	Long (longer) defects / gaps

- Open, **laparoscopic and robotic repair give equal results**; vaginal reimplantation reported. **Success ≥90% (90% or more).**
- The vaginal opening is left to heal; no separate vaginal repair is needed.

```
Suspected UVF / Leak + normal voiding -> exclude VVF (dye test, cystoscopy) -> CTU (CT
urography) / RGP
  -> stent passes (wire passes)? Yes -> double-J 4-6 wk -> re-image -> healed / persistent
                  No  -> PCN + antegrade stent -> fails
  -> ureteroneocystostomy + psoas hitch / Boari flap
```

Complications and follow-up

- Hydronephrosis, infection, ischaemic stricture.
- Follow-up ultrasound 1–2 weeks after stent removal; some add a renal scan at 6 weeks. Stricture usually appears within a year but may appear **up to 30 months** later, so follow beyond this.

Diagram to draw: ureter passing under the uterine artery; double dye test; ureteroneocystostomy with psoas hitch.

Key points to remember:

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Distal ureteric injury at hysterectomy; leak within 4 weeks, normal voiding (continuous leak with normal voiding).
- UVF: orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if it fails; success ≥90%; follow 30 months.

Female pelvic floor, continence mechanism, classification and management of SUI

Asked as:

- Discuss continence mechanism in females . classify SUI and principles of different surgical options — *Summer 2017 Paper III Q2 (LAQ)*
- Describe anatomy of pelvic floor in females and describe surgeries for SUI in females. — *Winter 2017 Paper III Q2 (LAQ)*
- Management of stress urinary incontinence. — *Winter 2024 Paper III Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction (definition and mechanism)

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, coughing or sneezing (ICS, International Continence Society).
- Continence needs **urethral pressure greater than bladder pressure** at rest and during stress.
- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.
- Two mechanisms: **urethral hypermobility** (loss of the suburethral "hammock") and **ISD (intrinsic sphincter deficiency)**.

A. Anatomy of the female pelvic floor

1. Pelvic diaphragm

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction	Anorectal angle
Pubococcygeus (pubovaginal)	Pubis, ATFP (arcus tendineus fasciae pelvis) → viscera, raphe	Slings urethra and vagina
Iliococcygeus	Arcus tendineus levator ani, ischial spine → coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine → sacrum, coccyx	Lies on sacrospinous ligament

- Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, fascia then carries the load.

2. Perineal membrane and perineal body

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

3. Endopelvic fascia and ligaments

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

4. DeLancey levels (1992)

Level	Structures	Defect
I suspension	Cardinal–uterosacral	Apical prolapse, enterocele
II attachment	ATFP, levator fascia	Cystocele (central or paravaginal), rectocele

Level	Structures	Defect
III fusion	Perineal membrane, perineal body	**Urethral hypermobility**, perineal descent

B. Continence mechanism in the female

1. Intrinsic urethral factors (smooth and striated sphincters, vascular submucosa)

- **Mucosal seal**: folded urothelium coapts. The **oestrogen-sensitive (oestrogen-dependent) submucosal vascular plexus** (oestrogen-dependent vascular submucosa) provides up to **30%** of closure force (much of closure pressure).
- **Smooth muscle** (internal sphincter, bladder neck and proximal urethra): involuntary tone. The female bladder neck is weak and often open in continent women.
- **Striated urogenital sphincter**: the rhabdosphincter, mostly **slow-twitch** fibres, gives tonic mid-urethral closure. Distal parts are the **compressor urethrae** and **urethrovaginal sphincter**.
- The **continence zone is the proximal and mid-urethra.**

2. Support (extrinsic)

- The urethra lies on the **anterior vaginal wall and endopelvic fascia**, attached laterally to the **ATFP (arcus tendineus fasciae pelvis)** and levator ani.
- **Pubourethral ligaments** anchor the mid-urethra; **levator ani** (pubococcygeus) tone keeps the backing firm.

3. Neural

- **Pudendal nerve (S2–4)** drives the sphincter (**guarding reflex**). Urethral pressure rises **just before** a cough reaches it, so closure is active, not just transmitted.

4. Theories of continence and SUI

Theory	Mechanism	Surgical implication (answer)
Pressure transmission (Enhorning 1961)	Hypermobile urethra drops out of (leaves) the abdominal pressure zone	Restore position (Burch)
Hammock (DeLancey 1994)	Abdominal pressure compresses the urethra against a firm suburethral layer (urethra compressed against it); a lax layer fails	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament laxity at the mid-urethra	Tension-free MUS (mid-urethral sling)
ISD (intrinsic sphincter deficiency; McGuire & Lytton 1978)	Urethra itself fails to close (VLPP (Valsalva leak point pressure) <60, MUCP (maximum urethral closure pressure) <20 cmH_2O)	Sling, bulking, sphincter (AUS (artificial urinary sphincter))

- Most SUI has some ISD. **Hypermobility alone never causes SUI.**

C. Classification of SUI

1. Blaivas & Olsson (1988), on video-urodynamics

Type	Bladder neck at rest	On stress
0	Closed, above lower border of symphysis	Descends and opens, but no leak seen (history of SUI)
I	Closed, above symphysis	Descends **<2 cm**; leaks; little or no cystocele
IIA	Closed, above symphysis	Descends **>2 cm** with cystourethrocele; leaks
IIB	Closed but **at or below** lower border of symphysis	Proximal urethra opens; leaks with or without further descent
III	**Open at rest** without detrusor contraction	Leaks with minimal stress = **ISD**

2. Functional: hypermobility versus ISD

	Urethral hypermobility	ISD
Q-tip test	Angle change **>30°**	Often fixed urethra
VLPP (Valsalva leak point pressure)	**≥90 cmH_2O**	**<60 cmH_2O** (60–90 equivocal)
MUCP (maximum urethral closure pressure)	Normal	**≤20 cmH_2O**
Cause	Childbirth, ageing, obesity	Previous surgery, radiation, neuropathy, ageing

3. Other

- **Severity (Stamey):** grade 1 leak on cough or sneeze; grade 2 on minimal effort (walking, rising); grade 3 at rest.
- **Index (uncomplicated)** versus **complicated** SUI (recurrent, prolapse, neurogenic, high PVR (post-void residual), radiation); pure versus mixed incontinence.

D. Evaluation (before surgery)

- Bladder diary, pad test, **cough stress test**, PVR (post-void residual), urinalysis; Q-tip >30° = hypermobility.
- **Urodynamics not needed** in the uncomplicated (index) patient (VALUE trial); use if SUI not demonstrated, or complicated: mixed, recurrent, neurogenic or with prolapse.

E. Conservative (first line)

- **PFMT (pelvic floor muscle training)**, weight loss, pessary first.
- Weight loss, fluid and caffeine adjustment.
- **Supervised PFMT** for at least 3 months (Cochrane: more cure than no treatment).
- **Pessary**; vaginal oestrogen for atrophy.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): increases Onuf's nucleus drive to the sphincter; approved in Europe, not USA.

F. Principles of surgical options (surgeries for female SUI)

Operation	Principle (how it works)	Best for	Key facts
MUS (mid-urethral sling): retropubic (TVT, tension-free vaginal tape, Ulmsten 1995), transobturator (TOT; Delorme 2001; inside-out de Leval), single-incision	Loose (tension-free) type I polypropylene tape under the **mid-urethra** acts as a new pubourethral ligament; urethra kinks against it on (during) stress	First-line index SUI	**First line** (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) strong recommendation). **TOMUS**: retropubic 80.8% vs transobturator 77.7% cure (equal, equivalent); retropubic often preferred in ISD. Retropubic: more bladder perforation; transobturator: more groin pain
SIMS (single-incision mini-sling)	Short tape, vaginal only	Option	Non-inferior to transobturator (TOT) for subjective cure (in most studies); conditional recommendation (2023)

Operation	Principle (how it works)	Best for	Key facts
Autologous pubovaginal sling (PVS) (rectus fascia or fascia lata)	At the **bladder neck**, minimal tension; restores the hammock and compresses (bladder-neck support and compression)	**All types, including ISD**, recurrent SUI, neurogenic SUI, failed MUS, after mesh removal, mesh-averse	Gold standard. **SISTEr**: sling beat Burch (47% vs 38%) but more voiding dysfunction and UTI (urinary tract infection)
Burch colposuspension (open, laparoscopic or robotic)	Paravaginal fascia beside the bladder neck (at mid-urethra and bladder neck) sutured to **Cooper's ligament**; lifts the bladder neck back into the pressure zone (restores bladder neck position) and gives a stable backing	Hypermobility; women declining mesh; at abdominal surgery	85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD
Paravaginal repair, VOS (vagino-obturator shelf)	Vagina to ATFP or obturator fascia		Flatter, less obstructive
MMK (Marshall–Marchetti–Krantz)	Bladder neck to pubic (symphysis) periosteum	Historical	Osteitis pubis 0.7–2.5%; abandoned
(Urethral) bulking agents	Submucosal injection at bladder neck or proximal urethra improves coaptation	ISD with fixed urethra; frail (patients); minimal surgery	Low durability (less durable); repeat injections
AUS (artificial urinary sphincter)	Cuff round the bladder neck (bladder-neck cuff); constant circumferential closure	Severe recurrent (refractory) ISD, fixed urethra	Second-line; erosion, infection, revision

- **Also know:** needle suspensions abandoned (sutures pull through; pull-through); anterior colporrhaphy is not an SUI operation; mesh complications (exposure, erosion into bladder or urethra, pain) need excision; FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.

Burch steps: Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

Retropubic vs transobturator MUS

	Retropubic	Transobturator
Bladder perforation	4.5%	0.6%
Voiding dysfunction	More (TOMUS surgery 2.7% vs 0%)	Less
Groin pain	1.3%	6.4%
ISD, long-term durability	Often preferred	Possibly more repeat surgery

G. Complications

- **MUS complications:** perforation, retention, de novo urgency, mesh exposure (~2%), pain.

- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal) → urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS (International Urogynecological Association/International Continence Society) terms): exposure, extrusion, perforation into bladder or urethra, pain → excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.
- **Also know:** CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension).

H. Male SUI (brief)

- Usually after radical prostatectomy. PFMT first; no surgery for 6–12 months, as continence keeps improving.
- **AUS** for moderate–severe or irradiated men (gold standard, ~80% continent); **male sling** for mild–moderate leakage without radiotherapy. Bulking agents not advised.

Diagram to draw: DeLancey levels and hammock cross-section; sites of Burch sutures, bladder-neck sling and mid-urethral tape; sagittal pelvis showing retropubic and transobturator tape paths and Burch sutures.

Key points to remember:

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD (VLPP <60, MUCP ≤20 cmH_2O).
- Burch corrects position; slings and AUS correct function.
- VALUE: no urodynamics in index SUI.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.

SUI: evaluation in females, role of UDS, and surgical therapy in males

Asked as:

- Describe evaluation of SUI in females. Discuss surgical therapy of SUI in Males. — *Summer 2019 Paper III Q2 (LAQ)*
- Role of UDS in SUI — *Practice Set 1 Paper III Q3b (SAQ)* (SUI evaluation also answered in 2019-Summer Paper 3, Q2)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Part A: Evaluation of SUI (stress urinary incontinence) in women

- **Definition (ICS [International Continence Society]):** involuntary leakage on effort, exertion, sneezing or coughing.
- **Mechanisms:** urethral **hypermobility** (loss of suburethral support, DeLancey hammock) and **ISD (intrinsic sphincter deficiency)**. Most women have some ISD.

History

- Leakage with cough or activity means SUI; after urgency, UUI (urgency urinary incontinence); both, MUI (mixed). A positive SUI history predicts urodynamic SUI in about three-quarters.
- Severity (pads/day), bother; voiding symptoms, prolapse, obstetric history, prior surgery or radiation, neurological disease, drugs.
- **Questionnaires:** ICIQ-UI SF (International Consultation on Incontinence Questionnaire–UI Short Form); UDI-6 (Urogenital Distress Inventory: bother); IIQ-7 (Incontinence Impact Questionnaire: impact). **Bladder diary** (3 days).

Examination (comfortably full bladder)

- Neurological (S2–S4 sensation, anal tone, bulbocavernosus reflex); atrophy; prolapse by **POP-Q (Pelvic Organ Prolapse Quantification)**.
- **Cough stress test:** cough supine, then standing if negative. Leakage synchronous with cough confirms SUI. Reduce prolapse to unmask **occult SUI**.
- **Q-tip test:** swab to bladder neck, Valsalva; rotation **>30°** = hypermobility. Visual mobility examination is now routine.

Tests

Test	Method and use
Urinalysis	Required; exclude infection, haematuria
PVR (post-void residual)	Required before surgery; ultrasound; repeat if high (no agreed cut-off)
1-hour pad test (ICS)	500 mL fluid, standard exercises; >1.4 g positive; 1–10 g mild, 11–50 g moderate, >50 g severe
24-hour pad test	Normal activities; more reproducible
Cystoscopy	Haematuria, irritative symptoms, previous mesh

Role of urodynamics (UDS) in SUI

- **Introduction:** UDS (urodynamic studies) assess storage and voiding. In SUI the key question is: **will the result change the operation?** Pure, demonstrable SUI gains little. **Urodynamics (UDS): not routine.**

What UDS shows in SUI

Parameter	Method and meaning
USI (urodynamic stress incontinence)	Leak with raised abdominal pressure, no detrusor contraction
ALPP/VLPP (abdominal/Valsalva leak point pressure)	Standing, from 150 mL, Valsalva or cough; **<60 cmH2O = ISD (intrinsic sphincter deficiency)**, 60–90 equivocal, ≥90 = little ISD (McGuire 1993)
MUCP (maximum urethral closure pressure)	Urethral pressure profile; a low MUCP (<20 / ≤20 cmH2O) also suggests ISD; does not diagnose or grade SUI
Filling cystometry	DO (detrusor overactivity), compliance, capacity; cough-induced DO mimics SUI
Pressure–flow study	Voiding dysfunction, poor contractility (retention risk after sling)
Prolapse reduction	Occult SUI

- If no leak with the catheter, repeat with it removed.

Evidence

- **VALUE / ValUE trial (2012), 630 women** with uncomplicated SUI: office evaluation alone was **non-inferior** to office evaluation plus UDS before surgery (success 77.2% vs 76.9%).
- **Dutch VUE study:** immediate midurethral sling was non-inferior to UDS-tailored treatment.
- **SISTEr:** UDS did not predict outcome.

Guidelines

- **AUA [American Urological Association]/SUFU [Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2023:** UDS may be omitted in the index patient with demonstrable SUI.
- **Indications / UDS advised in non-index patients (AUA/SUFU 2023):** unclear diagnosis or SUI not shown (not demonstrated); mismatch between symptoms and findings; urgency-predominant MUI (mixed incontinence); neurogenic bladder; high (raised) PVR or voiding dysfunction; prolapse stage ≥3 without demonstrable SUI; failed previous surgery (previous anti-incontinence or prolapse surgery).
- **EAU (European Association of Urology):** no routine UDS for uncomplicated SUI; do not use leak point or urethral pressure to grade severity.

Men

- Before AUS (artificial urinary sphincter) or sling for post-prostatectomy incontinence: confirms USI, finds DO, poor compliance or weak contractility. MASTER trial recruited men with USI.

Part B: Surgical therapy of male SUI

- **Aetiology:** commonest after **radical prostatectomy (RP)**, which removes the internal sphincter, leaving continence to the rhabdosphincter. Also TURP (transurethral resection of the prostate), radiation, pelvic fracture.
- **Work-up:** 24-h pad weight (mild <150 g, moderate <400 g, severe above); standing cough test; **cystoscopy** (anastomotic stricture, sphincter coaptation); UDS if mixed or poor contractility.
- **Timing:** PFMT (pelvic floor muscle training) first; continence improves for up to 2 years, but do not delay surgery beyond 6 months for severe leak.

Devices

Device	Mechanism	Use and results
Transobturator sling (AdVance XP)	Repositions the bulb 3–4 cm proximally; needs residual sphincter	Mild–moderate, no radiation; cure about 60%
Quadratic sling (Virtue)	Compression plus elevation	Mild; limited data
Adjustable sling (ATOMS, adjustable transobturator male system)	Cushion adjusted via scrotal port	Mild–moderate; about two-thirds dry
ProACT balloons	Bladder-neck compression	More erosion
Bulking agents	Submucosal bulk	Small leaks, unfit men; rarely durable
AUS (artificial urinary sphincter)	Circumferential compression	**Gold standard**; all severities, after radiation

AMS 800 AUS

- **Components:** urethral **cuff** (bulbar, usually 4.0–4.5 cm), **PRB (pressure-regulating balloon) 61–70 cmH2O** (51–60 high risk; 71–80 bladder neck) in a preperitoneal pocket, scrotal **pump**. Filled with 23 mL.
- **Technique:** perineal incision; cuff on proximal bulbar urethra; **deactivate for 4–6 weeks**.
- **Needs** dexterity and cognition. Social continence (0–1 pad) 55–77%; satisfaction about 90%; lifespan 7–10 years.
- **Complications:** infection 1–3% (explant; reimplant after 3–6 months); **erosion** 2–15% (explant, catheter about 3 weeks); urethral **atrophy** (late recurrent leak); mechanical failure; retention.
- **Erosion risks:** radiation, prior erosion, urethroplasty; use **transcorporal cuff** (tunica adds bulk) or lower PRB. Atrophy: smaller or tandem cuff.

Algorithm (AUA/SUFU 2019)

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Severe SUI or any radiotherapy -> AUS
Moderate, no radiotherapy      -> AUS or male sling
Mild, no radiotherapy          -> sling (or PFMT, AUS)
Sling failure                  -> AUS
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- **MASTER RCT (randomised controlled trial):** sling was **non-inferior** to AUS for continence at 12 months; AUS remains preferred for heavy leak (>250 g/day) or radiation.

Diagram to draw: AMS 800 (cuff, pump, PRB) and transobturator sling path; cystometrogram trace with abdominal pressure spike and leak (ALPP).

> **Key points to remember:**
> - Q-tip >30° = hypermobility; VLPP / ALPP <60 cmH2O = ISD; MUCP ≤20 cmH2O.
> - VALUE (ValUE) trial: UDS not needed in uncomplicated SUI.
> - Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
> - AUS PRB 61–70 cmH2O; deactivate 4–6 weeks.
> - Radiation or severe SUI: AUS first; MASTER: sling non-inferior.

Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management

Asked as:

- Etio-pathogenesis, diagnosis and management of Recto-urethral fistula? — *Summer 2020 Paper III Q1 (LAQ)*
- Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula — *Winter 2025 Paper III Q1 (LAQ)* (also answered in 2020-Summer Paper 3, Q1, as recto-urethral fistula)

Type: LAQ (25 marks)

Assumed meaning (recto-prostatic): rectourethral fistula at the level of the prostatic urethra or the vesicourethral anastomosis.

Definition

- **RUF (rectourethral fistula):** a communication between the posterior urethra / prostatic urethra (or the vesicourethral anastomosis after prostatectomy) and the distal rectum.
- Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

Aetiology

Group	Causes
Congenital	Anorectal malformation in boys (rectobulbar or rectoprostatic fistula)
Surgical (commonest acquired)	**RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate) and open (simple) prostatectomy
Radiation and energy	**EBRT (external beam radiotherapy) plus brachytherapy** (commonest), then brachytherapy alone, then pelvic radiotherapy; cryotherapy; HIFU (high-intensity focused ultrasound): low after primary, up to about 9% after salvage HIFU
Others	Trauma (penetrating injury, pelvic fracture); Crohn disease; advanced rectal or prostate cancer

Pathogenesis

- **Surgical:** the rectum is torn during apical or posterior dissection behind Denonvilliers fascia, especially with small glands and salvage surgery. An unrecognised tear (near the apex) leaks urine (urine leaks into it), an abscess forms and the tract epithelialises.
  - Rectal injury at RP: laparoscopic about 1%, open and robotic under 0.5%, salvage RP about 7%. (Another source: rectal injury is commonest with retropubic and laparoscopic RP, lowest with robotic RP, and was over 15% in historical open salvage series.)
- **Radiation or ablation:** obliterative endarteritis makes the tissue hypoxic and fibrotic. Necrosis follows months to years later. These fistulae are large and heal poorly.
- **Congenital:** abnormal cloacal development leaves the rectum ending in the posterior urethra.

Classification: Martini (after prostate cancer treatment)

Axis	Category	Definition
Stage (size on CT (computed tomography) urography or cystography)	I / II / III	≤1.5 cm / >1.5 cm / any size with sphincter damage
Grade (tissue)	0	Surgical injury, no prior non-surgical treatment

Axis	Category	Definition
	1	Prior radiation, brachytherapy, HIFU or other energy
	2	Salvage RP or salvage ablation

- **Tissue quality (grade), not size, chiefly decides the repair.**

Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding.
- Recurrent UTI (urinary tract infection), dysuria, urosepsis; coexisting SUI (stress urinary incontinence).

Diagnosis

- **DRE (digital rectal examination):** may feel the defect on the anterior rectal wall.
- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.
- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; gives size and site for planning.
- Upper-tract imaging to exclude ureteric injury; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: fistula repair alone will not cure severe SUI; an AUS (artificial urinary sphincter) may be needed later; a later AUS erodes more often.

Prevention

- At RP: a recognised tear without gross spillage → two-layer (multilayer) closure, air test, **omental or peritoneal interposition**, drain, antibiotics; colostomy if (previously) irradiated.
- Radiotherapy: a **hydrogel rectal spacer** lowers rectal dose and late rectal toxicity (randomised trial: late rectal toxicity 2.0% vs 7.0%).

Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close urethra and rectum separately in **tension-free, non-overlapping layers with vascularised (interposition) tissue interposed**.
- **Conservative:** no sepsis and no faecaluria → catheter (urethral or suprapubic), low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; choice depends on site, radiation and prior repair. **A diverting colostomy usually comes first.**

Approach	Essentials	Strengths	Weaknesses
Perineal (Young; Goodwin)	Separate and close urethra and rectum separately; **gracilis flap** between (or dartos; buccal graft)	High closure; most stomas reversed	Urinary incontinence common
Transsphincteric (York-Mason)	Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure	Avoids scarred plane; spares sexual function	Poor membranous exposure; no flap; faecal incontinence up to 20%
Abdominal (transvesical, extravesical, O'Conor-like) / robotic	Omental flap	Large or irradiated fistulae, extensive resection	Morbid; poor deep-pelvic view; small robotic series

Approach	Essentials	Strengths	Weaknesses
Laparoscopic / robotic	Open pouch of Douglas; close rectum in two layers; omental or peritoneal flap; can combine with perineal gracilis	Deep-pelvic vision	Small series only
Salvage	Cystoprostatectomy, ileal conduit ± permanent colostomy	Unreconstructable irradiated field	Major surgery

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RUF → sepsis control, catheter ± colostomy, biopsy
  small, non-irradiated, no faecaluria → conservative ~12 weeks
  persistent, grade 0 → perineal (gracilis) or York-Mason
  irradiated or large → abdominal/robotic (omentum) or perineal (gracilis)
  unreconstructable → urinary + faecal diversion
Congenital → newborn colostomy → posterior sagittal anorectoplasty
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- **PSARP (posterior sagittal anorectoplasty)** separates the rectum from the urethra and places it within (in) the sphincter complex.

Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, urethral stricture, ED (erectile dysfunction), wound infection, permanent stoma.

Diagram to draw: sagittal pelvis showing (with) the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy to exclude cancer.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Urological complications of OBGY surgery and urethrovaginal fistula

Asked as:

- Describe in brief urological complications of OBGY surgery. Describe in detail, management of urethrovaginal fistula — *Practice Set 3 Paper III Q1 (LAQ)*
- Urethrovaginal fistula — *Winter 2025 Paper III Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(urethrovaginal fistula also answered in 2025-Winter Paper 3, Q3a; iatrogenic ureteric injury in Practice Set 2 Paper 3, Q1)

Assumed meaning: OBGY = obstetric and gynaecological surgery.

Part A: Urological complications of OBGY surgery

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding, prolapse or mesh surgery.

Complication	Key facts
Bladder injury (commonest organ injured)	Caesarean 0.27–0.3%; hysterectomy about 0.3–0.75% (laparoscopic highest); retropubic tape 3–9%. Repair: debride, 2-layer closure, fill test, catheter 7–14 days; check ureters if trigonal
Ureteric injury	Gynaecology causes over half of iatrogenic injuries; over 90% distal third. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery 1.7–4.3%. **Ligation** is the commonest gynaecological mechanism
Fistulae	VVF (vesicovaginal fistula), about 0.8 per 1000 hysterectomies; UVF (ureterovaginal fistula); vesicouterine fistula after caesarean (Youssef syndrome); urethrovaginal fistula
Voiding dysfunction	Radical hysterectomy (pelvic plexus injury) 16–80% historically: retention, poor compliance; CIC (clean intermittent catheterisation) and wait. Post-partum or post-epidural overdistension
Mesh and sling	Exposure, erosion into bladder or urethra, outlet obstruction, pain, dyspareunia
Radiation	Radiation cystitis; VVF in 0.6–2% after cervical cancer radiotherapy; ureteric stricture

- **Ureteric danger sites:** pelvic brim (infundibulopelvic ligament), under the uterine artery, beside the uterosacral ligament, anterolateral vaginal fornix.
- **Prevention and recognition:** haemostasis and anatomy; prophylactic stents only selectively (they aid recognition but do not prevent injury); cystoscopy with dye (IV indigo carmine or oral phenazopyridine) after minimally invasive hysterectomy and prolapse or continence surgery.
- Only about one-third of ureteric injuries are recognised at open surgery. Missed injury presents with fever, ileus, flank pain, urinoma or leakage; do CTU (computed tomography urography) with delayed films, then retrograde stenting. Distal repair: ureteroneocystostomy ± psoas hitch or Boari flap (see Q3d).

Part B: Urethrovaginal fistula

Aetiology

- Developing world: **obstructed labour**, often with a VVF (vesicovaginal fistula); also trauma, radiation, catheter injury.

- Developed world (iatrogenic): **urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of urethrovaginal fistulae (UVFs)), slings, bulking agents, anterior repair or mesh, indwelling catheters (catheter pressure necrosis), pessaries, radiation, trauma; also gender-affirming and paediatric genital surgery.

Clinical features

- Depend on position (site) relative to the sphincter. **Distal** fistula: often asymptomatic, split stream, vaginal voiding, post-void dribble. **Proximal** (bladder neck): continuous leakage or SUI (stress urinary incontinence).
- Dyspareunia, recurrent UTI (urinary tract infection), perineal irritation; partner discomfort with exposed mesh.

Evaluation (diagnosis)

- **Speculum examination** usually suffices; **methylene blue test** if rugae hide it (distal stain on tampon). Assess prolapse, SUI, atrophy.
- **VCUG (voiding cystourethrogram), lateral view;** VUDS (videourodynamics) for coexisting (associated) SUI/incontinence or detrusor overactivity.
- **Cystourethroscopy** (flexible is better, outflow at the lens; flexible scope distends the urethra better): size, site, bladder neck, exposed mesh, other fistulae. **About 20% have a separate VVF.** Image the upper tract.
- Goh location/classification (distance from external meatus): type 1 >3.5 cm, 2: 2.5–3.5, 3: 1.5–2.5, 4: <1.5 cm from meatus. Closer means worse closure and continence.
- Optimise: treat UTI, vaginal oestrogen, nutrition; biopsy if previous cancer.

Management

- Asymptomatic distal fistula: may need no repair. Timing: early if tissue healthy; otherwise after inflammation settles.
- **Standard: transvaginal rotational-flap repair:**

Step	Details
1. Exposure	Lithotomy (prone gives better anterior view/exposure); cystourethroscopy, cannulate tract; suprapubic catheter (Lowsley tractor), urethral catheter (catheter ± suprapubic)
2. Incision	Hydrodissect; circumscribe tract with 1–2 mm cuff, minimal excision; inverted **J or U** (J- or U-shaped vaginal flap) incision
3. Flaps	Raise vaginal flap; mobilise periurethral tissue widely to bladder neck; open retropubic space if sling planned
4. Closure	**Close urethra** over a urethral sound: **4-0 absorbable** sutures, transverse or longitudinal; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test
5. Interposition	**Martius flap** (labial fat pad, external pudendal pedicle) most often; omentum, gracilis, rectus fascia occasionally
6. Cover	Advance vaginal flap beyond closure (2-0)

- **Autologous pubovaginal sling** for (if) SUI or bladder-neck involvement; **never synthetic** (erodes ischaemic urethra). Remove all (any) mesh in the field.
- **Near-total urethral loss:** tubularised anterior bladder flap neourethra. Irreparable cases: diversion.
- **After repair:** urethral catheter out at 2–3 days; suprapubic catheter ≥14 days; lateral VCUG before removal; nothing per vagina for 6 weeks.

Outcomes and complications

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI in up to half. Sling then or staged later.

- Urethral stricture (may follow), recurrence, dyspareunia.

Diagram to draw: inverted-U vaginal flap, two perpendicular urethral layers (layered closure), Martius flap.

Key points to remember:

- Bladder is the commonest organ injured; ureteric injury is mostly distal and missed.
- Diverticulectomy is the commonest Western cause; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

Vesicovaginal fistula (VVF): management options and early vs delayed repair

Asked as:

- Discuss management options of VVF — *Practice Set 3 Paper III Q2 (LAQ)*
- Early vs. delayed repair in VVF — *Practice Set 5 Paper III Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(timing in 2021-Summer Paper 2, Q3b; recurrent VVF in 2025-Winter Paper 2, Q3f; early vs delayed repair also answered in 2021-Summer Paper 2, Q3b)

Introduction

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract causing continuous leakage; commonest acquired urinary fistula (>75% of genitourinary fistulae).
- **Developing world (low-resource settings):** obstetric (obstructed labour) (>90%); fetal head impacted for days causes pressure necrosis. **Developed world (high-resource settings):** iatrogenic (>75% after hysterectomy; mainly hysterectomy), usually supratrigonal at the vaginal cuff. Also radiation, malignancy, mesh, foreign body.
- Mechanism (common mechanism): **ischaemic necrosis**; continuous urine flow keeps the tract open until it epithelialises.

Classification

Goh (distance from meatus)	Size	Tissue
1: >3.5 cm; 2: 2.5–3.5; 3: 1.5–2.5; 4: <1.5 cm	A <1.5 cm; B 1.5–3 cm; C >3 cm	i mild fibrosis; ii moderate/severe; iii radiation, ureteric, circumferential, previous repair

- **Also know:** Waaldijk (closing mechanism involvement); WHO (World Health Organization) simple vs complex.

Evaluation (guides choice)

- Speculum; fluid creatinine. **Dye tests:** single (blue backfill, tampon); **double** (oral phenazopyridine plus blue): orange = UVF (ureterovaginal fistula), blue = VVF.
- **Cystoscopy and vaginoscopy:** number, size, ureteric orifices, bladder neck, mesh; **biopsy if previous cancer.**
- **Upper tract (CTU, computed tomography urography) is mandatory:** coexisting ureteric fistula (up to 12–25%).
- Urodynamics: capacity, compliance, SUI (stress urinary incontinence) in 10–15%.
- Prepare: vaginal oestrogen for atrophy; counsel about residual incontinence, dyspareunia and possible reoperation.

Principles

- Optimise nutrition, treat infection, drain continuously, relieve obstruction, exclude malignancy.
- Surgery: tension-free, multilayer, **non-overlapping**, watertight closure; remove foreign material; consider interposition; drain afterwards. **The first repair has the best chance.**
- At repair: treat infection; ureteric stents if within 1 cm of a ureter; vaginal or abdominal route (open, laparoscopic, robotic); **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter drainage for about 2 weeks.

Option 1: Conservative and endoscopic (conservative trial while waiting)

- **Catheter drainage** cures about 15%. Favourable (best if): small, <1 cm (best <3 mm), fresh, started (catheterised) within **3 weeks**, non-irradiated. Unlikely after 6 weeks (once epithelialised, ≥6 weeks, closure is unlikely). EAU (European Association of Urology): if tried, ≥12 weeks (at least 12 weeks).
- **Fulguration** plus 2 weeks' drainage for tiny tracts <3.5 mm.
- **Sealants** (fibrin glue, cyanoacrylate, collagen): tiny series; not recommended by EAU.

Timing: early vs delayed repair (the debate)

	Early repair	Delayed repair
Definition (varies)	Days, <6 weeks or <3 months	Traditionally ≥3 months
Rationale	Clean iatrogenic tissue heals well; ends wetness and distress early	Inflammation settles, necrosis sloughs, margins become clear
Evidence	**Comparable closure** in most series; Waaldijk: immediate obstetric repair gave very high first-attempt closure	One small series: more recurrence when repaired <6 weeks
Drawback	Friable tissue if inflamed	Prolonged distress, no better outcome

- Some authors avoid the **4–12 week** window (active healing).
- **EAU (European Association of Urology) consensus:** no minimum delay if there is **no active infection and no recent irradiation (radiation)**.
- **Timing by cause:**
  - **Clean iatrogenic / post-hysterectomy:** early repair.
  - **Obstetric with necrosis (necrotic edges):** catheter trial, then repair at about 3 months.
  - **Radiation:** wait **≥12 months** after radiotherapy; biopsy to exclude recurrence; interposition flap; consider diversion if the bladder is poor.

Option 2: Surgical route

Vaginal	Abdominal (open, laparoscopic, robotic)
Low, trigonal, accessible apex	Supratrigonal, narrow vagina, no descent
Ureters distant	Ureter involved, reimplant needed
Less pain, blood loss, stay	Augmentation, Tanagho flap, diversion possible
Risk: vaginal shortening	Large (≥4 cm primary, ≥2 cm recurrent), radiation

- First-attempt closure about **90% either route**; choose the route the surgeon knows best.

Vaginal techniques

- **Flap-splitting (Raz):** ureteric stents if within 1 cm; Foley through tract for traction; inverted J/U incision; raise flaps; layer 1 3-0/4-0 including tract edge; layer 2 perpendicular; fill test 200–300 mL; Martius or peritoneal flap; advance opposite flap ≥2 cm beyond.
- **Latzko (high partial colpocleisis):** denude vagina 1–2 cm around tract and close vaginal walls over it; bladder not opened. ≥90% in non-irradiated VVF; favoured for radiation VVF and tracts near the ureters; vaginal shortening.
- **Also know:** vaginal cuff excision technique.

Abdominal techniques

- **O'Conor (1951):** extraperitoneal; bivalve bladder from dome down to fistula; excise tract; separate bladder from vagina 2–3 cm beyond; close vagina; **omental or peritoneal flap**;

close bladder; suprapubic catheter.

- **Transvesical** (limited cystotomy) and **extravesical** (no cystotomy; fewer LUTS [lower urinary tract symptoms]).
- **Laparoscopic (1994) and robotic (2005):** same closure (85–100%) with less blood loss and shorter stay; vaginal stent, backfill, ICG (indocyanine green) perfusion check.

Interposition flaps (for radiation, recurrent, ischaemic or large VVF)

Flap	Supply	Use
Martius (labial fat pad)	Internal and external pudendal	Mid–distal VVF; not apex
Peritoneal (vaginal)	Pouch of Douglas fold	Apical post-hysterectomy
Omentum (J-flap)	Gastroepiploic arteries	Abdominal; radiation, large
Gracilis, Singapore, rectus	Myocutaneous	Refractory, tissue loss

- RCT (randomised controlled trial): Martius added nothing in simple VVF, with donor-site pain in nearly a third.

Postoperative care

- Catheter ≥10–14 days (urethral and/or suprapubic); antimuscarinic or β3-agonist for spasm; cystogram before removal; stents out at 4–6 weeks.

Option 3: Urinary diversion

- Multiple failures, severe radiation, massive tissue loss: conduit or continent reservoir. Unfit: ureteric occlusion with nephrostomies. Resource-poor: ureterosigmoidostomy as last resort.

Outcomes

- Residual urgency or incontinence despite closure (Goh 3–4 predicts); retention (vertical bladder closure); dyspareunia (narrow vagina, fistula ≥3 cm). ICS (International Continence Society): pelvic floor physiotherapy.

Diagram to draw: flap-splitting repair: perpendicular bladder layers, Martius flap, advanced vaginal flap. For timing: timeline showing early window, 4–12 week zone, 3 months (obstetric) and 12 months (radiation).

Key points to remember:

- Image upper tracts; exclude ureteric fistula and cancer.
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- Interposition for complex VVF; radiation repair after ≥12 months.

Section 6 · Female Urology — Short answer questions (SAQ)

Female sexual dysfunction

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 3, Q3f

Definition and prevalence

- FSD (female sexual dysfunction): persistent desire, arousal, orgasm or pain problems causing **personal distress**.
- About **40%** of women in their lifetime; **HSDD (hypoactive sexual desire disorder) commonest**. Almost half of women with UI (urinary incontinence) have FSD.

Physiology and response models

- Arousal: pudendal afferents, pelvic parasympathetic efferents → **NO (nitric oxide)–cGMP (cyclic guanosine monophosphate)** → engorgement and lubrication. Oestrogen maintains tissues; testosterone drives desire.
- **Masters and Johnson** (linear): excitement, plateau, orgasm, resolution; Kaplan added desire. **Basson** (circular): desire is often responsive.

Classification

DSM-5 (Diagnostic and Statistical Manual, 2013)	ICSM (International Consultation on Sexual Medicine)/ISSWSH (International Society for the Study of Women's Sexual Health)
Female sexual interest/arousal disorder (≥3 of 6 criteria)	HSDD and arousal disorder kept separate (cognitive vs genital arousal)
Female orgasmic disorder	Orgasm disorders
Genito-pelvic pain/penetration disorder	Genital-pelvic pain dysfunction
–	PGAD (persistent genital arousal disorder)

- ≥6 months with distress; lifelong or acquired; generalised or situational.

Causes

- Hormonal (menopause, GSM [genitourinary syndrome of menopause]), drugs (SSRIs [selective serotonin reuptake inhibitors], antihypertensives), diabetes, depression, relationship factors.
- Urological: incontinence, prolapse, sling dyspareunia, cystectomy nerve or clitoral injury.

Evaluation

- Clinician must ask. **PLISSIT** (permission, limited information, specific suggestions, intensive therapy); BSSC-W (Brief Sexual Symptom Checklist for Women) screen; **FSFI (Female Sexual Function Index, 19 items, six domains)**.
- Examination: atrophy, lichen sclerosus, cotton-swab test, pelvic floor tone. Laboratory tests rarely help.

Management

- Treat causes; education; sex therapy (sensate focus, mindfulness).

- **HSDD (premenopausal, FDA [US Food and Drug Administration] approved): flibanserin** 100 mg at bedtime (5-HT1A agonist/5-HT2A antagonist; alcohol caution); **bremelanotide** 1.75 mg subcutaneously on demand (melanocortin agonist). Postmenopausal: transdermal testosterone (off-label).
- **Arousal:** testosterone; PDE5 (phosphodiesterase type 5) inhibitors only with neurovascular disease.
- **Orgasm:** education, directed masturbation.
- **Pain/vestibulodynia:** pelvic floor physiotherapy, dilators, biofeedback, topical lidocaine, vaginal oestrogen; vestibulectomy if refractory.
- **Also know:** ospemifene, intravaginal prasterone (menopausal dyspareunia).

Diagram to draw: Basson circular sexual response model.

> **Key points to remember:**
> - About 40% prevalence; HSDD commonest; distress required.
> - DSM-5 merges desire and arousal, dyspareunia and vaginismus.
> - Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

Duloxetin

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 3, Q3e

Read as: duloxetine, mainly its role in stress urinary incontinence.

Introduction

- **Duloxetine** is an SNRI (serotonin–noradrenaline reuptake inhibitor), also an antidepressant.
- Licensed in Europe (not USA) for **moderate to severe SUI (stress urinary incontinence) in women**; off-label in men.

Mechanism

- During storage, bladder afferents drive glutamate excitation of pudendal motor neurons in **Onuf's nucleus (S2–S4)**, which keep the striated urethral sphincter closed.
- Serotonin (5-HT2) and noradrenaline (α1) amplify this glutamate drive. Duloxetine blocks their reuptake, so **sphincter tone and the sneeze-induced continence reflex increase**.
- In voiding, glutamate drive switches off, so retention does not occur.
- Central effects also reduce bladder activity.

Dose

- **40 mg twice daily**; starting at 20 mg twice daily for 2 weeks reduces nausea. Taper on stopping.

Evidence

- Phase III RCTs (randomised controlled trials) in women: about 50% median fall in incontinence episodes versus about 30–40% with placebo; better quality of life; **cure rare**.
- Benefit is temporary and lasts only while taken.
- Male PPI (post-prostatectomy incontinence): small, short-term benefit.

Guideline position

- **EAU (European Association of Urology):** may be offered to women for **temporary** improvement, not cure; warn about side effects. Men: offer for SUI (weak), and **inform of off-label use and adverse effects** (strong).
- **NICE (National Institute for Health and Care Excellence):** not first line; not routinely second line; only if surgery is unwanted or unsuitable.
- Adding PFMT (pelvic floor muscle training): evidence conflicting.

Adverse effects

- **Nausea** (commonest, early), dry mouth, dizziness, insomnia, constipation; **high discontinuation**.
- Raised blood pressure, hyponatraemia (elderly), suicidal ideation, discontinuation syndrome.

Contraindications and interactions

- MAOIs (monoamine oxidase inhibitors), including **methylene blue** (serotonin toxicity); CYP1A2 inhibitors (fluvoxamine, ciprofloxacin); hepatic impairment; severe renal impairment; uncontrolled hypertension or narrow-angle glaucoma.

Diagram to draw: Onuf's nucleus with 5-HT and noradrenaline terminals driving the pudendal nerve.

Key points to remember:

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA.

Rapid Revision Sheet – Section 6 · Female Urology

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Urological complications in OBS-GYN and uretero-vaginal fistula (LAQ)

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Distal ureteric injury at hysterectomy; leak within 4 weeks, normal voiding (continuous leak with normal voiding).
- UVF: orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if it fails; success ≥90%; follow 30 months.

Female pelvic floor, continence mechanism, classification and management of SUI (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD (VLPP <60, MUCP ≤20 cmH_2O).
- Burch corrects position; slings and AUS correct function.
- VALUE: no urodynamics in index SUI.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.

SUI: evaluation in females, role of UDS, and surgical therapy in males (LAQ)

- Q-tip >30° = hypermobility; VLPP / ALPP <60 cmH2O = ISD; MUCP ≤20 cmH2O.
- VALUE (ValUE) trial: UDS not needed in uncomplicated SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- AUS PRB 61–70 cmH2O; deactivate 4–6 weeks.
- Radiation or severe SUI: AUS first; MASTER: sling non-inferior.

Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management (LAQ)

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy to exclude cancer.
- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Urological complications of OBGY surgery and urethrovaginal fistula (LAQ)

- Bladder is the commonest organ injured; ureteric injury is mostly distal and missed.
- Diverticulectomy is the commonest Western cause; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

Vesicovaginal fistula (VVF): management options and early vs delayed repair (LAQ)

- Image upper tracts; exclude ureteric fistula and cancer.
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.

- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- Interposition for complex VVF; radiation repair after ≥12 months.

Female sexual dysfunction (SAQ)

- About 40% prevalence; HSDD commonest; distress required.
- DSM-5 merges desire and arousal, dyspareunia and vaginismus.
- Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

Duloxetin (SAQ)

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA.

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

Sacral neuromodulation

Asked as:

- Sacral neuromodulation — *Summer 2015 Paper III Q3a (SAQ)*
- Sacral Neuromodulation. — *Winter 2017 Paper III Q3a (SAQ)*
- Sacral Neuromodulation — *Winter 2025 Paper III Q3e (SAQ)* (also answered in 2017-Winter Paper 3, Q3a)

Type: SAQ (10 marks)

Definition and mechanism

- **SNM (sacral neuromodulation)** (Tanagho and Schmidt): continuous low-amplitude **S3** stimulation through (via) a tined lead and IPG (implantable pulse generator).
- It acts on **afferents** to the forebrain (not motor nerves), which modulate forebrain awareness. In OAB (overactive bladder) this damps the PMC (pontine micturition centre) firing; in retention it restores filling awareness (sensation), so the PMC switches on and the sphincter relaxes. Hence it treats **both**.

Indications (third line after behavioural and drug therapy; AUA (American Urological Association) and EAU (European Association of Urology))

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997	Bladder pain syndrome
Urgency–frequency 1999	Neurogenic bladder (small studies positive; low-level evidence)
Non-obstructive urinary retention (e.g. Fowler's syndrome; Fowler syndrome responds well) 1999	Pelvic pain
Faecal incontinence 2011	

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography) activity; responds well to SNM.
- Retention needs some detrusor function; success falls with age.

Technique

Test phase	PNE (percutaneous nerve evaluation)	Staged implant (staged tined lead)
Lead	Temporary, office, local anaesthesia (dislodges)	Permanent tined lead, theatre, fluoroscopy
Duration	5–7 days	Up to 4 weeks
Use	OAB	More reliable; higher conversion; used for retention

- Prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

Root	Pelvic response	Foot response
S2	—	Calf and foot (external) rotation
S3 (target)	**Bellows**, perineal sensation	**Great-toe flexion**

Root	Pelvic response	Foot response
S4	Bellows/perineal only	None

- **Implant if ≥50% (at least 50%) improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral no better.

Outcomes

- **InSite**: success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years (another source: about half keep benefit at ≥5 years).
- **ROSETTA**: onabotulinumtoxinA 200 U slightly better for UUI (−3.9 vs −3.3 episodes/day) and higher satisfaction, but more UTI (urinary tract infection) (24% vs 10%) and self-catheterisation; SNM removal 9% at 2 years.

Complications and contraindications

- Pain at IPG site, lead migration, loss of efficacy (effect), infection; **revision over 30% at 5 years**; costly (costliest option); battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

Recent advances

- Rechargeable devices / IPGs (≥15 years); **full-body (whole-body) MRI (magnetic resonance imaging)-conditional** systems; bilateral stimulation no better. **Also know:** PTNS (percutaneous tibial nerve stimulation) and pudendal stimulation as alternatives (PTNS alone as alternative in another source).

Diagram to draw: lead in S3 foramen with IPG in the buttock.

> **Key points to remember:**
> - Afferent modulation treats both OAB and retention.
> - S3: bellows plus great-toe flexion.
> - Implant if test improves ≥50% (at least 50%); revision >30% at 5 years.

Urinary bladder compliance

Type: SAQ (10 marks)

Source: 2017-Summer Paper 3, Q3e

Definition

- **Compliance** is the change in volume for a change in detrusor pressure: **C = ΔV / ΔPdet**, in **mL/cmH_2O**.
- It reflects **low-pressure storage** during filling.

Physiological basis

- **Viscoelastic wall**: smooth muscle, collagen (about 50% of the wall) and elastin (about 2%). Coiled collagen straightens as the bladder fills.
- More collagen or less elastin makes the wall stiff.

Measurement (filling cystometry)

- ICS (International Continence Society) points: Pdet at **start of filling** and at **cystometric capacity** (or before a leak-causing contraction), **excluding detrusor contractions**.
- **No agreed normal value**; compliance depends on capacity. Below **20 mL/cmH_2O** (some use 10–20) suggests poor accommodation. In children, normal is often taken as **>15 mL/cmH_2O**.

Pitfalls

Pitfall	Effect	Fix
Fast filling	False low compliance	Slow or stop filling; pressure falls
Sustained low-amplitude detrusor contraction	Mimics low compliance	Stop filling: pressure returns to baseline
Pop-off: VUR (vesicoureteric reflux), diverticulum, leaking outlet	Falsely good compliance	Video-urodynamics; occlude outlet

- A small bladder can give a low number without danger (**compliance paradox**). **Absolute storage pressure matters more than the ratio.**

Causes of low compliance

- **Neurogenic**: spina bifida, spinal cord injury, lower motor neuron and decentralisation injury (radical pelvic surgery).
- **Outlet obstruction**: PUV (posterior urethral valves), long-standing BPO (benign prostatic obstruction).
- **Structural**: radiation cystitis, tuberculosis, schistosomiasis, multiple bladder operations.

Consequences and DLPP

- **DLPP (detrusor leak point pressure)**: lowest Pdet at which leakage occurs without contraction or strain.
- **McGuire (1981)**: DLPP **>40 cmH_2O** in myelodysplastic children led to hydronephrosis or VUR in 85%. Not a validated safe limit (20 cmH_2O is more sensitive); aim as low as possible.
- A strong outlet keeps the patient dry but transmits high pressure to the kidneys.

Management

- **Clean intermittent catheterisation + antimuscarinics** (first line).

- **Intradetrusor botulinum toxin**.
- **Augmentation cystoplasty** if refractory; vesicostomy in infants.
- Repeat urodynamics to confirm safety.

Diagram to draw: cystometrogram: normal flat versus steep rising filling curve.

> **Key points to remember:**
> - C = ΔV/ΔPdet; <20 mL/cmH_2O suggests poor compliance.
> - McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
> - Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

Botulinum toxin in urology

Asked as:

- Botulinum toxin — *Winter 2017 Paper III Q3b (SAQ)*
- Botox in uro — *Practice Set 4 Paper III Q3a (SAQ)* (also answered in 2025-Winter Paper 2, Q3d); *Read as: botulinum toxin in urology.*

Type: SAQ (10 marks)

Introduction

- *Clostridium botulinum* neurotoxin, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**; type B is short-acting.
- 150 kDa: heavy chain (100 kDa; binding, entry) plus zinc-dependent light chain (50 kDa).

Mechanism

1. Heavy chain binds the cholinergic terminal; toxin is internalised.
2. Light chain **cleaves SNAP-25 (synaptosomal-associated protein 25)**, a SNARE (vesicle-docking) protein, so vesicles cannot fuse and **acetylcholine release stops**.

- **Afferent action**: less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) receptors, so urgency falls.
- Effect wanes as nerve terminals sprout (6–9 months); repeat injections work.

Technique

- Cystoscopic injection into detrusor at **20 sites (100 U) or 30 sites (200 U)**, usually sparing trigone (trigone usually spared); sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

Urological uses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK): incontinence −59%, about 25% dry. **ABC** trial: equal to antimuscarinic, more complete dryness. **ROSETTA**: 200 U ≥ SNM (sacral neuromodulation) (another source: slightly better than SNM), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY): less incontinence, lower detrusor pressure, larger capacity
Paediatric NDO (spina bifida)	About **10 U/kg (maximum 360 U)** in Campbell series; FDA (Food and Drug Administration) label (2021, age ≥5) is 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation
DSD (detrusor–sphincter dyssynergia), underactive detrusor	Intrasphincteric injection 50–100 U; men unable to do CISC (clean intermittent self-catheterisation)
IC/BPS (interstitial cystitis/bladder pain syndrome)	100 U, often trigonal; AUA (American Urological Association) option, off-label
BPH (benign prostatic hyperplasia)	Intraprostatic: early small trial positive, but larger (large) placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo** (no better than placebo); not recommended

- **Also know:** paediatric dysfunctional voiding (sphincter); pelvic floor spasm; chronic pelvic pain; ED (erectile dysfunction) trials. Units of different toxin products are not interchangeable.

Complications and contraindications

- **Retention needing CISC**: 0–12% in OAB trials; real-world about 20% women, up to 35% men; higher (commoner) in NDO. Each 10 mL rise in baseline PVR raises risk about 27%. **Patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.
- Contraindications: active UTI, myasthenia gravis, aminoglycosides, pregnancy. Repeat injections stay effective.

Diagram to draw: cholinergic synapse (nerve terminal) showing SNAP-25 cleavage.

> **Key points to remember:**
> - Cleaves SNAP-25; efferent and afferent action (acts on efferent and afferent nerves).
> - 100 U idiopathic OAB; 200 U NDO; children 10 U/kg (max 360 U) in series, FDA label 6 U/kg (max 200 U).
> - Main risks: retention needing CISC, UTI.

OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron

Asked as:

- Darifenacin vs. solifenacin vs. tolterodine — *Practice Set 2 Paper III Q3e (SAQ)*
- Mirabegron in over active bladder. — *Winter 2024 Paper III Q3d (SAQ)*

Type: SAQ (10 marks)

(also answered in Practice Set 1 Paper 1, Q3e)

Introduction

- OAB (overactive bladder: urgency, with or without urgency incontinence, usually with frequency and nocturia).
- Antimuscarinics treat OAB after behavioural therapy. Acetylcholine on **M3** receptors contracts the detrusor; M2 (commoner) adds to it. Blockade reduces involuntary contractions and afferent urgency signals.
- Side effects: **M1** (brain) cognition; **M2** heart rate; **M3** (glands, gut) dry mouth, constipation.
- Selectivity is by **receptor subtype** (M3) or **tissue** (bladder over salivary gland); tissue selectivity matters more.
- **Mirabegron** is the first **β3-adrenoceptor agonist** for OAB. FDA (Food and Drug Administration) approval 2012.

Antimuscarinics: comparison

Feature	Darifenacin	Solifenacin	Tolterodine
Selectivity	**Most M3-selective (about 30-fold)**; not tissue-selective	M3 about 10-fold over M2; bladder-selective	Non-subtype-selective; **bladder-selective**
Dose	7.5–15 mg daily (ER (extended-release))	5–10 mg daily	2 mg twice daily (IR (immediate-release)); 4 mg daily (ER)
Half-life	Moderate (ER once daily)	**Long** (about 2–3 days): once daily	Short; active metabolite 5-HMT (5-hydroxymethyl tolterodine)
Metabolism	CYP3A4 (a liver enzyme) and CYP2D6	CYP3A4	CYP2D6 (and 3A4)
Dry mouth	Dose-related	Dose-related	Lower with ER
Constipation	**Highest** (gut M3)	Moderate–high	Lower
CNS (central nervous system)	Spares M1: **preferred in elderly**	Better persistence	High dose: QT prolongation
Dose limit	Lower dose with CYP3A4 inhibitors or liver disease	5 mg in severe renal or moderate liver impairment	Halve with renal or liver impairment

Common points

- **Efficacy is similar**; response is individual, so switch agent or dose.
- **STAR trial:** flexible solifenacin beat tolterodine ER for urgency and incontinence.
- **Contraindications:** urinary or gastric retention, uncontrolled narrow-angle glaucoma.
- Men: add to α-blocker; retention up to 3%; check PVR (postvoid residual).

Mirabegron (β3 agonist)

Mechanism of action

- **β3 receptors** are the most highly expressed adrenoceptor in human detrusor.
- Activation → adenylyl cyclase → **↑ cAMP (cyclic adenosine monophosphate)** → detrusor relaxation during **storage**.
- Reduces non-voiding contractions and afferent firing. The voiding contraction (muscarinic M3) is spared, so PVR (post-void residual) barely rises.

Pharmacology and dose

- Oral, extended release, once daily: start **25 mg**, increase to **50 mg** after 4–8 weeks.
- **25 mg maximum** in severe renal impairment (eGFR [estimated glomerular filtration rate] 15–29) or moderate hepatic impairment; avoid in end-stage renal disease and severe hepatic impairment.
- **Moderate CYP2D6 inhibitor:** raises levels of metoprolol, desipramine, flecainide; monitor digoxin.

Efficacy

- Phase III trials (SCORPIO, ARIES, CAPRICORN): fewer incontinence episodes and micturitions than placebo. **TAURUS** (12 months): efficacy similar to tolterodine, with less dry mouth.
- **Combination with solifenacin** (SYNERGY, SYNERGY II, BESIDE): solifenacin plus mirabegron better than either alone (beats monotherapy), without extra cardiovascular effects.

Adverse effects and contraindications

- Hypertension, tachycardia, headache, nasopharyngitis, UTI (urinary tract infection).
- **Dry mouth and constipation near placebo rates**, far below antimuscarinics.
- **Contraindicated:** severe uncontrolled hypertension (≥180/110 mmHg). Caution with arrhythmia and BOO (bladder outlet obstruction) with antimuscarinics (retention).

Place in therapy

- After behavioural therapy, as first-line pharmacotherapy alongside antimuscarinics (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]).
- **Dementia:** antimuscarinic use >3 months raises risk 10–65% (is linked to 10–65% more new-onset dementia; SUFU white paper). Prefer **β3 agonist** (mirabegron **preferred in the elderly and cognitive impairment:** no anticholinergic burden), or ER trospium, darifenacin or fesoterodine, in the elderly. Beers criteria: avoid in dementia.
- Mirabegron: add-on to α-blocker in men with BPH (benign prostatic hyperplasia); neurogenic detrusor overactivity.
- **Also know:** vibegron 75 mg (no CYP2D6 interaction).

Diagram to draw: muscarinic subtypes M1 (brain), M2 (heart), M3 (bladder, gland, gut) with each drug's selectivity; detrusor cell: noradrenaline → β3 → cAMP → relaxation, versus acetylcholine → M3 → contraction.

Key points to remember:

- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- Similar efficacy; STAR: solifenacin beat tolterodine ER.
- β3 → cAMP → storage-phase relaxation; voiding spared.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: darifenacin, trospium or β3 agonist; mirabegron preferred over antimuscarinics in the elderly (dementia risk 10–65% with >3 months' use).
- Add-on to solifenacin (SYNERGY) works.

Uroflowmetry

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 3, Q3a (also answered in 2018-Summer Paper 2, Q3f)

Definition

- Non-invasive urodynamic test recording **voided volume per unit time** (mL/s). It shows detrusor contraction and outlet resistance together.

Method

- Comfortably full bladder; private void in the preferred position. Ask whether the void was typical; compare with the frequency–volume chart.
- **Voided volume ≥150 mL** (Qmax unreliable below 125–150 mL). Repeat; then measure **PVR (postvoid residual)** by ultrasound.
- Flowmeters: weight transducer (load cell), rotating disc, capacitance dipstick.

Parameters

Parameter	Note
Qmax (maximum flow)	Key value; read from smoothed curve, ignoring spikes
Qave (average flow)	Less accurate for obstruction
Voided volume	≥150 mL
Flow time vs voiding time	Voiding time includes interruptions
Time to Qmax	Normally early in the void
Voiding efficiency	VV ÷ (VV + PVR) × 100

Curve patterns (ICCS (International Children's Continence Society))

Pattern	Suggests
Bell	Normal
Tower (high, short)	Overactive bladder
Plateau (flat, prolonged)	Fixed obstruction: stricture, PUV (posterior urethral valves)
Staccato (never reaches zero)	Sphincter overactivity, dysfunctional voiding
Interrupted (reaches zero)	Detrusor underactivity with straining, or DSD (detrusor-sphincter dyssynergia)

Interpretation

- **Qmax <10 mL/s:** obstruction likely (more specific). **<15 mL/s:** more sensitive, less specific.
- **Volume correction:** corrected Qmax = Qmax ÷ √VV; **Siroky** (men) and **Liverpool** (men and women) nomograms.
- **Key limitation:** cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity); only a pressure–flow study can.
- Children: valid if voided volume ≥50% of expected capacity; Qmax only if the peak lasts ≥2 s.

Uses

- BPH (benign prostatic hyperplasia): before active treatment and to monitor outcome.

- After urethroplasty or urethrotomy (stricture recurrence).
- Children: BBD (bladder and bowel dysfunction), after valve ablation; with patch EMG (electromyography).
- Women with voiding dysfunction.
- **AUA (American Urological Association):** pressure–flow study may be done before BPH surgery if Qmax >10 mL/s.

Diagram to draw: bell, plateau, tower and interrupted curves.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = fixed obstruction; staccato = sphincter; interrupted = weak detrusor.
> - Cannot distinguish BOO from DUA.

Rapid Revision Sheet – Section 7 · Voiding Dysfunction

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Sacral neuromodulation (SAQ)

- Afferent modulation treats both OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves ≥50% (at least 50%); revision >30% at 5 years.

Urinary bladder compliance (SAQ)

- C = ΔV/ΔPdet; <20 mL/cmH_2O suggests poor compliance.
- McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

Botulinum toxin in urology (SAQ)

- Cleaves SNAP-25; efferent and afferent action (acts on efferent and afferent nerves).
- 100 U idiopathic OAB; 200 U NDO; children 10 U/kg (max 360 U) in series, FDA label 6 U/kg (max 200 U).
- Main risks: retention needing CISC, UTI.

OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron (SAQ)

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Uroflowmetry (SAQ)

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = fixed obstruction; staccato = sphincter; interrupted = weak detrusor.
- Cannot distinguish BOO from DUA.

Section 8 · Nocturia & Male Incontinence — Short answer questions (SAQ)

Desmopressin (DDAVP)

Asked as:

- Desmopressin. — *Winter 2023 Paper III Q3b (SAQ)*
- DDAVP — *Practice Set 3 Paper III Q3c (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 3, Q3b)

Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**, desmopressin), a synthetic analogue of AVP (arginine vasopressin/antidiuretic hormone).
- Deamination and D-arginine substitution prolong action (give longer action) and make it a **selective V2 agonist (selective V2 agonism) without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells raises cAMP (cyclic adenosine monophosphate), inserting **aquaporin-2**; water is reabsorbed, so urine is less and concentrated (V2 receptor → cAMP → aquaporin-2 insertion → water reabsorption → less, concentrated urine).
- Endothelial V2 action releases von Willebrand factor and factor VIII (haemostatic use).

Formulations

Form	Dose
Tablet	0.2–0.4 mg at bedtime, about 1 h before bed (enuresis)
Oral lyophilisate (melt)	120–240 µg, about 1 h before bed
Low-dose nasal spray (nocturia)	0.83 or 1.66 µg, 30 min before bed
Sublingual ODT (orally disintegrating tablet)	27.7 µg women, 55.3 µg men
IV / subcutaneous	0.3 µg/kg (bleeding)

Uses (in urology)

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. **Rule of thirds**: one-third full, one-third partial, one-third no response. Acts (works) at once but relapses on stopping. Restrict fluids in the evening (avoid evening fluid); add oxybutynin if partial.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer below (under) 65; low dose above (over) 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia** (multiple sclerosis, MSA [multiple system atrophy]).
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired, in Wilms tumour). Tachyphylaxis after about 3 (repeated) doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, baseline sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women are more sensitive.

- Most events occur in the **first 2–3 weeks**.

Monitoring and contraindications

- Sodium at baseline, then within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

Diagram to draw: V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

> **Key points to remember:**
> - Selective V2 agonist; no pressor effect.
> - Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
> - Hyponatraemia: age ≥65, low sodium; check sodium early.

Rapid Revision Sheet – Section 8 · Nocturia & Male Incontinence

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Desmopressin (DDAVP) (SAQ)

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early.

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

Priapism: Diagnosis and Management

Asked as:

- Diagnosis and Management of Priapism — *Winter 2025 Paper III Q2 (LAQ)*
- Priapism — *Practice Set 4 Paper III Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(also answered as an SAQ in 2015-Summer Paper 2, Q3a; full LAQ answer in 2025-Winter Paper 3, Q2)

Definition

- Full or partial erection lasting **≥4 hours** beyond sexual stimulation, or unrelated to it (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2021–2022).

Classification (types)

	Ischaemic (low flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Veno-occlusion	Repeated (recurrent) ischaemic episodes	Cavernosal artery–sinusoid fistula (arterio-sinusoidal fistula)
Pain, rigidity (penis)	Painful, fully rigid	Painful, recurrent, often nocturnal	Painless, partial
Typical cause	ICI (intracavernosal injection), drugs, SCD (sickle cell disease)	SCD	Perineal or needle trauma
Urgency	**Emergency**	Emergency per episode + prophylaxis	Not an emergency (not urgent)

Aetiology

- **Ischaemic:** ICI (intracavernosal injection: papaverine, phentolamine, PGE1 (prostaglandin E1)), including recreational use; α-blockers, trazodone, antipsychotics, cocaine; heparin or warfarin withdrawal; **SCD** (29–42% of affected men in their lifetime; 63% of paediatric ischaemic priapism); leukaemia (1–5%, high white cell count); penile metastases; scorpion and spider venom.
- **Non-ischaemic:** straddle or coital injury, pelvic fracture, needle injury. Onset is often delayed, when the clot over the arterial tear lyses.

Pathophysiology

- **Ischaemic:** failed outflow acts like a compartment syndrome. Hypoxia, hypercarbia and acidosis cause sinusoidal thrombosis, smooth-muscle necrosis and TGF-β (transforming growth factor beta)-driven fibrosis, so ED (erectile dysfunction) follows (hence ED). Irreversible damage starts within hours.
- **Stuttering (molecular):** low basal NO (nitric oxide) from eNOS (endothelial NO synthase) dysfunction downregulates PDE5 (phosphodiesterase 5). When an erection starts, cGMP

(cyclic guanosine monophosphate) is not broken down, so relaxation is unopposed (cGMP rises unopposed). Low Rho-kinase and high adenosine add to this. This explains why daily PDE5 inhibitors prevent episodes.
- **Non-ischaemic:** unregulated, well-oxygenated arterial inflow, so fibrosis is unusual.

Diagnosis

- **History:** duration, pain, drugs, trauma, SCD or blood disease, prior episodes, baseline erectile function.
- **Examination:** rigid tender corpora with a soft glans and spongiosum (ischaemic); partial, non-tender (high flow); perineal bruising; vital signs before phenylephrine.
- **Corporal blood gas** (cornerstone); high flow resembles arterial blood:

	Colour	pO2 (mmHg)	pCO2 (mmHg)	pH
Ischaemic	Dark red	**<30**	**>60**	**<7.25**
Flaccid (mixed venous)	Medium red	40	50	7.35
Non-ischaemic	Bright red	>90	<40	7.40

- **PDUS (penile duplex ultrasound)** when the gas is equivocal: ischaemic = absent cavernosal flow or PSV (peak systolic velocity) <50 cm/s with no diastolic flow; high flow = PSV >50 cm/s, may show the fistula (high flow with fistula). Angiography only for embolisation.
- Blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology: in parallel, never delaying treatment (without delaying treatment).

Management of ischaemic priapism (AUA/SMSNA 2021)

```
Counsel on ED risk → dorsal penile block
<36 h:
  aspiration (19 G, penoscrotal junction, 3 or 9 o'clock) ± saline irrigation
  + phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h
    (monitor blood pressure, pulse: reflex bradycardia)
  → fails → distal shunt (Winter, T-shunt, Al-Ghorab) ± tunnelling
  → proximal shunt (Quackels) or penoscrotal decompression
>36 h (beyond 36 h) or refractory → early penile prosthesis (or analgesia and counselling)
```

- Aspiration alone succeeds in under 1 in 4. Cold (10 °C) saline irrigation was better than body temperature in an RCT (randomised controlled trial); avoid chilled saline in SCD.
- Ice, oral agents and IV (intravenous) fluids are adjuncts only; in SCD, intracavernosal treatment still comes first.

Shunt	Method
Winter (distal)	Tru-Cut needle through glans; least effective
Ebbehoj / T-shunt (distal)	Scalpel through glans; T-shunt rotates blade 90°
Al-Ghorab (distal, open)	Dorsal glans incision; excise corporal tips
Snake tunnelling	7/8 Hegar dilator passed proximally through distal shunt
Quackels / Sacher (proximal)	Cavernosum to spongiosum at bulb; stagger ≥1 cm
Grayhack / Barry	Cavernosum to saphenous / dorsal vein
Penoscrotal decompression	Proximal corporotomy, clot suction; allows prosthesis

- **ED by duration** (Bennett and Mulhall, SCD series): ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- **Early prosthesis** (Zacharakis 2014): less shortening and fewer revisions than delayed insertion — shortening 3% vs 40%, revision 9% vs 27%, satisfaction 96% vs 60% compared

with delayed insertion.

Stuttering priapism

- An episode lasting ≥4 hours is treated as ischaemic; home self-injection of phenylephrine.
- **Prophylaxis:** daily PDE5 inhibitor (Burnett 2014 RCT: primary endpoint missed, but promising), pseudoephedrine, etilefrine. **Also know:** GnRH (gonadotropin-releasing hormone) agonists, antiandrogens, finasteride (cost: libido); digoxin; baclofen.

Non-ischaemic priapism (AUA/SMSNA 2022)

- Observation, perineal compression and ice; about 60% resolve. Reassess at 4 weeks.
- Persistent → PDUS → **selective cavernosal artery embolisation** (selective arterial embolisation; resolution ≥75%, recurrence 30–40%). For recurrence, repeat embolisation with non-resorbable PVA (polyvinyl alcohol), not surgery. Surgical ligation is a last resort (ED about 50%).

Complications

- ED, corporal fibrosis, penile shortening; shunt-related oedema, haematoma, urethral injury or fistula, skin necrosis, infection.

Diagram to draw: distal (glans) and proximal shunt sites on a penile outline.

Key points to remember:

- Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling next.
- ED risk 0% ≤12 h, 100% ≥36 h (beyond 36 h); consider early prosthesis beyond 36 h.
- High flow: observe, then embolise; repeat embolisation rather than surgery.

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

Types of penile prosthesis. What are biofilms in penile prosthesis?

Type: SAQ (10 marks)

Source: 2015-Summer Paper 3, Q3d

Introduction

- For ED (erectile dysfunction) failing less invasive treatment; satisfaction **80–90%**. Repairs: about 15% at 5 years, 30% at 10.

Types

Type	Examples	Features
Malleable	Tactra (nitinol), Genesis	Simple, cheap, no mechanical failure; always rigid
2-piece inflatable	Ambicor (discontinued)	No reservoir; poor flaccidity
3-piece inflatable	AMS 700, Titan	**Best rigidity and true flaccidity**; cylinders, scrotal pump, reservoir

- Penoscrotal or infrapubic approach; reservoir in Retzius or **ectopic submuscular** after pelvic surgery.
- **Also know:** erosion, pump migration, glans hypermobility, cylinder aneurysm; modelling in Peyronie's disease.

Biofilm

- **Definition**: a structured microbial community in a self-made **polysaccharide matrix**, attached to the implant.
- **Stages**: (1) conditioning film of host proteins coats the device within minutes; (2) reversible, then irreversible bacterial adhesion; (3) microcolonies; (4) maturation into a 3D structure with water channels and quorum sensing; (5) dispersal of free bacteria, causing clinical infection.
- **Organisms**: coagulase-negative staphylococci commonest; also S. aureus, Gram-negatives, Candida.
- **Why antibiotics fail**: the matrix blocks penetration, bacteria are slow-growing (persisters), and they survive up to **1000–1500 times** normal killing levels.
- **Relevance**: organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

Prevention (infection 1–3%)

- Glucose under 200 mg/dL; treat skin and urine; decolonise MRSA (methicillin-resistant S. aureus) carriers.
- **Chlorhexidine-alcohol** skin preparation; vancomycin or cephalosporin plus gentamicin for up to 24 hours.
- **Coatings halve infection**: InhibiZone (rifampin–minocycline); hydrophilic coating dipped in antibiotic.
- **No-touch technique**: under 0.5%.

Infected implant

- Pus, fixed pump or exposed parts: remove **all** components.
- **Mulcahy salvage**: remove all parts; wash with dilute povidone-iodine (3 minutes) then saline or antibiotic; change gloves, drapes and instruments; insert a new device at once. Success **84%**; fails with early onset, aggressive organisms, cellulitis.

Diagram to draw: 3-piece device; biofilm life cycle.

> **Key points to remember:**
> - 3-piece inflatable gives best rigidity and flaccidity.
> - Biofilm: microbes in a matrix; antibiotics fail.
> - Infection 1–3%; Mulcahy salvage 84%.

Oligospermia and oligoasthenozoospermia

Asked as:

- Oligoanaesthospermia. — *Summer 2016 Paper III Q3a (SAQ)*
- Oligospermia — *Practice Set 4 Paper III Q3c (SAQ)* (also answered in 2018-Summer Paper 2, Q3a)

Type: SAQ (10 marks)

Assumed meaning (oligoanaesthospermia): oligoasthenozoospermia (low sperm count with poor motility).

Definition (WHO (World Health Organization) 2021, 6th edition, 5th centiles)

- **Oligozoospermia:** sperm concentration below the lower reference limit: **WHO 2021 (6th edition) 16 million/mL** (5th centile; total count 39 million). The legacy cut-off was 20 million/mL.

Parameter (other WHO 2021 5th centiles)	Lower reference (value)
Concentration	16 million/mL (total 39 million)
Progressive motility	30%
Total motility	42%
Volume	1.4 mL
Normal forms (strict)	4%
Vitality	54%

- These are reference values (these centiles come) from fertile men, not pass/fail cut-offs; they are a reference, not a fertility test. Define oligospermia in context (testis size, FSH (follicle-stimulating hormone), repeated values).
- **Severe oligozoospermia: <5 million/mL (under 5 million/mL)**; cryptozoospermia: sperm only in a centrifuged pellet.

Causes

- **Varicocele** (commonest correctable; heat and oxidative stress) and **idiopathic** (largest group).
- **Pre-testicular / Hormonal:** hypogonadotropism (hypogonadotropic hypogonadism), **exogenous testosterone or anabolic steroids**, obesity (high / raised oestradiol), hyperprolactinaemia, opioids.
- **Testicular: varicocele**, cryptorchidism, mumps orchitis, torsion, chemotherapy or radiation, gonadotoxins; genetic (Klinefelter, Klinefelter mosaic, AZF (azoospermia factor) c deletion / **AZFc (azoospermia factor c) deletion**).
- **Post-testicular / Motility:** infection or **leucocytospermia** (over 1 million/mL), antisperm antibodies, partial ejaculatory duct obstruction (EDO; low volume), PCD (primary ciliary dyskinesia).
- Lifestyle: smoking, alcohol, cannabis, heat, gonadotoxins.

Evaluation

- **At least two semen analyses, 2–3 weeks apart**, same abstinence (WHO 2–7 days abstinence; parameters peak at 1–2 days); history and examination (testis size / volume, varicocele, vasa).
- Morning **testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone)**; oestradiol and prolactin if testosterone is low.

- Under 5 million/mL (especially with raised FSH or small testes): **karyotype and Y-microdeletion** (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine) 2020 guideline).
- Selected: vitality test (eosin or hypo-osmotic swelling), antisperm antibodies, semen culture, scrotal Doppler, TRUS (transrectal ultrasonography / ultrasound) if volume low (<1.0 mL), DNA fragmentation.

Management

- **General**: stop smoking, weight loss, avoid heat; **stop exogenous testosterone** and gonadotoxins; treat infection.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen; improves parameters and pregnancy. Not (never) for subclinical ones.
- **Medical**: hypogonadotropic → hCG (human chorionic gonadotropin) with or without (±) FSH. Idiopathic with low testosterone → off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if testosterone:oestradiol ratio under 10 (<10). Antioxidants: weak evidence.
- **ART (assisted reproductive technology)**: mild → IUI (intrauterine insemination); severe, or failed IUI → **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc deletion.
- Spermatogenesis takes about 64 days, so judge response (reassess) after 3–6 months.

Diagram to draw: hypothalamic–pituitary–testis axis with sites of treatment (treatment sites).

> **Key points to remember:**
> - WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
> - Karyotype and Y-microdeletion if under 5 million/mL.
> - Repair palpable varicoceles only; never give testosterone.

Sperm retrieval techniques

Type: SAQ (10 marks)

Source: 2016-Summer Paper 3, Q3b

Indications

- **OA (obstructive azoospermia)**: CBAVD (congenital bilateral absence of the vas deferens), unreconstructable obstruction, failed or declined vasectomy reversal, or at the time of reconstruction (back-up).
- **NOA (non-obstructive azoospermia)**: testicular failure; only testicular retrieval works.
- Anejaculation (spinal cord injury): vibratory stimulation or electroejaculation first.
- All need **ICSI (intracytoplasmic sperm injection)**; cryopreserve spare sperm.

Techniques

Technique	Method	Best for	Drawback
PESA (percutaneous epididymal sperm aspiration)	Needle into caput under local	OA	Few sperm, may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened at 16–25×; micropipette by capillary action	OA (best)	Microsurgical skill
TESA (testicular sperm aspiration)	23 G needle, suction, many passes	OA	Immotile sperm, small yield; poor in NOA
TESE (testicular sperm extraction)	Open biopsy, one or several sites	OA, NOA	Blind sampling
Micro-TESE	Testis bivalved; larger opaque tubules picked under the microscope	**NOA (best)**	Long operation

- **MESA points**: move proximally until motile sperm appear (most motile in efferent ductules); 4–12 pipettes, frozen in aliquots.
- **Micro-TESE logic (Schlegel 1999)**: spermatogenesis in NOA is **focal**; tubules with sperm are larger and plumper than thin white Sertoli-cell-only tubules. Incision through an avascular plane, then the other testis if negative. It removes the least tissue for the most sperm.

Results

- OA: retrieval almost always succeeds.
- NOA micro-TESE: sperm in **about half**; better than conventional TESE.

Condition	Retrieval
Post-orchidopexy	74%
AZF (azoospermia factor) c deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
Sertoli-cell-only	30%
AZFa or AZFb deletion	**0% (do not operate)**

- NOA pregnancies have more miscarriage (sperm aneuploidy and DNA damage).

Complications

- Haematoma, infection, epididymal obstruction (PESA), testicular atrophy or transient fall in testosterone (TESE).

Also know: testicular sperm may beat ejaculated sperm when DNA fragmentation is high; postmortem retrieval only with prior consent; microfluidic sperm sorting.

Diagram to draw: micro-TESE with dilated tubules beside thin Sertoli-cell-only tubules.

> **Key points to remember:**
> - OA: epididymal (MESA); NOA: micro-TESE.
> - Micro-TESE finds sperm in about half of NOA.
> - AZFa/AZFb deletion: no sperm, do not operate.

Genetic causes of male infertility. Enumerate and classify.

Type: SAQ (10 marks)

Source: 2017-Summer Paper 3, Q3d

Introduction

- Genetic defects underlie many cases of **azoospermia** and **severe oligozoospermia**. They decide **sperm retrieval odds** and **transmission to offspring** through ICSI (intracytoplasmic sperm injection).

Classification

Group	Condition	Key features
1. Chromosomal, numerical	**Klinefelter syndrome (47,XXY)**	**Commonest genetic cause**; 1 in 500–1000; small testes, high FSH (follicle-stimulating hormone), low testosterone, usually azoospermia. Non-mosaic in 80–90%. Testicular sperm in about 50% (micro-TESE, microdissection testicular sperm extraction)
	47,XYY; **46,XX male** (SRY gene translocated to X)	XX male: azoospermia, no sperm retrievable
2. Chromosomal, structural	**Robertsonian and reciprocal translocations**, inversions, ring Y	Oligozoospermia; recurrent miscarriage
3. Y chromosome microdeletions (Yq, AZF (azoospermia factor))	**AZFa** (USP9Y, DDX3Y), **AZFb**, **AZFc** (DAZ genes)	Complete AZFa or AZFb deletion: **no sperm at retrieval**. AZFc: sperm often in ejaculate or testis; sons inherit the deletion
4. Single-gene: excurrent ducts	**CFTR (cystic fibrosis transmembrane conductance regulator) mutations**: CBAVD (congenital bilateral absence of vas deferens); idiopathic ejaculatory duct obstruction	Low-volume azoospermia, normal spermatogenesis; **test both partners** (ΔF508)
5. Single-gene: hypothalamic–pituitary	**Kallmann syndrome** (KAL1, anosmia); GnRH (gonadotropin-releasing hormone) receptor mutations	Low LH (luteinising hormone) and FSH; treat with hCG ± FSH
6. Androgen action	**Androgen receptor** mutation (long CAG repeat); 5α-reductase type 2 deficiency	High testosterone and LH, poor sperm
7. Sperm structure and motility	**Globozoospermia** (SPATA16, DPY19L2: no acrosome); macrocephalic sperm (AURKC); **PCD (primary ciliary dyskinesia)**/Kartagener (DNAH5, DNAI1: immotile sperm)	ICSI; macrocephalic sperm: ICSI not advised (aneuploidy)

Genetic testing indications

- **Karyotype and Y microdeletion:** non-obstructive azoospermia and severe oligozoospermia (**<5 million/mL**, ASRM/AUA (American Society for Reproductive Medicine/American Urological Association) 2021; EAU (European Association of Urology) uses <10 million/mL for karyotype).

- **CFTR:** CBAVD, idiopathic obstructive azoospermia; screen the partner; counsel before ICSI.
- **Also know:** Noonan syndrome, myotonic dystrophy.

Diagram to draw: Y chromosome long arm showing AZFa, b, c.

> **Key points to remember:**
> - Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
> - AZFa or AZFb complete deletion: do not attempt retrieval.
> - CBAVD = genital form of cystic fibrosis; test both partners.

Premature ejaculation.

Type: SAQ (10 marks)

Source: 2023-Winter Paper 3, Q3a

Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed:
  1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a clinically significant, bothersome fall in latency, often to **≤3 minutes** (acquired).
  2. Inability to delay ejaculation on all or nearly all penetrations.
  3. Negative personal consequences: distress, frustration, avoidance of intimacy.
- **AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2020:** lifelong PE (premature ejaculation) within about **2 minutes**.
- **Types:** lifelong, acquired, variable and subjective (normal latency).

Aetiology

- **Lifelong:** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.
- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

Evaluation

- Diagnosis is by **history**: latency, control, bother; lifelong or acquired.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials. Population median 5.4 minutes.
- **PEDT (Premature Ejaculation Diagnostic Tool):** 5 items; score ≥11 suggests PE, 9–10 probable PE. Also PEP (Premature Ejaculation Profile) and IPE (Index of Premature Ejaculation).
- IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Treat the cause in acquired PE (ED, prostatitis, thyroid).

Drug	Dose and timing
Dapoxetine (short-acting SSRI [selective serotonin reuptake inhibitor]; only licensed oral drug)	30–60 mg, 1–3 h before intercourse
Paroxetine (strongest delay), sertraline, fluoxetine, citalopram	Daily, off-label; full effect 2–3 weeks; do not stop abruptly
Clomipramine	12.5–50 mg daily or 3–4 h before
Topical lidocaine–prilocaine (spray licensed in Europe 2014)	5–30 min before; condom avoids partner numbness
PDE5Is (phosphodiesterase type 5 inhibitors)	With ED, alone or with SSRI
Tramadol	Low evidence; addiction risk

- SSRIs raise synaptic serotonin, which delays ejaculation. Avoid in bipolar disorder.

Diagram to draw: office algorithm (confirm PE, then treat cause or classify lifelong/acquired).

Key points to remember:

- ISSM: about 1 min (lifelong), about 3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Spermatogenesis

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 3, Q3d (also answered in 2017-Winter Paper 1, Q2)

Definition

- Production of haploid spermatozoa from diploid spermatogonia in the seminiferous tubules: about **64 days** (42–76); about 123 million sperm a day.

Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- Pulsatile **GnRH (gonadotropin-releasing hormone)** → **LH (luteinising hormone)** → Leydig cells → testosterone. **FSH (follicle-stimulating hormone)** → Sertoli cells.
- Feedback: testosterone on LH; **inhibin B** (Sertoli) on FSH.
- **Intratesticular testosterone** is about 50 times serum, held by ABP (androgen-binding protein). Exogenous testosterone suppresses LH and so stops spermatogenesis.

Sertoli cells and blood–testis barrier

- Nurse cells: nourish germ cells; secrete ABP, inhibin, fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** form a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). This protects immunologically "foreign" post-meiotic cells.

Phases

Phase	Events
1. Proliferation	**Ad (dark type A)** stem cells self-renew (GDNF [glial cell line–derived neurotrophic factor] niche); **Ap (pale type A)** divide every 16 days into type B → primary spermatocytes. Daughters remain joined by cytoplasmic bridges
2. Meiosis	Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; synapsis and crossing-over) → secondary spermatocytes → meiosis II → 4 haploid round spermatids
3. Spermiogenesis	No division: acrosome from Golgi, flagellum from centriole, nucleus compacts to 10%, mitochondria form the midpiece, cytoplasm shed
4. Spermiation	Sertoli cell releases sperm into the lumen

- **Cycle, stage, wave:** six stages in humans; cycles arranged as a helical wave, so production is continuous.

Post-testicular maturation

- Epididymal transit 2–12 days: sperm gain **progressive motility** and fertilising ability; stored in the cauda. **Capacitation** and acrosome reaction occur in the female tract.

Clinical relevance

- A semen sample reflects events about 3 months earlier; reassess 3 months after treatment.
- Disruptors: heat (testis normally 2–4 °C below core; varicocele, cryptorchidism), exogenous testosterone, chemotherapy, radiation.
- Defective meiotic recombination causes maturation arrest.

Diagram to draw: seminiferous tubule cross-section with basal and adluminal compartments.

Key points to remember:

- About 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; FSH → Sertoli, LH → Leydig.
- Tight junctions between Sertoli cells form the blood–testis barrier.

Ejacultory duct obstruction

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 3, Q3b

Assumed meaning: ejaculatory duct obstruction (EDO).

Anatomy

- Intraprostatic continuations of the seminal vesicles, joined by the vasal ampullae; they lack muscle and open at the **verumontanum**. Seminal vesicles give about 60% of the ejaculate and its fructose.

Aetiology

- **Congenital (commonest):** midline (Müllerian/utricular) cyst, Wolffian duct cyst, atresia. Idiopathic EDO carries CFTR (cystic fibrosis transmembrane conductance regulator) mutations in about half.
- **Acquired:** infection, instrumentation, duct stones, compression.
- **Zinner syndrome:** seminal vesicle cyst, ipsilateral renal agenesis, EDO.
- Complete bilateral EDO causes under 5% of azoospermia. Partial EDO gives oligoasthenospermia.

Clinical features

- Infertility, painful ejaculation, haematospermia, perineal pain.

Diagnosis

Semen (complete bilateral EDO)

- Volume **<1.0 mL**, **acidic pH**, low or absent **fructose**, azoospermia.
- Palpable vas, normal FSH (follicle-stimulating hormone) and testes.
- Exclude retrograde ejaculation and CBAVD (congenital bilateral absence of the vas deferens).

Investigations

Test	Method / finding
TRUS (transrectal ultrasound)	Seminal vesicle AP diameter **>1.5 cm**, duct >2.3 mm, midline cyst. Low specificity
Seminal vesicle aspiration	**≥3 sperm per high-power field** confirms obstruction
Vesiculography	Contrast into vesicles; invasive
Chromotubation	Indigo carmine into vesicles; dye at the veru on cystoscopy

Management

TURED (transurethral resection of the ejaculatory ducts)

- For azoospermic or severely oligoasthenospermic men, after the couple declines IVF (in vitro fertilisation).
- **Steps:** TRUS-guided aspiration (cryopreserve sperm) → instil indigo carmine with contrast → 24-Fr resectoscope, finger in rectum → resect the veru until dye effluxes. Spare bladder neck, sphincter and rectum. Catheter overnight.
- **Results:** volume rises in most; sperm appear in about half. Cold-knife incision alone re-obstructs.

Complications

- **Urine reflux** into the ducts, **epididymitis**, retrograde ejaculation, rectal injury, re-obstruction.

Alternatives

- **Seminal vesiculoscopy** (4–6 Fr via utricle): avoids reflux; restenosis unknown. Sperm retrieval with ICSI (intracytoplasmic sperm injection).

Diagram to draw: sagittal prostate with midline cyst compressing the ducts at the veru.

Key points to remember:

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

Tadalafil

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 3, Q3b

Pharmacology

- Selective **PDE5 (phosphodiesterase type 5) inhibitor**; FDA (Food and Drug Administration) approved 2003.
- **Mechanism:** sexual stimulation releases NO (nitric oxide), raising cGMP (cyclic guanosine monophosphate), which relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; tadalafil blocks this. It **augments, not initiates,** erection.
- Structurally distinct from sildenafil; cross-reacts with **PDE11**, little with PDE6 (retina).

Property	Tadalafil (vs sildenafil)
Time to peak	2 h (0.8 h)
Half-life	**17.5 h** (3–5 h)
Fatty food	No effect (reduces absorption)
Window	Widest (up to 36 h)

Uses and doses

1. **ED (erectile dysfunction):** 10–20 mg on demand, or 2.5–5 mg once daily. Intercourse success about 70%; 40–50% in diabetes and after prostatectomy.
2. **BPH (benign prostatic hyperplasia) with LUTS (lower urinary tract symptoms), with or without ED: 5 mg daily**, the only PDE5 inhibitor licensed. IPSS (International Prostate Symptom Score) falls about 6 points (placebo 3.6) **without Qmax (peak flow rate) change**; mechanism is NO–cGMP relaxation, perfusion and reduced afferent activity, not de-obstruction. Equal to tamsulosin for IPSS. EAU (European Association of Urology) 2022 and AUA (American Urological Association) 2021: option for moderate–severe LUTS.
3. **Also know:** fixed finasteride–tadalafil (FDA 2021); penile rehabilitation (benefit unproven); prevention of stuttering priapism (PDE5I paradox); pulmonary arterial hypertension.

Adverse effects

- Headache, dyspepsia, **back pain and myalgia** (characteristic), flushing, nasal congestion. Visual effects rare.
- NAION (non-arteritic anterior ischaemic optic neuropathy): no proven link; stop if sudden visual loss.

Contraindications and interactions

- **Nitrates in any form: absolute.** After tadalafil, no nitrates for **48 h** (24 h for sildenafil).
- Caution: α-blockers (start lowest dose of the added drug), aortic stenosis, hypotension. CYP3A4 inhibitors (ketoconazole, ritonavir) raise levels.
- Not studied: MI (myocardial infarction) or stroke within 6 months, unstable angina, BP <90/50 or >170/100 mmHg, retinitis pigmentosa.

Diagram to draw: NO–cGMP–PDE5 pathway in cavernosal smooth muscle.

> **Key points to remember:**
> - Half-life 17.5 h; food-independent; back pain.
> - 5 mg daily for LUTS: IPSS improves, Qmax does not.
> - No nitrates for 48 h after a dose.

Retrograde ejaculation

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 3, Q3f

Definition

- RE (retrograde ejaculation): semen passes backwards into the bladder at orgasm because the **bladder neck fails to close**; orgasm is preserved, ejaculate absent or low volume.

Physiology

Phase	Event	Control
Emission	Vas, seminal vesicles, prostate deliver semen to posterior urethra	Sympathetic T10–L2
Ejection	**Bladder neck closes**, external sphincter relaxes, bulbospongiosus contracts	Somatic S2–S4 + sympathetic reflex
Orgasm	Cerebral perception	Cortical

Causes

- **Anatomical: TURP (transurethral resection of prostate)** 42–100%, TUIP (incision) 5–45%; most other resections and enucleations; previous bladder neck reconstruction. Ejaculation-sparing techniques, PUL (prostatic urethral lift) and Aquablation preserve ejaculation.
- **Neurogenic: diabetic autonomic neuropathy** (commonest medical), RPLND (retroperitoneal lymph node dissection; less with nerve sparing), colorectal surgery, sympathectomy, aortic surgery, spinal cord injury, multiple sclerosis.
- **Drugs:** α-blockers (silodosin, tamsulosin), antipsychotics, antidepressants, thiazides.
- **Congenital:** Müllerian duct cyst, Wolffian duct anomaly, prune belly syndrome.

Evaluation

- Low-volume or absent ejaculate with orgasm; infertility.
- **Post-ejaculatory urinalysis:** void, ejaculate, void again; centrifuge. **>5–10 sperm per HPF (high-power field)** confirms RE; significant if urinary sperm approach or exceed antegrade sperm.
- Sperm and fructose in urine separate RE from **failure of emission**; TRUS (transrectal ultrasound) if EDO (ejaculatory duct obstruction) suspected.

Treatment

- Reassure if fertility not desired; stop offending drug.
- **Sympathomimetics** (close bladder neck): **pseudoephedrine** 60–120 mg (120 mg 6-hourly the day before and twice on the day); midodrine, ephedrine, phenylpropanolamine. Side effects: dry mouth, hypertension.
- **Imipramine** 25 mg twice daily; antihistamines (brompheniramine) add to sympathomimetics.
- **Bladder neck reconstruction:** poor results; collagen injection experimental.
- **Sperm retrieval:** alkalinise urine (sodium bicarbonate), void or catheterise after orgasm, wash sperm in buffer → IUI (intrauterine insemination), IVF (in vitro fertilisation) or ICSI (intracytoplasmic sperm injection).
- **Prevention:** nerve-sparing RPLND, ejaculation-preserving procedures for BPH (benign prostatic hyperplasia).

Diagram to draw: bladder neck open at ejaculation with semen entering the bladder.

Key points to remember:

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

Immunobead test for semen

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 3, Q3d

Introduction

- The IBT (immunobead test) detects **antisperm antibodies (ASA)** bound to sperm (direct) or free in body fluids (indirect).
- **Pathogenesis:** Sertoli tight junctions form the blood–testis barrier. If it is breached, sperm antigens meet the immune system. ASA cause **agglutination, poor motility, poor cervical-mucus penetration and impaired fertilisation**.
- **Causes:** vasectomy and other obstruction; orchitis, prostatitis, STI (sexually transmitted infection); varicocele, cryptorchidism, heat; trauma, torsion; genetic (HLA-B28).

Indications

- Sperm agglutination or isolated asthenozoospermia; unexplained infertility; after vasectomy reversal.

Principle and method

- Polyacrylamide beads coated with rabbit anti-human **IgG, IgA or IgM**.

Test	Method	Use
Direct IBT	Patient's motile sperm washed free of seminal plasma; mixed with beads; motile sperm with attached beads counted	Sperm-bound antibody, class and site
Indirect IBT	Antibody-free donor sperm incubated with heat-inactivated patient serum, seminal plasma or female serum/cervical mucus; then direct IBT done	Free antibody; patient sperm too few or immotile; female partner
MAR (mixed antiglobulin reaction)	Fresh unwashed semen + IgG (or IgA)-coated latex beads + anti-IgG; mixed sperm–bead clumps counted	Quick, cheap screen; IgG/IgA only

- Report: **% of motile sperm with beads**, immunoglobulin class, and **site** (head, midpiece, tail). Tail-tip binding is ignored.
- **WHO 2010 (5th edition) threshold: ≥50% of motile sperm bound is clinically significant.**
- **Head binding** and **IgA** matter most (IgA is produced locally and blocks mucus penetration; IgM is too large to enter semen).
- Both direct tests need motile sperm. **Also know:** ELISA (enzyme-linked immunosorbent assay) and flow cytometry; serum tests correlate poorly with sperm-bound antibody.

Management

- Condoms and sperm washing lack good evidence (washing removes unbound, not bound, antibody).
- Corticosteroids: two controlled trials conflict.
- IUI for mild binding; **IVF with ICSI (intracytoplasmic sperm injection)** is the usual treatment, bypassing antibody effects.

Diagram to draw: motile sperm with beads attached at head and tail.

Key points to remember:

- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Treatment of choice: ICSI.

CBAVD

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 3, Q3f

Definition

- **CBAVD (congenital bilateral absence of the vas deferens):** both vasa absent, causing **obstructive azoospermia** with normal spermatogenesis.
- Often a **genital form of CF (cystic fibrosis)**: nearly all men with CF lack both vasa.

Genetics

- **CFTR (cystic fibrosis transmembrane conductance regulator)**: chloride channel gene; more than 1600 mutations.
- Severe mutation on both alleles: CF. **Mild mutation on one or both alleles: CBAVD.** Commonest severe mutation **ΔF508**; the **5T variant** (intron 8) is a common mild allele.
- A minority have other mesonephric duct gene defects instead, with more **renal agenesis**.

Clinical features

- **Vasa not palpable** (Meacham: a presumed vas slipping away three times is absent).
- Caput epididymis present and distended; body and tail often absent.
- Normal testis size, FSH (follicle-stimulating hormone) and testosterone.
- Seminal vesicles absent or hypoplastic: **semen volume <1.0 mL, acidic, fructose-negative**, azoospermic.
- Sinus or chest symptoms may reveal mild CF.

Evaluation

- Semen analysis; FSH; **TRUS (transrectal ultrasound)** for seminal vesicles.
- **Renal ultrasound** (renal agenesis in about 1 in 10).
- **CFTR testing of the man and his female partner** (carrier rate about 1 in 20 in northern Europeans). Panels cover 25–40 mutations, so a negative panel leaves residual risk.
- Testis biopsy unnecessary when FSH and testis size are normal.

Management

- **Genetic counselling**: if the partner carries a mutation, offspring may have CF; PGT (preimplantation genetic testing) or donor sperm can be offered.
- Reconstruction is impossible, so **surgical sperm retrieval + ICSI (intracytoplasmic sperm injection)**:

Technique	Points
MESA (microsurgical epididymal sperm aspiration)	Best: many motile sperm; cryopreserve several vials
PESA (percutaneous epididymal aspiration)	Local anaesthetic; fewer sperm
TESA/TESE (testicular aspiration/extraction)	If epididymal retrieval fails

- ICSI results are good; one study reported more miscarriage and stillbirth.
- **Also know:** CUAVD (unilateral absence) is a Wolffian duct defect with ipsilateral renal agenesis.

Diagram to draw: absent vas and seminal vesicle with intact testis and caput epididymis.

Key points to remember:

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

Sertoli cell only syndrome

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 3, Q3a (non-obstructive azoospermia also answered in 2025-Winter Paper 2, Q3a)

Definition

- SCOS (Sertoli cell only syndrome), **germ cell aplasia** (Del Castillo syndrome): seminiferous tubules lined only by Sertoli cells, with **complete loss of spermatogonia**. Leydig cells usually normal.
- The worst histological pattern of NOA (non-obstructive azoospermia).

Types

Type	Feature
Pure (complete, congenital)	All tubules lack germ cells
Mixed (focal, acquired)	Some tubules keep foci of spermatogenesis, the basis of sperm retrieval

Aetiology

- **Developmental:** primordial germ cells fail to migrate into the testis.
- **Genetic: complete AZFa (azoospermia factor a) deletion** gives pure SCOS; **AZFc** may give SCOS with foci; Klinefelter (47,XXY).
- **Acquired:** cryptorchidism, chemotherapy or radiation, mumps orchitis, torsion; idiopathic.

Clinical features

- Normal virilisation; **azoospermia** with normal ejaculate volume; small soft testes.

Evaluation

- Semen: azoospermia on two centrifuged samples.
- **FSH (follicle-stimulating hormone) high**, **inhibin B low** (Sertoli marker); LH (luteinising hormone) normal or high; testosterone normal or low.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- **Karyotype and Y microdeletion** before surgery.
- **Histology** confirms, usually at sperm retrieval.

Pattern	Meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages, fewer	Best
Maturation arrest	Stops at one stage	About 40%
SCOS	No germ cells	**About 30%**

Management

- **Counsel** on genetic findings.
- **Complete AZFa or AZFb deletion:** retrieval essentially nil; do not operate; donor sperm or adoption.
- **Micro-TESE (microdissection testicular sperm extraction):** at ×20–25, **larger, opaque, yellow tubules** (foci of spermatogenesis) are selected; SCOS tubules are thin, white and stringy. Live births are reported even in SCOS.
- Sperm used for **ICSI (intracytoplasmic sperm injection)**.
- Optimise first: stop exogenous testosterone; hormonal stimulation evidence weak.

- **Also know:** spermatogonial stem cell transplantation and in-vitro spermatogenesis are experimental.

Diagram to draw: tubule cross-section: normal layered epithelium versus Sertoli cells only.

Key points to remember:

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

Oxidative stress and male infertility

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 3, Q3b

Definition

- **Oxidative stress (OS):** imbalance where ROS (reactive oxygen species) exceed the antioxidant capacity of semen. The term **MOSI (male oxidative stress infertility)** has been proposed.
- Low ROS levels are **physiological**: needed for capacitation, hyperactivation, acrosome reaction and sperm–oocyte fusion.

Sources of ROS

Source	Examples
Leucocytes (main)	Infection, prostatitis, epididymitis, leucocytospermia
Immature sperm	Excess residual cytoplasm
Varicocele	Heat and hypoxia
Lifestyle	**Smoking**, alcohol, obesity, heat
Others	Ageing, diabetes, pollutants, radiation

Why sperm are vulnerable

- Membrane rich in **polyunsaturated fatty acids** → lipid peroxidation.
- Little cytoplasm, so few antioxidant enzymes; limited DNA repair.
- **Seminal plasma antioxidants:** superoxide dismutase, catalase, glutathione peroxidase; taurine, vitamins C and E, zinc.

Effects

- Lipid peroxidation → loss of **motility** and membrane fluidity.
- **Sperm DNA fragmentation** → poor embryo quality, miscarriage, lower ART (assisted reproductive technology) success.
- Mitochondrial damage and apoptosis.

Evaluation

- Semen analysis: peroxidase-positive WBC (white blood cells) **>1 million/mL** = leucocytospermia.
- **Direct ROS:** chemiluminescence (luminol).
- **Indirect:** total antioxidant capacity, malondialdehyde (lipid peroxidation), ORP (oxidation-reduction potential).
- **DNA fragmentation tests:** TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labelling), SCSA (sperm chromatin structure assay), SCD (sperm chromatin dispersion), Comet.
- Use selectively (unexplained infertility, recurrent miscarriage, failed ART); not routine.

Management

- **Lifestyle:** stop smoking, reduce weight and alcohol, avoid heat.
- **Treat cause:** antibiotics for infection; **varicocelectomy** for clinical varicocele (reduces DNA fragmentation).
- **Antioxidants** (vitamins C and E, CoQ10 [coenzyme Q10], zinc, selenium, carnitine): Cochrane review, low-quality evidence; **MOXI trial (2020)** showed no gain in semen

parameters, DNA fragmentation or live birth. Avoid excess: **reductive stress** also harms.
- High DNA fragmentation with failed ICSI (intracytoplasmic sperm injection): **testicular sperm** has less fragmentation.

Diagram to draw: balance of ROS sources against seminal antioxidants, with effects on sperm.

Key points to remember:

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

Dhat syndrome

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 3, Q3c

Definition

- **Dhat syndrome:** a **culture-bound syndrome** of South Asia. The man believes he is losing **semen ("dhat")** in urine or nocturnal emissions, and attributes weakness, fatigue and sexual problems to it.
- Term coined by **Wig** (India); listed in ICD-10 (International Classification of Diseases, 10th revision) and in the DSM-5 (Diagnostic and Statistical Manual, 5th edition) glossary of cultural concepts of distress.
- Related syndromes: **Shen-kui** (China), **Koro** (genital retraction).

Cultural basis

- Traditional belief that semen is a **precious vital fluid** made from many drops of blood; its loss is felt to drain body and mind.

Epidemiology

- Young men, often unmarried or newly married, rural, lower education; common in urology and sexual medicine clinics.

Clinical features

- **Somatic:** fatigue, weakness, aches, poor appetite.
- **Psychological:** anxiety, low mood, guilt (about masturbation), poor concentration, sleep loss.
- **Sexual:** fear of ED (erectile dysfunction) or PE (premature ejaculation), worry about penis size and fertility.
- Whitish urine is usually phosphaturia, oxaluria or normal secretions.
- **Comorbidity: depression** and **anxiety** in many; somatisation.

Evaluation

- Non-judgmental sexual history and belief system.
- Examination of genitalia; urine microscopy (phosphates, oxalates, sperm, pus cells).
- Exclude organic disease: **UTI (urinary tract infection), prostatitis, STI (sexually transmitted infection), diabetes**, true spermatorrhoea.
- Screen for depression and anxiety.

Management

1. Counselling (main)

- **Sex education:** semen is continuously produced; nocturnal emission and masturbation are normal and harmless.
- Reassure without dismissing.

2. Psychological therapy: CBT (cognitive behavioural therapy) for misbeliefs and anxiety.

3. Drugs

- **SSRIs (selective serotonin reuptake inhibitors)** for depression, anxiety or associated PE; short anxiolytic course if needed.
- Treat genuine ED or PE.

4. Avoid harm

- Avoid unnecessary "tonics", antibiotics and repeated tests; they reinforce the belief.
- Refer to psychiatry when depression is significant.

Prognosis

- Good with counselling; drop-out is common.

Diagram to draw: flowchart: complaint of semen loss → exclude organic cause → counsel → treat depression or anxiety.

> **Key points to remember:**
>
> - Culture-bound semen-loss anxiety of South Asia.
> - Depression and anxiety common; exclude UTI, prostatitis, STI.
> - Counselling and sex education first; SSRIs for comorbidity.

Rapid Revision Sheet – Section 9 · Andrology & Sexual Medicine

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Priapism: Diagnosis and Management (LAQ)

- 36 h (beyond 36 h) or refractory → early penile prosthesis (or analgesia and counselling)
- Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling next.
- ED risk 0% ≤12 h, 100% ≥36 h (beyond 36 h); consider early prosthesis beyond 36 h.
- High flow: observe, then embolise; repeat embolisation rather than surgery.

Types of penile prosthesis. What are biofilms in penile prosthesis? (SAQ)

- 3-piece inflatable gives best rigidity and flaccidity.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; Mulcahy salvage 84%.

Oligospermia and oligoasthenozoospermia (SAQ)

- WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
- Karyotype and Y-microdeletion if under 5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

Sperm retrieval techniques (SAQ)

- OA: epididymal (MESA); NOA: micro-TESE.
- Micro-TESE finds sperm in about half of NOA.
- AZFa/AZFb deletion: no sperm, do not operate.

Genetic causes of male infertility. Enumerate and classify. (SAQ)

- Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
- AZFa or AZFb complete deletion: do not attempt retrieval.
- CBAVD = genital form of cystic fibrosis; test both partners.

Premature ejaculation. (SAQ)

- ISSM: about 1 min (lifelong), about 3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Spermatogenesis (SAQ)

- About 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; FSH → Sertoli, LH → Leydig.
- Tight junctions between Sertoli cells form the blood–testis barrier.

Ejacultory duct obstruction (SAQ)

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

Tadalafil (SAQ)

- Half-life 17.5 h; food-independent; back pain.
- 5 mg daily for LUTS: IPSS improves, Qmax does not.
- No nitrates for 48 h after a dose.

Retrograde ejaculation (SAQ)

- TURP and diabetic neuropathy are the commonest causes.

- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

Immunobead test for semen (SAQ)

- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Treatment of choice: ICSI.

CBAVD (SAQ)

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

Sertoli cell only syndrome (SAQ)

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

Oxidative stress and male infertility (SAQ)

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

Dhat syndrome (SAQ)

- Culture-bound semen-loss anxiety of South Asia.
- Depression and anxiety common; exclude UTI, prostatitis, STI.
- Counselling and sex education first; SSRIs for comorbidity.

Section 10 · Testis, Penis & Urethra — Long answer questions (LAQ)

Urethral stricture: classification, DVIU and surgery after failed VIU

Asked as:

- Classify urethral strictures. Discuss in brief, surgical management of stricture after failed VIU — *Summer 2017 Paper III Q1 (LAQ)*
- DVIU (Direct Visual Internal Urethrotomy). — *Winter 2017 Paper III Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Definition

- A **urethral stricture** is a fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar with variable **spongiofibrosis**.
- **VIU (visual internal urethrotomy)**, also called DVIU (direct vision internal urethrotomy), cuts (incision of) the stricture scar into healthy tissue under vision (Sachse cold-knife urethrotome), so the lumen heals enlarged, and lets it heal (healing is) by **secondary intention**: a **race between epithelialisation and wound contraction**. It fails (the stricture recurs) when contraction wins, which is usual when spongiofibrosis is deep. Deep spongiofibrosis cannot be cured endoscopically.

Classification

1. By site (WHO (World Health Organization) 2004 / ICUD (International Consultation on Urological Diseases) 2010 terminology)

Segment	Term
Anterior: meatus, fossa navicularis, penile (pendulous), bulbar	**Stricture**
Membranous urethra after pelvic fracture	**PFUI (pelvic fracture urethral injury)** / distraction defect
Posterior urethra after prostatectomy or radiotherapy; bladder neck	**Stenosis or contracture**

2. By aetiology

Category	Examples
Idiopathic (commonest, with iatrogenic)	No cause found
Iatrogenic	Catheters, TURP (transurethral resection of prostate), cystoscopy, hypospadias repair, radiotherapy
Traumatic	Straddle injury (bulbar); pelvic fracture (PFUI)
Inflammatory	**Lichen sclerosus (LS)**; gonorrhoea (now rare)
Congenital	Rarest; infants only

3. By length: short (up to about 2 cm), long, multifocal, **panurethral** (whole anterior urethra).

4. By depth (Jordan): (A) mucosal fold, (B) iris constriction, (C) full-thickness with minimal spongiofibrosis, (D) full-thickness spongiofibrosis, (E) fibrosis beyond the spongiosum, (F) complex stricture with fistula or abscess.

5. By complexity: simple (short, bulbar, first treatment) versus complex (long, multiple, LS, radiation, failed previous treatment, fistula).

- **PFUI** (Ch 125): major distraction, minor distraction, partial injury.
- **Also know: LSE system** (Lumen calibre, Segment involved, Etiology) for standard reporting.

DVIU: indications (favourable)

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis, any non-bulbar
Short (<1.5–2 cm)	Longer
Superficial, no dense spongiofibrosis	Dense scar (straddle, LS (lichen sclerosus))
First procedure	Repeat; luminal narrowing >74%

DVIU: technique

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define length and site.
2. Saline irrigation; **guidewire** through the stricture.
3. **Cold-knife incision traditionally at 12 o'clock** through scar to healthy spongiosum. Campbell notes the spongiosum is thinnest from 10 to 2 o'clock; deep cuts enter the corpora (ED (erectile dysfunction), fistula).
4. Catheter **3–7 days**; longer catheterisation adds no benefit.

- **Laser** (holmium, Nd:YAG (neodymium:yttrium–aluminium–garnet), others): **no advantage over cold knife**.

DVIU: adjuncts and complications

- **CISC (clean intermittent self-catheterisation)/self-dilation**, tapering over 3–6 months: delays recurrence, which returns when stopped.
- Mitomycin C, steroid injection (limited data); UroLume withdrawn.
- **Recurrence** (commonest); bleeding (often with erections); **irrigant extravasation**; UTI (urinary tract infection), sepsis; ED; spongio-cavernous fistula; incontinence if sphincter incised.

Why repeat VIU is futile after failure (outcomes)

- VIU succeeds only in **bulbar strictures <1.5 cm without dense spongiofibrosis, at the first attempt**. Recurrence shows deep spongiofibrosis, which an incision cannot remove.
- **Steenkamp 1997, Heyns 1998 (randomised trials):** dilation = urethrotomy; a second urethrotomy helps few, a third none. **A second urethrotomy adds virtually nothing.**
- **Santucci & Eisenberg (2010):** about **8% stricture-free** long term after a first DVIU; repeat procedures almost always fail.
- Each (repeat) dilation or urethrotomy lengthens the scar and lowers later urethroplasty success; urethroplasty is more **cost-effective** after one failure. The "reconstructive ladder" is abandoned.
- Guideline position: AUA (American Urological Association) guideline (2016): DVIU or dilation may be offered for **short (<2 cm) bulbar** strictures; offer **urethroplasty, not repeated endoscopy**, after failed dilation or DVIU; no permanent stents.

Pre-operative evaluation

- Define the **4 Ls: location, length, depth, density**.
- **Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)**, flexible urethroscopy, sonourethrography; MRI (magnetic resonance imaging) for PFUI.
- **Urethral rest**: suprapubic catheter for 6–8 weeks lets the true length show. Avoid instrumentation before surgery.
- Treat infection; look for LS at the meatus; examine the mouth; record erectile function.

Surgical options after failed VIU

Stricture	Operation	Principle
Bulbar, short (1–2 cm, up to 3–5 cm proximally)	**EPA (excision and primary anastomosis)**: gold standard, about 90–95% success	Excise all scar, spatulate both ends widely, tension-free anastomosis. **Non-transecting** or **vessel-sparing** variants keep bulbar arteries
Bulbar, longer	**BMG (buccal mucosal graft) onlay**: dorsal (**Barbagli**), ventral, dorsal inlay (**Asopa**); Kulkarni one-sided	Graft spread-fixed on a well-vascular bed (corpora or spongiosum)
Bulbar with an obliterated segment	**Augmented anastomosis**	Excise the narrowest part, anastomose the floor, graft the roof
Penile	One-stage BMG inlay or onlay; **penile skin flap** (Orandi, Quartey, circular)	Avoid EPA here (chordee)
LS, failed hypospadias, infection	**Staged repair** (Johanson): graft, then tubularise after about 6 months	Oral mucosa, not genital skin, in LS
Panurethral	Kulkarni one-stage dorsal graft, or staged	—
Elderly, unfit, multiple failures	**Perineal urethrostomy** (BMG-augmented)	Simple, durable bypass
PFUI after failed VIU	Delayed **perineal anastomotic urethroplasty**	Bulbar mobilisation, then intracrural separation, inferior pubectomy, rerouting

- **Onlay is better than tube.** Graft and flap results are equal; grafts are simpler.
- **Not fit or declines surgery: Optilume** paclitaxel-coated balloon (FDA (Food and Drug Administration) 2023) (**ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after repeated failed endoscopy); intermittent self-dilation; long-term suprapubic catheter.
- **Urethral stents** (UroLume) are withdrawn.

Complications

- Recurrence, chordee, ED (erectile dysfunction; similar to circumcision overall), post-void dribble, graft-site pain and numbness.

Follow-up

- Voiding cystourethrogram at catheter removal (about 3 weeks); then uroflow, PVR (post-void residual) and symptom score; urethroscopy if flow falls. EPA recurrences occur mostly in the first 6 months; graft results decline with time.

Recent advances (recent advance)

- Tissue-engineered oral mucosa; robotic posterior repair.

- **Optilume** paclitaxel-coated balloon: **ROBUST III** stricture-free 74.6% vs 26.8% at 1 year after failed endoscopy.

Diagram to draw: dorsal versus ventral onlay cross-section; EPA with spatulated ends; urethrotome cutting at 12 o'clock over guidewire.

Key points to remember:

- VIU (DVIU) cures only (is best for) short bulbar strictures (<1.5–2 cm) at first attempt; do not repeat it.
- Santucci: ~8% long-term success; repeat DVIU futile.
- AUA: urethroplasty after failure; laser = cold knife.
- EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
- Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
- ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

Classify testicular tumors and describe management of seminoma

Type: LAQ (25 marks)

Source: Practice Set 1 Paper 3, Q1 (stage I seminoma also answered in 2018-Summer Paper 2, Q2)

Introduction

- Over 95% of testicular tumours are GCTs (germ cell tumours). Seminoma is 52–56% of GCNIS (germ cell neoplasia in situ)-derived GCTs, peaks in the fourth–fifth decade.
- **Risk factors:** cryptorchidism, previous contralateral GCT (twelvefold), family history, GCNIS, White race, infertility.
- All postpubertal GCTs except spermatocytic tumour carry **isochromosome 12p**.

WHO (World Health Organization) classification (2016, retained 2022)

Group	Tumours
1. GCNIS-derived GCTs	GCNIS; **seminoma**; **NSGCT (non-seminomatous GCT)**: embryonal carcinoma, postpubertal yolk sac tumour, choriocarcinoma and other trophoblastic tumours, postpubertal teratoma, teratoma with somatic-type malignancy; mixed GCT
2. Non-GCNIS-derived GCTs	**Spermatocytic tumour** (formerly spermatocytic seminoma); prepubertal teratoma, dermoid and epidermoid cyst; prepubertal yolk sac tumour
3. Sex cord–stromal tumours (<5%)	Leydig cell, Sertoli cell (including large cell calcifying), granulosa cell (adult, juvenile), fibroma–thecoma, mixed
4. Mixed germ cell–sex cord stromal	Gonadoblastoma (DSD [difference of sex development])
5. Others	Lymphoma (commonest testicular tumour in older men), rete testis and collecting duct tumours, adnexal and paratesticular tumours, metastases

- Any NSGCT element, however small, makes the tumour an NSGCT and is treated as such.
- **Seminoma pathology:** sheets of clear cells with lymphocyte-rich fibrovascular septa. Immunohistochemistry: CD117 and PLAP (placental alkaline phosphatase) positive, CD30 negative. Syncytiotrophoblasts (hCG [human chorionic gonadotropin]) in about 15%. "Anaplastic seminoma" is no longer recognised.
- **Spermatocytic tumour** (<1%): older men, no GCNIS, no i(12p), benign unless sarcomatous.

Evaluation and staging

- Scrotal ultrasound; AFP (alpha-fetoprotein), β-hCG and LDH (lactate dehydrogenase) before and after orchidectomy. **A raised AFP means NSGCT**, whatever the histology.
- **Radical inguinal orchidectomy** with early cord control at the internal ring; never trans-scrotal. Offer sperm banking and prosthesis.
- CT (computed tomography) of chest, abdomen and pelvis; TNMS (tumour, node, metastasis, serum markers) staging.
- About 80% of seminomas are clinical stage (CS) I; 15–20% CS II (nodes IIA ≤2 cm, IIB 2–5 cm, IIC >5 cm).

Management of seminoma by stage

CS I (cure by orchidectomy alone 80–85%; cancer-specific survival about 100% with any option)

	Surveillance (preferred)	Carboplatin	Radiotherapy
Method	Examination, markers, CT for at least 5 years	1 cycle **AUC (area under the curve) 7**, dose = 7 × (GFR [glomerular filtration rate] + 25) mg	**20 Gy** para-aortic strip
Relapse	14–20%	About 5% (1 cycle)	3–5%
Drawback	Late relapses (10–20% after 4 years)	Underdosing; late effects unknown	Second cancers, GI (gastrointestinal) toxicity

- **Relapse risk factors:** tumour **>4 cm** and **rete testis invasion** (relapse 12%, 16%, 32% with 0, 1, 2 factors). LVI (lymphovascular invasion) is not a factor in seminoma.
- **Trials:** MRC TE19/EORTC (Medical Research Council/European Organisation for Research and Treatment of Cancer) 30982: one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE18: 20 Gy equals 30 Gy.
- Surveillance relapses are 84–100% retroperitoneal and almost all salvaged.

CS IIA–B

- **Non-bulky (≤3 cm):** dog-leg radiotherapy with nodal boost (25–30 Gy in the chapter; EAU [European Association of Urology] uses 30 Gy for IIA and 36 Gy for IIB), or chemotherapy, or **primary RPLND (retroperitoneal lymph node dissection)** (SEMS trial).
- **Bulky (>3 cm) or multiple nodes:** BEP (bleomycin, etoposide, cisplatin) ×3 or EP (etoposide, cisplatin) ×4.

CS IIC–III: chemotherapy by IGCCCG (International Germ Cell Cancer Collaborative Group)

Risk	Seminoma definition	5-year survival	Regimen
Good (90%)	Any primary, **no NPVM (non-pulmonary visceral metastases)**, normal AFP, any hCG/LDH	86%	BEP ×3 or EP ×4
Intermediate (10%)	NPVM present	72%	BEP ×4
Poor	**None for seminoma**	–	–

- Carboplatin alone is inferior in advanced disease.

Post-chemotherapy residual mass

```
Residual mass after chemotherapy
 ├─ <3 cm ──────────────► observe (necrosis about 90%)
 └─ >3 cm ─► FDG-PET (fluorodeoxyglucose positron emission tomography) at ≥6 weeks
              ├─ negative ► observe
              └─ positive ► resection (full template PC-RPLND) or salvage
```

- Surgery is difficult because of desmoplasia; complete resection only 58–74%.

Relapse and follow-up

- After surveillance or radiotherapy: radiotherapy for small retroperitoneal relapse, otherwise first-line BEP. After chemotherapy: VeIP (vinblastine, ifosfamide, cisplatin) or high-dose chemotherapy.

- Surveillance schedule: review every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly. CT detects >85% of relapses.

Recent advances

- Serum **miR-371a-3p** microRNA: sensitive marker of residual disease and relapse.
- MRI (magnetic resonance imaging) for surveillance to cut radiation; primary RPLND for low-volume stage II.

Diagram to draw: WHO classification tree (GCNIS-derived vs non-GCNIS-derived vs sex cord–stromal) and the residual-mass flowchart.

Key points to remember:

- WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
- CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
- Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
- No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

Uses of buccal mucosa in urology

Type: SAQ (10 marks)

Source: 2015-Summer Paper 3, Q3b

Introduction

- **BMG (buccal mucosal graft)** is the workhorse graft for urethral reconstruction. **Humby 1941** (hypospadias); **Bürger 1992** and **El-Kasaby 1993** (stricture).

Why it works

- Thick, **non-keratinised**, hairless epithelium used to a wet environment.
- **Thin, highly vascular lamina propria**: rapid imbibition and inosculation, so take is excellent. Little contraction; low donor morbidity.
- A graft lives on its bed, so it does best on thick vascular beds.

Harvest

- Nasotracheal intubation; about 4–6 × 2–2.5 cm per cheek, **1 cm from the commissure**, **avoiding Stensen's duct** (opposite the upper second molar).
- Defat, cut about 10% oversize, fenestrate, quilt to the bed. Alternatives: **lingual**, **lip** (thinner, for the glans).
- Donor problems: pain, numbness, mouth tightness.

Uses

Technique	How	Use
Dorsal onlay (Barbagli)	Urethra rotated, dorsal urethrotomy; graft spread-fixed on the corpora	Bulbar; reliable bed
Ventral onlay	Ventral urethrotomy; graft covered by spongioplasty	Bulbar with healthy spongiosum
Dorsal inlay (Asopa)	Ventral urethrotomy, dorsal plate incised, graft inlaid	Bulbar and penile, no mobilisation
Kulkarni	One-sided dissection keeps ventral blood supply; dorsal onlay	Panurethral, one stage
Augmented anastomosis	Excise the narrowest part, anastomose the floor, graft the roof	Long dense bulbar stricture
Staged (Johanson, Bracka)	Stage 1 lay open and graft; stage 2 tubularise after 6 months or more	**Lichen sclerosus**, failed hypospadias

- **Others**: redo hypospadias, BMG perineal urethrostomy, ureteroplasty, bladder neck contracture, female urethral stricture.

Outcomes

- Bulbar BMG urethroplasty about 85–90%; dorsal equals ventral; graft equals flap; onlay beats tube.
- Results fall with time and in penile, redo and lichen sclerosus cases. **Avoid skin in lichen sclerosus** (it recurs in skin).

- **Also know:** other grafts: full-thickness skin, bladder mucosa (meatal prolapse), meshed split skin (staged only).

Recent advances

- Tissue-engineered oral mucosa.

Diagram to draw: bulbar cross-section with dorsal onlay, ventral onlay and Asopa inlay.

> **Key points to remember:**
> - Thin vascular lamina propria gives excellent take.
> - Staged BMG for lichen sclerosus.
> - Bulbar success about 85–90%.

Varicocele: controversies and present status of treatment

Asked as:

- Controversies in management of varicocele. — *Summer 2018 Paper III Q3f (SAQ)*
- Present status of varicocele treatment — *Practice Set 6 Paper III Q3e (SAQ)* (related: 2018-Summer Paper 3, Q3f)

Type: SAQ (10 marks)

Background

- Dilated pampiniform veins: about **15% of men, 35% with primary and up to 80% (another source: 75–81%) with secondary infertility**; mostly left-sided (taller venous column to the left renal vein; vein enters renal vein 8–10 cm higher; valve failure / valve incompetence).
- Isolated right or non-decompressing varicocele → scan (image) for renal tumour (a mass).
- Harm / **Injury: heat** (lost countercurrent exchange), oxidative stress, **sperm DNA fragmentation**; progressive, duration-dependent atrophy; lower testosterone.
- **Grades:** subclinical (Doppler only), I (palpable with Valsalva), II (palpable), III (visible).

The controversies / current indications

Question / Situation	Evidence	Current (present) position
Does repair improve fertility? Infertile couple, **palpable** varicocele, abnormal semen	Madgar RCT (randomised controlled trial, 1995): pregnancy 44% versus (vs) 10% at 1 year; meta-analyses mixed	**Repair clinical** varicocele with **abnormal semen** in an infertile couple with normal female partner (AUA/ASRM (American Urological Association / American Society for Reproductive Medicine))
Subclinical varicocele?	No benefit for semen or pregnancy	**Do not treat**
Normal semen?	No proven benefit	Not for fertility
Adolescent varicocele	Testis volume is a surrogate; link to adult fertility unproven	Observe; repair for **testis asymmetry above 20%** (testicular hypotrophy over 20%), pain, or abnormal semen (Tanner 5 or 18 years)
NOA (non-obstructive azoospermia)	Small series: sperm returns to ejaculate in some	No definitive evidence; may be considered before micro-TESE (microsurgical testicular sperm extraction)
Before IVF/ICSI (in vitro fertilisation / intracytoplasmic sperm injection)	Repair may improve live-birth rate (improves outcomes) or allow less invasive ART (assisted reproduction / assisted reproductive technology)	Discuss, especially with oligozoospermia
High DNA fragmentation	Repair lowers fragmentation	Reasonable indication
Hypogonadism (low testosterone)	Repair raises testosterone in hypogonadal men (repair improves both)	Emerging indication
Pain	Dull ache on standing; relief in most after microsurgery	Repair after other causes excluded

Choice of technique (Techniques)

Technique	Recurrence	Hydrocele	Comment / Note
Microsurgical subinguinal/inguinal	Lowest (about 1%; **0.5–1%**)	Near 0 (**About 0**)	**Best / Preferred**: artery and lymphatics preserved
Retroperitoneal (Palomo)	High with artery spared (15–25%)	High with mass ligation (about 7%)	Children (artery ligation acceptable)
Laparoscopic	3–15%	Up to 12%	Bilateral; serious morbidity possible
Embolisation / sclerotherapy	4–25% (another source: 15–25%)	Nil (0)	Outpatient; technical failure

- **Microsurgical steps:** micro-Doppler maps the artery; ligate all veins except vasal; preserve artery and lymphatics; deliver testis to ligate gubernacular and external spermatic veins.
- **Complications:** hydrocele (lymphatic ligation, commonest after non-microsurgical repair); artery injury (atrophy); recurrence (diagnose by upright colour Doppler).
- **Predictors of poor result:** recurrence, artery ligation, hydrocele. Earlier repair recovers more.
- **Also know:** isolated right or non-decompressing varicocele → scan for renal tumour; recurrence diagnosed by upright colour Doppler.

Diagram to draw: cord cross-section at microsurgery: artery, lymphatics and vas preserved, veins ligated.

> **Key points to remember:**
> - Treat (repair) only palpable varicocele with abnormal semen; never subclinical.
> - Madgar 1995: 44% versus (vs) 10% pregnancy.
> - Adolescents: observe; repair for asymmetry above (over) 20%.
> - Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).

Inguinal Block Dissection?

Type: SAQ (10 marks)

Source: 2020-Summer Paper 3, Q3f

Taken as ILND (inguinal lymph node dissection), mainly for penile cancer. For VEIL see 2019-Summer Paper 4, Q3d; for nodal strategy see 2017-Summer Paper 2, Q1.

Indications

- Penile SCC (squamous cell carcinoma): **proven cN+**; **cN0 high risk** (pT1b or more, G3) if DSNB (dynamic sentinel node biopsy) is positive or unavailable. Also vulval cancer, melanoma.
- **Within 3 months**; aim for **at least 8 nodes per side**. Early ILND: 3-year disease-specific survival about 84% vs 35% if delayed (Kroon).
- Bulky or fixed nodes: chemotherapy first.

Surgical anatomy

- **Femoral triangle:** inguinal ligament, sartorius (lateral), adductor longus (medial); floor fascia lata.
- **Superficial nodes** (4–25) in **Daseler's five zones**; sentinel nodes mostly **superomedial**.
- **Deep nodes** (1–3) medial to femoral vein; **Cloquet's node** highest. Spread is stepwise, without skips.

Templates

Template	Extent
Radical	Superficial + deep; saphenous often ligated; sartorius transposed
Modified (Catalona)	Nothing lateral to femoral artery or below fossa ovalis; saphenous spared; positive frozen section → radical
Superficial	Above fascia lata only

Technique (open radical)

- Frog-leg; incision **2 cm below and parallel** to the ligament.
- Flaps **deep to Camper's fascia** (skin capillaries).
- Divide saphenous at apex; open fascia lata; clear along femoral vessels; **never lateral to the femoral artery** (femoral nerve).
- **Sartorius transposition**; suction drains until <30 mL/day.
- Pelvic dissection if more than 2 inguinal nodes or ENE (extranodal extension).

Complications

- **Wound infection, flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis).
- Reduced by correct flap plane, modified template, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.

Diagram to draw: femoral triangle with Daseler's zones.

> **Key points to remember:**
> - Ligament, sartorius, adductor longus.
> - Flaps deep to Camper's; stay medial to femoral artery.
> - Within 3 months; at least 8 nodes per side.

Minimal invasive retroperitoneal lymph node dissection

Type: SAQ (10 marks)

Source: 2025-Winter Paper 3, Q3f (cross-referenced from 2025-Winter Paper 4, Q3e)

Introduction

- **MI-RPLND (minimally invasive retroperitoneal lymph node dissection)** for testicular GCT (germ cell tumour): laparoscopic (1992) or robotic (2006). **It must duplicate open RPLND exactly.**

Indications

- **Primary:** marker-negative stage I NSGCT (non-seminomatous GCT), especially high-risk IB; small-volume stage II NSGCT or seminoma (1–2 nodes ≤3 cm in the landing zone).
- **Postchemotherapy:** only a small solitary residual mass with little desmoplasia; **open** for large masses or vessel involvement.

Templates and nerve sparing

- Boundaries: renal hila, ureters, ureter crossing the common iliacs; plus spermatic cord.
- **Primary:** extended **unilateral modified template** with nerve sparing (right: paracaval, precaval, interaortocaval; left: para-aortic, preaortic; both take nodes above the IMA (inferior mesenteric artery)). Enlarged nodes → bilateral.
- **Postchemotherapy:** bilateral full template.
- **Nerve sparing:** identify the sympathetic chains and L1–L4 postganglionic fibres to the hypogastric plexus below the IMA.

Steps (robotic)

1. Positioning: **modified lateral decubitus** (unilateral) or **supine steep Trendelenburg with peritoneal hammock** (any template).
2. Mobilise colon; Kocherise duodenum (right); divide gonadal vein.
3. **Split-and-roll:** split over the great vessel from common iliac to renal vein; roll packets off.
4. **Divide lumbar vessels** to clear retrocaval and retroaortic tissue.
5. Clip lymphatics (left renal vein); bag each packet.

Feature	Laparoscopic	Robotic
Retraction	Assistant	Fourth arm
Instruments	Straight	Wristed
Suturing, vessel control	Difficult	Easier

Outcomes

- Stay 1–3 days, blood loss mostly ≤200 mL; conversion about 5%; **antegrade ejaculation ≥90%**; chylous ascites as open. Node yields approach open (about 30).
- **Major bleeding:** pressure, raise pneumoperitoneum, convert if uncontrolled (open set ready).
- Concerns: staging-only dissections, frequent adjuvant chemotherapy, unindicated surgery, catastrophic bleeding, follow-up <5 years; no randomised trials.
- **Guidelines:** AUA (American Urological Association): experienced surgeons may offer it; EAU (European Association of Urology): specialist centres only.

Also know: serum miR-371a-3p for selection; RPLND for stage II seminoma (SEMS, PRIMETEST).

Diagram to draw: right and left modified templates.

Key points to remember:

- Same template, split-and-roll, lumbar division and nerve sparing as open.
- Best for chemotherapy-naive low-volume disease.
- Conversion about 5%; ejaculation preserved ≥90%.

Eponymous urethroplasty techniques and perineal urethrostomy

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 3, Q3d (perineal urethrostomy also answered in 2021-Summer Paper 2, Q3c)

Introduction

- Options: **EPA (excision and primary anastomosis)**, the gold standard for short bulbar strictures; **graft or flap onlay**; **staged repair**.

Eponymous techniques

Eponym	Technique	Use
Johanson	Two-stage: stricture laid open, edges sutured to skin (urethrostomy); tubularised later	LS (lichen sclerosus), failed hypospadias, infection
Blandy	Inverted-U scrotal or perineal skin flap into a ventral urethrotomy	Bulbar strictures; classic perineal urethrostomy (PU)
Turner-Warwick	Two-stage scrotal inlay; transpubic approach, omental wrap	Complex, posterior
Webster	Progressive perineal approach: mobilisation, crural separation, inferior pubectomy, rerouting	PFUI (pelvic fracture urethral injury)
Barbagli	**Dorsal onlay** BMG (buccal mucosal graft) spread-fixed to corpora	Bulbar
Asopa	**Dorsal inlay** graft through a ventral urethrotomy, no mobilisation	Penile and bulbar
Kulkarni	One-sided dorsolateral onlay; panurethral repair via perineum with penile invagination	Panurethral (LS)
Orandi	Ventral longitudinal penile skin island flap on dartos	Pendulous
Quartey	Showed skin island is a passenger on a fascial (dartos) flap	Basis of all genital skin flaps
McAninch	Circular fasciocutaneous penile skin flap	Long anterior strictures (up to 15 cm)
Jordan	Vessel-sparing EPA; ventral transverse island for fossa	Proximal bulbar; fossa navicularis

- **Also know:** Monseur dorsal urethrotomy; Andrich–Mundy non-transecting anastomosis; Bracka two-stage BMG.
- **Principles:** grafts need a vascular bed; onlay beats tube; graft equals flap.

Perineal urethrostomy (PU)

- **Definition:** bulbar urethra opened onto perineal skin; the patient voids sitting. It **bypasses** a diseased anterior urethra.
- **Indications:** LS panurethral disease, failed urethroplasties, elderly or high-risk patients, after penectomy.
- **Techniques:**
  - **Classic (Blandy):** inverted-U perineal flap into the opened bulbar urethra.
  - **BMG PU:** dorsal urethrotomy to healthy mucosa; ventral urethra to skin; calibrated to **30 Fr**; BMG on corpora completes the dorsal wall, preventing circumferential scarring.
- **Complications:** stenosis (main), spraying, perineal wetness.

Diagram to draw: bulbar cross-section showing dorsal onlay (Barbagli), dorsal inlay (Asopa) and ventral onlay.

Key points to remember:

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU; BMG PU at 30 Fr.

Prognostic index score in carcinoma penis

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 3, Q3e

Assumed meaning: prognostic factors, risk-group systems and nomograms in penile SCC (squamous cell carcinoma). No single universally used "prognostic index score" exists.

Introduction

- Prognosis rests on primary histology and stage and, above all, **inguinal node status**.

1. Nodal status (most important factor)

Nodal finding	5-year survival
Node negative	About 73%
≤2 nodes, unilateral, no ENE (extranodal extension)	About 80%
>2 nodes	0–50%
ENE or **pelvic nodes**	Poor (pelvic about 14%)

- TNM (tumour-node-metastasis) 8th edition: pN1 (1–2 unilateral nodes) about 90% 3-year DSS (disease-specific survival); pN3 (ENE or pelvic) about one-third.

2. Primary tumour factors (predict occult nodes in cN0)

- **Depth:** corporal invasion → about 60% node positive.
- **Grade:** any grade III component = grade III.
- **LVI (lymphovascular invasion)** and **PNI (perineural invasion)**: strongest predictors; define pT1b.
- **Subtype:** verrucous and PeIN (penile intraepithelial neoplasia) do not metastasise; basaloid and sarcomatoid aggressive.

3. Risk-group systems

System	Low	Intermediate	High
Earlier EAU (European Association of Urology), after Solsona	Tis, Ta, T1 G1	T1 G2	T2 or more, or G3
Current EAU–ASCO (American Society of Clinical Oncology) / TNM	PeIN, Ta, **T1a** (nodes 0–18%)	—	**T1b or more** (one-third to over half)

- Low risk → groin surveillance; high risk → **DSNB (dynamic sentinel node biopsy)** or ILND (inguinal lymph node dissection).

4. Nomograms and indices

- **Ficarra nomogram:** predicts nodal involvement (thickness, growth pattern, grade, LVI, corporal invasion, clinical node status).
- **Kattan nomogram:** predicts cancer-specific survival.
- A pathological **prognostic index** (grade + anatomical level of invasion + PNI; Cubilla group) has been proposed.
- **Status:** none externally validated; not used routinely.

5. Molecular markers

- **HPV (human papillomavirus)/p16 positive:** better DSS (not node status).

- Adverse: p53, PD-L1 (programmed death-ligand 1), Ki-67, high **NLR** (neutrophil-to-lymphocyte ratio), raised CRP (C-reactive protein).

6. Other

- Inguinal recurrence after observation: only about one-third survive 5 years. Size >5 cm worse.

Diagram to draw: flowchart: primary pathology → low or high risk → surveillance or DSNB/ILND.

> **Key points to remember:**
> - Node status is the strongest prognostic factor.
> - LVI, PNI, grade III define high-risk T1b.
> - Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

Rapid Revision Sheet – Section 10 · Testis, Penis & Urethra

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Urethral stricture: classification, DVIU and surgery after failed VIU (LAQ)

- VIU (DVIU) cures only (is best for) short bulbar strictures (<1.5–2 cm) at first attempt; do not repeat it.
- Santucci: ~8% long-term success; repeat DVIU futile.
- AUA: urethroplasty after failure; laser = cold knife.
- EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
- Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
- ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

Classify testicular tumors and describe management of seminoma (LAQ)

- WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
- CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
- Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
- No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Uses of buccal mucosa in urology (SAQ)

- Thin vascular lamina propria gives excellent take.
- Staged BMG for lichen sclerosus.
- Bulbar success about 85–90%.

Varicocele: controversies and present status of treatment (SAQ)

- Treat (repair) only palpable varicocele with abnormal semen; never subclinical.
- Madgar 1995: 44% versus (vs) 10% pregnancy.
- Adolescents: observe; repair for asymmetry above (over) 20%.
- Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).

Inguinal Block Dissection? (SAQ)

- Ligament, sartorius, adductor longus.
- Flaps deep to Camper's; stay medial to femoral artery.
- Within 3 months; at least 8 nodes per side.

Minimal invasive retroperitoneal lymph node dissection (SAQ)

- Same template, split-and-roll, lumbar division and nerve sparing as open.
- Best for chemotherapy-naive low-volume disease.
- Conversion about 5%; ejaculation preserved ≥90%.

Eponymous urethroplasty techniques and perineal urethrostomy (SAQ)

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU; BMG PU at 30 Fr.

Prognostic index score in carcinoma penis (SAQ)

- Node status is the strongest prognostic factor.
- LVI, PNI, grade III define high-risk T1b.
- Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Type: LAQ (25 marks)

Source: 2015-Summer Paper 3, Q1

Introduction

- A stent keeps the upper tract draining past obstruction, injury or an anastomosis. **Zimskind 1967** placed the first indwelling splint; **Finney 1978** introduced the self-retaining **double-J (double-pigtail)** stent.

Types and materials

Feature	Details
Size	3–8.5 Fr (adult usually 6 Fr); adults 20–30 cm by height, children about age + 10 cm
Polymers	Polyurethane, silicone (least encrusting, soft); hydrocoated silicone causes less pain
Metallic (Resonance)	Resists extrinsic compression: malignant obstruction, retroperitoneal fibrosis
Others	Hybrid metal-reinforced, thermo-expandable nitinol (Memokath), biodegradable (experimental)
Designs	Single-pigtail diversion stent; open-ended catheter to identify the ureter; modified bladder ends (Polaris loop: no benefit)

Also know: anatomy: three narrowings (PUJ (pelviureteric junction), iliac vessels, intramural ureter, the narrowest at 3–4 mm). Placement: retrograde pyelogram first, stent over a wire under fluoroscopy; antegrade route if retrograde fails.

Diagram to draw: double-J stent in situ with both coils and side holes.

Indications

- Obstruction: infected obstructed kidney (stent equals PCN (percutaneous nephrostomy)), malignancy, stricture, pregnancy.
- After URS (ureteroscopy), pyeloplasty, reimplantation, transplant; injury, leak, fistula; pre-stenting a tight ureter.

Stent symptoms and complications

- About **4 in 5** have urgency, frequency, haematuria or flank pain, measured by the **USSQ (Ureteral Stent Symptom Questionnaire)**. Cause: **intravesical material irritating the bladder and reflux up the stent**. Diameter and length matter little.
- Relief: **alpha-blockers and antimuscarinics**; intraureteral placement; shortest dwell; opioid-free ERAS (enhanced recovery after surgery) analgesia.
- **Misplacement** (submucosal) and **migration** (too short).
- **Stent failure** from compression, debris or encrustation: worst with malignant or irradiated ureter, bilateral obstruction; then PCN.
- **Encrustation** rises with dwell time, infection and stone-forming urine. **Forgotten stent** causes sepsis and renal loss; keep a register.

- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear encrustation; II circular encrustation; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.
- **Encrusted stent**: never pull by force; NCCT (non-contrast computed tomography); clear the **bladder coil**, then **ureter**, then **renal coil** (flexible URS or PCNL (percutaneous nephrolithotomy)).

Ureteroscopy

	Semirigid	Flexible
Tip	7 Fr or less	4.5–7.5 Fr; 3.6 Fr channel
Use	Distal and mid ureter	Upper ureter, kidney
Deflection	None	Over 270°

- Segura's dictum: always a safety wire. Tight ureter: stent and return after **2–4 weeks**; routine pre-stenting is unnecessary.
- **UAS (ureteral access sheath)**, usually 12/14 Fr: lowers intrarenal pressure and eases repeated passes, but **severe injury in about 1 in 8**; less after pre-stenting.
- **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm)** or **TFL (thulium fibre laser, about 1940 nm)**. **Dusting** (low energy, high frequency) is faster but leaves fines; **fragment and basket** confirms stone clearance and gives stone for analysis. Keep pressure and temperature low with outflow.

Steps of URS: (1) sterile urine, antibiotic dose; (2) cystoscopy and **safety guidewire**; (3) semirigid inspection, dilate only if needed; (4) access sheath over a working wire under fluoroscopy (for flexible URS); (5) low-pressure irrigation, laser, basket; (6) inspect the ureter on withdrawal; (7) decide on stent.

- **Also know:** anti-retropulsion devices; digital and single-use scopes; URS suits patients who cannot stop anticoagulants.
- Complications: sepsis about 5% (treat positive urine culture first), stricture about 3%, bleeding under 1%, perforation (stop and stent), avulsion under 1% (PCN, delayed repair).

Stent after URS?

- **Denstedt 2001** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: stent may be omitted when anatomy and renal function are normal, there is no ureteric injury and no second stage is planned. **EAU (European Association of Urology)** agrees.
- **Stent needed**: perforation, stricture or tight ureter, infection, planned second stage, solitary kidney, impacted stone.
- Removal at 3 days gave more unplanned visits than at 7 days; a string allows home removal. Stenting at SWL (shock wave lithotripsy) does not improve stone passage.

```
Uncomplicated URS -> no stent (or short string stent)
Complicated       -> double-J: injury 2-6 wk; second stage until re-look
```

Recent advances

- Suction sheaths, intrarenal pressure sensors, biodegradable stents.

> **Key points to remember:**
> - Finney 1978 double-J; symptoms from intravesical stent and reflux.
> - Encrusted stent: bladder coil, ureter, then kidney.
> - Denstedt 2001: no routine stent after uncomplicated URS.

Laparoscopic pyeloplasty.

Type: LAQ (25 marks)

Source: 2016-Summer Paper 3, Q2

Introduction

- **PUJO (pelviureteric junction obstruction)** is impaired flow from pelvis to ureter that, if untreated, causes progressive renal damage (Koff).
- Causes: **intrinsic** adynamic segment (abnormal circular muscle), **extrinsic** crossing lower-pole vessel, high insertion; secondary (failed repair, stone).
- **Anderson–Hynes (1949) dismembered pyeloplasty** is the gold standard by any route. First laparoscopic pyeloplasty **1993 (Schuessler)**; robotic **2002**. Success **over 90–95%**, equal to open, with less morbidity.

Indications

- Symptoms (pain, Dietl's crisis), infection, stones, hypertension (rare).
- Impaired split function or **fall of 10% or more** on serial renography; rising APD (anteroposterior diameter); obstructed solitary kidney or bilateral disease.
- Children: most antenatal hydronephrosis resolves; about one-third need surgery.
- **Contraindications**: uncorrected coagulopathy, untreated UTI (urinary tract infection), cardiopulmonary disease barring pneumoperitoneum.
- **Nephrectomy instead** if split function under 15% in adults with a normal other kidney.

Preoperative evaluation

- **USG (ultrasonography)**: large pelvis, small calyces, ureter not seen.
- **MAG3 (mercaptoacetyltriglycine) diuretic renogram**: split function and an obstructive curve that does not wash out after furosemide. Usual teaching: washout half-time under 10 minutes normal, 10–20 minutes equivocal, over 20 minutes obstructed (Ch 93 quotes over 40 minutes). Split function under 40% supports repair.
- **CT (computed tomography) angiography**: anatomy, **crossing vessels**, stones.
- **Retrograde pyelogram** at surgery: confirms the level, excludes ureteric stricture.
- **Whitaker test** if equivocal: perfusion 10 mL/min; pelvic pressure over 15–20 cm H2O means obstruction.
- Sterile urine. Drain beforehand (stent or nephrostomy) only for infection, azotaemia or severe pain.

Approaches

	Transperitoneal	Retroperitoneal
Space	Large, familiar landmarks	Small, created by balloon
Bowel	Colon reflected (left: **transmesocolic** window)	Not entered; leak stays extraperitoneal
Use	Commonest; easier suturing	Avoids bowel; harder orientation

- One paediatric randomised trial favoured retroperitoneal (shorter stay); overall outcomes similar.

Steps (transperitoneal Anderson–Hynes)

1. Cystoscopy, retrograde pyelogram, catheter.
2. 45–60° flank; pneumoperitoneum 12–15 mmHg in adults (under 12 mmHg in children); 3–4 ports, camera at umbilicus.
3. Reflect colon medially, or open the mesocolon over the pelvis.

4. Find the ureter; trace it to the PUJ with its periureteric tissue; minimal diathermy. Look for a **crossing vessel**.
5. Hitch stitch on the pelvis; transect the pelvis above the PUJ; **reduce** a redundant pelvis if needed.
6. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a handle (no-touch).
7. **Transpose** the PUJ in front of any crossing vessel.
8. Anastomose the spatulation apex to the most dependent pelvis with 4-0 absorbable (5-0/6-0 in infants): back wall, stent, front wall, then close the pelvis. Watertight, tension-free, funnel-shaped.
9. **Antegrade double-J** over a wire; clamp the catheter: urine from the stent side holes confirms the bladder coil.
10. Excise the narrow segment; drain near (not on) the suture line.

Crossing vessel

- It lowers endopyelotomy success (42% vs 86%), so pyeloplasty is preferred. **Never divide** a lower-pole vessel (segmental infarction).
- **Vascular hitch (Hellström)**: vessel moved up onto the pelvis, only if the PUJ is intrinsically normal and peristalsing.

Stenting and postoperative care

- Double-J for **4–6 weeks** (children 1–4 weeks: no difference). Externalised stent in infants avoids a second anaesthetic.
- Foley for 24–48 hours (bladder pressure refluxes up the stent); drain out when dry. If drain output rises after Foley removal, replace the Foley.
- **MAG3 renogram** after stent removal, then ultrasound; follow for years.

Robotic pyeloplasty

- 3D vision, wristed suturing, tremor filtration, **shorter learning curve**. Now first-line at major centres.
- Success 94.7–100%, the same as laparoscopy; advantage is ergonomic, at higher cost.

Outcomes

- Laparoscopic: operating time 2–4 hours; complications 2–16%; conversion 0–5.5%; stay about 2.7 vs 4.2 days (open); success 87–99%.
- **Up to 30% of failures appear after 2 years**, so follow-up must be long.

Complications

- Urine leak and urinoma, stent symptoms or migration, UTI, haematuria, ileus, bowel or vessel injury, port-site hernia, recurrent obstruction.

Management of failure

```
Early (clot, oedema)        -> nephrostomy, wait
Persistent obstruction      -> CT angiography (missed vessel?)
  short, no vessel          -> endopyelotomy (about 70%)
  otherwise                 -> redo laparoscopic/robotic pyeloplasty
                               (about 89% after failed open; buccal graft onlay)
  small intrarenal pelvis   -> ureterocalicostomy
  poor function             -> nephrectomy
```

Also know: Y-V plasty (high insertion); Culp spiral flap (long narrowing); stones removed through the pyelotomy with a flexible scope; safe in horseshoe and pelvic kidneys.

Recent advances

- Single-port robot (2018); indocyanine green fluorescence for redo cases; buccal mucosa ureteroplasty.

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition over the crossing vessel, dependent anastomosis.

Key points to remember:

- Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
- Transpose over a crossing vessel; never divide it.
- Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
- Robotic equals laparoscopic in outcome, with easier suturing.

Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis.

Type: LAQ (25 marks)

Source: 2023-Winter Paper 3, Q1 (medical management also answered in 2019-Summer Paper 2, Q3e)

Introduction

- RPF (retroperitoneal fibrosis) is a rare inflammatory fibrosis encasing the aorta, IVC (inferior vena cava) and ureters.
- Incidence about 1 per 200,000–500,000 a year; age 40–60; male:female 2–3:1.

Etiology

Idiopathic (about 70%)	Secondary (about 30%)
Part of **chronic periaortitis**, a large-vessel vasculitis	**Drugs** (commonest identifiable cause): methysergide and other ergots, β-blockers, phenacetin
Proposed antigen: **ceroid** (oxidised lipid in plaque)	**Malignancy:** lymphoma (commonest), myeloma, carcinoid, pancreas, prostate, sarcoma
Probably autoimmune: membranous glomerulonephritis, ankylosing spondylitis, granulomatosis with polyangiitis; ANA (antinuclear antibody) positivity predicts relapse	Radiotherapy; asbestos
Part of multifocal fibrosclerosis: sclerosing mediastinitis and cholangitis, orbital pseudotumour, Riedel thyroiditis	Infection: tuberculosis, actinomyces, gonorrhoea, schistosomiasis

- Many idiopathic cases are now classed as **IgG4 (immunoglobulin G4)-related disease**.

Pathology

- **Gross:** flat, tan, dense plaque centred on the **distal aorta at L4–L5**, wrapping the ureters; may invade ureter or psoas.
- **Histology changes with stage:**
  - Early (active): collagen with capillary proliferation and lymphocytes, plasma cells and fibroblasts.
  - Late (burnt-out): acellular, avascular sheets of collagen.
- **Trap:** malignant RPF looks identical except for small islands of tumour cells, so take **deep, multiple biopsies**.
- **Mechanism of obstruction:** extrinsic compression and loss of ureteric peristalsis; the lumen stays patent.

Clinical features

- **Dull lower back or flank pain** (commonest); weight loss, anorexia, nausea, malaise, fever, hypertension.
- About half present with **uraemia** or oliguria/anuria; symptoms last 4–6 months before diagnosis.
- **Extension:** IVC (leg oedema, DVT [deep venous thrombosis]); renal vein (gross haematuria); spermatic cord and scrotum. Examination is usually unrevealing.

Diagnostic evaluation

Laboratory

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein) in half to two-thirds; leucocytosis; anaemia; raised creatinine; ANA; serum IgG4.

Imaging

Test	Findings and role
CT (computed tomography) / CT urography	Homogeneous soft-tissue mass enveloping aorta, IVC and ureters; **medial deviation** of proximal and mid-ureters with smooth tapering
Retrograde pyelography	When renal function is poor
MRI (magnetic resonance imaging)	Low T1; **high T2 signal means active disease**; enhancement falls with treatment
PET-CT (positron emission tomography)	**Most sensitive test of activity**; guides biopsy and follow-up
Renography	Split function (repair or nephrectomy)

- **Biopsy** (CT-guided or at ureterolysis) if malignancy is suspected; may be omitted with no malignancy history, no lymphadenopathy and typical imaging.

Treatment

```
RPF with hydronephrosis
  -> Relieve obstruction (stents; nephrostomy if anuric or septic)
  -> Treat post-obstructive diuresis
  -> Stop culprit drug; exclude malignancy
  -> Active disease: steroids (+/- second-line drugs)
  -> Failure or burnt-out fibrosis: ureterolysis
```

1. Initial management

- Emergent **stents or PCN (percutaneous nephrostomy)**; nephrostomy if critically ill with little urine. Replace fluid during **post-obstructive diuresis**.

2. Medical therapy

Agent	Role and regimen
Prednisone (first line)	60 mg daily (about 1 mg/kg), tapered to 5 mg, up to 2 years. About 80% respond; relapse on tapering 25–50%
Azathioprine, mycophenolate mofetil, cyclophosphamide, ciclosporin, colchicine	Second line or steroid-sparing
Tamoxifen; medroxyprogesterone	Thought to inhibit fibroblasts
Rituximab	Refractory or IgG4-related disease

- Steroids work on the cellular, active phase: pain settles in days and the mass shrinks.
- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen.
- Drainage plus medical therapy clears obstruction in about two-thirds; only about **15% need ureterolysis**.

3. Surgical ureterolysis

- For failed medical therapy, burnt-out fibrosis or tissue diagnosis. **Bilateral**, even if hydronephrosis is unilateral; open, laparoscopic or robotic.
- Steps:
  1. Pre-place stents to identify the ureters.

2. Mobilise the colon; take deep biopsies for frozen section.
3. Start at the distal, non-dilated ureter.
4. "Split and roll": right-angle clamp between ureter and plaque; incise the fibrosis over it.
5. Close any ureterotomy with absorbable suture.
6. **Separate the ureter from the fibrotic bed so it cannot be re-encased.**

Protection method	How
Intraperitonealisation	Ureter lateralised; peritoneum closed behind it
Lateral retroperitoneal placement	Ureter displaced laterally (equivalent results)
Omental wrap	Omentum split on the gastroepiploic arteries, wrapped round each ureter; adds blood supply; best for extensive disease

- Postoperative steroids may reduce recurrence.
- **Outcomes:** open about 94% stent-free at 1 year; laparoscopic about 90%; robotic with omental wrap about 86%.

4. Other options: long-term metallic stents (unfit); autotransplantation if ureterolysis is impossible; nephrectomy for an unrecoverable kidney.

Follow-up

- Symptoms, ESR/CRP, creatinine, ultrasound; MRI or PET-CT for activity; remove stents once ureters drain; watch steroid side effects.

Diagram to draw: periaortic plaque at L4–L5 drawing both ureters medially, with an omental wrap.

> **Key points to remember:**
> - Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
> - Medially deviated ureters; PET-CT best shows activity; biopsy deep.
> - Drain first, then prednisone (about 80% respond; Vaglio 2011 beat tamoxifen).
> - About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury

Asked as:

- Iatrogenic ureteric injury — *Practice Set 2 Paper III Q1 (LAQ)* (upper ureteric injury also answered in 2023-Winter Paper 3, Q2)
- Discuss management of ureteric injuries occurring during gynec surgeries — *Practice Set 4 Paper III Q1 (LAQ)* (also answered in Practice Set 2 Paper 3, Q1)
- Discuss Etiology, Diagnosis and Management of Upper ureteric injury. — *Winter 2023 Paper III Q2 (LAQ)*

Type: LAQ (25 marks)

Assumed scope: surgical (iatrogenic) injury at any level of the ureter, in gynaecological, colorectal, vascular and urological surgery. Assumed meaning (upper ureteric injury): injury to the upper (proximal) ureter, from the PUJ (pelviureteric junction) to the pelvic brim, from external trauma or surgery.

Introduction

- Ureteric trauma is uncommon: 1–2.5% of genitourinary trauma. **Iatrogenic injury is commoner overall** than external trauma.
- **The proximal ureter is the commonest site of external injury. The distal ureter is the commonest (usual) iatrogenic site** in non-urological surgery.
- Gynaecological surgery has historically caused about half of all iatrogenic ureteric injuries; hysterectomy is the main culprit.
- It is mainly a problem of **recognition** (the core problem): thermal, ligation and devascularisation injuries are invisible at operation and declare themselves days to weeks later. Injury is often **missed** and declares itself later as urinoma, sepsis, stricture or fistula.
- Late diagnosis brings urinoma, sepsis, AKI (acute kidney injury), fistula and renal loss, and needs nephrostomy about ten times more often.

Aetiology and incidence (EAU (European Association of Urology) trauma guideline data)

1. External trauma

- **Penetrating** (most): gunshot wounds up to about 90%. Blast damages the intramural blood supply up to 2 cm from the wound.
- **Blunt** (up to about 40%): rapid deceleration avulses the ureter at the **PUJ**; linked to lumbar transverse process fractures.
- **Over 90% have associated injuries** (small bowel commonest, colon, vessels, kidney 10–28%). Mortality 6–9% reflects these.

2. Iatrogenic

Source	Key facts (upper-ureter relevance)
Gynaecological	Historically about half. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery highest (1.7–4.3%). Endometriosis and placenta accreta raise risk
Colorectal	0.15–10%; laparoscopic equals open
Colorectal, spinal, retroperitoneal surgery	Ligation, transection, cautery

Source	Key facts (upper-ureter relevance)
Urological / Ureteroscopy	Ureteroscopy is the commonest (iatrogenic) cause in some series: abrasion, perforation 0.2–2%, avulsion under 0.5% (another source: <0.3%) (proximal third most at risk); stricture about 3% (another source: 0.3–3%). Risk: proximal stone, impacted stone, long operation, large sheath, inexperience. Also radical prostatectomy, partial nephrectomy or ablation (late proximal stricture)
Partial nephrectomy, thermal ablation	Proximal stricture months later (lower-pole and hilar tumours)
Vascular (aortic) surgery	Aorto-iliac bypass: transient hydronephrosis 12–20%; up to 85% unrecognised (missed initially). Devascularisation. Rare ureteroarterial fistula (chronic stent plus pelvic surgery or radiation)

Gynaecological incidence (EAU trauma guideline data)

Procedure	Ureteric injury (%)
Caesarean section (emergency)	0.01–0.06
Vaginal hysterectomy	0.02–0.5
Abdominal hysterectomy	0.03–2.0
Minimally invasive hysterectomy	0.13–6.0 (laparoscopic = robotic, about 0.9)
Prolapse surgery	**1.7–4.3 (highest)**

- **Danger sites in the pelvis (applied anatomy):** pelvic brim near (at) the infundibulopelvic ligament; **under the uterine artery** ("water under the bridge"); beside the uterosacral ligament; vaginal fornix / vaginal cuff near the UVJ (ureterovesical junction).
- Up to a third are recognised during open surgery, fewer during laparoscopy; the uterosacral region is hardest to see.
- **Mechanisms:** ligation, crush, laceration, transection, avulsion, thermal injury, **devascularisation**. About 10% of women get a large share of distal ureteric blood from the uterine artery (the uterine artery supplies much of the distal ureter), so some (ischaemic) injuries are unavoidable.
- **Risk factors:** previous surgery, adhesions, malignancy, radiotherapy, heavy bleeding, long operation, learning curve; endometriosis (encasement, adhesions, medial deviation), placenta accreta.
- **URS (ureteroscopy) risks:** long operation, 12/14-Fr sheath, impacted or proximal stone, inexperience.

Classification: AAST (American Association for the Surgery of Trauma) ureter scale

Grade	Injury (description)
I	Haematoma or contusion, no devascularisation (without devascularisation)
II	Laceration <50% of circumference
III	Laceration ≥50%
IV	Complete transection, <2 cm devascularised (with <2 cm devascularisation)
V	Avulsion, >2 cm devascularised (with >2 cm devascularisation)

- Advance one grade for bilateral injury, up to grade III.

Prevention

- See (identify) the ureter at the **pelvic brim** (crossing the common iliac artery) and trace it before ligating pedicles. Good haemostasis and exposure.

- **Prophylactic stents** (or lighted stents) help recognition in high-risk (complex) cases, but do not clearly reduce (prevent) injury. **ICG (indocyanine green)** instilled into the ureter (intraureteric) makes it fluoresce (fluoresces) for hours.
- **Cystoscopy** after minimally invasive hysterectomy and prolapse surgery, with oral phenazopyridine or IV (intravenous) dye to show efflux.
- URS: safety wire; ≤14-Fr sheath in an unstented ureter; fragment before basketing; "stop, stent, stage".

Diagnosis and recognition

Suspicion (trauma)

- Penetrating flank or abdominal wound; high-energy deceleration.
- **Haematuria is absent in 25–45%**, so its absence does not exclude injury.

Intraoperative (best outcome)

- Inspect the ureter (at laparotomy); IV (intravenous) indigo carmine or methylene blue; cystoscopy for efflux (jets), with retrograde (on-table) pyelogram or stent. Check the other ureter and bladder. About 11% are missed at trauma laparotomy.
- Methylene blue is an MAO (monoamine oxidase) inhibitor: avoid with SSRIs/SNRIs (selective serotonin / serotonin–noradrenaline reuptake inhibitors; serotonin toxicity), G6PD (glucose-6-phosphate dehydrogenase) deficiency and pregnancy.

Delayed presentation (most cases) / postoperative

- Mean diagnosis about day 10: fever, ileus, flank pain, leucocytosis, rising creatinine, rising drain output, drain or wound urine leak, urinoma, anuria (bilateral), or **ureterovaginal fistula** (UVF): normal voiding plus constant leak, because the other kidney fills the bladder, usually within 4 weeks.

Imaging and tests

Test	Role
CT (computed tomography) urography with 10-minute delayed images (both ureters)	Best test in stable patients. Signs: extravasation, non-opacified distal ureter, delayed nephrogram, urinoma
Retrograde pyelography	Most sensitive; shows site; allows stenting at once
Antegrade nephrostogram	When retrograde fails or gap is wide
One-shot IVU (intravenous urography)	Non-diagnostic in 33–100%; never rely on it

- **Drain-fluid (or vaginal fluid) creatinine** above (far above) serum creatinine confirms urine.
- **Double-dye test** separates VVF (vesicovaginal fistula) from ureterovaginal fistula: oral phenazopyridine plus intravesical methylene blue. Orange pad = UVF; blue = VVF; both = both.

Management

Principles

- **Stability decides timing; level and length decide the operation.**
- Repair: mobilise sparing (preserve) adventitia; debride to bleeding tissue (edges); **spatulated, tension-free, watertight, stented** repair (anastomosis) with fine absorbable suture; drain; retroperitonealise; omentum (omental wrap) for thermal, radiation, blast or vascular injury; non-tunnelled reimplant.
- **No bowel interposition or autotransplantation acutely.**

By timing

```
Recognised at surgery -> repair at once (stable patient)
Unstable -> ligate + later nephrostomy, or exteriorised stent
Recognised late -> try retrograde stent (or nephrostomy + antegrade stent)
   |-- succeeds -> stent 4-8 weeks, then image
   |-- fails -> nephrostomy; early repair if <=7 days,
               otherwise delayed repair after about 6 weeks
Uncontrolled sepsis / collections -> drain; repair regardless of timing
```

```
Ureteric injury suspected
 |-- Unstable -> damage control: ligate + later PCN,
 |               or exteriorised stent / cutaneous ureterostomy
 |-- Stable, found at surgery -> immediate repair over stent
 |-- Found late / postoperatively
       -> Attempt retrograde stent (+/- PCN, drain)
       -> Fails -> PCN; delayed definitive repair
```

```
Suspected late injury
 -> retrograde stent (or nephrostomy + antegrade stent)
     |-- succeeds -> stent 4-8 weeks, then image
     |-- fails -> nephrostomy
          |-- found <=7 days -> early repair
          |-- later -> repair after about 6 weeks
Sepsis / uncontrolled collections -> drain; repair regardless of timing
```

At surgery, by injury

Injury	Action
Suture through wall, minor thermal	Remove suture; stent
Partial tear	Close transversely (transverse closure, 4-0/5-0) over stent, 4–8 weeks
Ligation	Remove ligature (deligate); stent if viable, excise and repair if not
Transection, lower ureter (usual)	**Ureteroneocystostomy ± psoas hitch** (95–100% success)
Long distal gap	**Boari flap** + hitch (gaps up to 12 cm)
Transection above pelvic brim	Ureteroureterostomy
Unstable patient	Ligate + later nephrostomy, or exteriorised stent

Repair by grade

- **Grade I–II (contusion, partial):** double-J stent for about 6 weeks; or close a laceration transversely over a stent. Excise non-viable contused ureter.
- **Grade III–V:** formal reconstruction.

By level (complete injury)

Level	Repair
Upper	Ureteroureterostomy; ureteropyelostomy (PUJ (pelviureteric junction) avulsion); ureterocalicostomy
Mid	Ureteroureterostomy ± downward nephropexy; TUU (transureteroureterostomy) when delayed
Lower (commonest iatrogenic)	**Ureteroneocystostomy, non-tunnelled ± psoas hitch** (95–100% success); **Boari flap** for gaps up to 12 cm
Long loss (delayed only)	Ileal ureter or Yang–Monti; buccal graft ureteroplasty; autotransplantation (last resort)

Options for the upper ureter

Procedure	Key points
Ureteroureterostomy	Commonest acute repair; short defects (2–3 cm); spatulate 5–6 mm, 180° apart; 5-0 absorbable over double-J; up to 90% success; leak 10–24%, stricture up to 12%
Ureteropyelostomy (reimplant into pelvis)	PUJ avulsion; spatulated
Ureterocalicostomy	Pelvis or PUJ destroyed; lower-pole parenchyma resected
Renal descensus	Mobilise kidney downwards; gains up to 8 cm
TUU (transureteroureterostomy)	Delayed; long loss when the bladder cannot reach; 90–97% success. Avoid with stones, RPF (retroperitoneal fibrosis), TB (tuberculosis), urothelial cancer or radiation
Ileal ureter / Yang–Monti	Long or panureteric loss; 81–100% success. Avoid if creatinine >2 mg/dL, bladder dysfunction, inflammatory bowel disease, radiation enteritis. Acidosis about 20%
Appendiceal interposition	Right side; about 92% short-term success
Buccal mucosa graft ureteroplasty	Delayed long strictures; onlay with omental wrap, often robotic; about 87% success
Autotransplantation	Last resort before nephrectomy; 5–10% renal units lost
Nephrectomy	Rarely; unstable with unreconstructable kidney, or poor function

Iatrogenic endoscopic injury

- **URS perforation:** stop the procedure and place a **stent** about 4 weeks (± nephrostomy), image later; later reconstruction if stricture develops. **Avulsion:** nephrostomy, delayed reconstruction; avoid nephrectomy.

Fistulae

- **Ureterovaginal fistula (UVF):** stent first (best / ideally within 2 weeks; most heal); if stenting fails, reimplant with psoas hitch or Boari flap (success ≥90%). Early and delayed repair do equally well.
- **Ureteroarterial fistula:** endovascular covered stent.
- **Also know:** TUU (transureteroureterostomy), ileal ureter, buccal graft ureteroplasty and autotransplantation for long delayed defects; robotic reconstruction (repair) for delayed repair matches open; avoid nephrectomy.

Complications and follow-up

- Urinoma, abscess, leak, fistula, stricture, loss of renal function.
- Persistent leak: check stent position, take drain off suction, catheterise the bladder, then nephrostomy.
- **Follow-up:** ultrasound or diuretic renogram at 4–12 weeks and (at) 1 year; strictures (late stenosis) can be silent and may appear up to 30 months.

Medicolegal

- Disclose and document; malpractice claims most often follow gynaecological surgery.

Diagram to draw: psoas hitch with non-tunnelled reimplant; ureter's relation to the uterine artery (ureter passing under the uterine artery); spatulated ureteroureterostomy over a double-J stent.

Key points to remember:

- Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
- Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
- AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
- Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
- Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Describe early and late management of PFUD injuries

Type: LAQ (25 marks)

Source: Practice Set 5 Paper 3, Q1

PFUD is taken as pelvic fracture urethral distraction defect, i.e. PFUI (pelvic fracture urethral injury).

Introduction and mechanism

- PFUI occurs in about 10% of male and up to 6% of female pelvic fractures. Highest risk: straddle fracture of all four pubic rami, unstable fractures, symphyseal diastasis.
- The posterior urethra is fixed to the pubis (puboprostatic ligaments, perineal membrane), so displacement shears it, usually at the **bulbomembranous junction**. In prepubertal boys the tear may extend into the prostate and bladder neck.
- The sphincter is avulsed but usually functional, so most men stay continent after repair.
- The gap fills with scar: an obliterative **distraction defect**, not a true stricture.

Classification

System	Types
Campbell-Walsh	**Partial**; **minor distraction** (a cystoscope passes retrogradely into the bladder); **major distraction** (it cannot). Need for urethroplasty rises from partial to major
Goldman	I stretched, intact; II tear above the perineal membrane; III tear of membranous urethra extending below the membrane into the bulb (commonest); IV bladder neck tear into the urethra; IVa bladder base injury mimicking IV; V pure anterior injury

Diagnosis

- Triad: **blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate is unreliable.
- **Blood at the meatus mandates RUG (retrograde urethrogram)** before catheterisation (oblique, 20 mL contrast). Extravasation with bladder filling means partial injury; without filling, complete.
- Females: vaginal examination and urethroscopy. Cystogram or CT (computed tomography) for bladder injury.

Early management

1. ATLS (Advanced Trauma Life Support) resuscitation; associated injuries take priority.
2. **SPC (suprapubic cystostomy): the standard of care.** Percutaneous (peel-away sheath, 16 Fr Foley; avoid small "punch" tubes) or open through a lower midline incision, which allows bladder inspection.
    - SPC is safe with pelvic plating (AUA [American Urological Association] Urotrauma guideline); tunnel it high, away from hardware.
3. **Primary realignment (PR):** a brief attempt in a stable patient, within days, by antegrade plus retrograde flexible cystoscopy over a guidewire. Catheter out at 4–6 weeks; keep the SPC until voiding is good.
4. Partial tear: one gentle catheter attempt allowed.
5. **Immediate open exploration** only for bladder neck or rectal injury, or laparotomy for other reasons.

- **Also know:** female urethral injuries are usually repaired early.

SPC versus primary realignment

	SPC + delayed urethroplasty	Primary realignment
Advantage	Simple; haematoma untouched	Partial tears may heal; aligned ends ease repair
Drawback	Complete injuries need urethroplasty	Most still form a 1–2 cm stenosis
Evidence	Prospective data: **no significant difference** in need for urethroplasty	Modern PR does not add ED (erectile dysfunction) or incontinence

Late management

Timing: after **3 months** (scar stable), once the patient is ambulant.

Evaluation

- **"Up-and-down-o-gram":** cystogram plus RUG while the patient tries to void, to measure the gap; antegrade flexible cystoscopy via the SPC tract if the bladder neck stays closed. Bladder neck appearance does **not** predict its later function.
- **MRI (magnetic resonance imaging):** gap length, prostatic displacement (cephalad displacement predicts pubectomy).

Endoscopic: DVIU (direct vision internal urethrotomy) only for short, incomplete stenosis. "Cut-to-the-light" for gaps ≥1 cm fails (false passages).

Perineal anastomotic urethroplasty (treatment of choice)

1. Exaggerated lithotomy (under 5 hours); sounds via meatus and SPC tract.
2. Midline or lambda incision; mobilise the bulbar urethra up to the penoscrotal junction (not beyond: chordee).
3. Divide the urethra at the scar; excise **all** fibrosis until healthy prostatic mucosa appears over the sound.
4. Spatulate both ends to 30 Fr; 8–10 interrupted 4-0 absorbable sutures; tension-free mucosa-to-mucosa anastomosis over a silicone catheter.

Progressive perineal approach (each step shortens the path to bridge a longer gap):

Step	Manoeuvre
1	Bulbar urethral mobilisation (enough for most)
2	Separation of the corporal bodies (intracrural space)
3	Inferior wedge pubectomy (Kerrison rongeur)
4	Supracrural rerouting of the urethra around one corpus

- **Abdominoperineal or transpubic approach:** urethrorectal fistula, failed repair, bladder neck injury, children.
- **Postoperative:** voiding study at 21–28 days; cystoscopy at about 6 months.

Outcomes and complications

- Success above 95%. Restenosis 5–15%; often DVIU-treatable because the scar is already excised.
- **ED** up to 82%, mostly from the injury (cavernous nerves, pudendal arteries). **Incontinence <4%** (bladder neck). Follow ≥1 year.
- **Also know:** chordee, penile shortening, ejaculatory dysfunction, lithotomy neuropathy, penile Doppler before repair in impotent men.

Diagram to draw: sagittal pelvis showing the bulbomembranous gap and the four progressive perineal steps.

Key points to remember:

- Blood at meatus → RUG; SPC is the standard initial step.
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Delayed perineal anastomotic urethroplasty after 3 months; bulbar mobilisation → crural separation → pubectomy → rerouting.
- ED mostly from the injury; incontinence <4%.

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

Accessories for RIRS

Type: SAQ (10 marks)

Source: 2019-Summer Paper 3, Q3c

Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy in the kidney. Its single **3.6 Fr channel** is shared by accessories and irrigation.

Guidewires (0.018–0.038 inch, 145–260 cm)

Type	Construction	Use	Weakness
Hydrophilic	Nitinol core, slippery when wet	Bypassing impacted stones, tortuous ureter	Displaces easily
PTFE (polytetrafluoroethylene)-coated	Steel or nitinol	Safety and working wire	Damages scope channel on backloading
Hybrid	Floppy hydrophilic tip, stiff PTFE shaft	Both roles	—

- **Safety wire** keeps access and allows stenting after injury (AUA [American Urological Association], EAU [European Association of Urology]).
- **Also know:** open-ended or dual-lumen catheter (pyelogram, second wire); stiff wire for the sheath; balloon dilators.

UAS (ureteral access sheath)

- Hydrophilic sheath with tapered obturator; usually **12/14 Fr** (inner/outer); tip below the PUJ (pelviureteric junction).
- **Pros:** lower intrarenal pressure and temperature; repeated passes; less scope wear.
- **Cons:** ureteric injury in nearly half, severe in about **1 in 8** (less if pre-stented); no proven fall in infection.
- **Suction sheaths and FANS (flexible and navigable suction access sheath):** aspirate dust and keep pressure low.

Retrieval devices

- **Nitinol tipless baskets** (about 1.5–2.4 Fr): kink-resistant, open flush in calyces; relocate lower-pole stones. Small baskets allow more flow but are fragile. Graspers release easily.

Laser and fibres

- **Ho:YAG (holmium:yttrium-aluminium-garnet)** or **thulium fibre laser**; small-core fibres (about 200 µm) keep deflection and flow; **≤10 W in the ureter**.

Irrigation and antiretropulsion

- Gravity, pressure bag, hand or foot pumps (hand pumps: more fever); pressure sensors emerging.
- Stone Cone, NTrap, BackStop gel: mainly ureteric stones.

Stents

- **Double-J**; AUA: may omit after uncomplicated URS (ureteroscopy).

Diagram to draw: sheath over wire, tipless basket in lower calyx.

> **Key points to remember:**
> - Hydrophilic wire passes; PTFE wire stays.
> - 12/14 Fr sheath lowers pressure; severe injury about 1 in 8.
> - Nitinol tipless basket for calyces; ≤10 W in ureter.

Management of lower ureteric stone of size less than 1 cm in pregnancy?

Type: SAQ (10 marks)

Source: 2020-Summer Paper 3, Q3a

Background

- About 1 in 200 to 1 in 1500 pregnancies; **renal colic is the commonest non-obstetric cause of admission**. Mostly second or third trimester; mainly **calcium phosphate** stones. Linked with preterm delivery and pre-eclampsia.
- Physiological hydronephrosis (up to 90% by third trimester) mimics obstruction.

Diagnosis

- Urine culture always; blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; **transvaginal US** for the distal ureter; Doppler RI (resistive index) rises in acute obstruction.
- Second line: **MRI (magnetic resonance imaging)**, stone as a filling defect; **low-dose CT (computed tomography)**. Total fetal dose under **50 mGy**.

Management

```
Lower ureteric stone <1 cm
 ├─ sepsis / solitary kidney / renal failure → urgent stent or nephrostomy
 ├─ pain controlled, no infection → observation (50–80% pass)
 └─ failed observation → ureteroscopy + laser
                         or stent/nephrostomy until delivery
```

- **Conservative:** hydration, paracetamol, opioids. **Avoid NSAIDs (non-steroidal anti-inflammatory drugs)** (ductus arteriosus closure in third trimester). **MET (medical expulsive therapy)** with tamsulosin is off-label.
- **URS (ureteroscopy) with holmium laser:** first-line intervention for a non-complex stone in any trimester; ideal for a distal stone. Neuraxial anaesthesia, lateral tilt, minimal fluoroscopy (US guidance, shield under the patient), fetal monitoring. Third-trimester URS may raise preterm labour risk.
- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid: change stents every **4–6 weeks**, nephrostomy about every 3 weeks; definitive treatment after delivery.
- **SWL (shock wave lithotripsy) contraindicated**. Second trimester is the safest window.

Diagram to draw: the flowchart above.

> **Key points to remember:**
> - US first; MRI or low-dose CT second.
> - Observation first; avoid NSAIDs.
> - URS with laser is definitive; SWL contraindicated.
> - Stents encrust: change every 4–6 weeks.

Type 2 Renal tubular acidosis?

Type: SAQ (10 marks)

Source: 2020-Summer Paper 3, Q3b

Definition

- **Proximal RTA (renal tubular acidosis):** failure of the proximal tubule to reclaim bicarbonate, causing **hyperchloraemic, normal anion gap metabolic acidosis**.

Pathophysiology

- The proximal tubule reclaims **85–90% of filtered bicarbonate** (luminal carbonic anhydrase; exit by the basolateral Na–HCO_3 cotransporter).
- **Self-limiting:** bicarbonate wasting → plasma HCO_3^- falls → filtered load drops below the lowered threshold → the intact distal tubule copes → **steady state at about 14–18 mEq/L with urine pH <5.5**.
- **Hypokalaemia** from distal sodium bicarbonate delivery; worsens on alkali.

Causes

Isolated	Generalised (Fanconi syndrome)
Carbonic anhydrase inhibitors (acetazolamide, **topiramate**)	**Cystinosis** (commonest in children), Lowe, Wilson, galactosaemia
Carbonic anhydrase II deficiency (mixed)	**Ifosfamide, cisplatin**, aminoglycosides, valproate
	Myeloma, amyloidosis, lead

- **Fanconi:** glycosuria, aminoaciduria, phosphaturia, uricosuria.

Clinical features

- Children: growth failure, **rickets**; adults: osteomalacia, weakness.
- **Stones uncommon**: urinary citrate stays near normal and urine is acid at steady state. Topiramate and acetazolamide are exceptions (alkaline urine, calcium phosphate stones).

Diagnosis

- Normal anion gap acidosis, low potassium; urine can acidify below 5.5 (unlike distal RTA).
- **Bicarbonate loading:** urine pH rises; **fractional bicarbonate excretion >15%**.
- Look for Fanconi features; drug history.

Treatment

- Treat the cause.
- **Large alkali doses (about 10–15 mEq/kg/day)** as potassium citrate or mixed citrate, replacing potassium.
- Thiazide; phosphate and vitamin D for rickets.

	Type 1 (distal)	Type 2 (proximal)	Type 4
Urine pH in acidosis	Always >6	<5.5 at steady state	Low
Potassium	Low	Low	High
Stones	Common (up to 70%)	Uncommon	Uncommon

Diagram to draw: proximal tubular cell with carbonic anhydrase.

Key points to remember:

- Bicarbonate wasting; HCO_3^- 14–18 mEq/L; urine pH <5.5 at steady state.
- Usually Fanconi; ifosfamide, cisplatin.
- Fractional HCO_3^- excretion >15%; stones uncommon.

Renal Autotransplantation

Asked as:

- Auto-transplantation of Kidney? — *Summer 2020 Paper III Q3c (SAQ)*
- Role of auto transplant of kidney in Urology. — *Summer 2021 Paper III Q3e (SAQ)*
- Renal Autotransplantation — *Winter 2025 Paper III Q3c (SAQ)*

Type: SAQ (10 marks)

(also answered in 2020-Summer Paper 3, Q3c and 2021-Summer Paper 3, Q3e)

Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** (bench (ex vivo) repair) if needed. First done (first reported) in **1963** for proximal ureteric injury.
- **Principle:** bringing the kidney to the bladder replaces the ureter without bowel; the bench gives a cold, bloodless field.

Role (indications)

Group	Examples
Ureteric loss (commonest)	Long-segment loss or stricture; long iatrogenic or traumatic loss; repeated failed repairs (**delayed setting only, never acute**); **RPF (retroperitoneal fibrosis)** when ureterolysis fails
RPF (retroperitoneal fibrosis)	When ureterolysis is impossible and function is satisfactory
Tumour	Total ureterectomy for UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy in a solitary kidney; left RCC (renal cell carcinoma) needing suprarenal caval resection (new drainage for the right kidney)
Vascular	Complex renal artery aneurysm or branch disease; complex branch renal artery aneurysm, stenosis or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome
Pain	Selected loin pain–haematuria syndrome

- **Chosen over ileal ureter** (preferred over ileal ureter) when the contralateral kidney (the other kidney) is absent or poor, or ileal ureter / bowel is unsuitable (creatinine >2 mg/dL, bowel disease: inflammatory bowel disease, radiation enteritis).
- **Contraindications:** poor function (poor renal function), severe iliac atherosclerosis, unfit patient.

Procedure

```
CT (computed tomography) angiography, renogram, ureteric study
→ harvest as for live donor nephrectomy / live donation (open/laparoscopic/robotic):
  maximal vessel length, preserve ureteric blood supply
→ cold perfusion on bench ± ex vivo repair (bench repair)
→ Gibson incision, extraperitoneal iliac fossa
→ renal vein to external iliac vein (end-to-side);
  renal artery to external iliac (end-to-side) or internal iliac (end-to-end)
→ drainage: pyelovesicostomy (± Boari flap), ureteroneocystostomy,
  or native ureter left in continuity
→ stent, catheter, drain
```

- Laparoscopic harvest plus Gibson implantation: less analgesia, faster recovery; **robotic autotransplantation** emerging.

Complications and outcome

- Arterial or venous **thrombosis**, bleeding, urine leak, stricture, lymphocele, infection.
- **Renal unit loss 5–10%** (lower in transplant centres); major (Clavien–Dindo ≥III) complications about 15–20%.
- **Long-term success about 90%.** Doppler US (ultrasound) early; renogram and blood pressure later.

Also know: alternatives: Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti, buccal ureteroplasty, renal descensus.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

> **Key points to remember:**
>
> - For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
> - Delayed setting only; bench surgery for complex vascular or tumour cases.
> - Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
> - Unit loss 5–10%; success about 90%.

Indication, procedure and complications of Uretero-Calycostomy?

Type: SAQ (10 marks)

Source: 2020-Summer Paper 3, Q3d

Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

Procedure (open, laparoscopic or robotic)

```
Flank position; pre-placed double-J stent
→ mobilise ureter with generous periureteral tissue
→ mobilise kidney; partial lower-pole nephrectomy
   to expose the calyx widely (not simple nephrotomy)
→ close the pelvic outlet; spatulate ureter laterally
→ apex stitch to lateral calyx wall; second at 180°
→ interrupted sutures, all placed then tied
→ capsule closed over cut surface, away from anastomosis
→ cover with perinephric fat, peritoneum or omentum; drain
```

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.
- Stent out at 4–6 weeks; renogram later. Later endoscopy needs **antegrade upper-pole access**.

Complications

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, renal unit loss.
- Success about 70% in the largest (mostly redo) series; better in small paediatric series.

Also know: alternatives are redo pyeloplasty, Davis intubated ureterotomy, ileal ureter, buccal graft, autotransplantation, nephrectomy.

Diagram to draw: kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection.
> - Stricture is the main failure; success about 70% in redo series.

Post traumatic bladder neck injury: acute and delayed management?

Type: SAQ (10 marks)

Source: 2021-Summer Paper 3, Q3c

Introduction

- The bladder neck is the **internal sphincter**. It is torn mainly in **pelvic fracture**, often with prostatic urethral, rectal or vaginal injury; **commoner in children**.
- Missed injury causes **incontinence, fistula and stricture**.

Diagnosis

- Pelvic fracture with gross haematuria, retention or blood at the meatus; **vaginal examination** in females.
- **RUG (retrograde urethrogram)**, then **retrograde CT or plain cystography** (350 mL).
- Grading: AAST (American Association for the Surgery of Trauma) bladder **grade V** = laceration into the bladder neck or ureteric orifice; Goldman type IV = bladder rupture extending into the proximal urethra.

Acute management

- Resuscitate; if unstable, damage control (drainage), repair later.
- **Bladder neck injury needs immediate repair** (AUA (American Urological Association) and EAU (European Association of Urology)), unlike isolated posterior urethral distraction (suprapubic tube, delayed repair).

```
Lower midline incision → anterior cystotomy
→ identify ureteric orifices (stent if near)
→ minimal debridement; re-approximate neck to prostatic urethra,
  layered absorbable sutures over a 22 Fr catheter
→ rectal or vaginal tear: separate layers, no overlapping suture
  lines, interpose omentum (colostomy for rectum)
→ remove bone spicules; drain ± suprapubic tube
```

- Catheter 2–3 weeks; cystogram before removal.

Delayed management (3–6 months)

Problem	Management
Associated urethral distraction	Cystogram + RUG + endoscopy; **anastomotic urethroplasty first**, not combined with bladder neck surgery
Bladder neck contracture	Endoscopic incision or resection; open Y-V plasty if refractory
Incontinence (both sphincters damaged)	Urodynamics; **AUS (artificial urinary sphincter)**; bladder neck reconstruction (Young–Dees–Leadbetter) or fascial sling; last resort **bladder neck closure + Mitrofanoff** ± augmentation
Fistula	Layered repair with tissue interposition

- Bladder neck appearance on imaging **does not predict** continence; restore continuity first, then treat incontinence.
- Continence is poor whenever the bladder neck is involved (paediatric series: incontinence up to 20% overall).

Diagram to draw: pelvic fracture with bladder neck tear and repair over a catheter.

Key points to remember:

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

Mini Perc.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 3, Q3c

See 2022-Summer Paper 1, Q1 (full answer).

Key points to remember (SAQ angle):

- Definition by sheath size: mini-PCNL (percutaneous nephrolithotomy) 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr (standard 24–34 Fr).
- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.

Boari flap

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 3, Q3d

Definition

- A **tubularised flap of anterior bladder wall**, based posterolaterally on the side of the lesion, swung up to bridge a distal or mid-ureteric gap. Described in dogs 1894, humans 1947; laparoscopic 2001, robotic 2008.

Reconstructive ladder (distal ureter)

Option	Gap bridged
Ureteroneocystostomy	Distal 4–5 cm
+ Psoas hitch	Adds up to 5 cm (total 6–10 cm)
Boari flap (+ hitch)	**10–15 cm**; spiral flap may reach pelvis (right)
Downward nephropexy	Adds up to 5–8 cm

- Beyond: TUU (transureteroureterostomy), ileal ureter, autotransplantation.

Indications

- Long distal or mid-ureteric loss: gynaecological injury (commonest), stricture, failed reimplantation, distal TCC (transitional cell carcinoma) ureter, tuberculosis.

Contraindications

- **Small, contracted or poorly compliant bladder** (radiation, tuberculosis); uncorrected outlet obstruction or neurogenic bladder. Not usually for acute unstable trauma.
- Preoperative: urodynamics or cystogram for capacity; antegrade and retrograde studies for length.

Technique

1. Mobilise bladder; divide contralateral superior vesical pedicle if needed.
2. Mark flap on the **non-distended** bladder: **base ≥4 cm, apex ≥3 cm, length:base ≤3:1**; length = gap + 3 cm if tunnelled.
3. Raise flap; **psoas hitch** its base (sutures superficial, parallel to fibres: spare genitofemoral and femoral nerves).
4. Anastomose spatulated ureter: **non-refluxing** 3 cm submucosal tunnel, or **refluxing** direct (apex about 3 times ureteric diameter).
5. Stent (double-J); tubularise in two layers 2-0 absorbable; drain, catheter 7–14 days.

- Laparoscopic and robotic: transperitoneal, refluxing anastomosis, results excellent.

Complications

- **Recurrent stricture** (commonest; ischaemia or tension), reduced capacity with frequency and urgency, leak, nerve palsy, iliac vessel injury, pseudodiverticulum.
- eGFR (estimated glomerular filtration rate) fell in 44% within 2 years after reimplantation: counsel.
- **Alternative:** bladder advancement flap, similar reach and simpler.

Diagram to draw: bladder outline with oblique flap (4 cm base, 3 cm apex), hitched and tubularised to ureter.

Key points to remember:

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

Newer methods (techniques) of stone analysis

Asked as:

- Newer methods of stone analysis — *Practice Set 6 Paper III Q3c (SAQ)*
- Newer techniques of stone analysis — *Practice Set 7 Paper III Q3d (SAQ)*
- Newer methods of stone analysis — *Practice Set 6 Paper III Q3c (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2024-Winter Paper 4, Q3a. Earlier version: **See Practice Set 6 Paper 4, Q3a (full answer).**)

Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former (analyse every first stone); **repeat if stones recur on treatment** (evolving metabolic condition) — on recurrence despite treatment or early recurrence after clearance.
- Composition points to cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.
- Composition can change over time (e.g. calcium oxalate converting to brushite), so re-analyse recurrent stones.

Why wet chemical analysis is obsolete (limits)

- Detects ions, not **crystal phases**: cannot separate COM (calcium oxalate monohydrate) from COD (dihydrate), or apatite from brushite. High error rates; misses rare stones; needs a large sample.

Newer methods

Method	Principle	Comment
FTIR (Fourier-transform infrared) spectroscopy	Each compound absorbs a characteristic infrared spectrum	Commonest reference method; tiny (small) sample; detects proteins, drugs
XRD (X-ray diffraction)	Crystal lattice diffracts X-rays in a unique pattern (gives a unique diffraction pattern)	Gold standard for crystals; misses amorphous material
Polarising microscopy	Crystal birefringence	Cheap; needs experience; observer-dependent
Morpho-constitutional analysis (Daudon)	Stereomicroscopy of surface and section, plus FTIR	Subtypes point to cause, e.g. Ic suggests primary hyperoxaluria
SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)	Surface (structure) and elements	Research
Micro-CT (micro-computed tomography)	3-D structure, non-destructive	Research
Raman spectroscopy	Scattered light (spectrum)	Research

In vivo prediction

- **DECT (dual-energy computed tomography):** two (tube) energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid** stones with near-perfect accuracy.

- **Hounsfield units:** SWL (shock wave lithotripsy) works best below about 900–1000 HU; >1000 HU predicts SWL failure; cystine and brushite are dense.
- **Endoscopic recognition** of morphology (lost when stones are dusted). Dusting loses the specimen: collect fragments.

Significance in management

- **Uric acid:** oral dissolution by alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite, cystine:** hard, SWL-resistant; prefer endoscopy. Brushite can convert from calcium oxalate after SWL or infection.
- **Brushite:** avoid over-alkalinisation.
- **2,8-DHA (dihydroxyadenine):** APRT (adenine phosphoribosyltransferase) deficiency; allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

Diagram to draw: FTIR spectrum outline; COM (dumbbell, dumb-bell) vs COD (envelope) crystals.

Key points to remember (including SAQ angle):

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemical analysis (wet chemistry) is obsolete.
- DECT (dual-energy computed tomography) separates/identifies uric acid stones in vivo; it estimates composition in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.
- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

Rapid Revision Sheet – Section 11 · Stones, Obstruction & Trauma

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

Laparoscopic pyeloplasty. (LAQ)

- Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
- Transpose over a crossing vessel; never divide it.
- Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
- Robotic equals laparoscopic in outcome, with easier suturing.

Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis. (LAQ)

- Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
- Medially deviated ureters; PET-CT best shows activity; biopsy deep.
- Drain first, then prednisone (about 80% respond; Vaglio 2011 beat tamoxifen).
- About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury (LAQ)

- Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
- Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
- AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
- Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
- Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Describe early and late management of PFUD injuries (LAQ)

- Blood at meatus → RUG; SPC is the standard initial step.
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Delayed perineal anastomotic urethroplasty after 3 months; bulbar mobilisation → crural separation → pubectomy → rerouting.
- ED mostly from the injury; incontinence <4%.

Accessories for RIRS (SAQ)

- Hydrophilic wire passes; PTFE wire stays.
- 12/14 Fr sheath lowers pressure; severe injury about 1 in 8.
- Nitinol tipless basket for calyces; ≤10 W in ureter.

Management of lower ureteric stone of size less than 1 cm in pregnancy? (SAQ)

- US first; MRI or low-dose CT second.
- Observation first; avoid NSAIDs.

- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change every 4–6 weeks.

Type 2 Renal tubular acidosis? (SAQ)

- Bicarbonate wasting; HCO_3^- 14–18 mEq/L; urine pH <5.5 at steady state.
- Usually Fanconi; ifosfamide, cisplatin.
- Fractional HCO_3^- excretion >15%; stones uncommon.

Renal Autotransplantation (SAQ)

- For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
- Unit loss 5–10%; success about 90%.

Indication, procedure and complications of Uretero-Calycostomy? (SAQ)

- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection.
- Stricture is the main failure; success about 70% in redo series.

Post traumatic bladder neck injury: acute and delayed management? (SAQ)

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

Mini Perc. (SAQ)

- Definition by sheath size: mini-PCNL (percutaneous nephrolithotomy) 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr (standard 24–34 Fr).
- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.

Boari flap (SAQ)

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

Newer methods (techniques) of stone analysis (SAQ)

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemical analysis (wet chemistry) is obsolete.
- DECT (dual-energy computed tomography) separates/identifies uric acid stones in vivo; it estimates composition in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.
- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

Immediate post TURBT intravesical Therapy for TCC

Type: SAQ (10 marks)

Source: 2019-Summer Paper 3, Q3d

Introduction

- After TURBT (transurethral resection of bladder tumour), NMIBC (non-muscle-invasive bladder cancer) recurs in 40–60%. Many early recurrences come from **floating cells implanting** on the raw bed.
- A single chemotherapy instillation (SI) kills them before implantation. TCC (transitional cell carcinoma) is now called urothelial carcinoma.

Rationale and timing

- Implantation occurs within hours, so instil **as early as possible, ideally within 2–6 h** (theatre or recovery). Guidelines allow up to 24 h, but MMC given at 24 h lost its benefit.

Agents

Agent	Dose	Note
MMC (mitomycin C)	40 mg in 20–40 mL	Alkylating; skin rash
Gemcitabine	2 g in 100 mL, 1 h	Standard in USA; mild toxicity
Epirubicin	50 mg	Chemical cystitis

- **Also know:** pirarubicin works; thiotepa does not.

Technique

- Exclude perforation; instil via catheter; retain 1–2 h; drain.

Evidence

- **Sylvester 2016 individual-patient meta-analysis:** about 35% relative reduction in recurrence; **no benefit** with >1 recurrence/year or EORTC (European Organisation for Research and Treatment of Cancer) recurrence score ≥5.
- **SWOG (Southwest Oncology Group) S0337:** gemcitabine versus saline in suspected low-grade NMIBC: 4-year recurrence 35% vs 47%; no excess toxicity.
- No effect on progression.

Who benefits

- **Low risk** (primary solitary low-grade Ta ≤3 cm): SI alone; greatest benefit.
- Intermediate risk: SI then adjuvant course. High risk: BCG (bacille Calmette-Guérin) follows.

Contraindications

- **Suspected perforation**, extensive resection, bleeding needing irrigation.
- **Never BCG immediately**: intravasation of live bacteria causes sepsis and death.

Complications

- Irritative symptoms (commonest), chemical cystitis, skin desquamation (MMC), contracted bladder; serious harm mainly after perforation.

- **Also know:** continuous saline irrigation may also reduce recurrence.

Diagram to draw: floating cells implanting on resection bed → killed by instillation.

Key points to remember:

- Instil within 6 h (at most 24 h); never BCG.
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Withhold if perforation; best for low-risk solitary tumours.

Management of Haemorrhagic cystitis.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 3, Q3b

Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after excluding other sources (cystoscopy, cytology, upper tract imaging).
- **Exclude bladder cancer first:** cyclophosphamide and radiation are also risk factors for it.

Causes and mechanism

Cause	Mechanism
Cyclophosphamide, ifosfamide	Liver makes **acrolein**, excreted in urine, which sloughs urothelium; onset within 48 hours
Pelvic radiation (5%, 6 months to 10 years later)	**Obliterative endarteritis** → ischaemia, fibrosis, fragile neovessels
Viral	**BK polyomavirus**, adenovirus 11 and 35; after bone marrow or renal transplant
Others	Bacterial infection; busulfan, thiotepa; penicillins; amyloidosis

Prevention

- **Mesna** binds and inactivates acrolein; hyperhydration with forced diuresis, CBI (continuous bladder irrigation).

Management (escalate by severity; stability decides the branch)

```
Resuscitate, transfuse; correct coagulopathy; treat infection
→ large-bore catheter, CBI
→ cystoscopy under anaesthesia: clot evacuation, fulguration, biopsy
→ intravesical alum
→ bleeding after 48 h → hyperbaric oxygen (+ nephrostomies if bleeding continues)
→ formalin 1% → 4% (cystogram first)
→ unstable: internal iliac embolisation
→ cystectomy + conduit
```

Intravesical agent	Mechanism and method	Caution
Alum 1% (first line)	Protein precipitation, vasoconstriction; 200–300 mL/h; no anaesthesia	Aluminium toxicity in renal failure
Aminocaproic acid	Antifibrinolytic CBI, 200 mg/L	Only in a clot-free bladder; thromboembolism
Prostaglandins (carboprost), silver nitrate, hyaluronate	Vasoconstriction, coagulation, GAG (glycosaminoglycan) repair	Silver nitrate: cystogram
Formalin 1–4%	Protein fixation; **80–90% success**	Anaesthesia; **cystogram to exclude reflux**; fibrosis, small bladder, ureteric stricture

- **HBOT (hyperbaric oxygen therapy):** 100% oxygen at 2–3 atmospheres, 30–40 sessions; neovascularisation; 80–90% response, but many relapse. **RICH-ART** trial: better urinary symptom score in radiation cystitis.

- **Angioembolisation:** anterior division of internal iliac; spare the posterior division (gluteal pain).
- **Nephrostomies:** divert urokinase-rich urine.
- **Cystectomy with diversion:** last resort; remove the bladder (pyocystis risk).
- **Viral:** supportive; cidofovir (intravenous or intravesical).

Also know: tranexamic acid and laser ablation are unproven; oral pentosan polysulfate in children.

Diagram to draw: stepwise escalation ladder.

> **Key points to remember:**
> - Acrolein (prevent with mesna); radiation endarteritis; BK virus.
> - Alum first; formalin only after cystogram.
> - Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

Methods of preoperative bowel preparation.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 3, Q3d

Rationale

- Bowel used in urinary diversion is opened into a sterile urinary tract. Faecal spillage causes wound infection, abscess, leak and sepsis.
- Flora: aerobes (*E. coli*, *Enterococcus faecalis*); anaerobes (*Bacteroides*, *Clostridium*).

Segment	Organisms per gram
Jejunum	$10–10^5$
Distal ileum	$10^5–10^7$
Ascending colon	$10^6–10^8$
Descending colon	$10^{10}–10^{12}$

Methods

1. Dietary: clear liquids the day before; clear drinks until 2 hours before surgery.

2. Mechanical bowel preparation (MBP) reduces faecal **amount**, so total bacteria, not concentration.

- **PEG (polyethylene glycol)–electrolyte** lavage: non-absorbable; preferred in elderly, renal or heart failure, cirrhosis.
- Sodium phosphate, magnesium citrate; enemas.
- Historical: 3-day purgation, whole-gut irrigation by nasogastric tube (9–12 L saline), mannitol (feeds bacteria).
- Harms: dehydration, electrolyte loss; liquid stool may spill more.

3. Oral antibiotic preparation reduces bacterial **concentration**.

- **Neomycin + erythromycin base** (Nichols–Condon, 1977: septic complications 43% with MBP alone vs 9% with added antibiotics); neomycin + metronidazole; kanamycin.
- Harms: diarrhoea, *C. difficile* colitis, candidiasis; useless in obstruction.

4. Intravenous prophylaxis (most important): anaerobic plus aerobic cover (e.g. cephalosporin + metronidazole) within 1–2 hours before incision; it must be in tissue before contamination.

5. ERAS (enhanced recovery after surgery): no routine MBP; prehabilitation, multimodal opioid-sparing analgesia, chewing gum, **alvimopan** (peripheral μ-opioid antagonist, 12 mg before surgery), early feeding and mobilisation.

Evidence

- Colorectal meta-analyses and Cochrane: MBP does not reduce leak or abscess; may raise SSI (surgical site infection).
- Oral antibiotics added to MBP reduce SSI and leaks in large colectomy series.
- Small cystectomy trials: no difference without preparation.

Who still needs full preparation

- Salvage cystectomy after radiation, or cystectomy for benign disease (radiation cystitis, fistula, neurogenic bladder); immunosuppression.

Also know: wound protectors reduce SSI; enema before transrectal biopsy.

Diagram to draw: bacterial load along the gut.

Key points to remember:

- Mechanical lowers amount; antibiotics lower concentration.
- Nichols–Condon: neomycin + erythromycin.
- Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder.

Type: SAQ (10 marks)

Source: 2023-Winter Paper 3, Q3e

Definition

- **CIS (carcinoma in situ, Tis):** flat, non-invasive, **high-grade** urothelial carcinoma. It is not "premalignant".
- Histology: severe atypia, loss of polarity, non-cohesive pleomorphic cells, **loss of umbrella cells** (dysplasia keeps them).
- **Types:** primary (isolated, about 10%), secondary or concurrent (with papillary tumour, about 90%).

Pathogenesis

- **Flat pathway:** dysplasia → CIS → high-grade invasive cancer.
- Driven by **TP53, RB and PTEN** loss; **no FGFR3** mutation (unlike low-grade papillary tumours).
- **Field change:** multifocal; spreads to prostatic urethra, distal ureters and upper tract.

Clinical importance

- Untreated, **40–83% progress** to muscle invasion. Primary CIS progresses in >50%.
- "CIS alone" hides invasion at cystectomy in up to 20%.
- **Second strongest prognostic factor after grade**; any CIS makes NMIBC (non-muscle-invasive bladder cancer) **high risk**.
- About 80% have **irritative symptoms**; it mimics cystitis.

Diagnosis

- **Urine cytology:** most useful in CIS, because the cells are poorly cohesive. Positive cytology with normal cystoscopy needs mapping.
- Cystoscopy: red velvety patch or normal-looking mucosa; **blue-light cystoscopy** (hexaminolevulinate) detects about one-third more CIS; NBI (narrow-band imaging).
- **Biopsies:** cold-cup from suspicious and random sites; prostatic urethra; upper tract imaging.

Management

- **BCG (bacillus Calmette–Guérin) is first line:**
  - Induction: 6 weekly instillations.
  - **Maintenance** (SWOG [Southwest Oncology Group] 2000: 3 weekly doses at 3 and 6 months, then 6-monthly for 3 years): median recurrence-free survival 76.8 vs 35.7 months.
  - Complete response up to 84%; durable in about half.
- **Persistent CIS at 3 months:** second BCG course; 30–50% more respond.
- **BCG-unresponsive** (persistent or recurrent CIS within 12 months of adequate BCG): **radical cystectomy**.
- Unfit or unwilling: **pembrolizumab** (KEYNOTE-057: complete response 41%), nadofaragene firadenovec, gemcitabine–docetaxel.
- Prostatic urethral CIS: TURP plus BCG; ductal involvement needs cystoprostatectomy.

Follow-up

- Cystoscopy and cytology 3-monthly for 2 years, then 6-monthly; upper-tract imaging.

Diagram to draw: flat versus papillary pathways of bladder cancer.

> **Key points to remember:**
> - Flat, high-grade; TP53/RB; no FGFR3.
> - Untreated 40–83% progress; always high risk.
> - BCG with maintenance; cystectomy if BCG-unresponsive.

Rapid Revision Sheet – Section 12 · Bladder Cancer & Diversion

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

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- Instil within 6 h (at most 24 h); never BCG.
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Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder. (SAQ)

- Flat, high-grade; TP53/RB; no FGFR3.
- Untreated 40–83% progress; always high risk.
- BCG with maintenance; cystectomy if BCG-unresponsive.

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management

Asked as:

- Discuss minimally invasive and endoscopic management of upper urinary tract malignancy. — *Summer 2016 Paper III Q1 (LAQ)*
- Discuss the diagnosis, staging and management of renal upper tract Urothelial tumor. — *Winter 2024 Paper III Q1 (LAQ)*
- Describe in detail TCC of ureter. — *Practice Set 5 Paper III Q2 (LAQ)*

Type: LAQ (25 marks)

(also answered in 2023-Winter Paper 2, Q1, with emphasis on aetiology; also answered for the whole upper tract in 2023-Winter Paper 2, Q1 and 2024-Winter Paper 3, Q1)

TCC (transitional cell carcinoma) of the ureter is taken as ureteric UTUC (upper tract urothelial carcinoma).

Introduction and epidemiology

- **UTUC (upper tract urothelial carcinoma)** is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Incidence is rising, **driven by ureteric tumours**.
- Peak age 70–90; male:female 2:1.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall (the upper tract wall is thin). Concurrent bladder cancer in 17%.
- Bladder recurrence follows in 22–47%, mostly from downstream seeding.
- **Site (ureter): distal ureter 70%, mid 25%, proximal 5%**, explained by downstream implantation.
- Standard is **RNU (radical nephroureterectomy) with bladder cuff**; kidney-sparing versus radical surgery is chosen by **risk**.
- URS (ureteroscopy) biopsy is small and shallow, so it **cannot stage**. Stage is inferred from grade, endoscopic appearance, imaging and cytology together.

Aetiology and pathogenesis

Factor	Key point
Tobacco	Commonest; risk 2.5–7-fold (aromatic amines)
Aristolochic acid	Balkan and Chinese herb nephropathy; A:T → T:A transversions in TP53; often bilateral
Lynch syndrome	Mismatch repair genes (MSH2 highest risk); screen if young or family history
Occupational	Aniline dyes, benzidine, β-naphthylamine
Phenacetin	Papillary necrosis (history in 11% of ureteric tumours)

- **Also know:** arsenic, cyclophosphamide, chronic stones (squamous carcinoma).
- **Intraluminal seeding** explains distal predominance and bladder recurrence (22–47%); **field change** explains multifocality.

Pathology and spread

- 85% papillary, 15% sessile; WHO (World Health Organization) 2004/2016 low or high grade. Ureteric tumours are low grade and low stage in 55–75%.
- **Why ureteric tumours do worse:** the renal parenchyma is a barrier for pelvic tumours, but periureteric fat holds a rich lymphovascular plexus. At equal stage, ureteric tumours have more local and distant failure.
- Spread: direct, lymphatic (**mid and distal ureter → pelvic nodes**; upper ureter → hilar and retroperitoneal), blood (lung, liver, bone).

Clinical features

- **Haematuria** (56–98%), **flank pain** (20–32%, dull from hydronephrosis/obstruction or colic from clots); lumbar mass about 10%; incidental about 15%.
- Weight loss, bone pain or a mass suggest advanced disease.

Diagnosis (investigations)

1. Cystoscopy: always, to exclude coexisting bladder cancer (concurrent bladder cancer 17%). Positive cytology with a normal bladder and urethra means the upper tracts must be examined.

2. Imaging

Test	Role and key points
CTU (computed tomography urography)	First line; best test; highest accuracy (sensitivity ~92%, specificity ~95%). Radiolucent **filling defect**, hydronephrosis, ureteric wall thickening, non-visualisation. Tumour ~46 HU (Hounsfield units) versus uric acid stone ~100 HU. Misses flat lesions (CIS, carcinoma in situ)
MRU (magnetic resonance urography)	If contrast or radiation is contraindicated; sensitivity ~75% for tumours <2 cm
Retrograde pyelography	Poorly filled calyces; accuracy ~75%; done with ureteroscopy. **Goblet (champagne-glass) sign**, the ureter dilated just below the tumour; **Bergman's sign**, the catheter coiling below it
FDG-PET/CT (fluorodeoxyglucose positron emission tomography)	Nodes and metastases when contrast is contraindicated
Renogram	Split function; contralateral kidney must be assessed

- **Hydronephrosis** predicts invasive disease (invasive ureteric tumour in up to 80%).

3. Urine cytology

- Sensitivity ~65%, grade-dependent (grade 1 ~20%, grade 3 ~75%). **Selective** ureteric washings (selective ureteric cytology) are more accurate than voided urine. Take it before contrast. FISH (fluorescence in situ hybridisation) is optional for surveillance.

4. Flexible URS (ureteroscopy) with biopsy: the key test

- Adds accuracy (75–85% → ~90%). Biopsy gives **grade in ~90%** (grade correct in ~90%), but **stage cannot be assessed** (shallow samples; a Ta biopsy is often T3 at surgery).
- Indications: diagnostic doubt, planned kidney-sparing treatment, solitary kidney.
- Diagnostic URS before RNU (radical nephroureterectomy / nephroureterectomy) raises bladder recurrence, without changing survival.

Staging (TNM [tumour, node, metastasis] 8th edition, 2017)

T	Definition	Ureter
Ta / Tis	Non-invasive papillary / carcinoma in situ	Papillary non-invasive / carcinoma in situ
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Muscularis	Muscularis
T3	Pelvis: **peripelvic fat or renal parenchyma**; ureter: periureteric fat	Periureteric fat
T4	Adjacent organs, or **through the kidney into perinephric fat**	Adjacent organs

- **N1** single node ≤2 cm; **N2** node >2 cm or multiple. **M1** distant metastasis.
- **AJCC (American Joint Committee on Cancer) grouping:** stage I = T1, II = T2, III = T3; **any N+, T4 or M1 = stage IV**.
- **Grade:** WHO (World Health Organization) 2004/2016: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade.
- Regional nodes: pelvis and upper ureter → hilar and retroperitoneal; lower ureter → pelvic.

Risk stratification

EAU (European Association of Urology)

Low risk (all present / all needed)	High risk (any one)
Unifocal, under 2 cm (<2 cm)	Multifocal or 2 cm or more (another source: >2 cm; ≥2 cm)
Low-grade cytology and biopsy	High-grade cytology or biopsy
No invasion on CTU (computed tomography urography)	Hydronephrosis; invasion on CT / imaging
—	Prior (previous) radical cystectomy; variant histology

- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023** splits each group into favourable and unfavourable (obstruction, multifocality, lower-tract involvement). Ablation is **preferred** in low-risk favourable, **offered** in low-risk unfavourable, **rare** in high-risk favourable, **palliative** in high-risk unfavourable. Endoscopic therapy suits tumours **under 1.5 cm**.
- **Prognostic factors:** stage (strongest), grade, LVI (lymphovascular invasion), nodes, sessile architecture, necrosis >10%, ureteric site, surgical delay >12 weeks.

Management

A. Kidney-sparing (minimally invasive and endoscopic) treatment

Indications

- **Elective**: low-risk disease, whatever the other kidney.
- **Imperative**: solitary kidney, bilateral tumours (bilateral disease), CKD (chronic kidney disease), unfit for RNU (unfit patients), after shared decision.
- The patient must accept frequent recurrence and **lifelong endoscopic surveillance**.

Ureteroscopic (retrograde) management

- Closed system, lowest morbidity; for small ureteric and renal tumours.
- **Steps**: cystoscopy → selective cytology and saline barbotage → no-touch semirigid then flexible URS → map all lesions → biopsy (3 Fr forceps or basket) into fixative → ablate the

base with laser (**ureteroscopic laser ablation**, holmium or thulium, after biopsy) → stent optional.

Laser	Penetration	Note
Ho:YAG (holmium:yttrium-aluminium-garnet)	Under 0.5 mm	Safe in thin ureter
Nd:YAG (neodymium:YAG)	Up to 5–6 mm	Coagulative necrosis; transmural injury risk
Thulium	Under 0.2 mm	Strong haemostasis

- Avoid circumferential fulguration / ablation (stricture); stage large tumours.
- Results: upper-tract recurrence 23–90%, bladder recurrence 15–45%, progression 0–20%. **Grade is the strongest predictor.**

Percutaneous (antegrade) management

- Percutaneous resection for large pelvic tumours, lower-pole or inaccessible calyces and urinary diversions.
- **Steps**: retrograde catheter → puncture a calyx away from the tumour (upper or middle calyx for pelvic tumours) → dilate to 30 Fr → cold-cup, bipolar resection or laser; base sent separately → nephrostomy left → **second-look nephroscopy**.
- Removes more tumour and can **stage**; the tract allows topical therapy. But it is an **open system**: tract seeding about 0.3% (tract seeding risk), bleeding.

Segmental ureterectomy

- **Distal ureterectomy with bladder cuff and reimplantation** (psoas hitch or Boari flap) for distal tumours, including high-grade ones when the kidney must be saved. It gives a staging specimen; add pelvic lymphadenectomy (pelvic node dissection) if high grade.
- First examine the whole tract above for multifocality. Lower-ureteric tumours with unfavourable features need resection, not endoscopy.

Site	Operation
Distal ureter	Distal ureterectomy with bladder cuff and reimplantation; psoas hitch (~5 cm extra) or Boari flap (10–15 cm)
Mid or proximal	Mid-ureteric resection: excision with 1–2 cm margins (frozen section margins) and ureteroureterostomy (rare)
Whole ureter	Ileal ureter or autotransplantation

- Refluxing reimplant eases surveillance; non-refluxing limits seeding (debated).
- Laparoscopic or robotic routes are feasible; outcomes equal RNU in selected patients.
- Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); UGN-101 mitomycin gel for low grade. Lifelong endoscopic surveillance.

B. RNU (radical nephroureterectomy) with bladder cuff: standard for high risk

- **Principles**: kidney outside Gerota's fascia; clip the ureter early; never open the collecting system (avoid opening the tract); remove the **whole intramural ureter and cuff** (stump recurrence 30–75% if left); bag extraction intact; watertight bladder closure; template lymphadenectomy for high risk.
- **Laparoscopic and robotic RNU steps**: modified flank, 3–4 ports → reflect colon → clip ureter → artery then vein → kidney outside Gerota's fascia (adrenal spared) → ureter traced under the superior vesical artery → cuff excised (extravesical or open) → two-layer closure.
- Abandoned (higher recurrence): transurethral "pluck", stripping, laparoscopic staple ("pluck" and stapling techniques are abandoned).

- Open, laparoscopic and robotic routes are equivalent: laparoscopic oncologically **equivalent to open** (AUA/SUO), with less blood loss and shorter stay; less favoured in bulky T3/T4. Robotic yields more nodes.
- **Template lymphadenectomy** for high risk (hilar plus paracaval or para-aortic); pelvic nodes for lower ureteric tumours.

C. Adjuvant topical therapy

- **Single postoperative intravesical dose / chemotherapy** (AUA/SUO strong): **ODMIT-C (2011)** MMC (mitomycin C) 40 mg cut bladder recurrence 28% → 16% at 1 year; **Ito (2013)** pirarubicin, 42% → 17% at 2 years.
- **Upper-tract instillation** via nephrostomy (low pressure, 25 cm H2O), ureteric catheter or reflux up a double-J. **BCG (bacillus Calmette–Guérin)** (or MMC): 6-week course for upper-tract CIS (carcinoma in situ), and after complete ablation of high-risk favourable Ta/T1. Risks: sepsis, stricture.
- **UGN-101 (Jelmyto)**: mitomycin reverse-thermal gel, **FDA (Food and Drug Administration) 2020** for low-grade UTUC not manageable endoscopically. **OLYMPUS**: complete response 59%, 56% disease-free at 12 months; ureteric stricture about 44%.

D. Perioperative (systemic) chemotherapy

- **POUT (2020)**: adjuvant platinum after RNU for pT2–T4 or N+ improved disease-free survival: 3-year disease-free survival 71% vs 46%.
- Neoadjuvant platinum (chemotherapy) is preferred if renal function is borderline / when post-RNU renal function would exclude cisplatin (RNU removes cisplatin eligibility).

E. Metastatic

- Enfortumab vedotin + pembrolizumab (EV-302) first line; gemcitabine–platinum with avelumab maintenance; erdafitinib for FGFR3 alterations.

Prognosis

- 5-year disease-specific survival: localised ~89%, regional ~63%, distant ~17%. Stage is the strongest factor.

Surveillance and follow-up (AUA/SUO)

- **Kidney-sparing**: cystoscopy and URS within 3 months, URS again at 6 and 12 months; imaging 3–6-monthly (high risk) or 6–9-monthly (low risk), then yearly to 5 years (URS and CTU at closer intervals).
- **After RNU**: cystoscopy and cytology from 3 months; CT / CTU yearly (more often if pT2 or more).

Recent advances

- Lynch syndrome screening; thulium fibre laser; single-use digital ureteroscopes.

Diagram to draw: ureteroscopic laser ablation of a pelvic tumour beside RNU with bladder cuff; TNM levels of a renal pelvic tumour (T3 parenchyma versus T4 perinephric fat), beside the

diagnostic flowchart CTU → cystoscopy → URS biopsy → risk group; distal ureterectomy with psoas-hitch reimplantation, beside the goblet sign.

Key points to remember:

- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU + cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C); adjuvant platinum (POUT).

Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy.

Type: LAQ (25 marks)

Source: 2019-Summer Paper 3, Q1

Definition and biology

- **SRM (small renal mass):** an enhancing solid or complex cystic renal mass **≤4 cm (cT1a)**, usually found incidentally.
- **Benign share falls with size:** 20–30% of masses ≤4 cm and about 40% of masses ≤2 cm are benign. At ≥7 cm few are benign and about half are high grade.
- Male sex triples the risk of malignancy. Under 2% of small cancers metastasise.

Differential diagnosis

Group	Entities	Pointers
RCC (renal cell carcinoma)	**Clear cell** (commonest), papillary (10–15%), chromophobe (3–5%), cystic RCC, rare molecular subtypes	Clear cell enhances strongly; papillary enhances weakly
Other malignant	Urothelial carcinoma, lymphoma, metastasis, sarcoma, epithelioid AML (angiomyolipoma)	Infiltrative growth
Benign solid	Oncocytoma, fat-poor AML, papillary adenoma, metanephric adenoma	See below
Cystic	Haemorrhagic or complex cyst, cystic nephroma	Bosniak category
Inflammatory, vascular	Abscess, focal pyelonephritis, tuberculosis; aneurysm, AVM (arteriovenous malformation), infarct, pseudotumour	Clinical picture

- **Oncocytoma:** commonest benign enhancing SRM (up to half of benign SRMs). Central stellate scar and segmental enhancement inversion suggest it but are not specific. A core cannot separate it from chromophobe RCC, which has oncocytic areas. Sestamibi SPECT (single-photon emission computed tomography) is under evaluation.
- **Fat-poor AML:** no visible fat (fat below −20 HU [Hounsfield units] diagnoses classic AML); homogeneous, T2-dark, mimics RCC; biopsy HMB-45 and Melan-A positive.
- **Also know:** metanephric adenoma (hypovascular, polycythaemia, WT1 positive); papillary adenoma (≤15 mm).

Narrowing the differential

- **Multiphase CT (computed tomography):** enhancement over 15–20 HU means solid tissue. MRI (magnetic resonance imaging) helps below 2 cm.
- **RMB (renal mass biopsy)** if it will change management (AUA [American Urological Association]); seeding almost unreported; non-diagnostic about 14%.
- Then surveillance, ablation or **PN (partial nephrectomy)**, preferred for cT1a.

Haemostasis in LPN (laparoscopic partial nephrectomy)

Principle: a clear field for excision, then secure closure of the bed. Plan with R.E.N.A.L. score and 3D (three-dimensional) CT; intraoperative ultrasound.

1. Vascular control

Technique	Method	Comment
Main artery ± vein clamp	Bulldog clamps, or laparoscopic Satinsky en bloc	Artery-only equals artery–vein. Aim warm ischaemia ≤25 min (trifecta)
Selective or super-selective	Clamp the tumour's segmental branch; 3D CT, NIRF (near-infrared fluorescence)	Avoids global ischaemia; no functional gain in an RCT (randomised controlled trial)
Off-clamp ("zero ischaemia")	No clamping; small exophytic tumours	More blood loss; same long-term eGFR (estimated glomerular filtration rate)
Early unclamping	Release after the inner suture layer	Shorter ischaemia; possibly fewer pseudoaneurysms
Cold ischaemia	Ice slush, cold saline via ureteric catheter, cold arterial perfusion	Solitary kidney, CKD (chronic kidney disease), long repair; benefit unproven

- Function depends mainly on parenchyma preserved. Mannitol: no benefit in an RCT. **Also know:** parenchymal clamp for peripheral tumours.

2. Excision: score the capsule, cut with cold scissors, clip visible vessels. Enucleation along the pseudocapsule bleeds less.

3. Renorrhaphy (most reliable step)

- **Inner layer:** running 3-0 or 2-0 absorbable suture closes cut vessels and collecting system (opened in about 60%).
- **Outer layer, sliding-clip technique:** mattress sutures anchored by a locking clip; clips slid down to the capsule give tension without knots. Pledgets prevent capsular tears; oxidised cellulose bolster fills the defect.
- **Barbed suture:** holds tension without knots, so ischaemia is shorter.
- Single layer gave better function and fewer pseudoaneurysms than two layers.

4. Energy: argon beam or bipolar coagulation of the bed. Argon can destroy collecting-system sutures and adds gas, so vent the ports.

5. Haemostatic agents (adjuncts, no proven benefit in RCTs): gelatin–thrombin matrix; fibrin sealant; oxidised cellulose; PEG (polyethylene glycol) hydrogel; cyanoacrylate. Sutureless closure only for selected small tumours.

6. Final check: unclamp, inspect, **lower pneumoperitoneum** to unmask venous bleeding; drain.

Postoperative bleeding

- Early (hypotension, falling haemoglobin): resuscitate, CT, angioembolisation.
- **Delayed haematuria (about day 12):** pseudoaneurysm or arteriovenous fistula; selective angioembolisation preserves function. Endophytic and posterior tumours carry higher risk.

Diagram to draw: sliding-clip renorrhaphy over a bolster, with bulldog clamp on the artery.

Key points to remember:

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

Nephron-sparing surgery for renal masses: methods, recent advances and neoadjuvant therapy

Asked as:

- Describe various methods of Nephron sparing surgery in relation to Renal masses? — *Summer 2022 Paper III Q1 (LAQ)* (also answered in 2017-Summer Paper 1, Q1 and 2017-Winter Paper 4, Q3f)
- Recent advances in nephron sparing surgery. Role of neoadjuvant therapy. — *Practice Set 6 Paper III Q3f (SAQ)* (NSS also answered in 2022-Summer Paper 3, Q1)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Definition and rationale (background)

- NSS (nephron-sparing surgery)/PN (partial nephrectomy): complete removal of the tumour with maximal preservation of functioning parenchyma.
- RN (radical nephrectomy) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN (partial nephrectomy) gives equal cancer control with better function.
- Function after PN depends mainly on the **parenchyma preserved**, less on ischaemia time. **Concept:** preserved parenchymal volume, not warm ischaemia time, predicts function.

Indications

Category	Situations
Imperative (absolute)	Solitary kidney; bilateral tumours; RN would mean dialysis
Relative	Contralateral kidney threatened (stones, renal artery stenosis, diabetes, hypertension); hereditary RCC (renal cell carcinoma)
Elective	cT1a with a normal contralateral kidney: **standard of care**

- Now extended to cT1b–T2 when feasible (better function).
- **AUA (American Urological Association) 2021:** RN only if all three: high complexity; no CKD or proteinuria; normal contralateral kidney with eGFR (estimated glomerular filtration rate) likely >45 after RN.
- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous tumour thrombus; cT4; cN1.

Preoperative planning

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging); chest imaging; eGFR, proteinuria.
- **R.E.N.A.L. score** (Radius, Exophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines): 4–6 low, 7–9 moderate, 10–12 high. It predicts complications, ischaemia time and conversion. PADUA (Preoperative Aspects and Dimensions Used for an Anatomical) is an alternative.
- RMB (renal mass biopsy) when it changes management, and before (or at) ablation. 3D models shorten ischaemia (randomised trial).

Methods

Group	Methods
Surgical PN	Open, laparoscopic, robot-assisted (RAPN)
Excision technique	Standard (wedge) resection with a rim; enucleation; polar resection or heminephrectomy (≥30% of parenchyma)

Group	Methods
Thermal ablation	Cryoablation, RFA (radiofrequency ablation), MWA (microwave ablation)
Developmental	SBRT (stereotactic body radiotherapy), HIFU (high-intensity focused ultrasound), irreversible electroporation

1. Open PN: steps

1. Flank extraperitoneal or transperitoneal approach.
2. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
3. Defat kidney; **intraoperative ultrasound** maps borders and depth.
4. Clamp; **ice slush** if ischaemia is expected ≥25–30 minutes (solitary kidney, CKD).
5. Excise with a thin rim; a negative margin is the priority, its width is not.
6. **Two-layer renorrhaphy:** inner layer closes vessels and collecting system; outer capsular sutures over a haemostatic bolster.
7. Unclamp, check haemostasis, close Gerota fascia, drain.

- Suits complex tumours in a solitary kidney.

2. Laparoscopic and robotic PN

- First laparoscopic PN 1993 (Winfield). **Transperitoneal** for anterior tumours (more space); **retroperitoneal** for posterior tumours (direct hilum, no bowel mobilisation).
- **Robotic vs laparoscopic:** shorter warm ischaemia, fewer conversions. **Robotic vs open:** less blood loss, shorter stay, fewer complications; same margins and eGFR.
- Haemostasis: sliding-clip renorrhaphy, barbed sutures, sealants.
- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes.

3. Enucleation vs standard resection

	Standard PN	Enucleation
Plane	Through normal parenchyma (thin rim)	Along tumour pseudocapsule
Pros	Fewer positive margins	Less bleeding, maximal parenchyma, eases zero ischaemia
Use	Default	Hereditary RCC (e.g. VHL [von Hippel–Lindau]), multifocal, solitary kidney, severe CKD

4. Vascular control strategies

Strategy	Verdict
En bloc or artery-only clamp	Equivalent; traditional warm ischaemia <25–30 minutes
Early unclamping (after inner layer)	May reduce pseudoaneurysm
Selective or super-selective segmental clamping	Avoids global ischaemia; no functional gain in a randomised trial
Off-clamp (zero ischaemia)	Small exophytic tumours; more blood loss; no long-term eGFR benefit in randomised trials
Parenchymal compression	Peripheral tumours
Hypothermia, mannitol	No benefit with a normal contralateral kidney

5. Thermal ablation

- Mostly **percutaneous** (CT or ultrasound guided). AUA 2021: alternative for **cT1a <3 cm**, especially elderly, comorbid, solitary kidney or VHL.
- **Cryoablation:** double freeze–thaw; cell death at ≤−40 °C; iceball 5–10 mm beyond the tumour. **RFA/MWA:** heat causes coagulative necrosis.

- Success = loss of enhancement on CT/MRI. Local recurrence is higher than PN but equal after repeat ablation.

Recent advances

Area	Advance and evidence
Planning	R.E.N.A.L. (Radius, Exophytic, Nearness, Anterior/posterior, Location) and PADUA (Preoperative Aspects and Dimensions Used for an Anatomical) scores; **3D (three-dimensional) models**: randomised trial showed shorter operation and ischaemia, less blood loss
Platform	**RAPN (robot-assisted PN):** shorter warm ischaemia than laparoscopic; less blood loss than open. **Single-port robot:** shorter stay, less opioid
Vascular control	Off-clamp, selective or super-selective clamping (guided by **NIRF (near-infrared fluorescence)** with indocyanine green), early unclamping. **Randomised trials: no long-term eGFR (estimated glomerular filtration rate) gain**
Excision	**Enucleation** along the pseudocapsule: less blood loss; hereditary, multifocal, solitary kidney
Reconstruction	Sliding-clip and **barbed suture** renorrhaphy; single-layer gives better function, fewer pseudoaneurysms; sutureless with sealants in selected
Outcome reporting	**Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes

Role of neoadjuvant therapy

- **Aim:** downsize a tumour so that PN becomes possible, mainly an **imperative indication** (solitary kidney, bilateral tumours) where radical nephrectomy means dialysis.
- **TKIs (tyrosine kinase inhibitors):** pazopanib, sunitinib, axitinib. Tumour volume falls **10–30%**; resection becomes possible in **40–45%**, including PN candidates. Best in clear cell RCC (renal cell carcinoma).
- **Toxicity:** serious adverse events about 20%, mainly **wound healing**. Stop TKIs several half-lives before and after surgery.
- **Checkpoint inhibitors:** nivolumab is safe but shrinks little; trials ongoing.
- **Status:** not standard (trials only); **upfront surgery whenever feasible**. Adjuvant pembrolizumab (KEYNOTE-564) suits high-risk resected disease.

Complications of PN

- Urine leak (drain, stent); haemorrhage, **pseudoaneurysm** or AV (arteriovenous) fistula (angioembolisation); acute kidney injury.
- Positive margin 0–7%: surveillance. Local recurrence 1–2% (elective), 3–5% (imperative).
- Hyperfiltration proteinuria when >50% of a solitary kidney is lost: ACE (angiotensin-converting enzyme) inhibitors.

Evidence

- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN, ≤5 cm: PN better function, similar oncology; 10-year overall survival 81% RN vs 76% PN; trial flawed.
- **Also know:** hereditary RCC treated at 3 cm; surveillance for frail patients.

Diagram to draw: clamped renal artery, tumour excised with rim, two-layer renorrhaphy; RAPN with selective segmental clamp and renorrhaphy.

Key points to remember:

- Imperative, relative, elective; elective PN is standard for cT1a.
- Preserved parenchyma, not clamp time, governs function; trifecta warm ischaemia ≤25 min (minutes).
- Enucleation for hereditary/multifocal; ablation for cT1a <3 cm after biopsy.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
- 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

Management of AML.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 3, Q3d

Assumed meaning: AML = renal angiomyolipoma (not acute myeloid leukaemia).

Introduction

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; AML in 55–90%: younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34) → unchecked **mTOR (mammalian target of rapamycin)** → growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis) → treat as RCC (renal cell carcinoma).

Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock). CKD (chronic kidney disease) in TSC.

Diagnosis

- **CT (computed tomography): macroscopic fat (about −15 to −20 HU) on non-contrast images is diagnostic.**
- US (ultrasound) hyperechoic (so are some RCCs); MRI (magnetic resonance imaging): fat-suppression signal loss, India-ink artefact.
- Fat-poor lesion → MRI ± biopsy (HMB-45 (human melanoma black-45), Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

Management algorithm (Restrepo 2022)

Situation	Management
Acute haemorrhage	Resuscitate → **SAE (selective arterial embolisation)** → nephrectomy if SAE fails or unavailable
Symptomatic (pain, haematuria) or poor access to care	SAE, NSS (nephron-sparing surgery), ablation in selected cases
Asymptomatic sporadic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy**	Active treatment
None of these	**Surveillance**: US or CT at 6 and 12 months, then yearly
TSC, asymptomatic, **>3 cm**	**Everolimus**; MRI every 1–3 years

- The old **4 cm** cut-off overtreats; EAU (European Association of Urology) 2023 favours surveillance for most.

Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis) common; abscess, infarction.
- **Partial nephrectomy:** preferred surgery, essential in TSC (multifocal, recurrent).
- **Thermal ablation** (radiofrequency, cryo): limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%** (placebo); stomatitis, acne, hyperlipidaemia.

Diagram to draw: management algorithm by bleeding, symptoms, size and TSC.

Key points to remember:

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Discuss the management of Metastatic Renal Cell Carcinoma.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 3, Q3b (also answered in 2020-Summer Paper 2, Q1; immunotherapy in detail in 2024-Winter Paper 2, Q3e)

Assessment

- CT chest, abdomen and pelvis; bone scan or brain MRI if symptomatic. **Biopsy** for histology.
- **IMDC (International Metastatic RCC Database Consortium) risk factors:** Karnofsky <80%; haemoglobin < normal; calcium > ULN (upper limit of normal); neutrophils > ULN; platelets > ULN; <1 year from diagnosis to systemic therapy.
- Favourable 0, intermediate 1–2, poor ≥3; median survival 43.2, 22.5 and 7.8 months.

First-line systemic therapy (clear cell RCC; NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

IMDC risk	Preferred	If IO (immuno-oncology) not possible
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib, pazopanib; active surveillance for slow, low-volume disease
Intermediate/poor	Same three, **or nivolumab + ipilimumab**	Cabozantinib, sunitinib, pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss → HIF (hypoxia-inducible factor) → VEGF (vascular endothelial growth factor), the target of TKIs (tyrosine kinase inhibitors). Checkpoint inhibitors release T-cell brakes.
- **Immunotherapy** trials (CheckMate 214, KEYNOTE-426, CheckMate 9ER, CLEAR) and toxicity: see 2024-Winter Paper 2, Q3e.

Later lines: switch mechanism

- After IO: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor; LITESPARK-005).
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR).

Cytoreductive nephrectomy (CN)

- **CARMENA (2018):** sunitinib alone non-inferior to CN + sunitinib in intermediate/poor risk. **SURTIME:** deferred CN safe.
- Now: immediate CN only for fit, low-burden patients; **deferred CN** in responders; avoid in poor risk. Palliative nephrectomy or embolisation for bleeding.

Metastasectomy and radiotherapy

- Complete resection of solitary or few metastases: 5-year survival 35–50%.
- **SBRT (stereotactic body radiotherapy)** for oligometastases; radiosurgery for brain; palliative radiotherapy for bone pain.

Supportive and special groups

- Zoledronic acid or denosumab for bone metastases. Non-clear cell: trials, cabozantinib or sunitinib. Adjuvant pembrolizumab after complete metastasectomy (KEYNOTE-564).

Diagram to draw: flowchart: biopsy + IMDC → IO combination → deferred CN in responders → switch class on progression.

Key points to remember:

- IMDC 6 factors; poor = ≥3.
- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor.
- CARMENA and SURTIME: no routine upfront CN.
- Belzutifan after IO and TKI.

Rapid Revision Sheet – Section 13 · Kidney & UTUC

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management (LAQ)

- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU + cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C); adjuvant platinum (POUT).

Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy. (LAQ)

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

Nephron-sparing surgery for renal masses: methods, recent advances and neoadjuvant therapy (LAQ)

- Imperative, relative, elective; elective PN is standard for cT1a.
- Preserved parenchyma, not clamp time, governs function; trifecta warm ischaemia ≤25 min (minutes).
- Enucleation for hereditary/multifocal; ablation for cT1a <3 cm after biopsy.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
- 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

Management of AML. (SAQ)

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Discuss the management of Metastatic Renal Cell Carcinoma. (SAQ)

- IMDC 6 factors; poor = ≥3.
- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor.
- CARMENA and SURTIME: no routine upfront CN.
- Belzutifan after IO and TKI.

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

Discuss the role of tissue typing and cross match in live and cadaveric

Type: LAQ (25 marks)

Source: 2018-Summer Paper 3, Q2

The question is incomplete; it is taken to mean "... live and cadaveric (deceased-donor) renal transplantation".

Introduction

- Two immune barriers decide compatibility: **preformed antibodies** (anti-A/B isoagglutinins and anti-HLA (human leukocyte antigen) antibodies) and **T-cell responses** to mismatched HLA.
- **Tissue typing** measures how well donor and recipient HLA match. **Antibody screening and crossmatch** detect preformed antibodies before the kidney meets them. Together they prevent **hyperacute rejection**, now practically eliminated.

The HLA system

- HLA is the human MHC (major histocompatibility complex), a gene cluster on **chromosome 6 (short arm)**. Genes are inherited as two **haplotypes** (one from each parent) and expressed codominantly.

	Class I	Class II
Loci	HLA-A, -B, -C	HLA-DR, -DQ, -DP
Expression	All nucleated cells	Antigen-presenting cells, endothelium
Presents to	CD8 (cytotoxic) T cells	CD4 (helper) T cells

- **Sibling** chance: 25% two-haplotype identical, 50% one-haplotype, 25% none. A parent or child always shares one haplotype.

Tissue typing methods

Method	Principle	Note
Serological (microlymphocytotoxicity)	Lymphocytes + typed antisera + complement; dead cells take up dye	Low resolution; largely replaced
Molecular (DNA-based)	PCR (polymerase chain reaction) with SSP (sequence-specific primers), PCR-SSO (sequence-specific oligonucleotide probes), sequence-based typing, **NGS** (next-generation sequencing)	High (allele-level) resolution; now standard

- **Matching:** usually reported as mismatches at **A, B and DR** (0–6). **DR matters most**, then B, then A; DQ increasingly typed. Fewer mismatches mean better graft survival and less sensitisation for a future transplant (Collaborative Transplant Study, Opelz). The effect is larger in deceased-donor grafts.

ABO compatibility

- Same rules as transfusion: **O universal donor, AB universal recipient**.
- **A2 (non-A1) donors** can go to O or B recipients with low anti-A titre.

- **ABO-incompatible** living transplant is possible: lower titres (plasmapheresis or immunoadsorption, rituximab, IVIG (intravenous immunoglobulin)) to a centre target before surgery.

Sensitisation and antibody screening

- Anti-HLA antibodies come from **pregnancy, blood transfusion, previous transplant** (and infection).
- **PRA (panel reactive antibody):** recipient serum tested against a panel of lymphocytes; result as % of panel killed.
- **Solid-phase (Luminex) single-antigen beads:** identify each antibody and its strength (MFI, mean fluorescence intensity). Antibodies against a donor are **DSA (donor-specific antibodies)**.
- **cPRA (calculated PRA):** % of donors the patient would be incompatible with, calculated from listed "unacceptable antigens". It replaced cell-panel PRA for allocation; very high cPRA gets allocation priority.

Crossmatch

Type	Method	Meaning if positive
CDC (complement-dependent cytotoxicity)	Donor T and B lymphocytes + recipient serum + complement; cell death scored. **DTT** (dithiothreitol) removes IgM autoantibody; AHG (antihuman globulin) increases sensitivity	**Positive T-cell CDC with IgG DSA = absolute contraindication** (hyperacute rejection; Patel and Terasaki, 1969)
Flow cytometry crossmatch	Antibody binding on donor cells detected by fluorescent anti-IgG	More sensitive; CDC-negative/flow-positive = higher risk of AMR (antibody-mediated rejection); desensitise or exchange
Virtual crossmatch	Compare recipient's bead-defined antibodies with donor HLA type	Speeds deceased-donor allocation; confirm by cell-based crossmatch

- **B-cell positive** alone: class II antibody or non-specific binding; interpret with bead results.
- Crossmatch uses a **recent serum** plus historical peak sera.

Living versus deceased donor

	Living donor	Deceased (cadaveric) donor
Typing	ABO and HLA early in donor work-up; choose best of several donors	ABO and HLA at donor referral
Crossmatch	At work-up and a **final crossmatch shortly before surgery**	At the time of offer, often virtual first; done against the list rapidly (cold ischaemia)
Role of matching	Less critical: even unrelated living donors do better than well-matched deceased donors (Terasaki, 1995)	Allocation uses HLA match, sensitisation (cPRA), waiting time and donor quality (KDPI, Kidney Donor Profile Index)
If incompatible	**Paired (swap) exchange**, desensitisation, ABO-incompatible protocol	Patient waits; acceptable-mismatch programmes

- **India:** swap (paired) donation is legal under the Transplantation of Human Organs (THOA) amendment, 2011.

Managing the incompatible or sensitised recipient

- **Kidney paired donation**, including chains with altruistic donors.
- **Desensitisation:** plasmapheresis + low-dose IVIG, or high-dose IVIG, ± rituximab; **imlifidase** (cleaves IgG) for highly sensitised deceased-donor recipients (conditional EU

approval). Desensitised HLA-incompatible living transplant gave better survival than waiting on dialysis (Montgomery, 2011).

After transplant

- Monitor for **de novo DSA** (especially class II) with dd-cfDNA (donor-derived cell-free DNA) and biopsy; de novo DSA predicts graft loss.

Recent advances

- **Eplet (epitope) matching**, high-resolution NGS typing, virtual crossmatch, imlifidase.

Diagram to draw: CDC crossmatch: donor lymphocytes + recipient serum + complement, dead cells take up dye (positive).

> **Key points to remember:**
>
> - HLA on chromosome 6; match A, B, DR; DR matters most.
> - Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
> - Flow and bead assays are more sensitive; cPRA guides allocation.
> - Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

Vascular grafts and vascular surgery in urology

Asked as:

- Vascular Grafts in Urology. — *Summer 2021 Paper III Q2 (LAQ)*
- Vascular Surgery in Urology. — *Winter 2024 Paper III Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(SAQ version answered in 2015-Summer Paper 2, Q3e) (vascular grafts answered in 2021-Summer Paper 3, Q2)

Assumed meaning: blood-vessel grafts and patches (conduits) used in urological surgery: transplantation, renal and IVC (inferior vena cava) tumour surgery, RPLND (retroperitoneal lymph node dissection), sarcoma, renovascular repair and autotransplantation, dialysis access and penile surgery. The chapters treat "vascular grafts" in this sense; tissue grafts (buccal mucosa) are not included.

Assumed meaning (vascular surgery): vascular procedures performed or met by the urologist (control and repair of vessels, caval surgery, transplant and renovascular surgery, penile vascular surgery and access), not vascular surgery as a specialty.

Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would leave tension or narrowing.
- **Patch**: widens a partly resected wall. **Interposition graft**: replaces a segment. **Bypass**: routes flow around a lesion.

Classification of graft materials

Type	Examples	Advantages	Disadvantages
Autologous vein	Great saphenous, gonadal, superficial femoral, donor external iliac vein	Resists infection; suits small vessels and contaminated fields	Limited length; valves; harvest wound
Autologous artery	Internal iliac (hypogastric), **Carrel aortic patch**	Best for renal artery repair	Limited supply
Allograft (homograft)	Cadaveric vein or artery	Off the shelf; avoids xenograft	Availability; degeneration
Xenograft (biological)	**Bovine or porcine pericardium**	Pliable, strong patch, low infection	Patch only
Synthetic	**PTFE (polytetrafluoroethylene, Gore-Tex)**, ringed PTFE, Dacron (polyethylene terephthalate), collagen-impregnated Dacron	Large calibre, any length	Infection needs removal; venous grafts thrombose

Principles

- **Control first:** proximal and distal control with vessel loops, Rummel tourniquets and atraumatic clamps (bulldog, Satinsky, DeBakey). Artery before vein.
- **Grafts: autologous** vein or artery (tissue) for small vessels and contaminated fields; **synthetic** (PTFE or Dacron) for large, high-flow vessels with no bowel contamination.
- **Venous (low-pressure) grafts thrombose** more than arterial ones, so use the **widest graft** possible.
- Patch: **smooth side luminal**, rough thrombogenic side outward; edges must not invert.

- Systemic heparin before clamping; flush air and debris before tying; tension-free, everting, spatulated anastomosis with fine monofilament polypropylene (Prolene).
- **Sutures:** polypropylene (Prolene), double-armed. IVC or aorta 3-0/4-0; renal vessels 4-0/5-0; transplant anastomoses 5-0/6-0.
- Wrap synthetic grafts in **omentum** or retroperitoneal fat, away from bowel and urine.

Applications in urology (setting → vascular procedure)

1. Renal transplantation

- Renal vein and artery end-to-side to external iliac vessels; Carrel patch; multiple arteries reconstructed on the bench.
- **Short right renal vein**: extend with donor IVC, free donor external iliac vein, or **spiral gonadal vein graft**.
- **Multiple arteries (18–43%)**: Carrel patch; on the back table, conjoined side-to-side or end-to-side reconstruction. **Preserve the lower-pole artery** (it supplies the ureter).
- **Atherosclerotic recipient iliac artery**: mount the graft artery on a **Gore-Tex or bovine pericardial patch**.
- **Children under 15 kg**: aorta and IVC anastomosis, transperitoneal.
- **Transplant renal artery stenosis (1–23%)**: angioplasty or stent first; surgical interposition or bypass risks graft loss.

2. RCC (renal cell carcinoma) with IVC tumour thrombus

- **Mayo levels 0–IV:** level I milking and partial caval clamp; II full caval control; III liver mobilisation ± venovenous bypass; IV cardiopulmonary bypass ± deep hypothermic arrest.

Situation	Repair
Residual lumen ≥50%	Primary closure
Residual lumen <50%	**Patch cavoplasty**: bovine or autologous pericardium, PTFE, Dacron or saphenous vein
Circumferential invasion (wall invaded)	**Caval replacement: PTFE 16–20 mm**; contralateral renal vein reimplanted into the graft
Infrarenal bland thrombus	Filter or caval interruption, sparing lumbar collaterals

- **Leibovich–Bower patch width** = normal caval diameter × % diameter resected × 3.
- Left tumour with suprarenal caval resection: the right kidney has no collaterals, so give it new drainage (autotransplantation, or **saphenous vein graft** to the splenic, portal or inferior mesenteric vein).
- After caval replacement: low-dose heparin, then **lifelong aspirin 81 mg**.

3. PC-RPLND (post-chemotherapy RPLND)

- Vascular procedures in **9–15%**; aorta most often.
- **IVC**: narrowing <25% → primary repair; >25% → patch or interposition graft. **Ligation is tolerated** (IVC may be ligated or patched; slow occlusion builds collaterals).
- **Aorta**: never ligated; must be repaired or grafted. Small injury → repair or pericardial patch; resection → **PTFE or Dacron** (about 1% need replacement). Graft also if the duodenum is injured, to prevent aortoduodenal fistula.

4. Retroperitoneal sarcoma

- Caval resection ± reconstruction (Dacron, PTFE, homograft); grafts **thrombose in 10–40%**; anticoagulate at least 3 months.

5. Renovascular repair (disease) and autotransplantation

- Ex vivo bench repair of branch aneurysm or stenosis (or ureteric loss) with internal iliac artery or saphenous vein, then **autotransplantation** (see Q3e). Angioplasty or stent first

for fibromuscular dysplasia.
- Traumatic renal artery repair (main renal artery: endovascular stent): attempt open repair only for solitary kidney or bilateral injury (warm ischaemia <120 minutes).

6. Dialysis access

- Arteriovenous fistula first; **AV (arteriovenous) graft (PTFE)** when no native fistula is possible (veins are poor): usable in 2–3 weeks; more stenosis, thrombosis and infection than AVF (arteriovenous fistula).

7. Iatrogenic vascular injury

- SMA (superior mesenteric artery): repair or graft bypass. SMV (superior mesenteric vein): autologous vein graft; Gore-Tex only if no vein.

8. Penile surgery (penis)

- Priapism shunts: **Grayhack** saphenous vein–cavernosal shunt (priapism). Vein grafts for Peyronie tunical defects (survive by endothelial perfusion); synthetic grafts not recommended.
- **Arterial revascularisation** (inferior epigastric artery to dorsal artery) for young men with post-traumatic pudendal stenosis; venous ligation investigational.

9. Endovascular

- Angioembolisation for trauma, AML (angiomyolipoma) bleeding, pseudoaneurysm and AVF (arteriovenous fistula) after partial nephrectomy.

Complications

- **Thrombosis**, anastomotic stenosis (intimal hyperplasia), bleeding, pseudoaneurysm.
- **Graft infection** (synthetic → usually removal); graft–enteric fistula.
- **Ureteric**: transient hydronephrosis after aorto-iliac grafts (12–20%), symptomatic stricture 1–2%, **ureteroarterial fistula** (treat with endovascular covered stent).

Recent advances

- Endovascular covered stent grafts; bioengineered acellular vessels (under study); robotic autotransplantation.

Diagram to draw: IVC patch cavoplasty and PTFE caval replacement with left renal vein reimplanted; Mayo IVC thrombus levels with the matching vascular control.

Key points to remember:

- Proximal and distal control, heparin, flush, everting Prolene anastomosis.
- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- Mayo levels decide caval control and bypass.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
- PC-RPLND: IVC may be ligated; aorta must be grafted.
- Transplant: Carrel patch; preserve the lower-pole artery.
- Bench surgery plus autotransplantation for complex renovascular lesions.

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

Marginal Donor

Type: SAQ (10 marks)

Source: 2016-Summer Paper 3, Q3f

Definition

- A donor whose kidney carries **more DGF (delayed graft function) and shorter graft survival** than a standard donor; used because of organ shortage.

Categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002: age **60 or over**, or **50–59 with 2 or more** of: hypertension, **creatinine over 1.5 mg/dL**, death from **CVA (cerebrovascular accident)**. Graft-loss risk about 70% higher.

2. KDPI (Kidney Donor Profile Index) over 85%: replaced ECD in the US (2014). **Longevity matching**: lowest-KDPI kidneys go to recipients with the best EPTS (Estimated Post-Transplant Survival).

3. DCD (donation after circulatory death), Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of support	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled

- Warm ischaemia raises DGF (roughly 40–50%); long-term survival is similar to DBD (donation after brain death).

4. Others: AKI (acute kidney injury) donors; **HCV (hepatitis C virus)-viraemic** donors with antivirals; **paediatric donors under 15 kg (en bloc)**; older living donors.

- **Excluded**: metastatic or haematological cancer, melanoma, untreatable viral infection.

Assessment and use

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, vessels, 0–3 each): **0–3 single**, **4–6 dual kidney transplant**, **7 or more discard**.
- **HMP (hypothermic machine perfusion)** reduces DGF versus cold storage (Moers, 2009), without long-term survival difference; normothermic perfusion tests viability.
- **ESP (Eurotransplant Senior Program)**: donors 65 or over to recipients 65 or over, with short cold ischaemia.
- Recipient: informed consent; induction and delayed CNI (calcineurin inhibitor) if DGF.

Urological relevance

- Older and ECD kidneys have more **urine leak and ureteric stricture** (ischaemic distal ureter), as do multiple arteries and DGF.
- **Ojo (2001)**: ECD transplantation improves survival over staying on dialysis.

Diagram to draw: Maastricht categories on a timeline of circulatory arrest.

> **Key points to remember:**
> - ECD: 60 or over, or 50–59 with 2 of hypertension, creatinine over 1.5, CVA death.
> - Remuzzi 0–3 single, 4–6 dual, 7 or more discard.
> - HMP reduces DGF; Maastricht III is the commonest DCD.

Tacrolimus.

Type: SAQ (10 marks)

Source: 2017-Winter Paper 3, Q3c

Introduction

- **Tacrolimus (FK506)**: macrolide CNI (calcineurin inhibitor) from *Streptomyces tsukubaensis*; about 10–100 times more potent than ciclosporin. With mycophenolate it is maintenance in **over 90%** of kidney recipients.

Mechanism

```
Tacrolimus + FKBP12 (FK506-binding protein 12)
   → complex inhibits calcineurin (Ca²⁺/calmodulin phosphatase)
   → NFAT (nuclear factor of activated T cells) stays phosphorylated, cannot enter nucleus
   → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)
```

- Ciclosporin binds **cyclophilin** instead; same end result.

Pharmacokinetics

- Poor, variable oral bioavailability; reduced by food. Binds red cells, so **whole-blood trough (C0)** levels are monitored (narrow therapeutic index).
- Metabolised by **CYP3A4/3A5**; effluxed by **P-gp (P-glycoprotein)**. CYP3A5 expressers need higher doses.

Formulations

- Immediate-release (twice daily); prolonged-release (Advagraf, once daily); LCP-tacrolimus (Envarsus, once daily, higher bioavailability); intravenous; topical. Once-daily forms aid adherence.

Dosing and monitoring

- Started about 0.1–0.2 mg/kg/day in two doses, then adjusted to trough. Targets vary by centre and risk; **ELITE-Symphony low-dose arm: 3–7 ng/mL**.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent vasoconstriction (reversible); chronic arteriolar hyalinosis, striped fibrosis; TMA (thrombotic microangiopathy)
Metabolic	**NODAT/PTDM (new-onset/post-transplant diabetes)**, more than ciclosporin; **hyperkalaemia**, **hypomagnesaemia**, hyperuricaemia
Neurological	**Tremor**, headache, paraesthesia, seizures, PRES (posterior reversible encephalopathy syndrome)
Others	Hypertension, GI (gastrointestinal) upset, alopecia, infection (CMV (cytomegalovirus), BK polyomavirus), malignancy (skin, PTLD (post-transplant lymphoproliferative disorder))

Drug interactions

- **Raise levels (CYP3A4 inhibitors):** azoles, clarithromycin/erythromycin, diltiazem, verapamil, protease inhibitors, grapefruit; diarrhoea.
- **Lower levels (inducers):** rifampicin, phenytoin, carbamazepine, St John's wort.
- Additive nephrotoxicity: NSAIDs (non-steroidal anti-inflammatory drugs), aminoglycosides; ACE (angiotensin-converting enzyme) inhibitors add to hyperkalaemia.

Versus ciclosporin

	Tacrolimus	Ciclosporin
Acute rejection	Lower	Higher
Diabetes, tremor, alopecia	More	Less
Hirsutism, gum hypertrophy, hyperlipidaemia, hypertension	Less	More

- **ELITE-Symphony (2007):** low-dose tacrolimus + mycophenolate + daclizumab + steroids gave the **lowest rejection, best GFR (glomerular filtration rate) and graft survival** at 1 year versus ciclosporin or sirolimus regimens.
- **Also know:** CNI minimisation with belatacept or everolimus; topical tacrolimus in genital dermatoses (lichen planus, Zoon balanitis).

Diagram to draw: T-cell showing FKBP12–calcineurin–NFAT–IL-2 pathway.

Key points to remember:

- FKBP12 → calcineurin → NFAT → IL-2 blocked.
- CYP3A4/P-gp; trough-guided; many interactions.
- Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
- ELITE-Symphony: low-dose tacrolimus best.

Assessment criteria for kidney donor.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 3, Q3a (also answered in 2017-Summer Paper 2, Q3b)

Assumed meaning: evaluation and acceptance criteria for a living kidney donor.

Principles

- **Donor safety first.** KDIGO (Kidney Disease: Improving Global Outcomes) 2017 is risk-based: lifetime ESKD (end-stage kidney disease) risk vs the centre's threshold.
- Multidisciplinary team with an **independent donor advocate**.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011:** near relatives; others need authorisation committee approval; swap donation allowed.

Assessment sequence

1. Compatibility: ABO group, HLA (human leukocyte antigen) typing, crossmatch, DSA (donor-specific antibody). Incompatible pair → paired exchange.

2. Kidney function (two methods, one measured, e.g. DTPA [diethylenetriamine pentaacetic acid] GFR)

Test	Accept	Individualise	Decline
GFR (glomerular filtration rate), mL/min/1.73 m²	≥90	60–89	<60
Albuminuria, ACR (albumin–creatinine ratio), mg/g	<30	30–100	>100

- Urinalysis; persistent microscopic haematuria → cystoscopy, imaging, ± biopsy.

3. Medical exclusions and limits

- **Diabetes:** type 1 excludes; HbA1c (glycated haemoglobin) and OGTT (oral glucose tolerance test) if at risk.
- **Hypertension:** controlled on 1–2 drugs without organ damage may be acceptable.
- **Obesity:** individualised risk.
- **Stones:** metabolic work-up; bilateral stones generally exclude.
- Active cancer, chronic infection, cardiovascular disease exclude.

4. Infection screen: HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), syphilis, tuberculosis.

5. Cardiopulmonary: ECG (electrocardiogram), chest X-ray, echo.

6. Age-appropriate cancer screening.

7. Anatomy: CT (computed tomography) angiography for vessel number, early branching, stones, masses. Split function if asymmetric: **the better kidney stays with the donor**; left preferred (longer vein).

8. Psychosocial: voluntary, no coercion or payment.

Counselling figures

- Function settles at 60–80% of baseline; mortality 3 per 10,000.
- **Muzaale (2014):** lifetime ESKD 90 per 10,000 donors vs 14 in matched healthy non-donors.

Also know: APOL1 (apolipoprotein L1) testing in African ancestry; more pre-eclampsia in later pregnancy.

Diagram to draw: donor work-up flowchart.

Key points to remember:

- KDIGO 2017: GFR ≥90 accept, <60 decline; ACR >100 decline.
- Type 1 diabetes and bilateral stones exclude.
- CT angiography; better kidney stays with the donor.
- THOA: near relatives, else authorisation committee.

Urological Complications post Kidney Transplantation.

Type: SAQ (10 marks)

Source: 2023-Winter Paper 3, Q3d

Introduction

- Urological complications are the **commonest surgical complications** (8–9.5%). Most arise from the **ureter and ureterovesical anastomosis**.
- The distal donor ureter depends on a single renal-artery blood supply, so ischaemia is the root cause. **Denervation** masks pain.
- A **routine stent** reduces leak and stricture; remove at **14–21 days** (earlier increases leak; later increases UTI [urinary tract infection]).

Complications

Complication	Key facts	Management
Urine leak (1.2–8.9%; commonest early)	Distal ureteric necrosis; lost lower-pole artery; technical	Fluid creatinine far above serum. Foley + PCN (percutaneous nephrostomy) + antegrade stent; early reimplant or native ureteroureterostomy if large
Ureteric stricture (0.5–9%; commonest beyond 3 months)	Ischaemia; long cold ischaemia, rejection, BK (BK polyomavirus) and CMV (cytomegalovirus)	Nephrostomy first; balloon or laser endoureterotomy; **open reimplant, native ureteroureterostomy or Boari flap best long-term**
Lymphocele (symptomatic 0.03–26%; peak 6 weeks)	Divided lymphatics; mTOR (mammalian target of rapamycin) inhibitors	Treat only if symptomatic; **laparoscopic peritoneal fenestration** best
VUR (vesicoureteric reflux)	50–86% radiological; <1% significant	Treat cause; endoscopic injection low grade; reimplant
Haematuria	Anastomotic; persistent after biopsy = AV fistula	Self-limiting; selective embolisation
Stones (about 1%)	Hyperparathyroidism; calcineurin inhibitors	As native kidney; watch silent obstruction
UTI (commonest infection)	Stent, reflux, steroids	Treat 14–21 days if pyelonephritis

- **Obstruction:** early is technical, oedema or compression (haematoma, lymphocele); late is stricture, stone or lymphocele.
- **Encrusted stent:** CT, nephrostomy, then PCNL (percutaneous nephrolithotomy), RIRS (retrograde intrarenal surgery) or ESWL (shock-wave lithotripsy).
- **Lower tract:** bladder capacity recovers; BPH (benign prostatic hyperplasia) treated by TURP (transurethral resection of prostate) after 3 weeks, never with a stent in place.
- **Also know:** ED (erectile dysfunction) persists in 20–50%; bladder cancer risk about threefold; RCC (renal cell carcinoma) mostly in native kidneys; renal artery stenosis and thrombosis.

Diagram to draw: transplant kidney in iliac fossa showing leak, stricture and lymphocele sites.

Key points to remember:

- Ureter is the weak point; ischaemia is the cause.
- Leak early, stricture late; open repair best.
- Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

Rapid Revision Sheet – Section 15 · Renal, Nephrology & Transplant

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Discuss the role of tissue typing and cross match in live and cadaveric (LAQ)

- HLA on chromosome 6; match A, B, DR; DR matters most.
- Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
- Flow and bead assays are more sensitive; cPRA guides allocation.
- Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

Vascular grafts and vascular surgery in urology (LAQ)

- Proximal and distal control, heparin, flush, everting Prolene anastomosis.
- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- Mayo levels decide caval control and bypass.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
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Urological Complications post Kidney Transplantation. (SAQ)

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- Leak early, stricture late; open repair best.
- Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

Section 16 · BPH — Short answer questions (SAQ)

TUR Syndrome.

Type: SAQ (10 marks)

Source: 2017-Winter Paper 3, Q3f

Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorption of hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). About 1–2% in large series; falling.

Pathophysiology

- Prostatic venous pressure is about **10 mmHg**; irrigant at higher pressure enters open sinuses, or the periprostatic space after perforation.
- Monopolar needs ion-free fluid: **glycine 1.5% (~200 mOsm/kg)**, sorbitol, water (plasma ~290).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water moves into brain → **cerebral oedema**.
- **Glycine toxicity**: ammonia → encephalopathy; inhibitory retinal neurotransmitter → **transient blindness**.
- Even routine resection absorbs 800–1000 mL.

Risk factors

- Gland **>45 mL**, resection **>90 minutes** (cooperative study), high irrigation pressure, capsular perforation, sterile water.

Clinical features

- Awake patient: restlessness, confusion, **nausea, vomiting**, **visual disturbance**.
- Cardiovascular: **hypertension and bradycardia**, later hypotension, tachypnoea, ECG (electrocardiogram) changes.
- Severe: seizures, coma, death. **Na <120 mEq/L** is dangerous.

Prevention

- **Bipolar TURP or laser with saline** (eliminates it; NNT (number needed to treat) about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous-flow resectoscope; low-pressure irrigation.
- Limit resection time (stage large glands); check sodium in long cases.
- Spinal anaesthesia for early detection; **1% ethanol tagging** (breath test) monitors absorption.

Treatment

1. Stop surgery, achieve haemostasis, insert catheter.
2. ABC (airway, breathing, circulation); oxygen; monitor; send **sodium, osmolality**, glucose.
3. Fluid overload: **furosemide**; restrict fluids.
4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes up to 300 mL. **Target rise 4–6 mEq/L.**

5. Correction limit: up to **12 mEq/L in 24 h** for acute hyponatraemia (8–10 mEq/L if symptoms begin >48 h after surgery), avoiding **osmotic demyelination (central pontine myelinolysis)**.
6. Seizures: benzodiazepines; ICU (intensive care unit) care.

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy) and hysteroscopy; sorbitol → hyperglycaemia; mannitol → osmotic diuresis.

Diagram to draw: flowchart: absorption → overload + hyponatraemia → cerebral oedema.

Key points to remember:

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

IPSS score

Type: SAQ (10 marks)

Source: 2022-Summer Paper 3, Q3e

Definition

- **IPSS (International Prostate Symptom Score):** validated, self-administered questionnaire grading male LUTS (lower urinary tract symptoms) over the past month.
- Based on the **AUA-SI (American Urological Association Symptom Index, 1992)**; adopted by WHO (World Health Organization) in 1993 by adding one quality-of-life question.

Items

No.	Symptom	Phase
1	Incomplete emptying	Voiding
2	Frequency (again within 2 hours)	Storage
3	Intermittency	Voiding
4	Urgency	Storage
5	Weak stream	Voiding
6	Straining	Voiding
7	Nocturia	Storage

- **Scoring:** items 0–5. 0 not at all; 1 less than 1 time in 5; 2 less than half the time; 3 about half; 4 more than half; 5 almost always. **Nocturia scored by number of times** (0 to ≥5).
- **Total 0–35:** voiding subscore 4 items, storage 3 (not validated).

Severity	Score
Mild	0–7
Moderate	8–19
Severe	20–35

- **QoL (quality of life)/bother question:** "If you were to spend the rest of your life with your urinary condition as it is now, how would you feel?" Scored 0 (delighted) to 6 (terrible); reported separately, outside the 35.

Uses

- First step in LUTS work-up (AUA and EAU [European Association of Urology]); grades severity and bother.
- Guides treatment: mild, non-bothersome → watchful waiting; bothersome → medical therapy or surgery.
- Monitors response: **a 3-point fall is the minimum perceptible change**.
- Standard end point in trials.

Limitations

- **Not disease-specific:** women score similarly; does not diagnose BOO (bladder outlet obstruction) or BPH (benign prostatic hyperplasia); correlates poorly with urodynamics, prostate size and flow.
- No incontinence, pain or post-micturition items.
- Needs literacy.

Alternatives

- **VPSS (Visual Prostate Symptom Score):** pictures, for low literacy.
- BPH Impact Index; ICIQ-MLUTS (International Consultation on Incontinence Questionnaire–Male LUTS); DAN-PSS (Danish Prostate Symptom Score); OAB-q (overactive bladder questionnaire); LURN (Lower Urinary Tract Research Network) SI-29 adds incontinence and pain.

Diagram to draw: IPSS chart with scoring columns.

> **Key points to remember:**
> - 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
> - QoL 0–6 is separate; IPSS = AUA-SI + QoL.
> - 3-point change is noticeable; IPSS does not diagnose BOO.

Laparoscopic Trans vesical Prostatectomy

Type: SAQ (10 marks)

Source: 2025-Winter Paper 3, Q3d

Assumed meaning: laparoscopic transvesical (suprapubic) simple prostatectomy for large BPH (benign prostatic hyperplasia).

Introduction

- Laparoscopic version of the Freyer suprapubic operation: the adenoma is enucleated through a **cystotomy** in the **surgical capsule plane** (compressed peripheral zone). First LSP (laparoscopic simple prostatectomy): Mariano, 2002.

Indications

- Gland **>80–100 g** with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or obstructive nephropathy.
- **Transvesical route preferred** for a large intravesical median lobe, bladder stones or a diverticulum needing surgery.
- Work-up: PSA (prostate-specific antigen) ± MRI (magnetic resonance imaging) to exclude cancer; TRUS (transrectal ultrasound) volume; cystoscopy.

Steps

1. General anaesthesia, supine, Trendelenburg. **Ports:** 10-mm umbilical camera, two 10-mm working ports, 5-mm and 12-mm assistant ports. Transperitoneal or extraperitoneal (Retzius).
2. **Cystotomy** with stay sutures; identify ureteric orifices.
3. **Circumferential bladder-neck incision** (start 3 to 9 o'clock posteriorly), away from orifices.
4. Develop the **posterior plane first** to the apex, then the lateral lobes.
5. Divide anterior commissure; **transect urethra just proximal to the verumontanum** (protects sphincter).
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries enter posterolaterally).
7. **Bladder-neck mucosal advancement** to urethral mucosa covers the fossa and reduces bleeding.
8. 22 Fr three-way catheter; two-layer cystotomy closure.

Outcomes versus open

- Less blood loss, shorter catheter time and stay, less analgesia; **same functional results** (IPSS (International Prostate Symptom Score), flow rate, residual).
- Limitations: steep learning curve, poor ergonomics; **robotic simple prostatectomy** has largely replaced it. **Single-port transvesical robotic enucleation** (2.7 cm suprapubic incision, pneumovesicum) allows same-day discharge.

Complications

- Bleeding (transfusion rare), urine leak, ureteric orifice injury, bladder neck contracture (<1%; open 2–6%).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence rare.

Diagram to draw: sagittal view of cystotomy, bladder-neck incision and enucleation plane.

Key points to remember:

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

Lasers in BPH, including HoLEP

Asked as:

- Various lasers in treatment of BPH — *Practice Set 6 Paper III Q3a (SAQ)*
- HOLEP — *Practice Set 7 Paper III Q3f (SAQ)*

Type: SAQ (10 marks)

(also answered in 2017-Winter Paper 2, Q3a; also answered in 2020-Summer Paper 4, Q3d)

HOLEP read as: holmium laser enucleation of the prostate for BPH (benign prostatic hyperplasia).

Principle

- LASER (light amplification by stimulated emission of radiation) turns light into local heat. The **chromophore** (water or haemoglobin) and wavelength decide **penetration depth**.
- About 60 °C coagulates (delayed sloughing); 100 °C or more vaporises. Coagulation lasers are obsolete.
- **HoLEP (holmium laser enucleation of the prostate)** removes the whole adenoma along the **surgical capsule**: the endoscopic equivalent of open simple prostatectomy.
- **Ho:YAG (holmium:yttrium–aluminium–garnet), 2120 nm**: water absorbs it; pulsed; **0.4 mm penetration**; coagulates vessels. Saline irrigation, so no TUR (transurethral resection) syndrome.

Lasers used in BPH (benign prostatic hyperplasia)

Laser	Wavelength	Chromophore	Penetration	Procedure
Ho:YAG (holmium:yttrium-aluminium-garnet), pulsed	2120 nm	Water	0.4 mm	**HoLEP** (holmium laser enucleation of the prostate)
Thulium:YAG (continuous) / **TFL** (thulium fibre laser)	2013 / 1940 nm	Water (TFL at absorption peak, about 4× holmium)	0.2 mm	ThuLEP (enucleation), ThuVEP (vapoenucleation), ThuVARP (vaporesection)
KTP (potassium titanyl phosphate) / LBO (lithium triborate)	532 nm	Haemoglobin	0.8 mm	**PVP** (photoselective vaporisation), 120–180 W
Diode	Various	Mixed	0.5–5 mm	Vaporisation
Nd:YAG (neodymium)	1064 nm	—	About 1 cm	Abandoned (sloughing)

- **KTP:** Nd:YAG beam halved to 532 nm by a crystal; haemoglobin absorbs it, so vaporisation is bloodless.

HoLEP

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without median lobe.
- **Anticoagulated or high bleeding risk** (AUA [American Urological Association]); alternative to open prostatectomy for large glands.

Steps

- 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr scope; **2.0 J × 40–50 Hz** for enucleation.

1. Bladder-neck incisions at 5 and 7 o'clock (median lobe) or one at 6 o'clock.

2. Perimontanal incision; **early apical release** protects the sphincter.
3. Develop the capsular plane to 12 o'clock; push lobes into the bladder (**en bloc, two- or three-lobe**).
4. Haemostasis with a defocused beam.
5. **Morcellation** with a full bladder; three-way catheter.

Limitation

- **Steep learning curve** (about 50 cases).

Evidence and outcomes

- **HoLEP:** endoscopic equivalent of open prostatectomy; **size-independent**. The only endoscopic procedure **superior to TURP** (versus TURP) (transurethral resection of the prostate) for IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate): **better IPSS and Qmax**, durable. Less bleeding, shorter catheter time and stay; longer operating time; steep learning curve. Retreatment rare (whole transition zone removed).
- **ThuLEP:** equals HoLEP.
- **GOLIATH** (180 W PVP vs TURP, about 270 men, ≤100 cc): PVP non-inferior at 2 years; shorter catheter time and stay. Retreatment high above 80 cc.
- **AUA (American Urological Association):** HoLEP or ThuLEP for any size; PVP for under 80 cc; HoLEP, ThuLEP or PVP for **higher bleeding risk** (anticoagulated men).

Complications

- Dysuria and storage symptoms (worst after PVP); transient storage symptoms and early / transient stress incontinence (SUI, stress urinary incontinence); retrograde ejaculation in most (**retrograde ejaculation about 75%** after HoLEP); bladder neck contracture (BNC) 0–3.2%; stricture; capsular perforation; ureteric orifice injury; **morcellator bladder injury**. No TUR (transurethral resection) syndrome (saline).
- Erectile function unchanged.
- **Safety:** wavelength-specific eyewear for all; laser safety officer.

Recent advances

- **Moses pulse modulation**; ThuLEP (thulium enucleation); median-lobe-only HoLEP (antegrade ejaculation about 90%); same-day discharge.

Diagram to draw: HoLEP: adenoma enucleated to the capsule, pushed into bladder, morcellated; sagittal prostate with enucleation plane along the capsule.

> **Key points to remember:**
>
> - Penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm.
> - HoLEP is size-independent and superior to TURP (only endoscopic method superior to TURP); suits anticoagulated men.
> - Retrograde ejaculation about 75%.
> - GOLIATH: PVP non-inferior to TURP; avoid PVP above 80 cc.

Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 3, Q3b (also answered in Practice Set 2 Paper 4, Q1)

Anatomy

- About 20 g; base at bladder neck, apex at urogenital diaphragm. **No true capsule**.

Zone (McNeal)	Glands	Relevance
Peripheral	70%	**70% of cancers**; posterolateral extension along nerves
Central	25%	Around ejaculatory ducts
Transition	5%	BPH (benign prostatic hyperplasia); anterior cancers
AFMS (anterior fibromuscular stroma)	None	Anterior margin

- **Fascial planes:** intrafascial (most nerves, more margins), **interfascial (standard)**, extrafascial (widest margin). Denonvilliers fascia excised if posterior extension suspected.
- **NVB (neurovascular bundle):** cavernous nerves posterolateral, **5 and 7 o'clock** at the apex; diffuse "hammock" distribution. Injured by heat and traction.
- **Accessory pudendal arteries:** preserve. **DVC (dorsal venous complex)** tamponaded by pneumoperitoneum.
- **Continence:** striated sphincter at apex; anterior support (puboprostatic ligaments, detrusor apron).
- **Lymphatics:** obturator, external and internal iliac, presacral nodes.

Newer surgical concepts

1. **RARP (robot-assisted radical prostatectomy):** Yaxley 2016 RCT (randomised controlled trial): function equal to open, less blood loss.
2. **Risk-graded nerve sparing** using mpMRI (multiparametric MRI) and biopsy, side by side.
3. **NeuroSAFE:** frozen section of posterolateral margin guides nerve sparing.
4. **Retzius-sparing** (Bocciardi 2010), through the pouch of Douglas: earlier continence, more margins anteriorly; **Hood technique** preserves anterior support.
5. **Apical dissection:** maximal urethral length, cold division (commonest margin site); posterior (Rocco) reconstruction.
6. **Extended PLND (pelvic lymph node dissection)** when nodal risk >5% (EAU [European Association of Urology]): captures about 63% of drainage; staging, not therapeutic.
7. **Single-port robot**, extraperitoneal; **focal therapy** of the index lesion (HIFU [high-intensity focused ultrasound], cryotherapy).

Diagram to draw: axial prostate with three fascial planes and NVB at 5 and 7 o'clock.

> **Key points to remember:**
> - Peripheral zone: 70% of cancers; no true capsule.
> - Interfascial plane standard; apex commonest margin site.
> - Retzius-sparing: early continence, more margins.

Rapid Revision Sheet – Section 16 · BPH

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

TUR Syndrome. (SAQ)

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

IPSS score (SAQ)

- 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
- QoL 0–6 is separate; IPSS = AUA-SI + QoL.
- 3-point change is noticeable; IPSS does not diagnose BOO.

Laparoscopic Trans vesical Prostatectomy (SAQ)

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

Lasers in BPH, including HoLEP (SAQ)

- Penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- HoLEP is size-independent and superior to TURP (only endoscopic method superior to TURP); suits anticoagulated men.
- Retrograde ejaculation about 75%.
- GOLIATH: PVP non-inferior to TURP; avoid PVP above 80 cc.

Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate (SAQ)

- Peripheral zone: 70% of cancers; no true capsule.
- Interfascial plane standard; apex commonest margin site.
- Retzius-sparing: early continence, more margins.

Section 17 · Prostate Cancer — Short answer questions (SAQ)

Management of newly Metastatic prostate cancer

Type: SAQ (10 marks)

Source: 2025-Winter Paper 3, Q3b

Definitions

- **mHSPC (metastatic hormone-sensitive prostate cancer):** no prior hormones. **Synchronous** (at diagnosis) or **metachronous** (after local therapy).
- **CHAARTED high volume:** ≥4 bone metastases with ≥1 outside the axial skeleton, or any visceral metastasis.
- **LATITUDE high risk:** ≥2 of Gleason ≥8, ≥3 bone lesions, visceral metastasis.

Work-up

- CT (computed tomography) and bone scan; PSMA (prostate-specific membrane antigen) PET (positron emission tomography) finds more, but trials used conventional imaging.
- **Germline and somatic testing** for all (>10% have DNA-repair mutations).

Systemic therapy

- **ADT (androgen deprivation therapy) is the backbone, never alone:** LHRH (luteinising hormone-releasing hormone) agonist with antiandrogen flare cover, or antagonist/orchidectomy (for impending cord compression).
- **Doublets:** ADT plus one of:

Agent	Trial	Overall survival HR (hazard ratio)
Docetaxel 75 mg/m², 6 cycles	**CHAARTED** (benefit in high volume only); STAMPEDE	0.61; 0.78
Abiraterone 1000 mg + prednisone 5 mg	**LATITUDE**; STAMPEDE arm G	0.62; 0.63
Enzalutamide	ARCHES (radiographic PFS (progression-free survival)); ENZAMET	0.67 (ENZAMET)
Apalutamide	**TITAN**	0.67

- Hormonal agents work **regardless of volume**; docetaxel needs high volume (GETUG-AFU 15 negative).
- **Triplets** (ADT + docetaxel + abiraterone or darolutamide): **PEACE-1** (abiraterone, synchronous; overall survival HR 0.75 with docetaxel) and **ARASENS** (darolutamide; overall survival HR 0.68). For chemotherapy-fit men, especially high volume.

Local and directed therapy

- **RT (radiotherapy) to the prostate in low-volume synchronous disease:** STAMPEDE arm H, survival HR 0.68 (3-year 81% vs 73%); HORRAD negative overall.
- Cytoreductive prostatectomy: trials only. **MDT (metastasis-directed therapy)** for oligometastases (STOMP, ORIOLE) delays ADT.

Follow-up

- PSA (prostate-specific antigen) and bloods 3–6-monthly; testosterone if PSA rises (rising PSA with castrate testosterone = CRPC (castration-resistant prostate cancer)); bone and cardiovascular health.
- *Guideline note:* the chapter lists ADT plus docetaxel as an option; current EAU (European Association of Urology) guidance gives docetaxel only within a triplet.

Diagram to draw: algorithm splitting by volume and fitness to doublet or triplet ± prostate RT.

Key points to remember:

- ADT alone is no longer standard.
- CHAARTED volume vs LATITUDE risk definitions.
- Triplets: PEACE-1, ARASENS.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).

Rapid Revision Sheet – Section 17 · Prostate Cancer

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Management of newly Metastatic prostate cancer (SAQ)

- ADT alone is no longer standard.
- CHAARTED volume vs LATITUDE risk definitions.
- Triplets: PEACE-1, ARASENS.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).

Appendix · Original papers: headings, notes and contents

Each source file's own title, note and contents list, exactly as printed, so every word of the originals is kept.

2015-Summer Paper 3 – Subspeciality of Urology (practice paper)

Contents

1. Q1. Ureteric stents; ureteroscopy with or without stenting (LAQ)
2. Q2. Medical management of VUR; 8-yr boy with bilateral grade IV VUR (LAQ)
3. Q3a. Sacral neuromodulation
4. Q3b. Uses of buccal mucosa in urology
5. Q3c. HSG (USG) in neonates
6. Q3d. Penile prosthesis types and biofilms
7. Q3e. Uretero-vaginal fistula
8. Q3f. Prune belly syndrome

2016-Summer Paper 3 – Subspeciality of Urology

Contents

1. Q1. Minimally invasive and endoscopic management of upper urinary tract malignancy (LAQ)
2. Q2. Laparoscopic pyeloplasty (LAQ)
3. Q3a. Oligoanaesthospermia (oligoasthenozoospermia)
4. Q3b. Sperm retrieval techniques
5. Q3c. Treatment strategies for bilateral Wilms' tumour
6. Q3d. Asymptomatic haematuria
7. Q3e. Prenatal and neonatal assessment of neurogenic bladder dysfunction
8. Q3f. Marginal donor

2017-Summer Paper 3 – Subspeciality of Urology (practice paper)

Practice paper compiled from the question bank: the actual 2017-Summer Paper 3 was not available.

Contents

1. Q1. Classification of urethral strictures; surgical management after failed VIU (LAQ)
2. Q2. Female continence mechanism; classification of SUI; principles of surgical options (LAQ)
3. Q3a. Bladder myoplasty
4. Q3b. Pathogenesis and management of uretero-vaginal fistula
5. Q3c. Differential diagnosis and treatment of frequency and urgency in a 3-year-old boy
6. Q3d. Genetic causes of male infertility
7. Q3e. Urinary bladder compliance
8. Q3f. Congenital ureteral anomalies

2017-Winter Paper 3 – Subspeciality of Urology

Contents

1. Q1. Embryology of the penis; management of proximal hypospadias (LAQ)
2. Q2. Anatomy of the female pelvic floor; surgeries for SUI in females (LAQ)
3. Q3a. Sacral neuromodulation
4. Q3b. Botulinum toxin
5. Q3c. Tacrolimus

6. Q3d. Retrocaval ureter
7. Q3e. DVIU (direct visual internal urethrotomy)
8. Q3f. TUR syndrome

2018-Summer Paper 3 – Subspeciality of Urology

Contents

1. Q1. Embryology and management of ectopic ureter (LAQ)
2. Q2. Role of tissue typing and cross match in live and cadaveric renal transplantation (LAQ)
3. Q3a. Treatment algorithm of undescended testis
4. Q3b. Collin's knife (Collings knife electrode)
5. Q3c. Autosomal recessive polycystic kidney disease
6. Q3d. Embryology and management of the posterior urethral valve patient
7. Q3e. CSIC (clean intermittent self-catheterisation)
8. Q3f. Controversies in management of varicocele

2019-Summer Paper 3 – Subspeciality of Urology

Contents

1. Q1. Differential diagnosis of small renal tumours; haemostasis during laparoscopic partial nephrectomy (LAQ)
2. Q2. Evaluation of SUI in females; surgical therapy of SUI in males (LAQ)
3. Q3a. Prepucioplasty (preputioplasty)
4. Q3b. Bacteriuria and antibiotic policy in pregnancy
5. Q3c. Accessories for RIRS
6. Q3d. Immediate post-TURBT intravesical therapy for TCC
7. Q3e. Endoscopic correction of VUR: indications and complications
8. Q3f. Management of DSD (differences of sex development)

2020-Summer Paper 3 – Subspecialities of Urology

Contents

1. Q1. Etio-pathogenesis, diagnosis and management of recto-urethral fistula (LAQ)
2. Q2. Development and clinical anatomy of the vesico-ureteric junction; grade 3 VUR in an adult (LAQ)
3. Q3a. Lower ureteric stone under 1 cm in pregnancy (SAQ)
4. Q3b. Type 2 (proximal) renal tubular acidosis (SAQ)
5. Q3c. Auto-transplantation of kidney (SAQ)
6. Q3d. Uretero-calycostomy (SAQ)
7. Q3e. Management of bilateral Wilms tumour (SAQ)
8. Q3f. Inguinal block dissection (SAQ)

2021-Summer Paper 3 – Subspecialities of Urology

Contents

1. Q1. Embryological development of the urinary bladder; management of Hutch diverticulum (LAQ)
2. Q2. Vascular grafts in urology (LAQ)
3. Q3a. Koyanagi repair: indication, procedure and complications (SAQ)
4. Q3b. Antenatal management of fetal uropathy (SAQ)
5. Q3c. Post-traumatic bladder neck injury: acute and delayed management (SAQ)
6. Q3d. Management of AML (angiomyolipoma) (SAQ)
7. Q3e. Role of autotransplantation of the kidney in urology (SAQ)
8. Q3f. Urinary biomarkers in the diagnostic evaluation of haematuria (SAQ)

2022-Summer Paper 3 – Subspecialities of Urology

Contents

1. Q1. Methods of nephron-sparing surgery for renal masses (LAQ)
2. Q2. Renal tuberculosis: etiopathogenesis, presentation, diagnosis and management (LAQ)
3. Q3a. Assessment criteria for a kidney donor (SAQ)
4. Q3b. Management of haemorrhagic cystitis (SAQ)
5. Q3c. Mini Perc (SAQ; cross-reference)
6. Q3d. Methods of preoperative bowel preparation (SAQ)
7. Q3e. IPSS score (SAQ)
8. Q3f. Isolated penopubic epispadias (SAQ)

2023-Winter Paper 3 – Subspecialities of Urology

Contents

1. Q1. Retroperitoneal fibrosis: pathology, etiology, clinical features, evaluation and treatment (LAQ)
2. Q2. Upper ureteric injury: etiology, diagnosis and management (LAQ)
3. Q3a. Premature ejaculation (SAQ)
4. Q3b. Desmopressin (SAQ)
5. Q3c. Snodgrass repair for hypospadias (SAQ)
6. Q3d. Urological complications post kidney transplantation (SAQ)
7. Q3e. Carcinoma in situ of the urinary bladder (SAQ)
8. Q3f. Laboratory diagnosis of genitourinary tuberculosis (SAQ)

2024-Winter Paper 3 – Subspecialities of Urology

Contents

1. Q1. Discuss the diagnosis, staging and management of renal upper tract Urothelial tumor. (LAQ)
2. Q2. Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria. (LAQ)
3. Q3a. Vascular Surgery in Urology. (SAQ)
4. Q3b. Discuss the management of Metastatic Renal Cell Carcinoma. (SAQ)
5. Q3c. Prognostic indicators in emphysematous pyelonephritis. (SAQ)
6. Q3d. Mirabegron in over active bladder. (SAQ)
7. Q3e. Management of stress urinary incontinence. (SAQ)
8. Q3f. Sting procedure. (SAQ)

2025-Winter Paper 3 – Subspecialities of Urology

Contents

1. Q1. Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula (LAQ)
2. Q2. Diagnosis and Management of Priapism (LAQ)
3. Q3a. Urethrovaginal fistula (SAQ)
4. Q3b. Management of newly Metastatic prostate cancer (SAQ)
5. Q3c. Renal Autotransplantation (SAQ)
6. Q3d. Laparoscopic Trans vesical Prostatectomy (SAQ)
7. Q3e. Sacral Neuromodulation (SAQ)
8. Q3f. Minimal invasive retroperitoneal lymph node dissection (SAQ)

Practice Set 1 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Classify testicular tumors and describe management of seminoma (LAQ)

2. Q2. Define algorithm for evaluation of antenatally diagnosed unilateral HN in male fetus. What are the current rate of prenatal fetal anomalies (LAQ)
3. Q3a. ANH (SAQ)
4. Q3b. Role of UDS in SUI (SAQ)
5. Q3c. Retrocecal ureter (SAQ)
6. Q3d. Spermatogenesis (SAQ)
7. Q3e. Enuresis (SAQ)
8. Q3f. Female sexual dysfunction (SAQ)

Practice paper compiled from the question bank.

Practice Set 2 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Iatrogenic ureteric injury (LAQ)
2. Q2. Management of bilateral VUR, including surgical techniques and their selection (LAQ)
3. Q3a. Uroflowmetry (SAQ)
4. Q3b. Ejaculatory duct obstruction (SAQ)
5. Q3c. Ureterocele (SAQ)
6. Q3d. Treatment of bilateral VUR: surgical techniques and criteria for their selection (SAQ; cross-reference)
7. Q3e. Darifenacin vs solifenacin vs tolterodine (SAQ)
8. Q3f. Primary obstructive megaureter (SAQ)

Practice paper compiled from the question bank.

Practice Set 3 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Urological complications of OBGY surgery; management of urethrovaginal fistula (LAQ)
2. Q2. Management options of VVF (LAQ)
3. Q3a. Diagnostic and operative paediatric laparoscopy in urology (SAQ)
4. Q3b. Tadalafil (SAQ)
5. Q3c. DDAVP (SAQ)
6. Q3d. Boari flap (SAQ)
7. Q3e. Single stage repair of exstrophy of urinary bladder (SAQ)
8. Q3f. Retrograde ejaculation (SAQ)

Practice paper compiled from the question bank.

Practice Set 4 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Management of ureteric injuries occurring during gynaecological surgery (LAQ)
2. Q2. Evaluation and treatment of a newborn with ambiguous genitalia (LAQ)
3. Q3a. Botox in urology (SAQ)
4. Q3b. Priapism (SAQ)
5. Q3c. Oligospermia (SAQ)
6. Q3d. Immunobead test for semen (SAQ)
7. Q3e. Duloxetine (SAQ)
8. Q3f. CBAVD (SAQ)

Practice paper compiled from the question bank.

Practice Set 5 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Describe early and late management of PFUD injuries (LAQ, 25)

2. Q2. Describe in detail TCC of ureter. (LAQ, 25)
3. Q3a. Early vs. delayed repair in VVF (SAQ, 10)
4. Q3b. Maldescended testes (SAQ, 10)
5. Q3c. Extended indications of Shodgrass repair in hypospadias (SAQ, 10)
6. Q3d. Snodgrass Repair for distal hypos. (SAQ, 10)
7. Q3e. Mixed gonadal dysgenesis. (SAQ, 10)
8. Q3f. Hemostasis in laparoscopic surgery. (SAQ, 10)

Practice paper compiled from the question bank.

Practice Set 6 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Management of bilateral VUR and its long-term outcome (LAQ)
2. Q2. Development of the urethra, classification of congenital urethral anomalies and management of epispadias (LAQ)
3. Q3a. Various lasers in treatment of BPH (SAQ)
4. Q3b. Telomerase (SAQ)
5. Q3c. Newer methods of stone analysis (SAQ; cross-reference)
6. Q3d. Newer diagnostic modalities for genitourinary TB, role of present tests and diagnostic algorithm (SAQ)
7. Q3e. Present status of varicocele treatment (SAQ)
8. Q3f. Recent advances in nephron sparing surgery; role of neoadjuvant therapy (SAQ)

Practice paper compiled from the question bank.

Practice Set 7 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Urological complications in obstetrics and gynaecology; treatment of ureterovaginal fistula (LAQ)
2. Q2. Developmental basis of exstrophy–epispadias complex; outline of management (LAQ)
3. Q3a. Energy sources in laparoscopic urology (SAQ)
4. Q3b. Anatomy of prostate and newer concepts in surgery for localised prostate cancer (SAQ)
5. Q3c. APOGEE (SAQ)
6. Q3d. Newer techniques of stone analysis (SAQ)
7. Q3e. Tissue engineering: principles and status in urology (SAQ)
8. Q3f. HoLEP (SAQ)

Practice paper compiled from the question bank.

Practice Set 8 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Normal development of bladder and ureter; management of a child born with ectopia vesicae (LAQ)
2. Q2. Therapeutic energy applications in urology and their present status (LAQ)
3. Q3a. Sertoli cell only syndrome (SAQ)
4. Q3b. Oxidative stress and male infertility (SAQ)
5. Q3c. Dhat syndrome (SAQ)
6. Q3d. Eponymous urethroplasty techniques and perineal urethrostomy (SAQ)
7. Q3e. Prognostic index score in carcinoma penis (SAQ)
8. Q3f. Paquin and Gil-Vernet techniques of ureteric reimplantation (SAQ)

Practice paper compiled from the question bank.

(Earlier version) Practice Set 6 Paper 3 – Subspecialities of Urology

Contents

1. Q1. Management of bilateral VUR and its long-term outcome (LAQ)
2. Q2. Development of the urethra, classification of congenital urethral anomalies and management of epispadias (LAQ)
3. Q3a. Various lasers in treatment of BPH (SAQ)
4. Q3b. Telomerase (SAQ)
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